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Report No: 48443-AR

PROJECT APPRAISAL DOCUMENT
ON A
PROPOSED ADAPTABLE PROGRAM LOAN
IN THE AMOUNT OF US\$ 840 MILLION
TO THE
ARGENTINE REPUBLIC
FOR THE
MATANZA-RIACHUELO BASIN SUSTAINABLE DEVELOPMENT PROJECT
PHASE 1 (APL1)
IN SUPPORT OF THE FIRST PHASE OF THE MATANZA-RIACHUELO BASIN
SUSTAINABLE DEVELOPMENT PROGRAM

April 27, 2009

Sustainable Development Department
Argentina, Chile, Paraguay, and Uruguay Country Management Unit
Latin America and the Caribbean Region

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CURRENCY EQUIVALENTS

(Exchange Rate Effective 23 April, 2009)

Currency Unit = AR Peso

AR\$3.70 = US\$1

FISCAL YEAR

January 1 – December 31

ABBREVIATIONS AND ACRONYMS

AASA	<i>Aguas Argentinas S. A.</i> (Former water and sanitation concessionaire in Buenos Aires)
ACUMAR	<i>Autoridad de Cuenca Matanza-Riachuelo</i> (Matanza-Riachuelo Basin Authority)
AGN	<i>Auditoría General de la Nación</i> (Office of the National Auditor General)
APL	Adaptable Program Loan
APLA	<i>Agencia de Planificación</i> (Planning Agency for the AySA concession area)
AR\$	Argentine peso
AySA	<i>Agua y Saneamientos Argentinos S.A.</i> (Argentina Water and Sanitation utility)
BOD	Biological Oxygen Demand
BP	Bank Policy
BPTP	Berazategui preliminary treatment plant
CARP	<i>Comisión Administradora del Río de la Plata</i> (La Plata River Administrative Commission)
CEAMSE	<i>Coordinación Ecológica Área Metropolitana Sociedad del Estado</i> (State Company for Waste Management in the Buenos Aires Metropolitan Area)
CFAA	Country Financial Accountability Assessment
CIJI	<i>Comisión Interjurisdiccional de Industria</i> (Interjurisdiccional Commission of Industry)
CMD	<i>Colector Margen Derecha</i> (Right Bank collector)
CMI	<i>Colector Margen Izquierda</i> (Left Bank collector)
CRI	<i>Convenio de Reconversión Industrial</i> (Industrial Conversion Agreement)
CTM	<i>Cloacas Troncales Máximas</i> (Trunk Sewage Pipes)
CY	Calendar Year
DA	Designated Account
EBITDA	Earnings Before Interest, Tax, Depreciation, and Amortization
EIA	Environmental Impact Assessment
EMF	Environmental Management Framework
EMP	Environmental Management Plan
ERAS	<i>Entidad Reguladora de Agua y Saneamiento</i> (Regulatory Entity for Water and Sanitation)
ERR	Economic Rate of Return
FAP	Fiduciary Action Plan
FM	Financial Management
FY	Fiscal Year
GIS	Geographical Information System
GoA	Government of Argentina
GP	Good Practice
HAB	Harmful Algal Blooms
IADB	Inter American Development Bank
IAU	Internal Audit Unit
ICB	International Competitive Bidding
IEA	Integrated Environmental Assessment

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IFC	International Finance Corporation
IFR	Interim Unaudited Financial Reports
JGM	<i>Jefatura Gabinete de Ministros</i> (Office of the Chief of the Cabinet of Ministers)
LCS	Least Cost Selection
MPFIPS	<i>Ministerio De Planificación Federal, Inversión Pública y Servicios</i> (Ministry of Federal Planning, Public Investment, and Services)
M-R	Matanza-Riachuelo
NGO	Nongovernmental Organization
NPV	Net Present Value
OM	Operational Manual
OP	Operational Policy
O&M	Operations and Maintenance
PAD	Project Appraisal Document
PDO	Project Development Objective
PFM	Public Financial Management
PGA	<i>Plan de Gestión Ambiental y de Manejo de la Cuenca Hidrica Matanza Riachuelo</i> (Matanza-Riachuelo River Basin Environmental Management Plan)
PISA	<i>Plan Integral de Saneamiento de la Cuenca Matanza-Riachuelo</i> (Integrated Cleanup Plan for the Matanza-Riachuelo Basin)
PMOEM	<i>Plan de Mejoras, Operación, Expansión y Mantenimiento</i> (Improvement, Operation, Expansion, and Maintenance Plan)
PRI	<i>Plan de Reconversión Industrial</i> (Plan for Industrial Conversion)
QCBS	Quality- and Cost-Based Selection
RPFM	<i>Rio de la Plata y su Frente Marítimo</i> (La Plata River and its Shoreline)
SAYDS	<i>Secretaría de Ambiente y Desarrollo Sustentable</i> (Secretariat of the Environment and Sustainable Development)
SBD	Standard Bidding Document
SEPA	<i>Sistema de Ejecución de Planes de Adquisiciones</i> (Procurement Plan Execution System)
SIDIF	<i>Sistema Integrado de Información Financiera</i> (Integrated Financial Information System)
SIGEN	<i>Sindicatura General de la Nación</i> (National General Audit Agency)
SMEs	Small and Medium Enterprises
SOE	Statement of Expenditures
SSRH	<i>Sub-secretaría de Recursos Hídricos</i> (Sub-secretariat for Hydrological Resources)
SWM	Solid Waste Management
TA	Technical Assistance
UCOFI	<i>Unidad de Coordinación para la Gestión Técnica y Financiamiento Internacional</i> (Coordination Unit for Technical Management and International Financing)
UEP	<i>Unidad de Ejecución del Proyecto</i> (Project Implementation Unit)
UEPEX	<i>Sistema de Administración y Control Financiero para las Unidades Ejecutoras de Préstamos Externos</i> (System of Financial Administration and Control for Agencies Executing Foreign Loans)
WHO	World Health Organization
WTP	Willingness to Pay

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MATANZA-RIACHUELO BASIN SUSTAINABLE DEVELOPMENT PROJECT
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ARGENTINA

MATANZA-RIACHUELO BASIN SUSTAINABLE DEVELOPMENT PROJECT

PROJECT APPRAISAL DOCUMENT

LATIN AMERICA AND CARIBBEAN

LCSEN

Date: April 27, 2009	Team Leaders: Menahem Libhaber and Franz Drees-Gross
Country Director: Pedro Alba	Sectors: General water, sanitation and flood protection (90%),
Sector Director: Laura Tuck	Central government administration (10%)
	Themes: Pollution management and environmental health (P),
	Environmental policies and institutions (P), Other urban
	development (S)
Project ID: P105680	Environmental screening category: Full assessment
Lending Instrument: APL	Safeguard screening category: A

Program Financing Data								
	Indicative financing plan					Estimated implementation period		Borrower
	IBRD US\$ m	%	Others US\$ m	%	Total US\$ m	Commitment date	Closing date	The Argentine Republic
APL 1 Loan	840	56	647	44	1,487	November 15, 2009	March 31, 2016	
APL 2 Loan	1,160	58	852	42	2,012	November 15, 2013	November 15, 2019	
Total	2,000	57	1,499	43	3,500			

Project Financing Data			
[x] Loan [] Credit [] Grant [] Guarantee [] Other:			
For loans/credits/other:			
Loan currency: United States Dollars			
Total Bank financing (US\$m.): 840			
Proposed terms: Fix-spread, commitment linked loan with level repayments			
Financing Plan (US\$m)			
Source	Local	Foreign	Total
BORROWER	645	2	647
INTERNATIONAL BANK FOR RECONSTRUCTION AND DEVELOPMENT	516	324	840
Total:	1,161	326	1,487
Borrower: The Argentine Republic			
Responsible agencies: Secretariat of the Environment and Sustainable Development (SAyDS) in cooperation with Agua y Saneamientos Argentinos (AySA) and the Matanza-Riachuelo Basin Authority (ACUMAR).			
Coordinator: Raul Vilariño (rvilarino@ambiente.gov.ar)			

Estimated Disbursements APL-1 (Bank FY/US\$m)							
FY	2010	2011	2012	2013	2014	2015	2016
Annual	50	50	100	210	210	200	20
Cumulative	50	100	200	410	620	820	840
Project implementation period: November 15, 2009 to December 15, 2015							
Expected effectiveness date: November 15, 2009							
Expected closing date: March 31, 2016							

Does the project depart from the CAS in content or other significant respects? <i>Ref. PAD A.3</i>	☐ Yes ☒ No
Does the project require any exceptions from Bank policies? <i>Ref. PAD D.7</i>	☐ Yes ☒ No
Have these been approved by Bank management?	☐ Yes ☐ No
Is approval for any policy exception sought from the Board?	☐ Yes ☒ No
Does the project include any critical risks rated "substantial" or "high"? <i>Ref. PAD C.5</i>	☒ Yes ☐ No
Does the project meet the Regional criteria for readiness for implementation? <i>Ref. PAD D.7</i>	☒ Yes ☐ No

Project development objectives *Ref. PAD B.2, Technical Annex 3*

The objectives of the project (APL-1) are to (i) improve sewerage services in the M-R River Basin and other parts of the Province and City of Buenos Aires by expanding transport and treatment capacity; (ii) support a reduction of industrial discharges to the M-R River, through the provision of industrial conversion grants to small and medium enterprises; (iii) promote improved decision-making for environmentally-sustainable land use and drainage planning, and to pilot urban drainage and land use investments, in the M-R River Basin; and (iv) strengthen ACUMAR's institutional framework for ongoing and sustainable clean-up of the M-R River Basin.

Project Description (Ref. PAD B.3.a, Technical Annex 4)

The proposed project has four components grouped under two broad macro-components:

Macro-component 1: Sanitation

(i) Sanitation: Investments in sanitation infrastructure including the Left Bank collector, the Baja Costanera bypass collector, the Riachuelo preliminary treatment plant, inflow and effluent pumping stations at the Riachuelo preliminary treatment plant, the Riachuelo outfall, the Industrial collector, and the Right Bank collector.

Independent technical supervision of the works listed above; data collection, analyses and modeling in support of engineering designs; and capacity development activities in support of the preparation of the engineering designs under this component.

Macro-component 2: Support for Industrial Pollution Abatement, Environmental Territorial Management, and Institutional Strengthening

(ii) Industrial Pollution Abatement: Technical assistance to improve environmental monitoring and enforcement of environmental targets for selected industrial activities, as well as financing of cleaner production investments to enhance and support environmental compliance among polluting small and medium size enterprises;

(iii) Environmental Territorial Management: Technical assistance for environmental land use planning and financing for pilot urban and drainage works; and

(iv) Institutional Strengthening and Project Management: Financing for technical studies to support project implementation (components 1, 2 and 3); a communication strategy to inform key stakeholders; project management activities; and support for the institutional strengthening of ACUMAR.
Which safeguard policies are triggered, if any? <i>Ref. PAD D.6, Technical Annex 10.</i> Environmental Assessment (OP/BP/GP 4.01), Natural Habitats (OP/BP 4.04), Physical Cultural Resources (OP 4.11), Involuntary Resettlement (OP/BP 4.12), and Projects on International Waterways (OP/BP/ GP 7.50)
Significant, non-standard conditions, if any: none.
<u>Conditions of Loan Effectiveness:</u> (i) The Borrower, through SAYDS, has adopted the Operational Manual; (ii) The AySA Concession Contract has been signed, and said contract is in full force and effect; (iii) The AySA Agreement (which sets forth AySA's responsibilities under component 1) has been signed by the parties thereto, and said agreement is in full force and effect; (iv) The ACUMAR Resolution (which sets forth ACUMAR's responsibilities under the project) has been issued, published in the Borrower's official gazette (<i>Boletín Oficial</i>), and said resolution is in full force and effect; and (v) The Borrower, through SAYDS, has established the UEP. <u>Main Loan Covenants:</u> (i) The Borrower, through SAYDS, shall cause AySA to carry-out a pre-feasibility study of alternatives to the construction of the Right Bank collector (<i>Colector Margen Derecha</i>) no later than March 31, 2010. (ii) The Borrower, through SAYDS shall cause AySA to carry out separate site environmental impact assessment acceptable to the Bank for the Right Bank collector, or a suitable alternative approved by the Bank, and for the industrial collector (including its pumping station), and thereafter prepare and implement an environmental management plan for these civil works. (iii) Resettlement is not expected under APL1. Considering the complex nature of works under Component 1 and that specific subprojects under Components 2 and 3 are not known (matching grants for SMEs and infrastructure investments, respectively), in case resettlement becomes necessary, the Borrower through SAYDS shall, or shall cause AySA to, prepare a resettlement plan acceptable to the Bank consistent with the Resettlement Policy Framework, as the case may be; and thereafter carry out the pertinent resettlement plan in accordance with its terms. (iv) The Borrower, through SAYDS, shall: (i) cause AySA to carry out component 1 of the Project in accordance with the provisions of the Project Environmental Management Plan, the AySA Environmental Management Plan, and the Operational Manual; (ii) carry out components 2 and 3 of the Project in accordance with the provisions of the Project Environmental Management Plan, the Environmental Management Framework for component 2 of the Project, the Environmental Management Framework for component 3 of the Project, and the Operational Manual; and (iii) carry component 4 of the Project in accordance with the provisions of the Operational Manual. (v) The Borrower, through SAYDS, shall cause AYSA to prepare and furnish to the Bank an environmental management plan for the Berazategui sewage treatment plant not later than six (6) months after the Effective Date. (vi) The Borrower, through SAYDS, shall cause AySA to, prior to the commencement of the main works under component 1 (Left Bank collector, the Baja Costanera bypass collector, the Riachuelo preliminary treatment plant, inflow and effluent pumping stations at the Riachuelo preliminary treatment plant, the Riachuelo outfall, the Industrial collector, and the Right Bank collector), hire one or more firms (with qualifications and experience, and under terms of reference, all acceptable to the Bank) to supervise construction.

A. STRATEGIC CONTEXT AND RATIONALE

Country and sector issues

1. The Matanza-Riachuelo (M-R), a tributary of the Río de la Plata (La Plata River), is the most contaminated river basin in Argentina¹ and the most visible environmental issue in the country. Over the past hundred years, the M-R River Basin has been used as a sewage sink for the entire city of Buenos Aires. Pollution levels have increased steadily with urbanization in the metropolitan area of Buenos Aires and accompanying industrial growth within the basin. It is estimated that more than 4,000 industrial facilities are located in the lower and middle sections of the basin. Many of these industries discharge untreated effluents into the drainage system or directly into the M-R River. In addition to high levels of organic pollution, these discharges contribute toxic contaminants such as heavy metals from petrochemical industries, tanneries, and meat processing facilities. The acute environmental and social degradation of the M-R River Basin is the result of limited public infrastructure investment, poor environmental management, and lack of adequate urban and industrial planning. This situation has created serious health risks, especially for the highly vulnerable social groups who have demanded government action to address these deteriorating conditions. The Fernández de Kirchner administration has publicly committed to making concrete progress in the clean-up of the basin.

2. The basin is home to Argentina's largest concentrations of urban poor. Of the approximately 3.5 million inhabitants in the basin, 1.2 million live below the poverty line. An estimated 10 percent of the total population in the M-R River Basin lives in informal settlements, often in flood-prone areas and/or near open garbage dumps. The poorest populations living alongside the river are in constant contact with numerous contaminants ranging from untreated organic waste to toxic industrial chemicals. The M-R River also floods frequently because of high flows in the rainy season and high water levels in La Plata River. This flooding spreads highly polluted waters into informal settlements, exposing the inhabitants to further contamination.

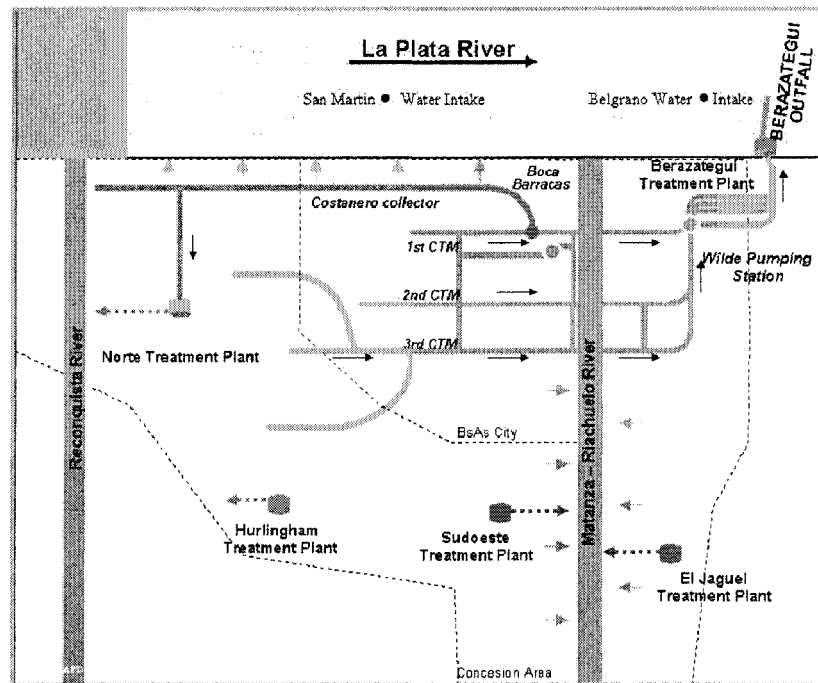
3. Currently, the Buenos Aires Metropolitan Area's sewage (domestic and some industrial) is collected and transported away from the city to the Berazategui short outfall on the La Plata River² downstream from the city of Buenos Aires and outside the city limits (see figure 1) in the province of Buenos Aires by way of three main sewage pipes (*cloacas troncales máximas*, or CTM). This conveyance system initially used to discharge directly to the M-R River. However, in the 1950s, the system was extended to Berazategui, crossing the M-R River and finally discharging to the La Plata River. Frequently, insufficient capacity or failures in the system generate an overflow of the CTMs leading to direct discharges to the M-R River. For these reasons, significant expansion of wastewater collection and treatment for final disposal into La Plata River is needed.

¹ The Matanza is the upper part of the river, and the Riachuelo the lower. The name changes when the Matanza reaches the City of Buenos Aires. The M-R River Basin discharges into La Plata River near the well known La Boca ("Mouth of the River") neighborhood.

² La Plata River is known to be the widest estuary in the world; the river's average flow is 17,000 m³/second and its average width is 50 km.

4. Preliminary data suggest that the industrial sector is the largest source of organic contamination³. In addition, industries are also a significant source of toxic pollutants, including heavy metals. While environmental standards, as well as maximum permissible limits for effluent and emission discharges exist on paper⁴, evidence shows that compliance is minimal as a consequence of unclear responsibilities across government jurisdictions, weak enforcement systems, and inadequate monitoring and control capacity in terms of human resources and technical expertise.

Figure 1. Schematic presentation of the Existing Buenos Aires sewerage system



5. Data on service coverage in the M-R River Basin⁵ indicate that 35 percent of the population does not have access to safe drinking water and 65 percent are not connected to the sewerage system.⁶ The situation is much more extreme in the poorer municipalities such as Esteban Echeverría, where, according to the last census in 2001, about 87 percent of the residents lacked access to sewerage and 63 percent lacked access to piped drinking water.⁷ Furthermore, 65 percent of the water supply for the service area of the Argentina Water and Sanitation utility (*Agua y Saneamientos Argentinos S.A.*, or AySA) is drawn from two intakes in the La Plata River, 1–2 km off the coast at Bernal and Palermo. In time, the Bernal intake could face

³ While ACUMAR's Modeling Report No. 5 suggests that industries account for about 73% of the total organic pollution load, the Government has emphasized the need for additional data collection and modeling to validate these results.

⁴ ACUMAR's Resolutions 1/2007, 2/2007 and 2/2008 list the admissible limits for discharge of effluents into sewage networks, surface waters, and soil absorption; as well as parameters for air quality relating to emissions.

⁵ Estimates are based on the percentage of the population of the 14 municipalities within the Basin boundaries. Total population of the 14 municipalities is 4.8 million while the total estimated population within the Basin is 3.5 million.

⁶ Buenos Aires is served by two water utilities: Agua y Saneamientos Argentinos S.A (AySA) and ABSA.

⁷ INDEC 2001 population census.

contamination risks from discharges of the M-R River and the existing Berazategui outfall into La Plata River (see Annex 10), raising significant health and environmental concerns that affect the entire population of the basin.

6. In recent decades, the Government of Argentina (GoA) has launched important initiatives to address flooding and reduce pollution levels in the M-R River Basin, but implementation has continually been postponed for legal, economic, or political reasons. In the mid-1990s, the Government completed a comprehensive M-R River Basin Environmental Management Plan (*Plan de Gestión Ambiental y de Manejo de la Cuenca Hídrica Matanza Riachuelo*, PGA) and received a US\$250 million loan from the Inter American Development Bank (IADB) through the *Environmental Recovery of the Matanza-Riachuelo* project to help finance its implementation. However, 12 years later, the IADB loan has disbursed only US\$10 million and is only now committing another US\$90 million for secondary networks and a treatment plant expansion (AySA's Sudoeste treatment plant). The GoA concluded that the lack of a clear mandate and accountability for action combined with an inadequate institutional and legal framework to coordinate the involvement of the relevant jurisdictions was a major obstacle to implementing the M-R PGA. Therefore, in spite of these early efforts, the social and environmental degradation of the M-R River Basin has worsened over the past decade.

7. Given worsening conditions in the M-R River Basin, national governments have made its clean-up a high priority over the past several years. At the same time, a high-profile lawsuit⁸ was filed by people affected by environmental degradation against the national government, Buenos Aires province, the Government of the City of Buenos Aires, and 44 industries in 2004. In 2006, the Supreme Court issued a ruling calling on the Government to put in place a concrete action plan to improve environmental conditions in the basin. As a result, the Integrated Cleanup Plan for the Matanza-Riachuelo Basin (*Plan Integral de Saneamiento de la Cuenca Matanza-Riachuelo*, PISA) was designed and agreed by all applicable jurisdictions and AySA. While the PISA builds on the earlier PGA, its scope is significantly wider (see table 1 below). A river basin authority (ACUMAR) was created by national law to implement this plan. In July 2008, the Supreme Court issued its final ruling establishing general criteria to ensure that the PISA is executed and setting a timeline for its main milestones.

⁸ The "Mendoza Case", Beatriz Silva and others against the National government and others.

Table 1: The Integrated Cleanup Plan for the Matanza-Riachuelo Basin (PISA) and the M-R River Basin Environmental Management Plan (PGA)

	PISA	PGA
Approval	Executive Summary of PISA approved by ACUMAR Resolution No. 8/2007	Never formally approved by resolution
Components	Includes 10 components including (i) institutional; (ii) infrastructure; (iii) environmental territorial development; (iv) integrated solid waste management; (v) clean-up of water bodies; (vi) pollution control; (vii) social participation; (viii) education; (ix) health; and (x) industry.	Includes 4 areas of action including (i) infrastructure; (ii) studies and pilot interventions for integrated solid waste management; (iii) education; and (iv) a pilot program for industries.
Institutional Setting	PISA built around ACUMAR	PGA built around an executing unit, the <i>Executing Committee for the Environmental Management Plan of the Matanza-Riachuelo River Basin (Comité Ejecutor del Plan de Gestión Ambiental y de Manejo de la Cuenca Hídrica Matanza Riachuelo (CEMR))</i>

8. **The new government strategy.** The new approach adopted by the GoA to solve the pollution issues in the M-R River Basin is different from previous clean-up attempts and is based on two concepts: (i) the establishment of a River Basin Authority that brings together all major actors in the basin and (ii) the progressive elimination of identified point source discharges of municipal and industrial wastes into the M-R River.

(i) *Establishment of a River Basin Authority.* The federal government concluded that one of the main reasons for the limited impact of previous clean-up attempts for the basin was the lack of an institutional framework to provide a consistent policy approach, interjurisdictional coordination⁹, and an enforcement plan. As a result, Law 26168 created a new river basin authority, *Autoridad de Cuenca Matanza-Riachuelo* (ACUMAR), which was designated to (i) implement a comprehensive action plan (*Plan Integral de Saneamiento, or PISA*) including structural and non-structural measures, (ii) coordinate and harmonize activities, and (iii) control and monitor environmental compliance. By law, the Secretary of SAyDS is the president of ACUMAR and ACUMAR's governing body also includes representatives from the city and province of Buenos Aires. (For a detailed description of ACUMAR, see Annex 11B).

(ii) *Progressive elimination of identified point source discharges into the M-R River.* The low flow characteristics of this urban river make it unsuitable for receiving residential or industrial discharges. Thus, a policy on the progressive elimination of point source discharges is being adopted for domestic and industrial effluents in both the lower and middle-upper basin which are inconsistent with the uses and objectives established by ACUMAR for the M-R River. The

⁹ The basin is covered by the jurisdictions of 14 municipalities and the City of Buenos Aires.

change to an approach of progressive elimination of point source discharges is fully justified given the low self-purification capacity of the M-R River and the severity of its pollution.

9. As part of its action plan to implement PISA, as mandated by the Supreme Court, the GoA has requested Bank support for parts of its long-term plan (PISA) through the proposed Matanza-Riachuelo Basin Sustainable Development Project. The proposed operation goes beyond a narrowly-focused sanitation infrastructure project by taking a more comprehensive urban environmental recovery approach, through components aimed at gradually upgrading environmental conditions as well as urban infrastructure in the basin. Integrated industrial and urban infrastructure with sufficient capacity is needed to alleviate pollution in the entire M-R River Basin and to adequately respond to the concomitant social and environmental problems.

10. The proposed Matanza-Riachuelo Basin Sustainable Development Project includes four basic components grouped under two broad macro-components:

Macro-component 1: Sanitation

(i) Investments in sanitation infrastructure including the Left Bank collector, the Baja Costanera bypass collector, the Riachuelo preliminary treatment plant, inflow and effluent pumping stations at the Riachuelo preliminary treatment plant, the Riachuelo outfall, the Industrial collector, and the Right Bank collector.

Independent technical supervision of the works listed above; data collection, analyses and modeling in support of engineering designs; and capacity development activities in support of the preparation of the engineering designs under this component.

Macro-component 2: Support for Industrial Pollution Abatement, Environmental Territorial Management, and Institutional Strengthening

(ii) Technical assistance to improve environmental monitoring and enforcement of environmental targets for industrial activities, as well as financing of cleaner production investments to enhance and support environmental compliance among polluting small and medium size enterprises;

(iii) Technical assistance for land use and urban planning and financing for pilot urban and drainage investments; and

(iv) Financing for technical studies to support project implementation; a communication strategy to inform key stakeholders; project management activities; and support for the institutional strengthening of ACUMAR.

2. Rationale for Bank involvement

11. The Matanza-Riachuelo river basin clean-up effort is large and complex, not only because of its scale but also because of the institutional coordination and capacity required for achieving its long-term objectives. The World Bank is well-positioned to assist in this type of project because of its convening power and capability to actively engage experienced specialists from its environmental, water, urban, and social units, recruit independent international expertise, and leverage additional resources (e.g., trust funds) to support project preparation.

12. In this capacity, the Bank has been collaborating with the GoA in defining the concept for the project. In November 2006, at the request of SAYDS, the Bank supported an international workshop to review the 1995 M-R PGA and evaluate independent technical alternatives for the river-basin clean-up. The workshop recommended the following: (i) redesigning the pollution management strategy in the M-R River Basin, taking into account the basin's dynamic interaction with the metropolitan area of Buenos Aires and La Plata River, (ii) revising the water quality objective proposed in the PGA and adopting an approach of progressive elimination of point source discharges for the M-R River, (iii) reviewing the priorities of investments in both the PGA and the city of Buenos Aires' pollution management program, and (iv) aggressively addressing the social problems of the poor in the M-R River Basin.

13. The long-term investment program envisaged under the proposed project is consistent with the Bank's strategy to establish a long-term partnership with the GoA focused on environmentally sustainable infrastructure. The Bank, which is already financing a major flood control project in the City of Buenos Aires and a multi-sectoral infrastructure project in the Province of Buenos Aires, is also preparing a new coordinated operation for the sustainable conversion of the industrial sector and is supporting an ongoing solid waste management project. The latter two projects may finance additional investments in the basin area. From an institutional point of view, the GoA believes that support provided by a multilateral institution like the Bank to the new basin authority, ACUMAR, and to the Secretariat of the Environment (SAYDS) will provide the necessary financing and continuity to help carry out the ambitious program proposed for the clean-up of the M-R River Basin.

14. Based on its knowledge and experience in dealing with large and complex contracts, the Bank can also contribute substantively to the planning and oversight of the highly challenging procurement plan (including pre-qualification and detailed technical specifications) designed for this operation.

3. Higher-level objectives to which the project contributes

15. The overall and specific objectives of this project are consistent with the Country Assistance Strategy (CAS) approved in June 2006, which seeks to reduce the extent and severity of poverty in Argentina by focusing on (i) the promotion of sustained economic growth with equity, (ii) social inclusion, and (iii) improved governance by strengthening institutions and increasing accountability and transparency.

16. The project is strongly aligned with all three CAS objectives, by (i) addressing a major environmental issue with substantial impact on poverty reduction (thereby contributing to the social inclusion pillar); and (ii) promoting a broad partnership between national and subnational governments, private sector firms and civil society as well the multi-jurisdictional basin agency, ACUMAR, (thereby contributing to the governance pillar). The two-stage APL program is expected to contribute to specific CAS outcomes, notably (a) expanding sewage coverage in areas of high sanitary and environmental vulnerability; (b) reducing flood risk in selected urban areas; and (c) deepening the emerging investment partnership in favor of institutional strengthening in selected areas of public sector management. The poorest and most vulnerable population will benefit from a better environment and improved coverage of public services, which in turn will lead to better health conditions. Looking forward, the proposed project is also

consistent with the proposed 2010-2012 Country Partnership Strategy for Argentina and the Bank's increasing emphasis on partnering with Argentine authorities on transformational operations that bring the Bank's experience and long-term commitment to bear in helping to solve complex problems.

B. PROJECT DESCRIPTION

17. The government's medium to long term objective under the PISA is the progressive elimination of point source discharges to the M-R River, which would result in water quality improvements in a 15 – 20 year timeframe. The proposed project is a building block towards this goal by supporting the following integrated activities: (i) the construction of a collector along the left bank of the M-R River (*Colector Margen Izquierdo*, or CMI), diversion of the wastewater collected by this collector to the Riachuelo preliminary treatment plant and discharging its effluent to the La Plata River via an effective sub-aquatic outfall, (ii) the construction of a collector from the upper M-R River Basin along the right bank of the M-R River (*Colector Margen Derecha*, or CMD) which will convey approximately 1.5 m³/s of wastewater from the upper basin as well as wastewater from the Sudoeste, El Jaguel, and Laferrere drainage basins in the southwest of the Buenos Aires Metropolitan area to the Third Trunk Sewage Pipe (Tercera Cloaca Troncal Máxima), (iii) technical assistance for designing an industrial control program reducing industrial (both organic and toxic) discharges to the M-R River by connecting these industries to the AySA sewerage network after suitable treatment of toxic and organic effluents in line with existing standards, inter alia by means of an industrial collector, and (iv) design and implementation of a comprehensive inter-jurisdictional land use plan including the provision of basic infrastructure services.

1. Lending instrument

18. The proposed operation will be structured as an Adaptable Program Loan (APL). The choice of an APL allows for flexibility in the implementation schedule within and between components to (i) enable adjustments in the pace of implementation depending on the evolution of the fiscal framework, (ii) take into consideration different preparation requirements and implementation times among different components of the project, and (iii) build the required institutional capacity to implement the project and strengthen coordination mechanisms among the different institutions involved in project execution.

19. **Project amount and duration.** The project comprises two overlapping phases to be executed over the 2009–2019 period, each phase to be implemented over 6 years. The first phase (APL-1) will extend from 2009 to 2015 for a total loan amount of up to US\$840 million. The second phase (APL-2), for a total loan amount of \$1,160 million, is expected to begin after achievement of defined targets, overlapping in time with APL-1 as new interventions are ready to be implemented.

2. Program objectives and phases

20. The overall development objective of the proposed APL program supports the Government's Integrated Basin Cleanup while simultaneously improving sanitary conditions along the banks of La Plata River and providing a long-term and cost-effective solution for safe

disposal of wastewater from the Buenos Aires Metropolitan Area (AySA's concession area). The two stage APL program that contributes to this objective and the allocation of works and activities under each APL has been specifically designed to ensure that APL-1 can be free-standing, with no stranded assets at the end of the first stage.

21. **Phase 1 (APL-1):** Phase 1 of the program includes four components under two macro-components which finance a subset of activities set out under the PISA.

Macro-component 1:

- (i) Investments in sanitation infrastructure including the Left Bank collector, the Baja Costanera bypass collector, the Riachuelo preliminary treatment plant, inflow and effluent pumping stations at the Riachuelo preliminary treatment plant, the Riachuelo outfall, the Industrial collector, and the Right Bank collector.

Independent technical supervision of the works listed above; data collection, analyses and modeling in support of engineering designs; and capacity development activities in support of the preparation of the engineering designs under this component.

Macro-component 2:

- (ii) Technical assistance to improve environmental monitoring and enforcement of environmental targets for selected industrial activities, as well as financing of cleaner production investments to enhance and support environmental compliance among polluting small and medium size enterprises;
- (iii) Technical assistance for environmental land use planning and financing for pilot urban and drainage works; and
- (iv) Financing for technical studies to support project implementation (components 1, 2 and 3); a communication strategy to inform key stakeholders; project management activities; and support for the institutional strengthening of ACUMAR.

22. **Phase 2 (APL-2):** Phase 2 of the program incorporates infrastructure investments under the sanitation component that are complementary to those initiated in Phase 1. These include (i) the Baja Costanera collector, (ii) Berazategui preliminary treatment plant, (iii) the new Berazategui outfall, (iv) the effluent pumping station at Berazategui, and (v) primary and secondary sewerage networks in various parts of the basin. The Berazategui preliminary treatment plant is functionally part of APL-2 as it will be connected to the new Berazategui outfall, but is temporally concurrent with APL-1 as the Government already contracted its construction.¹⁰ The investments for the industrial pollution and environmental territorial management components will incorporate the lessons learned from APL-1 and will scale-up successful activities.

23. **Trigger indicators.** Triggers and indicators for the Matanza-Riachuelo Basin Sustainable Development Project are shown in table 2.

¹⁰ This government action was initiated to comply with a 2006 ruling from Argentina's Supreme Court to clean-up discharges to La Plata River. Relevant safeguards information is included in Sections D4 and D5 and Annexes 10 and 14.

Table 2: Trigger indicators for APL-2

Trigger	Indicator
Overall Progress	– At least 65% of all APL-1 loan proceeds committed – At least 25% of all APL-1 loan proceeds disbursed
Sanitation Component	
• Evidence of satisfactory implementation of the sanitation works	– Final or advanced preliminary designs for the Right Bank collector (or other technical alternative agreed between the Borrower and the Bank) ready for Bank review – Construction contract for the Industrial collector signed
Industrial Pollution Abatement Component	
• Pollution load reduced in accordance with industrial conversion plans (PRIs), and effluents discharged to the sewerage network by small and medium enterprises (SMEs) participating in the matching grant program • Control and monitoring system of the M-R River Basin Authority (ACUMAR)	– Pollution loads, both organic and toxic, have been reduced to levels agreed in the PRIs in at least 20% of the SMEs participating in the matching grants program – ACUMAR's control and monitoring system is operational
Environmental Territorial Management Component	
• Evidence of satisfactory implementation of TA (institutional strengthening for territorial management)	– M-R River Basin macro territorial plan has been prepared and agreed with provincial and municipal governments
Institutional Strengthening and Project Management	
• Attainment of technical and managerial capacity within ACUMAR	– 30% of staff set out in staffing plan developed by ACUMAR is in place

3. Project development objective and key indicators

24. The **project (APL-1) development objectives** contribute to the overall program development objective by (i) improving sewerage services in the M-R River Basin and other parts of the Province and City of Buenos Aires by expanding transport and treatment capacity; (ii) supporting a reduction of industrial discharges to the M-R River, through the provision of industrial conversion grants to small and medium enterprises; (iii) promoting improved decision-making for environmentally-sustainable land use and drainage planning, and piloting urban drainage and land use investments, in the M-R River Basin; and (iv) strengthening ACUMAR's

institutional framework for ongoing and sustainable clean-up of the M-R River Basin. The key program indicators are listed below.

APL-1

- Percentage of all sewage that is currently discharged from AySA's concession area to the La Plata River without adequate treatment that is appropriately treated and disposed of by the end of APL-1;
- Percentage of regular monitoring for at least the 50 most polluting enterprises to track progress in reducing industrial pollution;
- Percentage of the enterprises in the matching grants program that have effectively reduced their discharge loads according to their PRIs by the end of APL-1;
- Both plans (Integrated Territorial Development Plan for the M-R River Basin and updated Drainage Master Plan) are agreed with key stakeholders and implementation arrangements are piloted; and
- ACUMAR has fully staffed against its new organigram, operates with its own operating budget and is fully able to fulfill the functions vested in it by law.

APL-2

- Increase in the percentage of the basin population connected to the sewerage network;
- Dissolved oxygen concentration along the M-R River higher than 2 mg/l;
- Percentage of treated wastewater in the basin and the AySA concession;
- Number of industries connected to the AySA sewerage network in line with standards in effect; and
- Percentage of drainage master plan¹¹ works completed.

Annex 3 contains the baseline and target values for selected APL-1 indicators, along with the results indicators for each component, and the data collection and reporting arrangements.

4. Project components

25. The project will contribute to the environmental recovery of the M-R River Basin through coordinated action under two macro-components: the first comprises a sanitation component, while the second includes components on industrial pollution abatement, environmental territorial management, and technical assistance and project management. A detailed project description is provided in Annex 4.

Macro-component 1: Sanitation

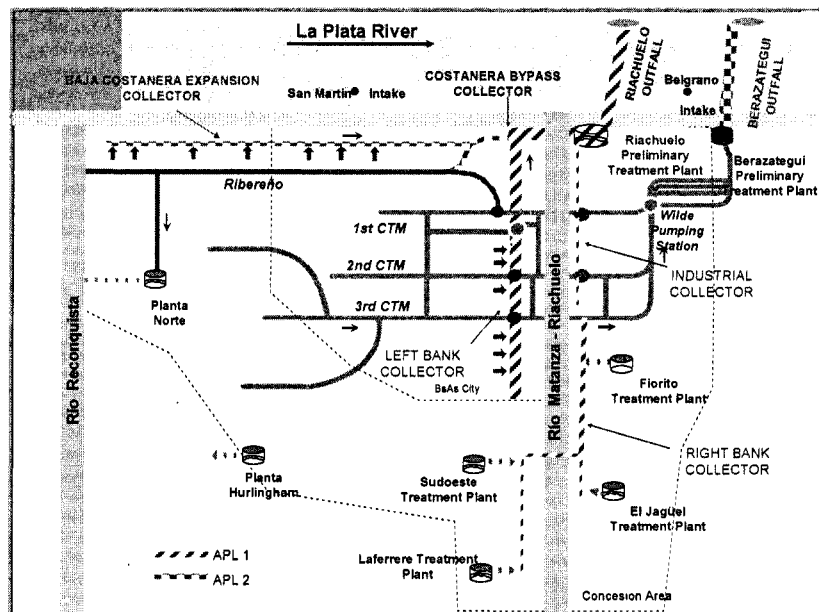
Component 1: Sanitation (US\$ 1,309 million total, US\$694 million in loan proceeds)

26. Over the course of APL-1 and APL-2, this component aims to progressively collect, convey, treat, and dispose of wastewater from the M-R River Basin along with wastewater from the AySA concession area. Sewage works included in this component are essential to improving the coverage and quality of sanitation services in the M-R River Basin and the entire AySA service area, to enhancing water quality, and to expanding the potential uses of the water bodies

¹¹ The existing drainage masterplan is known in Spanish as the *Plan Hidrico* for the M-R River Basin.

in the M-R River Basin and southern coastal strip of La Plata River. The infrastructure system selected is based on the principle of intercepting most of the domestic and industrial wastewater in the AySA concession area (except wastewater that is diverted to the North and Hurlingham treatment plants, which discharge their effluents to the Reconquista River), as well as some flows from the M-R River Basin outside AySA's concession area. This entire wastewater flow will be diverted to La Plata River through two new sub-aquatic outfalls, (the Riachuelo and Berazategui outfalls) after preliminary treatment. Specific sub-components financed under this project are detailed further below. Figure 2 schematically represents the proposed infrastructure works.

Figure 2: Schematic Presentation of Component 1



27. *Construction/extension of collectors and complementary works.* Under APL-1, waste water collection and conveyance works will include (i) the Left Bank collector, with a total length of approximately 12 km, in the lower part of the basin (the diameter varies between 0.8 m and 2.9 m); (ii) the Baja Costanera bypass collector with a length of 5 km and a diameter of 4.5 m; (iii) the Right Bank collector or a suitable alternative to convey the wastewater from the middle upper basin away from discharging into the M-R River; and (iv) an industrial waste conveyance pipe (collector) and pumping station on the right bank of the M-R River. Under APL-2, the Baja Costanera expansion collector (Radio Antiguo section and Capital Federal-Vicente Lopez section) as well as primary and secondary sewerage networks in Lomas de Zamora, Lanus, A. Brown, and other municipalities would be added.

28. *Treatment plants and pumping stations.* Prior to its discharge through each outfall to the La Plata River, the wastewater will be subjected to preliminary treatment in two plants namely the Riachuelo preliminary treatment plant located on the right bank of the M-R River at its confluence with La Plata River at Dock Sud, and the Berazategui preliminary treatment plant at Berazategui on the banks of La Plata River near the location of the current Berazategui outfall. The construction of the Berazategui preliminary treatment plant has already been contracted and is being financed by the Government using its own funds complemented by a loan from Brazil's

BNDES. The Riachuelo preliminary treatment plant, on the other hand, will be financed with loan proceeds under APL-1.

29. *Construction of new sub-aquatic outfalls.* This project will also finance the construction of two new sub-aquatic outfalls over the two APL phases. The Riachuelo outfall, financed during APL-1, will be located in the Dock Sud area. This outfall will measure 3.8 meters in internal diameter, and is expected to be 11.5 km long, of which the final 2.3 km are the diffuser section. The Berazategui outfall, expected to be financed under APL-2, will be located near the existing short outfall, measuring 3.8 meters in internal diameter and 7.5 km in length, of which the final 3.5 km will be the diffuser section¹².

30. *Construction of primary and secondary sewage networks (APL-2).* Included only under APL-2, is the construction of primary and secondary sewerage networks in the municipalities of the M-R River Basin which currently lack coverage and are home to some of the Basin's poorest people (for instance in Lomas de Zamora, Lanús, A. Brown, and other neighborhoods, but not in the city of Buenos Aires, which already has full coverage).

31. These infrastructure works, as well as other sewerage expansion investments executed by AySA under its master plan, will directly benefit approximately 2.4 million people in the 14 municipalities of the M-R River Basin, and about 1.2 million who live along the banks of La Plata River. Indirect beneficiaries include about 10 million people who will benefit from an improved and secure collection, conveyance and wastewater disposal system.

32. In addition to infrastructure works, APL-1 will also provide financing for (i) independent technical supervision of key works; (ii) data collection, analyses and modeling in support of the engineering designs for these works; and (iii) some capacity development activities for AySA in support of the preparation of the engineering designs of the works.

Macro-Component 2: Support for Industrial Pollution Abatement, Environmental Territorial Management and Institutional Strengthening

Component 2: Industrial Pollution Abatement (US\$ 75.4 million total, US\$ 60.3 million in loan proceeds)

33. Based on initial estimates, there are approximately 4100 industries in the M-R River Basin area that contribute a significant portion of the total organic load discharged to the M-R River in addition to virtually all toxic discharges. Of these, approximately 1600 are connected to the AySA network, while the remaining 2500 currently discharge effluents directly to the M-R River. ACUMAR will first focus on the 50 main industrial polluters that contribute approximately 95 percent of the total industrial organic load and 170 industries with the highest toxic loads that current discharge to the M-R River, using some of the technical assistance made available through this component. As part of the Government's strategy, plans for industrial restructuring (*Planes de Reconversión Industrial*, or PRIs), including indicators, targets, and deadlines will be developed and signed by ACUMAR with these industries.

¹² The dimensions presented are indicative. AySA is in the process of finalizing bidding-stage designs. Final designs will be prepared by construction contractor(s).

34. Activities under Component 2 will complement those under Component 1, thereby contributing to the clean-up of the M-R River. Technical assistance activities will be front-loaded in APL-1 to support ACUMAR in developing and signing PRIs with the highest organic and toxic industrial polluters, and in improving overall monitoring, control, and enforcement. These include (i) supporting pollution diagnostics and policy analysis; (ii) improving the monitoring system for industrial wastes by supporting pollution control institutional infrastructure; (iii) support for adjustments to the strategy for industrial discharges; and (iv) designing and implementing a program of corporate environmental responsibility, including for disclosure of information relating to compliance with PRIs.

35. Additionally, within APL-1, a matching grants program will support an initial batch of private sector investments in cleaner production processes. This program will target a limited number (20 - 50) of small and medium enterprises (SMEs) which have weaker financial and technical capacity to comply with environmental regulations and with whom SAYDS will sign CRIs (industrial conversion agreements). Firms signing CRIs with SAYDS would be a subset of those signing PRIs with ACUMAR. Enterprises that are eligible for participation in the matching grants program shall (i) be categorized as an SME¹³; (ii) have a signed PRI; and (iii) lack access to commercial credit. Detailed eligibility criteria for assistance to SMEs, as well as the specific disbursement modalities will be defined and disclosed in the project's Operational Manual to ensure transparency of resource allocation.

Component 3: Environmental Territorial Management (US\$81.9 million total, US\$65.5 million in loan proceeds)

36. This component's main objective is to improve public policies on environmental territorial management within the M-R River Basin. Towards this end, the component seeks to improve territorial planning and develop pilots of integrated investments in basic infrastructure services and flood control to improve urban living conditions in selected low-income areas of the basin.

37. The component includes a number of activities grouped under the following four sub-components: (i) Institutional development for environmental territorial planning, supporting technical assistance for ACUMAR with a comprehensive development perspective for the M-R River Basin to develop a regional planning system for the M-R Basin, and developing and implementing a geographic information system (GIS) to support ACUMAR's planning and decision-making capacity; (ii) Institutional development for flood control, supporting technical assistance to design and implement a flood control system, update of the hydrological master plan, improve the hydro-meteorological alert system and support municipal capacity

¹³ Under Article 1° of Title I of Law N° 25,300, as regulated by article 1 of Resolution 24/2001 of the Secretariat of Small and Medium Enterprises dated February 15, 2001 (as amended), companies are considered to be Micro, Small and Medium Enterprises according to the registered maximum annual sales, excluding value-added tax and internal tax, expressed in pesos in the table below:

Enterprise Size	Farming	Industry & Mineral	Commercial	Services	Construction
Micro	456.000	1.250.000	1.850.000	467.500	480.000
Small	3.040.000	7.500.000	11.100.000	3.366.000	3.000.000
Medium	18.240.000	60.000.000	88.800.000	22.440.000	24.000.000

strengthening in flood management (through the design of contingency flood emergency plans, the development of communication plans and design of environmental programs and disseminating information); (iii) Investments in basic infrastructure for territorial development, financing pilot interventions to improve infrastructure in low-income urban areas in the lower and middle sections of the basin and additionally to manage sustainable urban expansion in the middle and upper sections of the basin¹⁴; (iv) Investments in basic infrastructure for flood control, financing mainly micro-drainage systems and specific macro-drainage works associated with the selected micro system. The areas in which the investments will be carried out will be selected by ACUMAR in accordance with criteria acceptable to the Bank. Areas that will receive investments in drainage systems will be selected through a comprehensive socio-economic cost-benefit analysis in consultation with provincial and municipal governments. It is anticipated that two low-income settlements encompassing approximately 20,000 households will benefit from this sub-component. After construction, these systems will be operated by the respective municipal governments.

Component 4: Institutional Strengthening and Project Management (US\$18.2 million total, US\$18.2 million in loan proceeds)

38. In line with the development objectives, a separate TA component has been prepared to support the design and implementation of measures required in both the short and long term to establish the necessary institutional capacity for the new basin authority.

39. The component has been divided into five sub-components: (i) Institutional Strengthening of ACUMAR to adapt its current organizational structure to better respond to its institutional responsibilities. A new structure will be designed through a participatory process and using proven methodologies and strategies. This subcomponent will support the necessary activities such as workshops, consultants, and infrastructure investments to support the new institutional structure; (ii) a Communication Strategy to inform key actors and stakeholder groups in the basin about the project in a transparent manner; to strengthen the credibility of and support for the project; and to contribute to the overall consultation and participation framework of the project; (iii) Establishment of a public information office in ACUMAR supporting the necessary investments for the proper functioning of a public information office such as human resources, office equipment, infrastructure and logistic tools; (iv) Technical and monitoring studies encompassing both additional studies required during the course of project implementation and the provision of expert support (including a panel of recognized independent experts in the fields of environment, engineering, biodiversity, modeling etc.) to provide periodic advice in conjunction with project implementation. Studies will include additional data collection and analytical work relating to impact and water quality monitoring in the M-R River and La Plata River, as well as the control of high groundwater tables in the basin area. This subcomponent may also finance other studies required for the clean-up of the M-R River Basin as agreed with the Bank. Finally, (v) Project management, supporting human resources, physical space and equipment for UCOFI and the proposed UEP within SAYDS to strengthen purchasing, contracting and procurement processes as well as technical management of the project.

¹⁴ Basic infrastructure includes (i) secondary networks for water and sanitation and the connection of those networks to the main existing water and sanitation systems, (ii) micro drainage and the section of macro drainage required to operate the micro drainage system, (iii) secondary streets, and (iv) recreational areas.

40. **Project cost estimate.** The total project cost for APL-1 is estimated at US\$1487 million, including contingencies and the front-end fee. The loan amount for the APL-1 is US\$840 million.

41. **Financing plan.** Table 3 illustrates the proposed cost and financing of APL-1. See Annex 4 for a detailed description of project components and Annex 5 for a detailed cost breakdown.

Table 3: Project Cost and Proposed Financing

<u>Component</u>	<u>Category</u>	<u>Cost Incl. Contingencies (US\$M)</u>	<u>% of Total</u>	<u>Bank- financing (US\$M)</u>	<u>% of Bank- financing</u>
1. Sanitation Component	Physical	1309.1	88.1	693.9	53.0
2. Industrial Pollution Abatement Component	Physical	75.4	5.1	60.3	80.0
3. Environmental Territorial Management Component	Physical	81.9	5.5	65.5	80.0
4. Institutional Strengthening and Project Management	Institution Building and Project Management	18.2	1.2	18.2	100.0
<i>Front-end Fee</i>		<i>2.1</i>	<i>0.14</i>	<i>2.1</i>	<i>100.0</i>
	Total	1486.7	100.0	840.0	56.4

5. Lessons learned and reflected in the project design

42. The World Bank has acquired considerable experience in river basin clean-up projects, industrial pollution abatement and control projects, and a wide variety of urban rehabilitation projects. The following lessons learned from these projects, undertaken in Argentina and in other countries, have been reflected in the design of the Matanza-Riachuelo Basin Sustainable Development Project:

43. *Need for sound institutional coordination and strong political will.* The lack of an adequate institutional and legal framework to forge agreement among different government jurisdictions and assign accountability has clearly been a major obstacle to implementing a clean-up program for the M-R River Basin. While national governments have made clean-up a priority over the past several years, the high-profile Supreme Court lawsuit and ruling and recently passed legislation resulting in the creation of ACUMAR, have reinforced the framework to address the basin's challenges. Although the creation of ACUMAR represents significant progress over previous attempts to solve the M-R pollution problem, the agency is still in the process of consolidation and requires institutional strengthening. GoA efforts are also backed by strong political will: President Fernández de Kirchner publicly committed on her second day in office to make concrete progress in the basin within her 4-year mandate. Neither the interjurisdictional¹⁵ basin authority nor the Supreme Court ruling that forces the Government to implement actions to clean-up the M-R River were in place at the time the last major attempt was launched to clean-up the M-R River Basin under IADB's "Environmental Recovery of the Matanza-Riachuelo" project (1997). With the Supreme Court ruling, ACUMAR is politically and legally accountable.

¹⁵ The basin is covered by the jurisdictions of 14 municipalities and the City of Buenos Aires.

44. *Lessons learned from past environmental operations involving utilities.* Past operations have often ignored the additional financial burden that expanded sewage collection and treatment systems place on utilities. This project explicitly recognizes the importance of cost recovery – in terms of covering the operations and maintenance costs resulting from new sewage infrastructure -- and takes several concrete steps towards addressing it. While Law 26.221 provides a regulatory framework for AySA's operations, the utility has been operating without a concession contract since 2006, for example. Against this backdrop, the Government has committed to entering into a concession agreement with AySA before the Bank's loan is declared effective. Furthermore, this project also "piggy-backs" on a recently-approved IADB operation which includes conditions for AySA to increase tariffs to break even on O&M costs in 4 years. While these are important steps, the project team recognizes that AySA's financial position will remain vulnerable to any interruption of public financing as long as tariffs don't cover at least O&M costs. More generally, it is expected that a concession contract will help to enhance incentives to maintain AySA's relatively good operating efficiency. A detailed discussion of AySA's financial position and performance is provided in Annex 11B.

45. *Adopting an integrated basin-wide approach.* Experience with pollution control projects in river basins in Bangladesh, China, and elsewhere has shown the importance of adopting a basin-wide approach, with integrated components that fit together "naturally" and do not place an undue burden on borrowers and implementing agencies. This experience also highlights treating institutional and environmental objectives associated with integrated water quality management and urban upgrading projects as long-term program goals rather than short to medium-term project objectives. These projects have also pointed to the need for adequate monitoring of pollution sources through the creation of databases and for linking data to investment planning for wastewater treatment.

46. *Understanding obstacles to local capacity building.* Implementation experience from previous projects in Argentina, particularly in partnership with SAYDS, has shown the following main weaknesses: (i) lack of technical capacity, (ii) lack of awareness of project processes, and (iii) frequent changes in leadership and personnel. This situation has led to enormous delays in project implementation. Project design has sought to ensure efficient implementation by using AySA to implement component 1 (which accounts for approximately 88% of the APL-1 program) and SAYDS' experienced UCOFI unit (which has a good track record with internationally financed contracts) to manage the procurement, financial management and administrative aspects for components 2, 3 and 4.

47. *Need for public participation and communication.* The Bank's experience with pollution-control and sewerage infrastructure projects clearly demonstrates the need for a strong public participation and communication component to address public concerns related to project components. The proposed operation has already carried out consultation workshops to analyze the terms of reference for and draft of the Environmental Impact Assessment (EIA). ACUMAR has also agreed to establish and maintain more frequent dialogue with NGOs and other stakeholders through its Social Participation Commission. A dedicated communication strategy is being financed through the proposed project. Moreover, the Supreme Court's ruling in the Mendoza Case, specifically designates 5 NGO's/civil society organizations as part of a "collective body" coordinated by the Ombudsman to monitor compliance by national and subnational governments as well as ACUMAR with the Court's decision. The project has

maintained a dialogue with all of these entities during project preparation and is expected to deepen this dialogue further during implementation.

6. Alternatives considered and reasons for rejection

48. Various alternatives were considered during project design and preparation. The alternatives considered and the reasons for the rejection of those not adopted are discussed below.

49. ***Lending instrument alternatives.*** Lending instruments considered for this project were the Sector Investment Loan (SIL) and the Adaptable Program Loan (APL). Given the size of the project and its complexity, with multi-sectoral and multi-jurisdictional aspects, a longer-term commitment will be required from both the Government and the Bank. Under these circumstances, an APL affords greater flexibility to prepare and lay the groundwork for future investments. Because it establishes goals and triggers to move from one phase to the next, an APL is considered more appropriate than a SIL or a sequence of individual projects. For example, components 2 and 3 require several studies, additional analytical work, and institutional strengthening for appropriate investment decisions. These background studies will lay the groundwork for investments in subsequent phases. For component 1, which focuses on major infrastructure works, an APL design will permit the sequencing of works to be carried out in an orderly manner.

50. ***Technical alternatives.*** A variety of technical alternatives for the level of wastewater treatment, the length of outfalls, and the number and location of the wastewater treatment plants were considered for the sanitation component of the proposed project. Annex 11A describes the alternatives and the reasons for rejection of those that were not selected.

51. An issue of major importance for the project is the selection of preliminary treatment of wastewater coupled with discharge of the effluent to La Plata River through a long, effective outfall. The decision to adopt a low level of treatment rather than a higher level has been challenged by several NGOs. However, Buenos Aires and other riparian municipalities can be protected from the negative impacts of pre-treated effluent discharge to La Plata River by an effective outfall, making use of La Plata River's large oxygenation capacity. This conclusion was derived from mathematical modeling (see Annex 11A).

52. Choosing a higher level of treatment (secondary) and a shorter outfall while effective in removing organic matter would be unaffordable considering that disinfection to eliminate pathogens would still be required. This option would be much more expensive in terms of initial investment as well as operation and maintenance costs. The additional costs associated with secondary treatment would not provide additional benefits. Therefore, the alternative of secondary treatment in conjunction with an effective outfall was not selected because it is equivalent, in terms of risks to humans and the environment, to the selected alternative (preliminary treatment and an effective outfall), but much more expensive. These conclusions were reached with the support of international technical experts on subjects including outfall impacts on water bodies, hydrological modeling and integrated basin management. After the beginning of operations of the treatment and outfall system, comprehensive water quality monitoring will be undertaken in the La Plata River. If the water quality monitored is found to

deviate from what was foreseen by the modeling, and presents risks, additional treatment units will be installed in the treatment plant to mitigate these risks. Sufficient area within the site of the treatment plant has been secured to accommodate any additional treatment units that might be needed in the future.

53. For the industrial pollution abatement component (component 2), two alternatives were considered. The first was a high level of treatment for all industrial wastes and discharge of all effluents except those containing toxic matter into the M-R River. The second was a lower level of treatment of all industrial wastes and discharge of all effluents, except those containing toxic matter, to the municipal sewerage network for further processing with domestic wastewater in line with existing AySA standards. The first alternative was rejected because mathematical modeling showed that even after a high level of treatment, the discharge of industrial effluents into the M-R River would still render it anaerobic. The second alternative was found to be the only way to bring about aerobic conditions in the river and recover its water quality.

54. ***Phasing of investments.*** As currently designed, APL-1 includes the Right Bank Collector (CMD), a key piece of sewerage infrastructure that would receive and transport wastewater collected in the coverage area of the Southwest, Jagüel, Fiorito and Laferrere treatment plants as well as wastewater generated in the municipalities of the upper M-R River Basin located outside the AySA concession area. Initial water quality modeling during project preparation has identified the withdrawal of these discharges to the M-R River as a key enabler of improved water quality. Given the size of the proposed investments (estimated at about US\$275 million (without contingencies) based on preliminary designs), AySA will conduct a pre-feasibility study of alternatives to the CMD. AySA is considering, for example, whether side stream elevated pool aeration (SEPA) stations, being used in Chicago, Illinois, could be used -- possibly in conjunction with higher levels of treatment at existing and future treatment plants -- to meet water quality objectives established by ACUMAR. Water quality modeling of various alternatives is already underway in conjunction with ACUMAR. Should a better alternative emerge in terms of economic, environment and social aspects, the latter would need to be submitted to SAyDS and ACUMAR for approval before it would be presented to the Bank for review. A covenant has been added to the Loan Agreement requiring that the study of alternatives to the CMD be completed by March 31, 2010. Any alternative to construction of the CMD would need to be approved by the Bank and may need to be reflected in an amendment to the Loan Agreement. Given this uncertainty, the Bank and the Borrower initially included the CMD under APL-2. Subsequently, however, the Bank and the Borrower agreed to advance the CMD to APL-1 to provide the Government with an instrument for accelerated basin clean-up in the short and medium-term. Including the CMD in APL-1 also has the important advantage of concentrating all major trunk infrastructure works needed to remove contaminant flows from the M-R River Basin in a single phase over the next 6 years. On the other hand, construction of the CMD during APL-1 increases the size of the project's first phase to nearly US\$1.5 billion, meaning that the Bank's US\$840 million loan would, on average, finance only about 56% of APL-1 activities.

55. ***Institutional alternatives for project implementation.*** Several institutional arrangements for project implementation were considered. The first was to vest ACUMAR, the recently-created basin agency with direct implementation responsibility for the entire project. This alternative was rejected because ACUMAR lacks the capacity to construct sewage collectors,

networks, and treatment plants under component 1. These works are the natural responsibility of AySA, the entity in charge of water and sanitation service provision in Buenos Aires.

56. A second alternative considered was to split responsibilities by vesting the implementation responsibility for component 1 exclusively in AySA and the implementation responsibility for components 2, 3 and 4 exclusively in ACUMAR. Given that ACUMAR's organizational structure is still being finalized and ACUMAR will only gradually staff-up against its new organigram, the Government has opted for a more cautious approach whereby components 2, 3 and 4 are implemented by SAYDS with support from ACUMAR while ACUMAR reviews the technical consistency of any changes to the works proposed under component 1 that may impact PISA objectives.

57. *Alternative financing mechanisms for small and medium size polluting industries.* Component 2, on industrial pollution abatement, includes financing SMEs that are among the most polluting in the M-R River Basin, to support their investments in cleaner production measures. During preparation, the team considered providing commercial lending or matching grants for these SME investments. Given their weak financial capacity and limited access to commercial credit, it was decided to extend grants to SMEs on a matching basis, in order to help them improve environmental compliance (as spelled out in their PRIs). Large enterprises can access credit either through local commercial loans or through loans for the private sector from multi-lateral organizations (e.g. International Finance Corporation or Interamerican Investment Corporation). This financing mechanism is also consistent with the financial mechanism being developed for the *Sustainable Industrial Development Project*, a proposed complementary Bank-financed project in Argentina.

C. IMPLEMENTATION

1. Partnership arrangements (if applicable)

58. The Bank preparation team for the proposed project has coordinated with IADB's *Programa de Agua Potable y Saneamiento del Area Metropolitana y el Conurbano Bonaerense* project preparation team. The purpose of this collaboration was to share information and avoid duplication of work, especially in tasks related to AySA's financial assessment and the analysis of common socio-economic information in similar areas. It is expected that this partnership will continue.

2. Institutional and implementation arrangements

59. The Argentine Republic will be the Borrower for the proposed loan. SAYDS will be responsible for overall project coordination and will directly implement macro-component 2 together with ACUMAR (as the inter-jurisdictional entity). While ACUMAR has been created as an inter-jurisdictional entity with overall responsibility for implementing the PISA, the Government recognizes that ACUMAR currently has limited implementation capacity which will be reinforced and developed over the course of APL-1. The responsibilities for each entity are described in greater detail in Annex 6. Regarding macro-component 1, AySA will be the executing entity while ACUMAR, will ensure the consistency of project activities with the PISA in line with its legal mandate.

60. *Institutions.* ACUMAR: The governing body of ACUMAR has 8 members. It is headed by the Secretary of the SAYDS and comprises three other representatives from the national government,¹⁶ as well as four representatives from local governments -- two from the Province and two from the City of Buenos Aires. Through various resolutions, ACUMAR has been legally empowered to effectively coordinate the different agencies and manage and enforce regulations and norms relating to integrated basin management and the implementation of PISA. Details are summarized in Annex 11B. ACUMAR has the authority to enforce a wide range of preventive actions against polluters. Since its creation in 2006, it has used this authority widely. Through July 10, 2008, ACUMAR had issued 702 preventive actions (“*medidas preventivas*”) to prevent pollution, of which 110 have resulted in the temporary closure of polluting firms. Nonetheless, ACUMAR has not yet imposed preventive actions that would result in the definitive and total elimination of discharges or the closure of firms.

61. AySA: In March 2006, AySA was created as a private company -- with the Government as the controlling shareholder -- to take over the service area previously served by *Aguas Argentinas Sociedad Anonima (AASA)*, a private company controlled by the French group Suez and the Spanish group Aguas de Barcelona. The owners of AASA also included a minority group of shareholders comprising Bank of Galicia, Vivendi, the International Finance Corporation (IFC) and Anglian Water. After tariffs were frozen in the wake of the 2002 economic crisis under the Economic Emergency Law, the GoA and AASA were unable to reach agreement on a renegotiated concession agreement. In 2006, AASA’s concession contract was rescinded by the Government which alleged that AASA was not in compliance with its investment program. Suez and other foreign shareholders are currently suing the GoA before ICSID for economic compensation. The case is still pending resolution.

62. AySA is currently subject to controls mandated both for private companies and for public utilities and will be in substantial compliance with the operator eligibility criteria set out in the PAD for the *Buenos Aires Infrastructure Sustainable Development Project* once an appropriate concession contract is signed between the Government and AySA (see Annex 11B for details). The regulatory framework delegates primary oversight authority of the concession to the Subsecretaría de Recursos Hídricos (SSRH) – which exercises authority both as majority owner and primary regulator of AySA alongside the Regulatory Entity for Water and Sanitation (ERAS). The SSRH is part of the Ministry of Federal Planning, Public Investment and Services (MPFIPS) and plays a key role in the governance of AySA.

63. AySA’s capital structure is not typical of a fully-integrated utility. Its minimal level of debt and fixed assets are more akin to those of an operating company than a government-owned water utility with responsibility for financing all (or at least a portion of) its fixed assets. Details of AySA’s capital structure are presented in Annex 11B.

64. *Implementation arrangements.* The SAYDS has informed the Bank, that it is moving to a new management model for internationally financed projects under which procurement, financial management and general administrative functions are centralized in a single (existing) unit

¹⁶ These include SAYDS, the Sub-secretariat for Hydrological Resources (Subsecretaría de Recursos Hídricos, or SSRH), a dependency of the Ministry of Federal Planning, Public Investment, and Services (Ministerio de Planificación Federal, Inversión Pública y Servicios, or MPFIPS), and a representative of the Ministry of the Economy.

(UCOFI) while the technical leadership for each operation is vested in separate project implementation units (UEP). In line with this strategy, S AyDS will exercise its overall project coordination role for the present project through two units: UCOFI (*Unidad de Coordinación para la Gestión Técnica y Financiera Internacional*) and UEP (*Unidad de Ejecución del Proyecto*). UCOFI has experience with the management of projects financed by international financial institutions. UCOFI will manage the project's US Dollar denominated designated account as well as a Peso-denominated project account from which it will make all payments for goods, works, and services procured under components 2, 3 and 4. With technical support from UEP and ACUMAR (which will help produce terms of reference and technical specifications and support bid/proposal evaluation and contract supervision) UCOFI will also carry out all procurement for these components. For component 1, UCOFI will transfer funds to AySA which will contract all goods, works and consulting services and make all payments to consultants and contractors using its regular technical units. These units include AySA's procurement and legal departments (for the preparation of bidding documents) and AySA's planning department for technical aspects. The environmental, social relations and communications departments of AySA will also participate in project implementation. Details of the implementation responsibilities of S AyDS, ACUMAR and AySA can be found in Annex 6.

3. Monitoring and evaluation of outcomes/results

65. This project strengthens the monitoring and evaluation capacity of the newly created river basin authority, ACUMAR. During APL-1, financing allocated for component 2 includes technical assistance to support the development of an industrial pollution control and monitoring system including a GIS-based industrial pollution database. This database will include monitoring results of parameters relating to water consumption, as well as pollution loads from various industries in the basin. In addition, the project will also finance pollution control infrastructure, equipment, and testing in certified laboratories. Under component 3, technical assistance will support the development of a GIS database to aid planning processes; and to strengthen the meteorological and hydrological forecasting capacity in the M-R River Basin. Annex 4 presents details on these sub-components.

66. The recent Supreme Court ruling against the central government, Buenos Aires Province, the Government of the City of Buenos Aires, and 44 industries requires ACUMAR to adopt an internationally recognized system of indicators to assess the fulfillment of the PISA. While civil society participation in ACUMAR is provided under article 4 of Law 26.168 through a Social Participation Commission, the Supreme Court ruling provides explicit additional arrangements for civil society oversight of all basin clean-up measures, including the designation of a monitoring body ("*Cuerpo Colegiado*" or "*collective body*") headed by the national ombudsman, and composed of the five non-governmental organizations (NGOs) participating in the judicial case as "*terceros interesados*" (i.e., interested third parties). A Federal Court in Quilmes has been designated to monitor the Government's compliance with the terms of the ruling.

67. As mentioned previously, key program indicators have been developed for the Bank loan and will be monitored by AySA and ACUMAR with the help of local universities through loan financing at baseline and during project implementation to evaluate progress. The specific targets

for results monitoring are detailed in Annex 3, and include information on data collection and reporting.

4. Sustainability

68. While the GoA has made solving the M-R River Basin's environmental problems a high priority, the Supreme Court ruling requiring the central government, Buenos Aires province and the city of Buenos Aires to clean up the M-R River and its basin makes rehabilitation a matter of national (not merely government) policy. In addition, the Supreme Court ruling has also heightened the profile of ACUMAR which has been designated the lead agency responsible for clean-up of the M-R basin. Finally, given that the M-R River has been a source of national debate for several decades, and the recent Supreme Court ruling, future governments (both national and provincial) will almost certainly continue to consider the clean-up of the Basin a high priority. Even so, it is clear that implementation of PISA will require substantial political and financial commitment by the national government, at levels that may be difficult to sustain given global economic uncertainty. Moreover, the Supreme Court's ruling notwithstanding, ACUMAR remains dependent on national government financing and personnel.

69. The GoA will assure the adequate operation and maintenance of the main infrastructure investments under the sanitation component of the proposed project. The GoA will own the new assets, and AySA will be the infrastructure operator (See Box 1).

Box 1. International cases of government financing for water and sewerage services

Government financing for wastewater treatment and disposal investments through subsidies is common practice in developed and developing countries. The main reasons for supporting this practice include the following: (i) wastewater treatment and disposal is treated as a quasi-public good because of the wider public health benefits, (ii) the level of investment in the infrastructure involved is so high that it generally cannot be fully paid for by medium or low income municipalities; and (iii) the avoided environmental externalities cannot be directly charged to users.

Examples of subsidized investment are found in both developed and developing countries, including the USA where the federal government provided 75% of the investment costs for wastewater treatment and disposal until the beginning of the 1990s. Between 1972 and 1989 US\$56 billion in federal construction grants were provided to municipalities for secondary treatment. The federal government of Mexico provides subsidies for wastewater treatment and disposal amounting to about 50% of the investment through various federal programs. The subsidy can go up to 100% in the poorest municipalities. The government of Colombia has an innovative subsidy program which has provided some US\$150 million in investment subsidies during the past four years for investments in water and sanitation as part of the 2002-2006 National Development Plan, most of this for sewerage networks and wastewater treatment plants. The government plans to expand this program and allocate an additional US\$500 million to it over the next 4 years. EU programs for new member states include up to 100% subsidies for investments in wastewater treatment facilities.

In some cases investment subsidies are provided by governments paying for each cubic meter treated. Examples include (i) France, where it is a common practice to provide an "operator grant" to municipalities with sewage treatment plants, with subsidies that generally amount to 50% and provide an incentive to reduce contamination. If standards are not reached, the subsidy is not disbursed; and (ii) the German model, where all municipalities pay a sewage discharge fee to the state, with a 50% discount if discharge standards are reached.

70. In the wake of the economic and social crisis of 2002, one of the most severe losses of income and downturns in living standards on record, the GoA decided to tackle the imbalance between tariffs and the population's payment capacity by freezing the tariffs of all public

services (electricity, gas, water, and transport). Since then water tariffs have been frozen in Buenos Aires. In real terms they have declined substantially. Nonetheless, starting in 2008, the GoA has begun to increase public transport, electricity and gas tariffs. While AySA's tariffs have not increased, the Province of Buenos Aires recently approved¹⁷ a significant tariff increase for ABSA, another major public water and sanitation concessionaire operating just outside AySA's concession area.

71. In spite of this public service tariff policy, AySA was able to finance its O&M costs directly from tariffs until 2007. For 2008, however, AySA's O&M costs were on track to exceed tariff revenues by about 45 percent¹⁸ (see Annex 11B). The GoA, as the majority owner of AySA, is making budgetary transfers¹⁹ necessary for the normal operation of the company, as has been the practice since AySA's creation. In September 2008, Argentina's National Congress approved a AR\$1250 million (approximately US\$412 million) capital improvement budget for AySA, including transfers covering operating costs for CY2008. In addition to covering this O&M gap, the GoA through AySA and the Buenos Aires provincial government has increased investment levels during the last 2 years to improve service and expand coverage. Moreover, clause 4.11 of IADB's loan agreement (*Programa de Agua Potable y Saneamiento del Area Metropolitana y el Conurbano Bonaerense*) with Argentina for its November 2008 loan for AySA (which commits the Government to achieving operation, maintenance and administration cost recovery of AySA by year 4 of IADB's operation) is deemed to improve the sustainability of AySA. The government's commitment to AySA's financial sustainability is further underscored in the Development Policy Letter (DPL) (see Annex 11E) provided in conjunction with the proposed operation.

72. For industrial pollution abatement, clean-up investments will include those financed by the private sector within a framework of government-sanctioned economic incentives and penalties. The project proposes a US\$41 million fund that will provide financing to a number of SMEs for pollution control and cleaner production investments. This financing would be on a matching grant basis, as opposed to a targeted or revolving credit fund. The financial cost of future compliance would be assumed by the private sector.

73. SAYDS and ACUMAR will implement works and activities such as the infrastructure investments under the environmental territorial management component of the proposed project. For long-term sustainability, these investments rely on strengthening ACUMAR as the lead agency for the basin. Operation and maintenance functions will be transferred to the relevant jurisdictions, i.e., the relevant municipality for micro drainage investments and the province for macro-drainage investments.

¹⁷ Decree N° 3144/08 issued by the Government of the Province of Buenos Aires on 9 December, 2008. Under the decree, the base rate per m³ increased by about 120% from 0.276 Pesos/ m³ to 0.607 Pesos/ m³.

¹⁸ CASTALIA. 2008. Due Diligence of the Corporate and Regulatory Framework of Agua y Saneamiento Argentinos S.A. (AySA).

¹⁹ According to article 14 of Decree 304/06, the Ministry of Economy and Production has to implement all actions towards accelerating the economic and financial management of the service, creating a special account within the national budget.

5. Critical risks and possible controversial aspects

74. The proposed program is a highly complex operation with very substantial risks aiming at significant positive environmental and social impacts. The main risks involve: (i) delays in key investments essential to basin clean-up as the global economic downturn leads to slower growth in Argentina and competing priorities reduce the availability of government counterpart funds. This could lead to postponing those investments under the PISA (such as the Right Bank Collector) whose benefits would be felt only in the medium term. It could also complicate the willingness to enforce industrial clean-up in industries located in the M-R basin; (ii) special interest capture of the investment program proposed under the project; (iii) the weakness of ACUMAR as a new Basin Authority and (iv) that ACUMAR, subnational governments or private firms in the basin could use their own funds to carry out actions broadly in line with basin clean-up under the PISA without full consideration of social and environmental impacts. Table 3 shows the risks involved in the proposed project and measures to mitigate these risks.

Table 3: Critical risks and mitigation measures

Risk	Risk level	Mitigation measure	Residual risk level
As the global economic downturn leads to slower growth in Argentina, competing priorities reduce the availability of government counterpart funds and key investments are strung out.	H	• The government has publicly committed to cleaning up the M-R basin and has agreed to front-load an ambitious program of works in APL-1. • The Supreme Court ruling requiring the central government, the province of Buenos Aires and the Buenos Aires city government to clean up the M-R basin makes project execution a matter of national (not merely government) policy. • The Bank will increase the percentage of Bank financing for key works during the early years of implementation (which are expected to be the most difficult in light of the global economic downturn). • In case of a major decline in government revenue, the Government and the Bank are committed to restructuring the Bank's portfolio to realign Bank financing and government counterpart funds to protect social inclusion and key infrastructure projects.	H
Special interests capture the investment program under components 1, 2 and 3.	S	• Design of APL triggers keeps the project focused on the medium-term goal of basin clean-up. • The project will use transparent design of eligibility criteria for SME's wishing to participate under the matching grant program for industrial conversion. • The project will use clear prioritization criteria for selecting urban development and drainage investments.	M

Risk	Risk level	Mitigation measure	Residual risk level
Weak counterpart institutional capacity, particularly of ACUMAR, which has operated only a short time and has never been involved with large projects	H	- Procurement, financial management and general administration for components 2, 3 and 4 will be implemented by the SAyDS through the experienced UCOFI implementation unit which has committed approximately US\$100 million under an IADB operation for clean-up of the basin. ACUMAR will support project implementation, but its initial capacity limitations are not expected to hold up implementation of the components managed by SAyDS. - ACUMAR has a sound legal basis and will receive technical assistance as part of the project. Strengthening ACUMAR is one of the project development objectives. - AySA has adequate project implementation capacity and will implement the majority of the works.	S
ACUMAR, subnational governments or private firms in the basin could carry out actions broadly in line with basin clean-up under the PISA entirely with their own funds without full consideration of social and environmental impacts (including relocation or restoration of livelihoods), thereby creating reputational risks for the Bank	H	- The Bank team will conduct continuous consultation with government counterparts throughout project implementation. - There will be deepened social analysis as specific activities under components 2 and 3 are identified.	H
Project financial sustainability	H	- The GoA, as the majority owner of AySA, will assure any budgetary transfers necessary for normal operations, as has been the practice since AySA's creation and is in line with article 14 of Decree 304/06. The GoA has committed to the transfers through the development program letter. - Operations and maintenance responsibilities under the environmental territorial management component will be transferred to relevant jurisdictions (municipalities and province). - The Bank will continue to deepen its dialogue with the authorities in order to improve the water and sanitation tariff policy framework, based on the Government's commitment to achieve full O&M cost recovery through tariffs within 4 years.	H
General reputational risks to the Bank arising from the Supreme Court's ruling by association	S	The last Supreme Court (SC) ruling required a precise and comprehensive action plan to be followed by ACUMAR. Overall, the requirements are in line with the	S

Risk	Risk level	Mitigation measure	Residual risk level
with the Government's actions and pace in cleaning up the Matanza-Riachuelo River		activities included in the Integrated Cleanup Plan for the Matanza-Riachuelo Basin (<i>Plan Integral de Saneamiento de la Cuenca Matanza-Riachuelo</i>, or PISA) and the proposed project. • None of the activities under the proposed project will impose any delay in meeting SC requirements. • The proposed project provides support that will enhance the quality of the Government's response to the SC's requirements.	
General reputational risks arising from legal actions brought against the Government of Argentina by shareholders in the former <i>Aguas Argentinas</i> concession.	L	• One case is pending resolution before ICSID connected with the rescission of the <i>Aguas Argentinas</i> (AA) concession. Since former shareholders in AA are suing for monetary compensation (but not re-instatement of the former concession), the proceedings are expected to have little impact on the project.	L
NGO or community concerns, particularly regarding level of treatment and sub-aquatic outfalls due to a perception that the La Plata River cannot assimilate the treated effluent	H	• Consultations with NGOs and local communities have been held to explain the alternatives considered and the trade-offs analyzed. All parties recognize the urgency of taking action in this long-neglected river basin. • While NGO pressure continues for higher levels of wastewater treatment, such as the use of secondary wastewater treatment, the preliminary treatment proposed under this project is affordable and effective from health and environmental standpoints (see Annex 11A). This issue has been addressed through open and technically informed discussions at the workshop to present the results of the advanced drafts of the EIA as well as in individual discussions with many individual NGO's. • As part of the project, an environmental monitoring program will be set up to compare contamination levels in the riverbank areas before and after the construction of the outfall. This will help resolve the controversy. • AySA is in communication with its residential clients through its communications group and directly with its institutional clients and municipalities. • SAYDS and ACUMAR will implement a communication strategy for NGOs through its civil society and communications group.	S
Possible safeguards risks (as defined by the relevant Bank policies) relating to : • International waterways • Resettlement • Environmental impacts	S	• AySA has formally requested the approval of the Riachuelo outfall (APL-1) from the Administrative Commission of La Plata River (CARP), a bi-national commission made up of Uruguay and Argentina for the implementation of the Rio de la Plata Treaty. The information submitted to CARP contained a summary of the full scope of sewerage works to be undertaken under	M

Risk	Risk level	Mitigation measure	Residual risk level
		APL-1 and APL-2, including the Berazategui outfall. On 29 January, 2009, the Argentine government received a letter from the Commission, stating that the proposed works will not cause significant damage to the navigation interest of Uruguay or the regime of the river and that any impact of the works on water quality should be governed by Chapter IX of the CARP Treaty. In addition, AySA is expanding the data base for the existing water quality simulation model of La Plata River. • There are no cases of resettlement expected under activities financed by APL-1. A Resettlement Policy Framework has been developed in case any resettlement, as defined under OP 4.12, becomes necessary. When preparation of APL-2 begins, resettlement aspects will be addressed for the activities supported under APL-2. • EIAs have been consulted with relevant stakeholders. SAYDS, ACUMAR and AySA will implement a communications strategy defining methods for public disclosure of all relevant documents, including monitoring reports during implementation. The project will also support creation of a Public Information Center within ACUMAR.	
Social and environmental risks, caused by the works and the possible miscommunication or lack of participation of civil society in the project	H	• As a result of the consultation process and the social and environmental assessments, social and environmental mitigation plans will be carried out. A communication campaign will also accompany the implementation of the project.	S
Industries unwilling to invest in cleaner production technologies due to competitiveness concerns accentuated by the expected economic downturn.	H	• PRIs are legally binding for industries deemed to be contaminating; and, in some cases, SMEs will receive grant financing through industrial conversion agreements (CRIs). • The project team will work closely with industry associations to correct the misconception that cleaner production investments and profitability are mutually exclusive. • Regulatory and financial incentives will be piloted to motivate the private sector to participate.	S
Cost overruns in infrastructure works	H	• Physical and price contingencies have been included and major contracts are expected to attract large international contractors. The project's major bidding process has been structured to maximize competition. • Specialized engineering consulting firms will provide independent construction supervision.	S
Financial management risks due to large outlays involved.	M	• The flow of funds will be centralized.	M

Risk	Risk level	Mitigation measure	Residual risk level
		- Advances will be made to a separate bank account in AySA on the basis of eligible expenditures incurred and planned certifications of works as determined by the Bank. AySA will conduct monthly reconciliation of and reporting on uses of funds. - The small number of large contracts for the project's major works limits risks.	
Risks relating to procurement due to the size of contracts, timing of different activities, and institutional arrangements	H	- The Argentina country program has progressed substantially in addressing transparency and governance issues, including project-specific audits and follow-up, and through the procurement tracking system (SEPA). These tools will be aggressively applied. - International experts will be hired by AySA to provide advice on procurement and technical matters. - AySA will prepare a detailed time-bound plan for the different activities to be approved by the Bank before launching any pre-qualification process. - Legal instruments developed as part of the implementation arrangements for the project, as well as the Operational Manual, will clearly describe the roles, responsibilities, rules and reporting requirements on procurement.	H
Overall risk rate	H		S

Note: Rating of risks on a four-point scale: H – High, Substantial, Moderate, Low - according to the likelihood of occurrence and magnitude of potential adverse impact.

6. Loan/credit conditions and covenants

Conditions of Loan Effectiveness:

- (i) **The Borrower, through SAYDS, has adopted the Operational Manual;**
- (ii) **The AySA Concession Contract has been signed, and said contract is in full force and effect;**
- (iii) **The AySA Agreement (which sets forth AySA's responsibilities under component 1) has been signed by the parties thereto, and said agreement is in full force and effect;**
- (iv) **The ACUMAR Resolution (which sets forth ACUMAR's responsibilities under the project) has been issued, published in the Borrower's official gazette (*Boletín Oficial*), and said resolution is in full force and effect; and**
- (v) **The Borrower, through SAYDS, has established the UEP.**

Main Loan Covenants:

- (i) The Borrower, through SAYDS, shall cause AySA to carry-out a pre-feasibility study of alternatives to the construction of the Right Bank collector (*Colector Margen Derecha*) no later than March 31, 2010.
- (ii) The Borrower, through SAYDS shall cause AySA to carry out separate site environmental impact assessments acceptable to the Bank for the Right Bank collector, or a suitable alternative approved by the Bank, and for the industrial collector (including its pumping station), and thereafter prepare and implement an environmental management plan for these civil works.
- (iii) Resettlement is not expected under APL1. Considering the complex nature of works under Component 1 and that specific subprojects under Components 2 and 3 are not known (matching grants for SMEs and infrastructure investments, respectively), in case resettlement becomes necessary, the Borrower through SAYDS shall, or shall cause AySA to, prepare a resettlement plan acceptable to the Bank consistent with the Resettlement Policy Framework, as the case may be; and thereafter carry out the pertinent resettlement plan in accordance with its terms.
- (iv) The Borrower, through SAYDS, shall: (i) cause AySA to carry out component 1 of the Project in accordance with the provisions of the Project Environmental Management Plan, the AySA Environmental Management Plan, and the Operational Manual; (ii) carry out components 2 and 3 of the Project in accordance with the provisions of the Project Environmental Management Plan, the Environmental Management Framework for component 2 of the Project, the Environmental Management Framework for component 3 of the Project, and the Operational Manual; and (iii) carry component 4 of the Project in accordance with the provisions of the Operational Manual.
- (v) The Borrower, through SAYDS, shall cause AYSA to prepare and furnish to the Bank an environmental management plan for the Berazategui sewage treatment plant not later than six (6) months after the Effective Date.
- (vi) The Borrower, through SAYDS, shall cause AySA to, prior to the commencement of the main works under component 1 (Left Bank collector, the Baja Costanera bypass collector, the Riachuelo preliminary treatment plant, inflow and effluent pumping stations at the Riachuelo preliminary treatment plant, the Riachuelo outfall, the Industrial collector, and the Right Bank collector), hire one or more firms (with qualifications and experience, and under terms of reference, all acceptable to the Bank) to supervise construction.

D. APPRAISAL SUMMARY

1. Economic and financial analyses

75. The Matanza-Riachuelo Basin Sustainable Development Project (APL-1 and APL-2) is estimated to render a net economic benefit and, in particular, to favor unserved population groups. Detailed financial, economic, and distributive analyses carried out during preparation, complemented by sensitivity and risk assessments, are presented in Annex 9.

Economic Analysis

- Cost Benefit Net present value (NPV) = US\$0.5 billion
 Economic rate of return (ERR) = 22%
- Cost Effectiveness
- Other

76. The economic analysis for this project was carried out for the following activities: (i) sewerage expansion, (ii) improvement of water quality in the M-R River only, and (iii) improvement of water quality in both the M-R and La Plata Rivers. Economic benefits were estimated based on a combination of revealed and stated preference approaches. The revealed preference approach, which uses market information indirectly, was used to estimate (i) the benefits of sewerage expansion through avoided costs and hedonic pricing methods and (ii) the benefits of water quality improvement in the M-R River through hedonic pricing methods. The contingent valuation method (a stated preference approach), which is based on asking households how much they are willing to pay for an environmental quality improvement, was used to estimate the benefits of clean-up of the M-R River and of both the M-R and La Plata Rivers. Special attention was paid to ensure that these benefits were not double counted. The alternative combinations of methods that were used to estimate the total benefits of the project are detailed in Annex 9.

77. A *contingent valuation survey* was designed to evaluate the benefits expected from water quality modeling by ACUMAR, i.e., the elimination of unpleasant odors along the entire length of the M-R River, the possibility of using the M-R and La Plata Rivers for recreation without direct contact with water, and protection of the water intakes for the public water supply system. Results of this survey showed that the willingness-to-pay (WTP) for cleaning up both rivers was US\$3.12 per household per month, compared with US\$2.63 per household per month for the M-R River only. Also, as expected, the average WTP for households closer to the M-R River was significantly higher (US\$3.65 per household per month) than that for households farther (over 20 blocks) away (US\$1.44 per household per month). Similar figures for the clean-up of both rivers were US\$3.68 per household per month for houses nearer the river and US\$2.40 per household per month for houses farther away.

78. The *hedonic pricing approach* was used to estimate benefits relating to (i) sewerage connections and (ii) clean-up of the M-R River, using a series of attributes of properties and their location. For sewerage connections, results of the 2008 Real Estate Survey of 1,000 properties (empty lots, apartments, and houses) conducted as part of this economic evaluation revealed statistically significant higher average prices of houses (31 percent increase), apartments (24 percent increase), and empty lots (8.5 percent increase) with a sewerage connection. For cleaning up the M-R River, which would eliminate the river's bad odor, there is an important difference between the average prices of properties based on odor, and the presence of odor is strongly correlated with proximity to the river. Descriptive statistics from the survey data showed that property prices fall as proximity to the M-R River increases. Distance was used as a proxy for the presence of odor after controlling for other variables that are correlated with distance in the hedonic price function. The resulting average 9 percent increase—which varies from 3 to 18 percent depending on the type of property and distance from the river—was applied to the properties within 20 blocks of the M-R River to estimate the total benefits of removing the river's odor.

79. The *avoided cost method* was used to estimate the benefits relating to (i) the sewerage connection and (ii) the protection of water intakes. Avoided cost estimates associated with sewerage connections were based on current costs of different on-site sewage disposal systems. At present, 65 percent of households located in the M-R River Basin do not have access to a sewerage connection and use other solutions for sewage disposal: septic tanks (30 percent), holes with no septic chamber (17 percent), and other solutions (18 percent). The avoided costs associated with these private sewerage solutions range from US\$47 to US\$292 per household per year based on the data collected as part of this economic analysis. Avoided costs resulting from the protection of the Bernal water intake were provided by AySA.

80. *Health benefits.* The expansion of sewerage is expected to result in important health benefits, primarily from a reduction in diarrheal mortality among children under 5 and diarrheal morbidity of the total population. The cost-of-illness approach was used to estimate health benefits resulting from the provision of sewerage and the protection of drinking water quality at intake sites as an alternative for quantifying benefits under the avoided cost method. The annual benefits resulting from the reduction of diarrheal mortality in children under 5 and diarrheal morbidity in the total population are estimated at US\$29 per person per year from sewerage expansion and US\$6 per person per year from the protection of water at the Bernal water intake. While the health benefits from sewerage expansion and water intake protection are substantial for society as a whole, it should be noted that for the water utility (AySA), the with-project scenario will result in avoided costs from water intake protection. The health benefits were used to calculate total benefits in one of the alternatives presented in detail in Annex 9.

81. Among the different combinations of approaches used to calculate the total economic benefit of the project, the one selected for this project included (i) hedonic pricing for sewerage connections, (ii) hedonic pricing for households less than 20 blocks from M-R River, and (iii) contingent valuation for households in Greater Buenos Aires more than 20 blocks from the M-R River. The value of estimated health benefits was not included in the selected approach. The selected approach was considered the most robust as it uses hedonic pricing data that more fully capture the benefits of sewerage provision and the elimination of odor than the alternative revealed preference methods (avoided costs and cost of illness). The results obtained using this combination of valuation methods show a net benefit of US\$0.5 billion and an ERR of 22 percent.

82. Sensitivity and risk analyses confirm the robustness of the project, which has a 67 percent probability of having positive returns.

Financial Analysis of the Project

NPV = –US\$480 million

83. The financial viability of the program (APL-1+APL-2) was assessed from the perspective of the entity implementing the project (AySA). It is based on the actual prices that AySA will pay and receive for the project over a period of 30 years. Only O&M costs were taken into consideration and a 100 percent subsidy on investment was included in the evaluation, as a grant from the Federal Government. Revenues from tariff were considered constant in real terms over 30 years for the analysis. Taxes were also included.

84. The results show a net loss of the program of US\$480 million. The financial analysis of the program differs from the financial analysis of the entity (see Annex 11B) that operates the project, i.e. AySA; and hence the conclusions reached for the financial viability of the program do not necessarily apply to the financial viability of the company. The main differences between these two analyses are: (i) the financial evaluation of the program only calculates the incremental financial benefits and costs generated by each activity of the program (sewerage component and clean-up of the M-R and de La Plata Rivers); (ii) the financial evaluation of the program addresses only the cash flow generated by the program, without including other services provided by the company, for instance water supply.

2. Technical

85. Taking into account the existing sewerage system, which routes wastewater to the Berazategui outfall, and the geographical setup of the M-R River Basin in relation to the service area of AySA, the proposed project provides a combined solution for the treatment and disposal of the wastewater of the M-R River Basin and the entire concession area of AySA, which has a total population of about 10 million. Moreover, the project takes into consideration the liquid effluents of the approximately 4,000 industries located in the M-R River Basin.

86. Mathematical models for simulating the water quality of the M-R River system and the coastal area of La Plata River near Buenos Aires were developed during project preparation and were used for the technical assessment of the alternatives and their comparison. The main finding of the modeling analysis is that the M-R River is unable to absorb virtually any wastewater and effluent discharges because of its limited auto-purification capacity, especially in the dry season. Consequently, a policy of progressive elimination of point source discharges is needed, aimed at achieving with time the elimination of all identifiable wastewater discharges to the river, consistent with the uses and objectives defined by ACUMAR. With this goal in mind, the defined alternatives were compared using technical, economic, environmental, and social criteria. As a result of this comparison, the most favorable alternative was selected and its stages of execution determined.

87. The most suitable alternative, which from a combined standpoint offers lowest cost, technical advantages, simple operation and maintenance, the greatest positive environmental impacts, the lowest adverse impacts, and minimum negative social impacts, was found to be the distribution of the total collected wastewater flow to two preliminary treatment plants -- one located on the southern bank of the M-R River at its confluence with La Plata River and the other in Berazategui -- and the discharge of the effluents of these treatment plants to La Plata River through two long and effective sub-aquatic outfalls.

88. These conclusions are the result of mathematical modeling of the impact of the analyzed alternatives on the water quality of La Plata River. They are also compatible with the recommendation of the World Health Organization (WHO)²⁰ that, from the public health standpoint of human exposure to wastewater, additional treatment beyond preliminary treatment

²⁰ WHO. 2003. Guidelines for Safe Recreational Water Environments. Volume 12, Coastal and Fresh Waters. Geneva, p. 80. Available at http://www.who.int/water_sanitation_health/bathing/srwe1/en/.

prior to discharge to the receiving body through an effective outfall does not reduce health risks. That is, the health risk associated with a system composed of preliminary treatment followed by an effective outfall is identical to that associated with a system composed of secondary treatment followed by an effective outfall. The WHO report refers to risks related to human exposure through bathing, which is not exactly the risk in the case of discharge to La Plata River. However, because the main risk of effluent discharge in the La Plata River system comes from pathogenic organisms, the exposure risk referred to in the WHO report is a good proxy for the risk relating to La Plata River.

89. The comparison of the industrial pollution abatement alternatives demonstrated that the alternative of providing an appropriate level of treatment of all industrial wastes and discharging the effluents, except those containing toxic matter, into the municipal sewerage network for further processing with the domestic wastewater is the most suitable alternative. This alternative was coupled with the alternative selected for the sanitation component in the sense that the simulation model confirmed that the organic matter loads –including those of industrial origin – discharged to La Plata River through the two outfalls, can be absorbed by the river and will not generate water quality problems. Similarly, the conveyance and treatment systems were dimensioned to include all the industrial waste flows generated in the M-R River Basin.

90. In terms of execution stages, the first stage of the sanitation component of the project will include (i) the main collection system, consisting of the Left Bank collector and the Baja Costanera bypass collector, (ii) the Right Bank collector (or a suitable alternative for the same function), (iii) the industrial waste collector to Berazategui, (iv) the Riachuelo preliminary treatment plant including the inflow and effluent pumping stations, and (v) the Riachuelo sub-aquatic outfall. The second stage of the sanitation component will include (i) the Baja Costanera alleviation collector, (ii) the Berazategui preliminary treatment plant (being constructed by the Government), (iii) the Berazategui effluent pumping station, (iv) the Berazategui sub-aquatic outfall, and (v) additional primary and secondary sewerage networks in the M-R River Basin.

91. The system of preliminary treatment followed by an effective outfall is widely used in coastal cities around the world. About 140 such systems are in operation in Latin America, 40 of them in Chile. Montevideo, Uruguay, uses the same system, which also discharges its effluent to La Plata River through one existing outfall with a second one being planned.

92. A major part of the project consists of construction of the large pipes, including the outfall pipes, in diameters between 0.8 m and 4.5 m, using tunneling methods of construction. Most of the pipes will be constructed using segment-lined tunnels. The smaller-diameter pipes may be constructed using the pipe jacking method. Corrosion-resistant polymer concrete is a good option for the pipe material. Internally lined reinforced concrete or fiber-reinforced polymer pipes are also corrosion resistant and may be used. As stated in the section on analysis of technical alternatives, the selected alternative will discharge the effluent at a great distance from the shore so that it cannot reach the area used as a water supply source. This has been confirmed by mathematical simulation of the various alternatives (see details in Annex 11A). In addition, prior to the final design of the sub-aquatic outfalls, extensive hydrometeorological data collection will be undertaken and used for more detailed modeling. This detailed modeling will serve as the basis for the design of the outfalls and the diffusers.

3. Fiduciary

93. An assessment of the Financial Management (FM) arrangements for the proposed project was carried out in accordance with OP/BP 10.02 and applicable policies and guidelines.²¹ Results of the FM assessment are presented in Annex 7. The proposed financial management arrangements meet Bank requirements. UCOFI's FM staff performance in previous Bank-financed projects²² and a review of AySA financial management arrangements in place indicate a positive attitude by management toward financial control. However, there is a moderate element of risk due to the large amount of lending involved in the proposed operation. To avoid any additional complexity and reduce associated risk, the simplest flow of funds mechanism for Component 1 has been proposed, as follows:

(i) The SAYDS, through the Coordination Unit for International Technical and Financial Management (*Unidad de Coordinación para la Gestión Técnica y Financiera Internacional*), or UCOFI will have overall responsibility for FM aspects of the project, comprising accounting and financial reporting, disbursements and documentation of expenditures to the Bank, and external auditing arrangements. UCOFI has administrative capacity and experienced staff skilled in WB-financed activities;

(ii) Flow of funds for Component 1. UCOFI will submit withdrawal applications to the Bank and handle all disbursement issues. Loan proceeds will be transferred from the Bank to the Designated Account (DA) in dollars managed by UCOFI to be opened in Banco de la Nación Argentina (BNA). UCOFI will transfer loan proceeds to a "special account" in pesos opened by AySA in a bank acceptable to the Bank, from which AySA will make payments for eligible expenditures under Component 1. In order to monitor the uses of Loan funds, this bank account should not receive funds from other sources. In addition, transfers from UCOFI will be made solely on the basis of eligible expenditures incurred (certifications of works) or advances supported by expected expenditure reports prepared by AySA on the basis of signed contracts and planned certifications of works as determined by the Bank;

(iii) There will be no transfers of loan funds either to municipalities or to other subnational government agencies.

94. Procurement for this project will be carried out in accordance with the World Bank's Guidelines: Procurement under IBRD and IDA credits, May 2004, as revised October 2006 (the Procurement Guidelines) and Guidelines: the Selection and Employment of Consultants by the World Bank Borrowers, May 2004, as revised October 2006 (the Consultant Guidelines). Details are recorded in Annex 8. In August 2008, the Bank carried out an assessment of the capacity of the two agencies involved in project procurement, UCOFI and AySA to implement procurement actions for the project and prepared a detailed action plan to address all risks identified, as presented in Annex 8. The overall procurement risk has been rated as High for both implementation agencies. The proposed risk mitigation strategy for the project includes the following main actions: (i) agreement has been reached that additional experts be hired by AySA and UCOFI to strengthen local procurement teams; (ii) AySA is preparing a detailed plan and

²¹ Financial Management Practices in World Bank-financed Investment Operations, issued by the FM Sector Board on November 3, 2005

²² Pollution Management Project (P006050) Loan 4281-AR. Native Forest Project (P040808) - Loan 4085-AR

timetable for the different activities involved in the sanitation works (pre-qualification, preparation of basic designs, carrying out of bidding processes, and supervision) to be approved by the Bank before launching any pre-qualification processes; (iii) the detailed division of roles, responsibilities and reporting requirements for UCOFI, UEP and ACUMAR on the activities to be performed by each of the entities involved in the implementation of the Components 2, 3 and 4. The procurement section of the Operational Manual would also include flow charts detailing technical, administrative, and contract management tasks and responsibilities, as well as the expected timeframe for each of them; (iv) special Procurement Provisions will be included in the Loan Agreement aimed at increasing transparency, competition, and civil society monitoring while addressing the aspects of domestic legislation that are inconsistent with Bank procurement policy; (v) obtaining the commitment of the project entities to use the publicly accessible System of Implementation of Procurement Plans (*Sistema de Ejecución de Planes de Adquisiciones*, or SEPA) to manage procurement planning and execution; (vi) selecting prior review thresholds consistent with the assessed level of risks; (vii) ensuring that supervision of large sanitation works will be carried out using specialized consulting firms; and (viii) carrying out joint FM-Procurement supervision missions twice a year.

4. Social

95. The M-R River Basin is home to Argentina's largest concentrations of urban poor. The poorest populations live in low income settlements alongside the river, without adequate and safe water supply, sanitation, and garbage collection and are in constant contact with numerous contaminants ranging from untreated organic waste to toxic industrial chemicals. Frequent flooding of the M-R River spreads highly polluted waters into informal settlements, exposing the inhabitants to further contamination. High rates of health problems are the result of the poor environmental conditions of the basin.

96. The improvement in the quality of the river targeted by the proposed project will directly benefit the 3.5 million inhabitants in the basin, in particular the 1.2 million inhabitants below the poverty line. The main benefit will be the improvement of environmental and health conditions, through the introduction of sewage collectors which in turn, will enable water and sanitation expansion, especially to low income settlements.

97. The identified main adverse impacts of the project include the disruptions that sanitation construction would cause, proper disposal of solid waste generated by both treatment plants, and the possible reduction of jobs due to closure of industries or changes in technology. Given that most of the works financed by APL-1 will be done through tunneling, no displacement is expected. However, as a precaution, the project has developed a Resettlement Policy Framework in the event of any unexpected resettlement.

98. In order to identify the project's positive social impacts and avoid or mitigate the potential negative impacts, two social assessments were carried out during project preparation. These two assessments are complementary: one focuses on the specific potential impacts associated with the execution of the large sanitation infrastructure works, and the other focuses on the impacts of the comprehensive set of actions proposed by the project. The conclusions and recommendations emerging from these assessments were taken into account in the project design in order to improve the social benefits generated by the project.

99. The methodology adopted by these assessments ensured a meaningful consultation process, in which the broad range of stakeholders associated with the project fully participated in detailed discussions about project activities. The adopted methodology included focus groups, workshops, field visits, interviews and several meetings at the request of the stakeholders involved. Municipal and provincial governments, labor unions, private sector entrepreneurs, academia, community-based groups, and NGOs were among the stakeholders consulted.

100. The main finding of the SAs was that all stakeholders were in favor of the project, considering the present scenario whereby given by the intervention of the Supreme Court and with the assistance by the World Bank, the environmental improvement of the M-R River Basin becomes achievable. None of the interviewed stakeholders opposed the overall project. However, critical opinions about various aspects of the project were expressed, such as inadequate information sharing and consultation, concerns over industry closures, higher tariffs, etc. (See Annex 11C for a complete list of description of stakeholder opinions).

101. As part of the project preparation process, extensive stakeholder consultation has been undertaken using formal as well as informal methodologies (see Annex 10 for more details). The project EIAs have been made available through the ACUMAR and AySA websites as well as through the Bank's InfoShop. Information on the project has been made publically available through public disclosure events and workshops. Informal focus group discussions have been conducted with NGOs, local governments, affected people and beneficiaries at various points during preparation and appraisal. Some important events included: surveys and focus groups during the "willingness to pay" study for the economic analysis (see Annex 9), a workshop for the consultation of the terms of reference of the EIAs conducted on 10 July 2008, a workshop for the dissemination of the results of the EIAs conducted on 7 November 2008, AySA meetings with 11 NGOs prior to the EIA results workshop, World Bank meetings and exchanges of letters with several NGOs active in the basin, and meetings with stakeholders conducted by the 2 consultant teams which carried out the Social Assessment studies of the project (see Annex 11C).

102. In addition to the project specific consultation and disclosure efforts, it is important to recognize that both AySA and ACUMAR have social outreach units within their organizational structure. These teams have been established to be in constant contact with local communities, and are used to facilitate communication tools. ACUMAR, for example, has established a network of approximately 300 community and environmental NGOs that are part of its broader communications strategy (see Annex 11D) about the issues facing the M-R River Basin. This network provides an opportunity for information disclosure not just on the project but on a wide array of development issues facing the basin.

103. Through the consultation process, stakeholders have provided feedback and expressed their concerns on a range of topics including the TOR and scope of the EIA studies; the scope of the project including the levels of treatment selected and alternatives considered; the risks associated with pollution discharges to La Plata River; the sequencing of project works and their adequacy to address the scale of the problems faced by in the M-R basin; the licensing and clearance procedures that will apply to the project works; government capacity to implement their responsibilities under the project and how disruptions to daily life will be handled during the construction phase. For a more complete discussion of issues raised see Annex 11C, Social

Assessment and Public Consultation. As a result of the consultations, some important aspects of project design were modified. In particular, based on stakeholders feedback about the need to remove contaminant flow from the upper middle basin and right bank to the M-R River, the construction of the Right Bank Collector was advanced from APL-2 to APL-1 to provide the Government with an instrument for accelerated basin clean-up in the short and medium term.

104. ACUMAR will develop a Social Management Plan to integrate and coordinate the approaches suggested in the two social assessments and the project's communication strategy for addressing identified social impacts, risks and opportunities under the project. ACUMAR and AySA will confirm the institutional arrangements for coordination of the Social Management Plan to address: (i) the resettlement framework; (ii) the management and mitigation of impacts during the construction phase of Component 1; (iii) the management and mitigation of impacts during the operation phase of Component 1; (iv) the management and mitigation of impacts associated with industrial conversion of Component 2; (v) monitoring; and (vi) stakeholder consultation, communication and disclosure processes.

105. In particular, Communication Management Plan (see description of component 4 in Annex 4 as well as Annex 11D) has been designed with ACUMAR, in order to inform key actors and stakeholder groups in the basin about the project in a transparent manner, strengthen the credibility of and support for the project, and contribute to the overall consultation and participation framework for the project. The Communication Management Plan includes the creation of a Communication Unit in ACUMAR, which would be responsible for (i) providing adequate and timely information prior to consultation events; (ii) maintaining documentation on consultation processes and decisions taken; and (iii) providing feedback to concerned stakeholders. Staff in the new unit will coordinate with those in AySA's Communication Unit. This unit has already started to improve the communication of the project within the company by harmonizing project information among central and regional offices, and externally, by holding 10 workshops with NGOs which had highlighted the shortage of project information.

106. In response to the institutional weaknesses of ACUMAR's Social Participation Commission and lack of opportunities for stakeholder participation, component 4 will focus on their institutional strengthening. Close coordination between the Social Participation Component, Environmental Unit and Communications Unit will be key to ensuring that engagement with key stakeholders throughout the implementation of the environmental management plan and consultations on social issues remain a two-way dialogue.

107. Population resettlement: No resettlement is expected as a result of the project components financed under APL-1. The technology adopted to execute the sanitation works, which includes underground tunnels and construction of shafts to access the underground tunnels only on land owned/leased by AySA, as well as the design of the sewage collector routes has been conceived to minimize adverse impacts and avoid resettlement, and no displacement is expected regarding industrial and urban improvements. However, given the complexity of the works, and since the physical interventions of components 2 and 3 of the project have not yet been fully determined, a Resettlement Policy Framework has been prepared for the project as a precaution. As pilot and initial phase interventions for components 2 and 3 are selected, minimizing resettlement will be one of the selection criteria.

108. As described earlier, the Government has advanced the construction of the Berazategui preliminary treatment plant (BPTP). The World Bank team was recently informed (March 2009) that the construction site required resettlement of 18 families that had illegally occupied the BPTP site and erected precarious sheet-metal dwellings. In October 2008, families were relocated to new concrete houses approximately 2-3 km from the BPTP site. The Bank considers that the relocation of families from the BPTP site does not require application of OP 4.12 since construction of the BPTP is neither directly related to the project (APL-1) nor necessary to achieve its objectives. This conclusion notwithstanding, the Bank team conducted a preliminary assessment of the resettlement program and has found no evidence to indicate worsened living conditions. The Bank will continue to monitor living conditions going forward.

109. Mechanisms for mediation, conflict resolution, and redress. There are several alternative and robust mechanisms at various levels available to citizens who wish to seek information or complain if they feel the project does not adequately address their needs or concerns. At the lowest level, there are project consultation mechanisms through the Social Participation Commission (created within ACUMAR for channeling comments and proposals from civil society organizations to ACUMAR's management), as well as submissions to the specific administration unit or institution (ACUMAR, SAYDS, or AySA). At the next level, external mechanisms include the presentation of claims to a Collective Body ("Cuerpo Colegiado") led by the National Ombudsman. Alternatively, people can directly file "*amparo*" claims (restraining orders) before local, provincial or national courts, which may eventually be redirected to the Federal Judge of Quilmes. Claims related to AySA's conduct can be submitted to the company using different ways of communication (telephone, office visits or email); and in the case of non-response, users can file a claim with the Regulatory Entity for Water and Sanitation (*Entidad Reguladora de Agua y Saneamiento, or ERAS*) (see Annex 11C).

5. Environmental

110. *Environmental benefits.* The principal aim of the project is to improve the environmental conditions of the M-R River basin by reducing levels of contamination from various domestic and industrial discharges. Specifically, the sanitation component is designed to eliminate sewage discharges to the M-R River through the provision of sewage collection, treatment, and disposal infrastructure. The industrial pollution abatement component will help improve environmental compliance and monitoring and will finance investments in cleaner production technologies for SMEs in order to reduce toxic industrial pollutants discharged to the M-R River. The environmental territorial management component is designed to address urban land use and flood management problems. Finally, the technical assistance and project management component will finance additional studies and overall project management activities to improve the efficiency and effectiveness of the main counterpart agencies.

111. Overall, the environmental benefits of the project are substantial and are expected to outweigh environmental risks. Expected benefits include: (i) the improved quality of life for the 3.5 million inhabitants of the M-R River Basin through reduced exposure to domestic and industrial pollution, (ii) improved sewerage and drainage services in the M-R River Basin and other parts of Buenos Aires by increasing the capacity of the sewerage network enabling AySA to expand service to 1.7 million people in the basin (from 35 percent to about 75 percent), especially in poor marginal areas, (iii) improved environmental quality through investments in

pollution control for rivers and tributaries crossing the urban area and for La Plata River, and (iv) improved urban living conditions associated with territorial development and reduced flooding in the M-R River Basin.

112. *Environmental Risks of Component 1.* Through the EIA process, a number of potential environmental risks were identified for the construction and the operational phases of Component 1. During the construction phase, the main environmental risks relate to the management and control of the construction processes involved in the large civil works. These risks include: (i) impacts typically associated with large-scale construction projects (e.g., dust and noise control), (ii) management of hazardous materials, (iii) control of erosion and sedimentation, (iv) disposal of tunnel spoil and large earth movement, (v) re-suspension of potentially hazardous sediments during river dredging, and (vi) management of work sites as well as worker and public health and safety concerns.

113. During the operational phase of the project, the potential environmental risks of component 1 include: (i) the human and ecological health risks associated with contamination of the Rio de la Plata resulting from the operation of the treatment plants and associated sub-aquatic outfalls (ii) nuisance noise and odor associated with the operation of the treatment plants and pumping stations and (iii) disposal of the bio-solid waste generated by the treatment plants. The benefits and risks have been described and evaluated in the project's EIA and supporting technical analyses (see Section 6 below). A summary of the potential environmental impacts, risks, and mitigation measures is presented in Annex 10 of the PAD.

114. *Environmental Risks of Components 2 and 3.* Environmental risks associated with components 2 and 3 are less significant than component 1 and are expected to be offset by positive benefits. Potential adverse impacts could be associated with the construction of civil works for flood control and drainage under component 3 but these works are expected to be small scale with limited potential for negative impacts. Under component 2 there may also be risks associated with legacy environmental issues at industrial sites that may be beneficiaries of the project and in the demolition and disposal of outdated industrial processing and manufacturing equipment and stockpiles of hazardous materials.

115. Because most of the specific works investments to be made under these components have yet to be identified it is not possible to ascertain with absolute certainty the nature or extent of the risks. Thus, for both of these components an Environmental Management Framework (EMF) has been developed which describes, among other issues (i) the eligibility criteria for works to be funded under the project and (ii) procedures for screening and scoping potential issues for all proposed investments and, where appropriate, developing management and mitigation measures.

6. Safeguard policies

116. The proposed project is categorized as an Environmental risk category "A" based on the expected long-term impacts of the large-scale sanitation civil works investments under Component 1. Works to be carried out under Components 2 and 3 are expected to have impacts that are mostly positive and where adverse impacts may occur they are expected to be small-scale, temporary, local in nature, and can be addressed with known mitigation measures. Table 5 summarizes the applicable Bank safeguard policies.

Table 5. Safeguard policies triggered by the project

Safeguard policies	Yes	No
Environmental Assessment (<u>OP/BP</u> 4.01)	[X]	[]
Natural Habitats (<u>OP/BP</u> 4.04)	[X]	[]
Pest Management (<u>OP</u> 4.09)	[]	[X]
Physical Cultural Resources (<u>OP/BP</u> 4.11)	[X]	[]
Involuntary Resettlement (<u>OP/BP</u> 4.12)	[X]	[]
Indigenous Peoples (<u>OP/BP</u> 4.10)	[]	[X]
Forests (<u>OP/BP</u> 4.36)	[]	[X]
Safety of Dams (<u>OP/BP</u> 4.37)	[]	[X]
Projects in Disputed Areas (<u>OP/BP</u> 7.60) ²³	[]	[X]
Projects on International Waterways (<u>OP/BP</u> 7.50)	[X]	[]
Piloting the Use of Borrower Systems to Address Environmental and Social Safeguard Issues in Bank-Supported Projects (<u>OP/BP</u> 4.00)	[]	[X]

117. *Environmental Assessment (OP/BP/GP 4.01).* Environmental assessments have been prepared for the project at two levels of analysis. At the overall project and river basin level, an Integrated Environmental Assessment has been prepared which describes the regional environmental issues facing the M-R River Basin, the baseline environmental situation, the applicable legal and institutional framework guiding the management of the basin, the principal management strategies considered, and alternatives evaluated for different basin management approaches. In addition, the Integrated Environmental Assessment (IEA) provides a comprehensive description of the overall project components, their main environmental risks and benefits, and the measures to be employed during project implementation to ensure management of adverse environmental risks.

118. A seven-volume Environmental Impact Assessment (EIA) for Component 1 describes AySA's strategic corporate vision, the Integrated Plan developed for its concession area, baseline environmental conditions in the project's area of influence, and the applicable legal and institutional framework guiding AySA's sanitation works. The EIA report from AySA considers realistic alternatives to each proposed work and evaluates the proposed works in the context of AySA's five year investment plan (*Plan Director*). A comprehensive EMP has been developed, spelling out institutional responsibilities for implementation of mitigation actions as well as supervision and monitoring of compliance with agreed mitigation measures. For example, all construction contractors will be required to follow site-specific management plans agreed on and incorporated into all construction contracts.

119. A panel of four independent experts²⁴ was hired by the Bank to review the draft EIA reports and to provide comments and recommendations to ACUMAR and AySA in order to improve the quality of their EIA reports. The draft reports of the panel of experts were shared

²³ By supporting the proposed project, the Bank does not intend to prejudice the final determination of the parties' claims on the disputed areas

²⁴ See Annex 11 for the names of the experts.

with AySA and ACUMAR and were used by these institutions to improve the final version of the EIA reports.

120. The AySA EIA includes site-specific EIAs for works included in APL1 for which final technical solutions have been identified. For those works at or before pre-feasibility stage (such as the right bank collector or a suitable alternative for the same function), site-specific EIAs will be prepared by AySA using the same assessment methodology. AySA will apply the stakeholder consultation approach used for the other civil works under APL-1. This EIA, addressing construction and other site-specific impacts, will be carried out once the final technical solution is determined. The analysis of impact on water quality of the right bank collector or other equivalent alternative was already included in the Integrated Environmental Assessment by modelling the environmental and water quality effects of the wastewater diverted from the M-R River.²⁵

121. For Components 2 and 3, an Environmental Management Framework (EMF) has been prepared describing the procedures for screening and evaluating any proposed sub-projects. Where needed, an EMP will be developed for sub-projects that will address specific issues identified during screening. The frameworks also describe the procedures and expectations with respect to disclosure of information and consultation with affected communities.

122. EIA documentation has been made available for public review and comment. During project implementation, ACUMAR and AySA will carry out a strategic communications plan for the project that will define the methods for public disclosure of all relevant documents, including monitoring reports during implementation.

123. *Natural Habitats (OP/BP 4.04)*. Within the project's area of influence there are five known terrestrial ecological reserves and no existing aquatic reserves. Based on studies conducted by the GEF funded project "*Proyecto de Protección Ambiental del Río de la Plata y su Frente Marítimo*" (commonly known as FREPLATA), priority areas for protection of aquatic biodiversity have been identified but these areas are far from the immediate impact zone and not likely to be affected by the project.

124. Because the project aims to contribute to the significant improvement of water quality conditions in the M-R River Basin and in the coastal areas of the Río de la Plata, it is expected to have significant positive impacts on the natural riverine habitats found in the project area. An area of the receiving waters in La Plata River will be degraded as a result of the pollution discharge from the wastewater treatment plant via the outfall. Models developed for the outfalls indicate, however, that the discharge plume is expected to be fully contained within a relatively small area of the receiving waters. The pollution is not expected to affect downstream coastal areas designated as conservation areas.

125. *Pest Management (OP 4.09)*. The project will not procure nor will it lead to increased use of pesticides. Therefore no action is required under this policy.

126. *Physical Cultural Resources (OP 4.11)*. Preliminary site evaluations for proposed civil works construction do not show any evidence of cultural property that will require mitigation or

²⁵ Refer to paragraph C.6 of this PAD

management measures. However, since many of the proposed works are underground or will require some excavation, all construction contracts will include a “chance finds” clause that will require contractors to halt construction if any underground cultural property sites are encountered during construction. Site specific management plans for any cultural property identified by chance finds procedures will be developed before construction can start up again.

127. *Involuntary Resettlement (OP 4.12)*. Because the vast majority of construction will take place underground or on sites that do not require any land acquisition or resettlement, the project expects no resettlement. However, given the complexity of works, and since the physical interventions of components 2 and 3 of the project have not yet been fully determined, a Resettlement Policy Framework has been prepared for the project as a precaution. Among other things, the Resettlement Policy Framework spells out eligibility criteria, forms of compensation, and dispute resolution mechanisms that will be available to potentially affected people.

128. *Indigenous People (OP 4.10)*. No indigenous people will be affected by this project, and therefore no action is required under this policy.

129. *Forests (OP 4.36)*. There are no forests affected by the project and there are no activities proposed which would promote improved forest management. Therefore no action is required under this policy.

130. *Safety of Dams (OP 4.37)*. The project does not finance the construction or rehabilitation of dams therefore no action is required under the policy.

131. *Projects in Disputed Areas (OP/BP/GP 7.60)*. The project is not being implemented in any area known to be the subject of international dispute as defined under the policy. Therefore, no action is required under the policy.

132. *Projects on International Waterways (OP/BP/GP 7.50)*. Since both proposed sub-aquatic outfalls (the Riachuelo outfall at Dock Sud during APL-1 and the Berazategui outfall in APL-2), discharge to the Rio de la Plata, a bi-national waterway shared with Uruguay, OP 7.50 is triggered. Notification to Uruguay has been made through the bi-national “Comision Administradora del Rio de la Plata” (CARP). On 13 Aug, 2001, Argentina's then Ministry of Infrastructure and Housing submitted a request to CARP to authorize construction of an outfall at Dock Sud. CARP replied on 23 May, 2002, that it had “no objections” to the works, as long as several conditions were met (analysis of possible algae blooms in the diffuser area, establishment of an environmental monitoring & control program, determination of the types of wastewater that would be discharged and definition of pre-treatment standards for industries, etc.). On Sept 11, 2008, AySA (through the Sub-secretariat of Water Resources in the Ministry of Planning, Public Works and Services) presented an updated submission to CARP. Subsequently, on 28 January, 2009, Argentine representatives gave a presentation of the project before CARP and answered questions from Uruguayan representatives. The Argentine government has now received a reply from CARP that it does not express an objection to the proposed project.²⁶

133. *Disclosure*. As part of the project preparation process, a series of public disclosures, workshops, and focus group meetings were conducted. Others are being planned during

²⁶ See Acta 444 dated 27 January, 2009, from CARP available in the project files.

implementation in order to disseminate information on the project, solicit stakeholder concerns, and ensure an open process of debate and discussion about project objectives and goals. A major public consultation event organized by ACUMAR and AySA was held on 10 July, 2008, in Buenos Aires. The objective of the workshop was to present and discuss the terms of reference for the EIAs proposed for the project. This scoping workshop was attended by 140 participants representing a wide range of public and private stakeholders, including representatives of SAsyDS, ACUMAR, AySA, the Province of Buenos Aires, municipal governments, national and local environmental NGOs, national government agencies, community organizations, and professional associations. A second public workshop to review the findings and recommendations from the advanced drafts of the EIAs was held on 7 November 2008 in Buenos Aires. The revised EIAs document the range of meetings held, the issues raised, and the ways in which stakeholder views were taken into account and are available through the webpages of SAsyDS, ACUMAR, AySA and the Bank's InfoShop.

7. Policy exceptions and readiness

134. No policy exceptions are foreseen. The project is deemed ready for implementation for the following reasons:

- the proposed project has been thoroughly discussed with civil society and there is broad support for its implementation; and
- engineering designs for the first year of works are well advanced and the prequalification process for contractors is already being prepared.

135. Given that a draft of the Operational Manual has been presented to the Bank at negotiations, the finalization of the Operational Manual prior to effectiveness is not expected to delay project implementation.

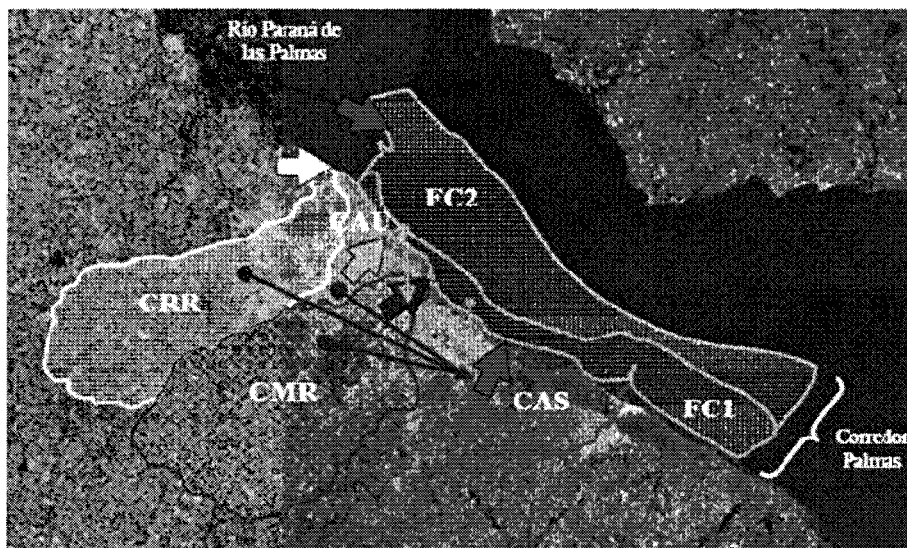
Annex 1: Country and Program Background

ARGENTINA: Matanza-Riachuelo Basin Sustainable Development Project

The Matanza-Riachuelo Basin

1. The Matanza-Riachuelo Basin lies in Buenos Aires Province and City and runs from the southwest to the northeast, discharging to Río de la Plata (La Plata River). Usage rights to the water in this estuarine river are shared between Argentina and Uruguay. La Plata River discharges to the Atlantic Ocean. The Matanza-Riachuelo River is known as the Matanza in the upper part of the basin and the Riachuelo in the lower part. The river's flow fluctuates between a minimum average of 2.9 m³/ second and a maximum average flow of 1,000 m³/second. Its mean slope is 0.04 percent, characteristic of this very flat region. The basin area is 2,240 km² and covers 14 municipalities and part of the City of Buenos Aires, with a total population of about 3.5 million. The Matanza-Riachuelo (M-R) and Reconquista river basins are the main river systems in the Metropolitan Area of Buenos Aires (Figure 1.1).

Figure 1.1. Interaction between different hydrological systems in the M-R River Basin



Notes: CRR = Reconquista River Basin, CAU = Urban Creek Basin, CM-R = Matanza-Riachuelo River Basin, CAS = South Creek Basin, FC1 = First Coastal Strip of La Plata River, and FC2 = Second Coastal Strip of La Plata River

2. The groundwater system in the basin comprises two aquifers. The shallow aquifer is called Pampeano or Epipelche and is extremely vulnerable to human activities. The deeper and less vulnerable aquifer is called Puelche. The Epipelche aquifer has suffered severe environmental damage, and the Puelche is used for productive activities and human water consumption.

3. The natural hydrological system has been altered by urban infrastructure including flood protection, drinking water networks, and sewerage systems, which import wastewater from the City of Buenos Aires into the basin. The M-R River's flow is also influenced by La Plata River.

These conditions, combined with the basin's topography, urbanization, groundwater system, and urban infrastructure, make the lower part of the basin very vulnerable to flooding.

4. The Matanza-Riachuelo River Basin should be seen as a territorial unit whose main development features result from the interaction of a typology of urban sprawl over a sensitive environmental area. Although various occupational patterns are found in the basin, in general the basin suffers from high levels of environmental degradation, especially in unregulated low-income urban areas.

5. Three different zones are clearly identified in the basin: (i) the lower basin, which is highly urbanized, (ii) the middle basin, which presents a combination of consolidated urban areas and peri-urban areas, and (iii) the upper basin, a rural area where environmental conditions are still adequate.

6. ***The lower basin.*** This is a medium-high-density urban area, with approximately 70 percent of the basin's 3.5 million inhabitants. It includes the southern part of the City of Buenos Aires and the municipalities of Avellaneda, Lanus, and Lomas de Zamora. Despite the generally high urban consolidation levels found in this part of the basin, there are still large vacant sections of land, zones where population density exceeds norms, and unregulated settlements. Land use is highly mixed, including residences, warehouses, and industries. The dominance of residential use increases in the upper portion of the lower basin, although this dynamic varies by area. Compared to the rest of the metropolitan area, this portion of the basin contains a large number of industries. The number of abandoned and dilapidated industrial buildings is striking, as are the numbers of illegal dumpsites and unregulated-low-income settlements along the riverbanks. The highest water contamination loads, resulting from raw domestic and industrial effluents, are found in this part of the basin. Nevertheless, good quality and planned residential settlements are also found here.

7. ***The middle basin.*** The urban density in this area varies from medium to low. The municipalities in this section of the basin are Almirante Brown, Esteban Echeverría, Ezeiza, La Matanza, and Merlo. Land use is predominantly residential, with the exception of commercial activities along the main transportation corridors and small and medium industries. Urban consolidated areas cover part of this portion of the basin, and urban sprawl is evident where low-density areas dominate and many large vacant land polygons are still found. There are also clubs and large green areas that offer recreational opportunities for the metropolitan population, as well as important logistical assets such as railroad warehouses and the largest food wholesale market in Buenos Aires. Illegal solid waste dumpsites are a major problem in the area, and the number of unregulated low-income settlements has increased significantly.

8. ***Upper basin.*** This area is predominantly rural, with some small and low-density urban centers. The municipalities in the area are Cañuelas, San Vicente, General Las Heras, and Marcos Paz.

B. Current Status of Water Quality

9. The Matanza-Riachuelo River has been used historically as the main drainage system of part of the City of Buenos Aires and some of its surrounding municipalities. The rapid increase

in population and accompanying regional economic development has significantly worsened the natural ecological environment of the basin. Currently the main environmental problems contributing to the deterioration of water quality are point and non-point source urban and industrial discharges into the river and groundwater system, as well as the uneven infrastructure development of potable water and sewerage networks, resulting in a net increase of wastewater inflows into the basin.

10. Figures 1.2 and 1.3 show the biological oxygen demand (BOD) concentration and dissolved oxygen (DO) concentration respectively, in the M-R River throughout its length. The figures show two situations with the low and medium flows that the river experiences more than 70 percent of the time. The figures show the high level of organic matter (measured in terms of BOD), one of the major causes of environmental degradation. This high level of organic matter, combined with low flow velocities, produces very low values of dissolved oxygen, resulting in an anoxic state, odors, methane emissions, unsightliness, limited use, and significant health risks.

Figure 1.2. Modeled and measured biological oxygen demand (BOD) concentrations along the M-R River.

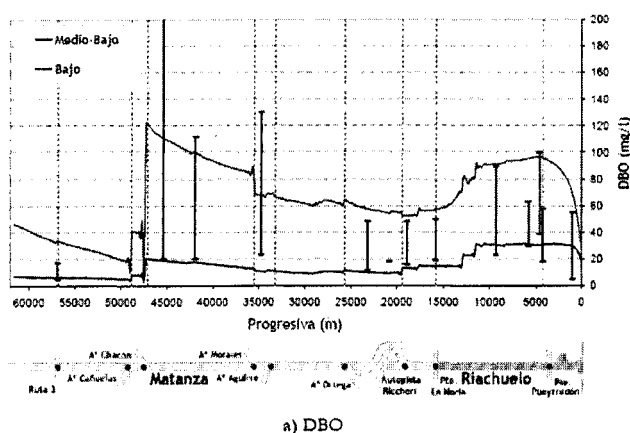
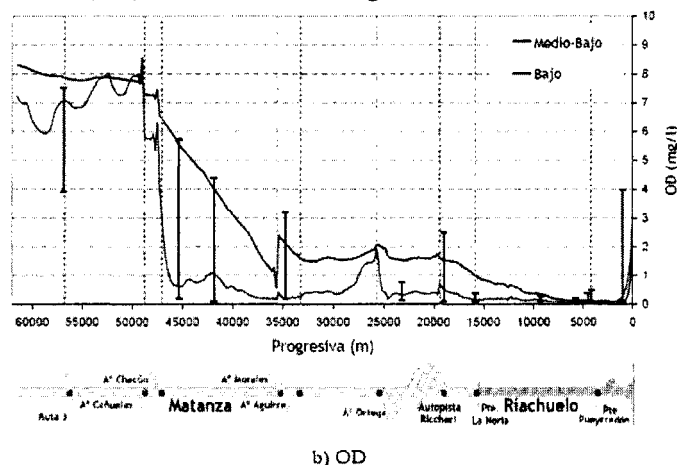


Figure 1.3. Modeled and measured dissolved oxygen (DO) concentrations along the M-R River



Source: ACUMAR M-R water quality modeling reports

11. Preliminary data suggest that the industrial sector is the largest source of organic contamination.²⁷ Table 1.1 shows a preliminary breakdown of the flows and characteristics of the inflows into the M-R River. As the table shows, most of the organic pollution comes from non-connected industrial discharges.

²⁷ While ACUMAR's Modeling Report No. 5 suggests that industries account for about 73% of the total organic pollution load, the Government has emphasized the need for additional data collection and modeling to validate these results.

Table 1.1. M-R River pollution sources and characteristics

Source	Flow (m ³ /s)	Total BOD (Tons/day)	% Total BOD
Point sources (Sudoeste, Jaguel and Cañuelas wastewater treatment plants and Cildañez stream)	3.9	12.0	12.9
Industrial non-connected sources (not in the sewage network)	0.7	67.9	72.8
Domestic non-connected sources (not in the sewage network)	1.6	13.5	14.5
TOTAL	6.3	93.3	100

Source: *Modelación Matemática de la Cuenca Matanza-Riachuelo, Informe #6*, Tabla 2.3.6, Universidad Tecnológica Nacional, Menéndez et al.

Causes of Water and Environmental Degradation

12. ***Institutional framework.*** The overlapping institutional responsibilities in this complex area have worsened the environmental conditions in the basin. A new Matanza-Riachuelo Basin Authority (*Autoridad de Cuenca Matanza-Riachuelo*, or ACUMAR) was created in 2006 with a broad mandate and enforcement authority to address the coordination issues in the basin. ACUMAR's board of directors includes representatives from the national, provincial, and municipal governments and is chaired by the Secretariat of the Environment and Sustainable Development (*Secretaría de Ambiente y Desarrollo Sustentable*, or SAyDS).

13. The institutions with some jurisdiction over the basin area include 11 institutions at the national level, 4 at the provincial level, 7 at the municipal level, and 14 municipalities and 4 institutions with a concession area larger than the City of Buenos Aires (but one that does not cover the whole basin).

14. ***Sewage management.*** The current sewerage system in the Argentina Water and Sanitation (*Agua y Saneamientos Argentinos S.A.*, or AySA) concession area is divided into four operational watershed systems: Norte, Sudoeste, el Jagüel, and Wilde-Berazategui. Each of these areas has a treatment plant. The average operational flow of the Norte watershed is 0.67 m³/s, that of the Sudoeste is 1.95 m³/s, that of El Jagüel is 0.15 m³/s, and that of the Berazategui is 23 m³/second. Most of the wastewater from the City of Buenos Aires is conveyed to the Berazategui District in the southwest and discharged to La Plata River through a sub-aquatic outfall.

15. Based on this situation, three main sewers were constructed, which convey an average flow of around 18 m³/second of wastewater to the existing outfall located in Berazategui. The Berazategui outfall—3.8 meters in diameter and 2.5 km in length—discharges the wastewater into La Plata River. The plan to construct a fourth main sewer to Berazategui, a preliminary treatment plant in Berazategui, and a longer outfall at this site with capacity to discharge the entire wastewater flow of the Buenos Aires metropolitan area, was never completed.

16. This system does not allow any flexibility or increase in network coverage with reasonable operating security. The existing metropolitan area wastewater evacuation system is clearly inadequate, as a significant portion of raw wastewater is currently discharged into the rivers and tributaries that cross the metropolitan area (principally to the M-R and Reconquista

rivers) or directly in the coastal zone of La Plata River. Consequently, the rivers and tributaries, as well as the coastal strip of La Plata River, are highly polluted. Moreover, the length and configuration (absence of diffusers) of the existing outfall at Berazategui do not allow adequate discharge to La Plata River.

17. AySA has designed a new plan that will allow it to incorporate the M-R River Basin's environmental objectives, expand its network coverage, and reduce the vulnerability of the system. For these purposes, AySA decided to divide the Wilde-Berazategui operational watershed system into two (Berazategui and Riachuelo), building a treatment plant and all the needed connection structures in this "new" operational watershed system and changing the sewage management structure in the basin.

18. **Industrial pollution.** The Matanza-Riachuelo is the most contaminated river basin in Argentina and one of the country's most visible environmental problems. Over the past hundred years, the M-R River Basin has been used as a sewage sink for the entire City of Buenos Aires. Pollution levels have steadily increased with urbanization in the metropolitan area of Buenos Aires and industrial growth within the basin. Based on estimations carried out in the M-R River Basin, there are more than 4,000 industrial facilities located in the lower and middle sections of the basin. Industries are typically located in clusters, but a significant number are also interspersed with residential areas (e.g., tanneries in Lanus). As a result, the pollution problem in the basin is compounded by the mixing of industrial and domestic wastewater, limited coverage of the sewerage network, and the absence of wastewater treatment. In areas without adequate sewerage, industrial and domestic wastewater is drained through open channels or infiltrates the groundwater.

19. The industrial growth in the basin has been rapid and includes both large enterprises and numerous small and medium enterprises (SMEs). Since 2002, the number of small and medium industries in the basin has grown significantly, with important industrial clusters in La Boca (in the City of Buenos Aires), Lanus, Avellaneda, and Lomas de Zamora. The main industries operating along the banks of the basin are food processing (dairy production and meat processing), tanneries, chemical, petrochemical, pulp and paper, textiles, and electroplating and metal production. Many of these industries dump their effluents directly into the river because of a lack of coordinated action among the various levels of government. In addition, the wide range of industrial types and sizes has made effective control more difficult.

20. **Existing capacity and pollution loads.** The current capacity of the sewerage system and the four existing wastewater treatment plants in AySA's concession area is far from sufficient to handle the existing domestic and industrial wastewater load. The vast majority of industries do not have any pre-treatment or process optimization before discharging wastewater into the sewerage system or surface water, resulting in significant increases in the organic and toxic loads in the existing sewerage network as well as the receiving water bodies.

21. Initial estimates suggest that the industrial sector contributes almost 68 tons a day of organic waste (measured by BOD), which accounts for 73 percent of the total BOD discharged into the M-R River Basin. A preliminary analysis was done of industrial contamination in the basin by industrial sector and pollutant type (organic as well as toxic substances). The analysis found that tanneries in the basin contribute about 26,000 tons of BOD per year, or 43 percent of

the total discharge into the basin, while meat processing and dairy industries together contribute about 10,300 tons of BOD annually, or 17 percent of the total. Levels of toxic effluents are also high, particularly for tanneries, which discharge 34 tons per year of chromium VI (100 percent of total discharge to the basin) as well as 20,000 tons/year of sulfates (100 percent of the total discharge to the basin). Toxic discharge from the metallurgical industry is also significant and includes chromium, zinc, and cyanide. The government has emphasized the need for additional data collection and modeling to validate these results.

22. Industrial effluents have different physical, biological, and chemical characteristics and flows, depending largely on the process and technology used. As a result, it is not possible to treat all industrial effluents in a single treatment plant as can be done for domestic wastewater, unless a group of homogenous industries is located in close proximity, as in some areas of Avellaneda and Lanus.

23. **Compliance.** Several laws and regulations in Argentina address issues relating to industrial pollution, with the Federal Government primarily responsible for environmental policy and regional governments for control and enforcement. Environmental standards, as well as maximum permissible limits for effluent and emission discharges exist on paper²⁸, but evidence shows that compliance is minimal as a consequence of unclear responsibilities across government jurisdictions, weak enforcement systems, and inadequate monitoring and control capacity in terms of human resources and technical expertise. The absence of a corporate culture of environmental responsibility, even within larger enterprises, compounds the problem.

24. **Public services coverage.** Water and sanitation services in the basin, as well as in the rest of Argentina, have been precarious since the 2001 economic crisis. Levels of coverage are very low in absolute terms and particularly in relative terms when compared with the national average, considering the importance of the area in economic terms (gross domestic product, or GDP). Table 1.2 shows the sewerage service coverage in each municipality in the basin.

Table 1.2. Sewerage service coverage of municipalities in the Matanza-Riachuelo Basin

Municipality	Without piped water access (%)	Without sewerage connection (%)	Total population	Estimated population living in the basin
Almirante Brown	53.0	83.1	514,622	138,948
Avellaneda	1.2	41.5	329,638	139,767
Cañuelas	66.9	69	42,475	41,201
Esteban Echeverria	63.3	86.5	244,622	244,622
Ezeiza	89.4	88.8	118,916	118,916
Gral. Las Heras	49.9	88.4	12,684	11,769
La Matanza	43.0	53.2	1,256,724	1,024,230
Lanus	1.2	71.4	452,512	273,317

²⁸ ACUMAR's Resolutions 1/2007, 2/2007 and 2/2008 list the admissible limits for discharge of effluents into sewage networks, surface waters, and soil absorption; as well as parameters for air quality relating to emissions.

Municipality	Without piped water access (%)	Without sewerage connection (%)	Total population	Estimated population living in the basin
Lomas de Zamora	6.2	73.2	590,677	425,287
Marcos Paz	20.4	77.1	43,236	43,236
Merlo	56.7	81.1	470,061	122,216
Moron	N/A	N/A	N/A	N/A
Pte. Peron	42.5	99.4	60,045	3,603
San Vicente	60.1	59.8	44,279	133
Ciudad de Buenos Aires Districts (III, IV, V, VI, XIII, XIX, XX, XXI)			837,355	719,085
Total				3,265,129

Source: INDEC 2001 Census.

25. Conventional sanitation services in the area have not expanded at the same rate as the population. Residents with no access to piped water have traditionally tended to obtain their water through wells, generally extracting directly from the Puelche. Low-income residents normally obtain water from the upper water table (Epipuelche) because it is closest to the surface and therefore less expensive to exploit, even though it is highly contaminated. These low-income groups also tend to concentrate in flood-prone areas, where the water table is very high.

26. In areas with no sewerage, the most common sanitation systems are septic tanks and pits. Their construction is frequently precarious, resulting in contamination of the upper water table. As cleaning and emptying these tanks costs approximately ARS50 per service, there is an additional incentive for illegal dumping in storm sewers (if available), streets, surface water bodies, or groundwater through infiltration.

27. **Flood control.** The area of interest is particularly vulnerable to urban and rural flooding. Increased runoff, clogged sewers, rising water tables, and the basin's location in a very flat and low-lying area increase the vulnerability.

28. Flooding affects the poorer segments of the population in particular. The lack of urban planning regulation or its enforcement has led to uncontrolled settlement of predominantly low-income dwellers in the more flood-prone locations. There are large unmet demands for drainage infrastructure, as urban centers have expanded without updating their storm water facilities.

Government Strategy

29. The idea of large-scale investments to improve the M-R River Basin is not new. In fact, governments for decades have discussed the importance of clean-up, and in the mid-1990s, the Government completed a comprehensive M-R Environmental Management Plan (PGA) and took a US\$250 million Inter American Development Bank (IADB) loan to help finance

implementation. However, 12 years later, the IADB loan had only disbursed US\$10 million, and the Government is only now committing another US\$90 million for urgent clean-up activities.

30. In response to these problems, a high-profile lawsuit²⁹ was filed against the national government, the Province of Buenos Aires, the City of Buenos Aires, and 44 industries in 2003 by people affected by environmental degradation. In 2006, the Supreme Court issued a ruling calling on the Government to put in place a concrete action plan to improve environmental conditions in the basin. As a result, the Integrated Cleanup Plan for the Matanza-Riachuelo Basin (*Plan Integral de Saneamiento de la Cuenca Matanza-Riachuelo, PISA*) was designed, and ACUMAR was created by national law to implement this plan. In July 2008, the Supreme Court issued its final ruling establishing general criteria to ensure that the PISA is executed, setting a timeline for its main milestones.

31. The PISA's objectives are (i) improvement of the quality of life, (ii) improvement of the quality of the air, water and soil, and (iii) prevention of future damage. The PISA represents an integrated approach to address the problems of the basin. The plan's main components are listed below.

- (i) ***Institutional strengthening.*** The plan highlights the need for more coordinated action among all institutional actors in the basin and states the need for a clearer regulatory and control framework.
- (ii) ***Cleanup and regional planning.*** The plan seeks proper management of all effluents from outside and within the basin and comprises actions on sewerage systems, industrial control, and treatment and disposal of wastewater.
- (iii) ***Social interventions.*** PISA addresses the needs of the more than 2 million people living in a situation of high environmental risk in an area where the poverty map, the flooding map, and high environmental impact industries map overlap. The social component comprises interventions in regional planning, including land use and urban planning.

32. The Supreme Court has also taken a more proactive role in the area of environmental compliance. The Supreme Court case and growing public pressure has forced SAYDS to update its policy framework and improve its dialogue and negotiation capacity with industries to ensure that their effluents, emissions, and waste management practices comply with environmental regulations and permissible limits for effluent discharges to sewage networks, within a specified timeframe. As a result of this process, Plans for Industrial Restructuring (*Planes de Reconversion Industrial*, or PRIs)³⁰ are being developed and signed, committing enterprises to reach mutually agreed discharge levels by certain dates and the regulatory agency to exempt these companies from non-compliance penalties before the agreed deadline.

33. The new strategy contemplated in the PISA tackles the main sources of pollution in the M-R River: (i) industrial discharges into the M-R River Basin, (ii) secondary treatment plant

²⁹ "Mendoza Case", Beatriz Silva and others against the National government and others.

³⁰ The environmental standards set under the PRIs are customized to individual enterprises and based on the capacity of the enterprises to comply with environmental standards and permissible limits for discharging effluents into AySA's sewage network as articulated in Resolutions 1/2007 and 2/2008.

effluents in the M-R River Basin, and (iii) non-point sources of domestic wastewater from urbanized areas without sewerage coverage.

34. Based on modeling simulations, ACUMAR developed different scenarios for managing wastewater from domestic and industrial origins in the basin in collaboration with the National Technological University (*Universidad Tecnológica Nacional*, or UTN). The simulations showed that the recuperation of the water quality in the M-R River Basin from its current anoxic state (consumption of oxygen from the degradation of organic waste) to an oxygenated condition (without emission of odors and allowing for recreational uses without direct water contact) requires the following action:

- (i) 90–95 percent reduction in organic loads of industrial origin with the implementation of PRIs and industrial effluents connected to AySA's sewerage network with discharge to the La Plata River;
- (ii) Elimination of the discharge from AySA's secondary treatment plants into the M-R River;
- (iii) Elimination of 80–90 percent of non-point domestic wastewater discharges through the provision of sewerage networks in urbanized areas that currently lack these services.

35. While these initial results are indicative of the size of the challenges ahead, the Government has emphasized the need for additional data collection and modeling to validate these results. To this end, funds have been set aside under component 4 of the project.

Annex 2: Major Related Projects Financed by the Bank and/or Other Agencies
ARGENTINA: Matanza-Riachuelo Basin Sustainable Development Project

Sector issue	Project	Objective/description	Latest Project Supervision Ratings (PSR) ³¹	
Bank financed			Development objective	Implementation Progress
Water resources	AR-Urban Flood Prevention and Drainage APL-1 (P088220)	Reduce the City of Buenos Aires' vulnerability to flooding through drainage investments and implementation of a risk management program.	Satisfactory	Satisfactory
	AR-Urban Flood Prevention and Drainage APL-2 (P093491)	Reduce vulnerability of the littoral provinces to flooding through drainage investments and implementation of a risk management program.	Moderately Unsatisfactory ³²	Moderately Unsatisfactory
	AR-Province of Buenos Aires Infrastructure APL-1 (P088032)	Improve water, sanitation, drainage, and transportation infrastructure in Buenos Aires Province.	Satisfactory	Satisfactory
	AR-Province of Buenos Aires Infrastructure APL-2 (P105288)	Improve water, sanitation, drainage, and transportation infrastructure in Buenos Aires Province.	Satisfactory	Satisfactory
	AR-Basic Municipal Services (P060484)	Improve the quality and coverage of water supply and sanitation, urban drainage, and road infrastructure services at the municipal level.	Moderately Satisfactory	Moderately Satisfactory
Pollution control	National Urban Solid Waste Management (P089926)	Develop environmentally and financially sustainable systems for solid waste management.	Moderately Unsatisfactory	Moderately Unsatisfactory ³³
	Sustainable Industrial Development Project (P110612)	Help small and medium enterprises with inadequate access to credit improve their compliance with environmental regulations while boosting their competitiveness.	Under preparation	
Inter American Development Bank (IADB)				
Sanitation	Sanitation Program for Buenos Aires Metropolitan Area	Seeks to contribute to: (i) the improvement/ expansion of the water supply and sanitation infrastructure in the Buenos Aires metropolitan area and (ii) the strengthening of AySA as a water utility.	Approved in 2008	

³¹ Bank-financed projects only

³² Development Objective (DO) and Implementation Progress (IP) are rated "MU" due to slow project start-up after effectiveness on 3 March, 2008.

³³ Implementation Progress (IP) is rated "MU" due to slow project start-up after effectiveness on 1 February, 2007.

Annex 3: Results Framework and Monitoring
ARGENTINA: Matanza-Riachuelo Basin Sustainable Development Project

Results Framework

Project development objective	Project outcome indicators	Use of project outcome information
<i>The project (APL-1) development objectives are:</i> a) Improving sewerage services in the M-R River Basin and other parts of the Province and City of Buenos Aires by expanding transport and treatment capacity; b) Supporting a reduction in industrial discharges to the M-R River, through the provision of industrial conversion grants to small and medium enterprises; c) Promoting improved decision-making for environmentally-sustainable land use and drainage planning, and piloting urban drainage and land use investments, in the M-R River Basin; d) Strengthening ACUMAR's institutional framework for ongoing and sustainable clean-up of the M-R River Basin.	a) 35% of all sewage (about 12m³/s) that is currently discharged from AySA's concession area to the La Plata River without adequate treatment will be appropriately treated and disposed of by the end of APL-1³⁴ b) (i) Regular monitoring for at least the 50 most polluting enterprises to track progress in reducing industrial pollution. (ii) 50% of the small and medium enterprises participating in the matching grants program have effectively reduced their discharge loads according to their PRIs by the end of APL-1 c) Integrated Territorial Development Plan for the M-R River Basin and updated Drainage Master Plan are agreed with key stakeholders and implementation arrangements are piloted d) ACUMAR has fully staffed against its new organigram, operates with its own operating budget and is fully able to fulfill the functions vested in it by law.	a) Inform stakeholders of advances in the improvement of wastewater management in the M-R River Basin. b) (i) Ensuring that progress is being made towards reducing industrial effluent discharges in the M-R River Basin (ii) Tested initial phase for scaling-up of the matching grants program for small and medium enterprises c) Tested pilots for replication and development of appropriate land use planning tools, taking into account the river basin protection perspective. d) Develop a strong basin authority capable of elaborating and implementing comprehensive and consistent policies and enforcing them.

³⁴ 35 percent is calculated as the ratio of 12 m³/s (average flow through the Riachuelo outfall by the end of APL-1) divided by the sum of 8 m³/s (average flow from the Baja Costanera Expansion Collector) + 12 m³/s (flow through Riachuelo outfall + 14 m³/s (expected flow through the Berazategui outfall by the end of APL-1).

Arrangements for results monitoring

		Target Values						Data Collection and Reporting	
	Baseline	YR1	YR2	YR3	YR4	YR5	YR6	Frequency and Reports	Responsibility for Data Collection
Project Outcome Indicators									
Component 1									
a) Percentage of all sewage (about 12m ³ /s) that is currently discharged from AySA's concession area to the La Plata River without adequate treatment that is appropriately treated and disposed of by the end of APL-1	0	0	0	0	0	35	35	Annually	AySA/ACUMAR
Component 2									
b) (i) Percentage of regular monitoring for at least the 50 most polluting enterprises to track progress in reducing industrial pollution.	0	0	50	100	100	100	100	Annually	SAYDS/ACUMAR
(ii) Percentage of the enterprises in the matching grants program that have effectively reduced their discharge loads according to their PRIs by the end of APL-1.	0	0	5	10	30	50	50	Annually	SAYDS/ACUMAR
Component 3									
c) Both plans (Integrated Territorial Development Plan for the M-R River Basin and updated Drainage Master Plan) are agreed with key stakeholders and implementation arrangements are piloted	0	0	0	2	2	2	2	Annually	SAYDS/ACUMAR

	Baseline	Target Values						Data Collection and Reporting	
		YR1	YR2	YR3	YR4	YR5	YR6	Frequency and Reports	Responsibility for Data Collection
Component 4 ACUMAR has fully staffed against its new organigram, operates with its own operating budget and is fully able to fulfill the functions vested in it by law.	On schedule (OS)	OS	OS	OS	OS	OS	OS	Annually	SAyDS/ACUMAR
Intermediate Outcome Indicators									
Component 1									
a) Linear kilometers of the Left Bank collector (including Baja Costanera bypass collector) completed	0	0	2	8	15	18	Works completed	Annually	AySA/ACUMAR
b) Linear kilometers of Right Bank collector completed	0	0	0	0	0	15	37	Annually	AySA/ACUMAR
c) Linear kilometers of the Riachuelo outfall completed	0	0	0	2	7.5	11.5	Works completed	Annually	AySA/ACUMAR
c) Percentage of the Riachuelo preliminary treatment plant construction completed	0	0	10	30	70	100	100	Annually	AySA/ACUMAR
Component 2									
b) ACUMAR conducts field audits, including sampling/lab tests of the 50 most polluting enterprises of the basin.	N/A		X	X	X	X	X	Annually	SAyDS/ACUMAR

	Baseline	Target Values						Data Collection and Reporting	
		YR1	YR2	YR3	YR4	YR5	YR6	Frequency and Reports	Responsibility for Data Collection
Component 3									
a) Satisfactory preparation of integrated regional and local land-use plans taking into account the river basin protection perspective	N/A		X	X				N/A	SAyDS/ACUMAR
b) Satisfactory update of the Drainage Master Plan	N/A		X	X				N/A	SAyDS/ACUMAR
c) Percentage of funds disbursed for activities to indicate satisfactory implementation of pilot integrated urban interventions, taking into account the river basin protection perspective	0		10	30	60	100	100	N/A	SAyDS/ACUMAR
d) Percentage of funds disbursed for activities to indicate satisfactory implementation of pilot drainage interventions	0		10	30	60	100	100	N/A	SAyDS/ACUMAR
Component 4									
Percentage of staffing completed on basis of staffing plan developed by ACUMAR	0	10	30	50	70	100	100	N/A	SAyDS/ACUMAR

Annex 4: Detailed Project Description

ARGENTINA: Matanza-Riachuelo Basin Sustainable Development Project

Macro-component 1: Sanitation

Component 1: Sanitation (US\$ 1,309 million total, US\$ 694 million loan)

1. This component includes the construction and external engineering supervision of sanitation infrastructure for the collection, treatment, and disposal of the wastewater of the Matanza-Riachuelo (M-R) basin and the AySA concession area. These works help meet two objectives including (i) extending and improving the level of coverage and quality of sewerage services in the M-R River Basin and in the entire AySA concession area, and (ii) contributing to the recovery and preservation of the quality of water bodies in the entire M-R River Basin and in the La Plata River.

2. Through a combination of sanitation works including the expansion and extension of collection systems, construction of treatment plants and pumping stations, and construction of new sub-aquatic outfalls, this project component aims to intercept most of the domestic and industrial wastewater in the AySA concession area³⁵ as well as the wastewater of the municipalities in the upper M-R River Basin, and divert this entire wastewater flow, after subjecting it to preliminary treatment, to La Plata River through long and effective sub-aquatic outfalls. The progressive elimination of point sources of contamination from the M-R River needs to be accompanied by an end to the dumping of municipal solid wastes into the river and on its bank. Solid waste management activities in the basin are undertaken by ACUMAR as part of the PISA using other financing sources.

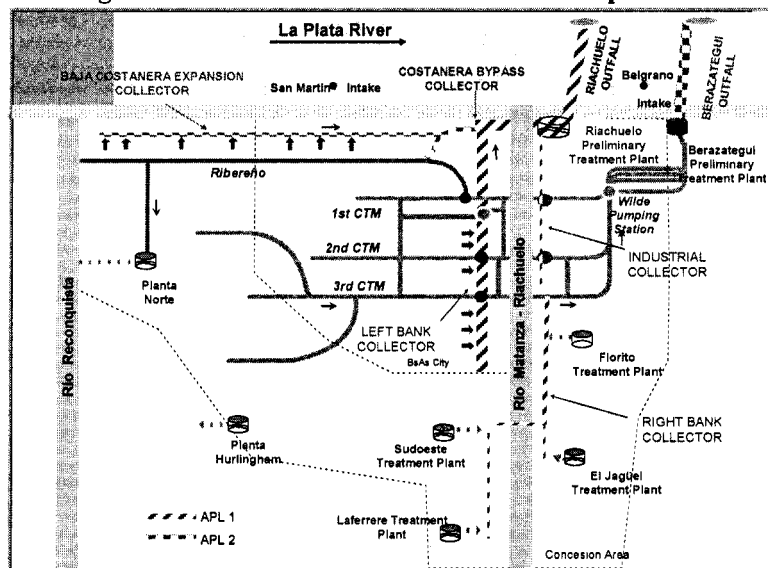
3. Among the various alternatives considered for the sanitation component, the selected option was chosen for the following reasons:

- It takes into account the proper dimensioning of the wastewater collection system, thereby limiting discharges of wastewater and effluents into inland rivers such as the M-R and its tributaries. These receiving bodies, which have limited auto-purification capacity, will be protected by progressively diverting all collected wastewater to the sewerage network and discharging it to La Plata River (which has ample auto-purification capacity) after appropriate treatment.
- It distributes the flow between the two preliminary treatment plants (Riachuelo and Berazategui) in order to provide greater flexibility and security in the main conveyance and treatment systems. The incorporation of the Riachuelo system makes it possible to divide wastewater flows between the two plants and perform treatment and effluent discharge independently in each one, thereby decentralizing the existing sewerage system and reducing the impact of the effluents discharged to La Plata River.
- It intercepts major dry season drainage flows in the area served by AySA's existing sewerage network in order to substantially reduce the pollution of the M-R and La Plata Rivers, since during the dry season the drainage system conveys mostly wastewater discharged to it through illegal or erratic connections.

³⁵ Except wastewater that is diverted to the North treatment plant, which discharges its effluent to the Reconquista River.

- It allows all industries located in the area served by AySA that pre-treat their wastewater to the level required by the norms to discharge their treated wastewater to the sewerage network, thereby reducing the required level of in situ treatment.
 - It increases the sewerage system capacity and allows the entire population located in the M-R River Basin to be connected to the sewerage network.
4. Because of the scale, complexity, and high cost of the Matanza-Riachuelo Basin Sustainable Development Project, the sanitation component will be implemented in two stages. Completion of sanitation construction works under both APL phases is required to attain the objective of cleaning up the M-R River Basin. The first stage of Component 1 (financed during APL-1) will consist of the following works: a collector on the left bank of the M-R River in the lower part of the basin, the Baja Costanera bypass collector, a collector on the right bank of the M-R River starting in the upper middle part of the basin and reaching its lower part, an industrial wastes collector and pumping station on the right bank of the M-R River, the Riachuelo preliminary treatment plant, the two Riachuelo pumping stations, and the Riachuelo outfall. While the left bank conveyance system (Left Bank collector and the Baja Costanera bypass collector) has been front-loaded in the early years of APL-1, the Right Bank collector (CMD) is scheduled for construction towards the end of APL-1. Moreover, given the size of the investments required, AySA will continue to study technical alternatives to the CMD. Any alternative solution would require approval from SAYDS and ACUMAR as well as the Bank.
5. The second stage of Component 1 (mostly financed through APL-2) will consist of the following works: the Baja Costanera expansion collector, the Berazategui effluent pumping station, the Berazategui outfall, and additional primary and secondary sewerage networks in the basin. The Berazategui preliminary treatment plant, which is functionally part of APL-2, is being financed and constructed separately by the Government. The investments in sewerage networks financed under the project are only part of the total investment required for networks, and it is estimated that additional funds will be mobilized by AySA to ensure meeting its coverage targets. The diagram of Component 1 is presented in Fig. 4.1, in which works under APL-1 and APL-2 can be identified.

Figure 4.1. Schematic Presentation of Component 1



The content and costs of Component 1 by project implementation stage is presented in Table 4.1

Table 4.1. Content and Costs of Component 1 by Project Implementation Stage

Item in APL-1	Cost (Million US\$)	Item in APL-2	Cost (Million US\$)
Left Bank collector	85	Baja Costanera expansion collector (first stage)	88
Baja Costanera bypass collector	65	Berazategui effluent pumping station	85
Complementary works	10	Berazategui preliminary treatment plant	148*
Right Bank collector	275	Berazategui outfall	228
Industrial wastes collector	34	Primary and secondary sewage networks	806
Riachuelo preliminary treatment plant	142		
Riachuelo inflow pumping station	76		
Riachuelo effluent pumping station	46		
Riachuelo outfall	274		
<i>Total costs of works under APL-1 (without contingencies)</i>	<i>1007</i>	<i>Total costs of works under APL-2: (without contingencies)</i>	<i>1355</i>
Total costs of Component 1 (APL-1 and APL-2): US\$ 2362 million (without contingencies)			

*Cost of contract (not financed by the World Bank) as awarded: 458 million pesos.

A detailed description of the works under Component 1 is presented below:

(i) Construction/extension of collectors and complementary works

6. The main wastewater collection network, and the conveyance and distribution system of wastewater to the two preliminary treatment plants are critical parts of the project, and alternatives for this system were considered. The selected collection and conveyance system consists of constructing the Left Bank collector (*Colector Margen Izquierda* or CMI) to convey the wastewater flows generated in the AySA concession area located in the lower part of the basin to the Riachuelo outfall and constructing the Right Bank collector (*Colector Margen Derecha* or CMD) to convey the wastewater generated in the AySA concession area in the middle and upper middle parts of the basin, as well as the wastewater generated in the upper part of the basin outside the concession area of AySA to the Berazategui outfall. Both collectors will be constructed during APL-1 thereby ensuring that the main conveyance infrastructure required for the clean-up of the M-R River Basin will be in place in a relatively short period of time. The Left Bank collector will be about 12 km long, with diameters ranging from 0.8 meters to 2.9 meters and will discharge into the Baja Costanera bypass collector – with a 4.5 m diameter and a length of 5 km – which in turn discharges through the Riachuelo outfall. The Right Bank collector is expected to be constructed with capacity to convey the wastewater of the upper and middle parts of the Basin and discharge it to the third *cloaca máxima*, which will convey the wastewater to the Berazategui preliminary treatment plant and outfall. The CMD length has been estimated to be about 37 km and its estimated diameter varies between 2.3 m in its upper part and 3.5 m in its lower part. The CMD collection system will mainly receive and transport wastewater collected in the coverage area of the Southwest, Jagüel, Fiorito, and Laferrere treatment plants, as well as wastewater

generated in the municipalities of the upper M-R River Basin located outside the AySA concession area.

7. The conveyance system also includes *complementary works* financed under APL-1 such as connections between the two operational watershed systems (Riachuelo and Berazategui), and the interception of storm water. Under APL-2, it includes the Baja Costanera Expansion collector (Radio Antiguo section), and the Baja Costanera Expansion collector (Federal Capital –Vicente López section). These two sections of the Baja Costanera Expansion collector constitute only the first stage of this collector. An additional stage will be financed by AySA from its own sources.

8. During the first stage, an industrial wastes conveyance pipe and pumping station will be constructed on the right bank of the M-R River to collect industrial effluents from about 23 industries, thereby diverting them from the M-R River. This system will significantly reduce industrial waste discharges into the lower part of the M-R River.

9. The aforementioned systems will collect the wastewater generated in the AySA concession area (excluding those generated in the Reconquista basin) and the wastewater generated in the upper basin and will distribute them between the two planned outfalls, so that a peak flow of 25 m³/s will be conveyed to the Riachuelo outfall and 33 m³/s will be conveyed to the Berazategui outfall. The treatment plants and outfalls will be designed and constructed to handle these peak flows. These systems are expected to provide sufficient capacity for approximately 20 years. AySA and ACUMAR plan to run, as part of the communication plan, campaigns for controlling water demand. In Buenos Aires, the water demand per capita is extremely high, reaching average values of over 400 liters/day/capita. This high water demand results in high flows of domestic wastewater. It is expected that through demand control the per capita water consumption in the city, and, consequently, the wastewater production per capita will decline with time so that the rate of increase in the generated wastewater flows declines and the systems constructed under the project will provide a capacity for much more than 20 years. Nonetheless, if the demand control strategy is less successful than expected over the next two decades, the treatment plants will need to be expanded, and a parallel outfall will have to be constructed at the Riachuelo site.

(ii) Treatment plants and pumping stations

10. Prior to discharging effluents through each outfall to the La Plata River, the collected wastewater will be subjected to preliminary treatment at two plants. The Riachuelo plant, which will be financed through this project, is to be located in an area near La Plata River in the Dock Sud area. The Berazategui preliminary treatment plant, which is being financed by the Government using its own funds complemented by a loan from Brazil's BNDES, is to be located at Berazategui near the inlet of the existing 2.5 km outfall. These preliminary treatment plants consist of coarse screens to retain heavy material, followed by milliscreens with 6 mm (or smaller) openings for removal of fine material, and then by aerated grit chambers to remove sand and floating material such as oil and grease. The Riachuelo preliminary treatment plant is designed to receive a peak flow of 25 m³/s of wastewater collected from the city of Buenos Aires and the lower part of the M-R River Basin. The Berazategui preliminary treatment plant was designed to receive a peak flow of 33.5 m³/s of wastewater collected from the right bank area of the M-R River, from the middle and upper parts of the M-R River Basin which will be collected by the Right Bank collector at a peak flow of about 12 m³/s, and from the upper part of the basin at a peak flow of about 1.5 m³/s,

also collected by the Right Bank collector. The peak conveyance capacity of the Right Bank collector will be designed for 13.5 m³/s.

11. The Riachuelo preliminary treatment plant site will include two pumping stations, a lifting station at the entrance to the treatment plant, and a second station to discharge the effluent to La Plata River through the outfall. The Berazategui plant site will have only one pumping station (the effluent discharge station), as the existing Wilde pumping station will pump the raw sewage into the treatment plant. While the efficiency of removing solids and organic matter by the preliminary treatment plants appears low, the plants play an important role in removing important waste material contained in the raw sewage. This waste material includes (i) coarse solids such as sand and grains, which, if not removed, tend to settle rapidly and accumulate around the outfall discharge points and are not carried away by the flows in the receiving water body, (ii) coarse floatable materials such as plastic bags, which, if not removed, float on the surface of the receiving water body, causing aesthetic problems, and (iii) oil and grease, which, if not removed, form an impermeable layer on the surface of the water, preventing re-aeration and oxygen transfer to the water which could result in oxygen depletion in the diffuser zone.

12. The Riachuelo preliminary treatment plant will generate approximately (i) 132 m³/day of solid waste residues, (ii) 29–74 m³/day of sand, and (iii) 48–96 m³/day of oil and grease, totaling about 200–300 tons/day (m³/day) of waste. The Berazategui preliminary treatment plant, during the first stage of its (full) operation, will generate approximately (i) 158 m³/day of solid waste residues, (ii) 35–88 m³/day of sand, and (iii) 57–128 m³/day of oil and grease, totaling about 249–374 tons/day (m³/day) of waste.

13. The quantity of solid waste generated at both treatment plants is expected to increase over time with increased wastewater flows. During the initial years of the operation of the plants, the quantity of solid waste will vary from 459 to 676 tons/day (m³/day) and will be removed by trucks or barges (or a combination of the two) and deposited in a sanitary landfill managed by the State Company for the Ecological Coordination of the Metropolitan Area (*Coordinación Ecológica Área Metropolitana Sociedad del Estado*, or CEAMSE)³⁶ along with the rest of the solid waste generated in Buenos Aires. Installations for the recovery of sand for reuse and generation of energy or production of biofuels from the oil are being considered for inclusion in the project at a later stage, but research is required before taking specific related actions. The costs related to the proper management of these solids will be included in the O&M costs of AySA and not subject to Bank financing.

(iii) Construction of new sub-aquatic outfalls

14. The project will finance the construction of two new sub-aquatic outfalls over the two APL phases. The Riachuelo outfall, to be financed during APL-1, is located in the Dock Sud area. This outfall measures 3.8 m in internal diameter and is 11.5 km long. The final 2.3 km are the diffuser zone.³⁷ The Berazategui outfall, expected to be financed under APL-2, is located near the existing short outfall, measures 3.8 m in internal diameter, and is 7.5 km long. The final 3.5 km are the diffuser zone.

³⁶ CEAMSE is the public company in charge of solid waste management in the City of Buenos Aires and 34 municipalities in Buenos Aires Province.

³⁷ Dimensions for the Riachuelo outfall are indicative. AySA is in the process of finalizing bidding stage designs. Final designs will be prepared by construction contractor(s).

15. The design peak flow of the Riachuelo outfall is 25 m³/s, and that of the Berazategui outfall is 33.5 m³/s. Because the Berazategui outfall is designed to convey a peak wastewater flow rate of 13.5 m³/s from the middle and upper sections of the M-R River Basin, the entire wastewater flow from municipal and industrial sources in the basin will reach the two outfalls. This is the principle of the clean-up of the basin and of the recovery of the water quality of its inland waters, which will enable the sustainable development of the basin. The clean-up will be achieved through progressive elimination of the point sources of contamination of the rivers and their discharge to the sewage collection system.

(iv) Construction of primary and secondary sewage networks

16. Included only under APL-2 is the construction of primary and secondary sewerage networks in the municipalities of the M-R River Basin that currently lack coverage (e.g., Lomas de Zamora, Lanús, Almirante Brown, and other neighborhoods but not the city of Buenos Aires, which already has full coverage). The investments in sewerage networks financed under the project are only part of the total investment required for networks and it is expected that additional funds will be mobilized by AySA to ensure its coverage targets are met.

(v) Supervision, data collection, analyses and modeling and capacity development

17. In addition to the infrastructure works listed above, APL-1 will also provide financing for (i) independent technical supervision of key works; (ii) data collection, analyses and modeling in support of the engineering designs for these works; and (iii) some capacity development activities for AySA in support of the preparation of the engineering designs of the works.

18. **Technical information.** Table 4.2 presents technical information regarding some works of Component 1, as well as the expected construction techniques and materials. The main conveyance system pipes and the outfall pipes will be installed using tunneling (i.e., trenchless methods). Costs of the safe management of dirt from tunneling will be incorporated in the respective construction contracts. Pipes up to 3 m in diameter may be installed by pipe jacking, while larger-diameter pipes will be constructed from pipe segments assembled inside the tunnel. For pipe installation by pipe jacking, shafts will be constructed every 1–1.5 km, and for pipe installation in tunnels by segments, shafts will be constructed every 4–5 km.

Table 4.2: Technical Information Regarding some Works of Component 1

Stage	Work	Section	Length (Km)	Internal diameter (m)	Construction Method	Material of construction
APL-1	Left Bank collector	First section (Gral Paz – Larrazabal)	1.7	0.8	Pipe Jacking tunneling	Concrete pipes
		Second section (Larrazabal – Goncalvez Dias)	9.4	2.9	Pipe Jacking tunneling	Pre-cast concrete lining segments
	Baja Costanera bypass collector	Left Bank collector- Riachuelo Pretreatment plant	5.0	4.5	Pre-cast lining segment tunneling	Pre-cast concrete lining segments
	Riachuelo outfall	Non diffuser segment	9.2	3.8		

Stage	Work	Section	Length (Km)	Internal diameter (m)	Construction Method	Material of construction
	Right Bank collector	El Jaguel	11.2	2.3	Pipe Jacking tunneling	Pre-cast concrete lining segments
	Right Bank collector	LaFerrere	11.4	2.2		
	Right Bank collector	Sudoeste	1.47	2.8		
	Right Bank collector	Inferior	17.8	3.5	Pre-cast lining segment tunneling	Pre-cast concrete lining segments
APL-2	Berazategui outfall	Non diffuser segment	4.0	3.8	Pre-cast lining segment tunneling	Pre-cast concrete lining segment

Macro-Component 2: Support for Industrial Pollution Abatement, Environmental Territorial Management and Institutional Strengthening

Component 2: Industrial Pollution Abatement (US\$75.4 million total, US\$60.3 million loan)

19. The objective of this component is to contribute to the progressive elimination of point-source industrial wastewater discharges (organic as well as toxic effluents) many of which currently enter the Matanza-Riachuelo (M-R) river. This process is expected to occur over the next 15–20 years and is essential to the overall environmental recovery of the river. This component of the project represents an additional tool to clean-up the M-R River by diverting treated industrial discharges to the Rio de la Plata through an improved, expanded and more secure system of sewage management (financed by Component 1).

20. Based on initial estimates, there are approximately 4,100 registered industries in the M-R River Basin area that contribute a significant part of the total organic load currently discharged to the M-R River, as well as virtually all toxic discharges. Of these, approximately 1,600 are connected to the AySA network, while the remaining 2,500 currently discharge effluents directly into the M-R River. ACUMAR will first focus on the 50 main industrial polluters that contribute approximately 95 percent of the total industrial organic load and 170 industries with the highest toxic loads that currently discharge to the M-R River, using technical assistance made available through this component. As part of the Government's strategy, plans for industrial restructuring (*Planes de Reconversion Industrial*, or PRIs), including indicators, targets, and deadlines will be developed and signed with these industries. While the Bank will help the Government monitor progress on PRIs signed with the largest industrial polluters (see sub-components (i) through (iv) below), it has little leverage for enforcing them unless firms are beneficiaries of the matching grants program financed under sub-component (v) of the loan. The absence of direct leverage notwithstanding, the design and implementation of a program for encouraging corporate environmental responsibility under sub-component (iv) is expected to create a public environment increasingly favorable to industrial clean-up.

21. At a broad level, ACUMAR will require industries currently discharging to the M-R River to instead connect to the AySA sewage network for ultimate discharge into the Rio de

la Plata after appropriate treatment. Additionally, the Government intends to also require identified contaminating industries (*agentes contaminantes*) to sign *Planes de Reconversión Industriales* (PRIs) committing them to making necessary investments to reduce effluent loads with targets and deadlines. The strategy for industrial discharges³⁸, which will be adjusted through this project, includes the following:

- Those industries that can currently connect to the sewage network will be required to reduce their organic load to under 200 mg/l and treat their toxic effluents before discharging to the network.³⁹
- For industries that cannot currently access the AySA network, ACUMAR will require that interim solutions (such as developing cleaner production plans in accordance with their PRIs) be developed while the construction/extension of the collectors (financed under Component 1) is undertaken. Subsequently, these industries will be expected to connect and discharge their treated effluents to the sewage network for transport to the Rio de la Plata.
- For industries that may not have access even after the collectors are constructed/extended (because AySA's concession area does not cover the entire M-R River Basin), ACUMAR envisions developing alternative solutions such as using trucks to collect treated effluents for discharge at points along AySA's network or building collectors to connect to AySA's existing sewerage network.

22. Activities under Component 2 will complement those under Component 1 in contributing to the clean-up the M-R River. This component combines technical assistance (TA) measures to improve environmental monitoring, control, and enforcement by ACUMAR with a matching grants program to support private sector investments in pollution control and cleaner production, as well as the construction of specific centralized treatment facilities included in the relevant plans for industrial restructuring (*Planes de Reconversion Industrial*, or PRIs).

23. Technical assistance activities will be oriented to increase the monitoring and implementation capacity of the Government through four sub components. Part of the TA

³⁸ This strategy is in line with the World Bank Group's Environmental Health and Safety Guidelines relating to Wastewater and Ambient Water Quality, which states that "*when wastewater treatment is required prior to discharge, the level of treatment should be based on: · Whether wastewater is being discharged to a sanitary sewer system, or to surface waters; · National and local standards as reflected in permit requirements and sewer system capacity to convey and treat wastewater if discharge is to sanitary sewer; · Assimilative capacity of the receiving water for the load of contaminant being discharged wastewater if discharge is to surface water; · Intended use of the receiving water body (e.g. as a source of drinking water, recreation, irrigation, navigation, or other)...*" ACUMAR is in the process of finalizing norms/ regulations relating to uses and quality of the receiving water bodies in the short, medium and long terms. Furthermore, water quality modeling has been the underlying basis of choosing the technical solution for the M-R project, and considers the assimilative capacity of Rio de la Plata.

³⁹ The admissible limits for organic and toxic substances that need to be reached before industries can discharge to a sewage network are in ACUMAR's Resolutions 1/2007 and 2/2008 and match those required by AySA for effluent discharges into their sewage networks. This is in line with the World Bank Group's Environmental Health and Safety Guidelines relating to Wastewater and Ambient Water Quality specifically on discharges to sanitary sewer systems, which states "*Discharges of industrial wastewater, sanitary wastewater, wastewater from utility operations or stormwater into public or private wastewater treatment systems should meet the pretreatment and monitoring requirements of the sewer treatment system into which it discharges.*"; as well as the EHS Guideline on Water and Sanitation, which states "*Treatment or pre-treatment to neutralize or remove toxic chemicals should ideally take place at the industrial facility itself, prior to discharge of the effluent to the sewer or water body...*".

will be targeted to the establishment of a registration system for industries located in the basin (see subcomponent 1). The information from this registry will serve as a basis for the management of PRIs and their implementation by ACUMAR.

24. The initial phase of the matching grants program will be launched during APL-1, to support private sector investments in pollution control and cleaner production measures. This initial phase of the program will target a limited number (20-50) of small and medium enterprises (SMEs) which have weaker financial and technical capacity to comply with environmental regulations. While the initial phase of the program will only reach a small sample of eligible SMEs, it aims to inform the project by (i) learning and sharing how SME-specific cleaner production plans are applied; (ii) encouraging associations of SMEs to adopt cost-effective solutions for reaching environmental compliance; and (iii) facilitating support for SMEs to gain access to commercial credit for cleaner production investments.

25. Table 4.3 lists the activities financed under this component as part of APL-1. Details about the activities follow further below.

Table 4.3 : Activities and costs for Component 2: Industrial Pollution Abatement

Sub-component	Estimated costs without contingencies (US\$ m)	Estimated loan amount without contingencies (US\$ m)
i. Support for pollution diagnostics and policy analysis	4	3
ii. Improvement of the monitoring system for industrial wastes	4	3
iii. Support for adjustments to the strategy for industrial discharges	2	2
iv. Design and implementation of program for encouraging corporate environmental responsibility	1	1
v. Support for pollution control and cleaner production investments by small and medium enterprises (SMEs)	47	41
Total	58	50

(i) Support pollution diagnostics and policy analysis

26. This sub-component will support pollution diagnostics carried out by SAyDS/ACUMAR using the baseline data in the GIS database (financed under component 3) and the application of water quality modeling to design a framework to establish appropriate effluent loads for industries located in the M-R River Basin. The database will include baseline information on about 20 parameters relating to water consumption and pollution loads (organic and toxic) discharged by each of the industries. In addition, this sub-component will support an analysis of possible social impacts of industrial conversion within the M-R River Basin.

27. The information from this centralized database, among other inputs, will help in evaluating the effluent loads from polluting agents discharged to the M-R, along with receptive capacity of the Rio de la Plata to accept these effluent discharges. Based on this baseline information, and corroborated by rapid assessments, all industries will be ranked by each polluting substance (organic as well as toxic) relative to the total effluents discharged to the M-R River basin and La Plata River by all the industries in the basin. The resulting information from these activities will serve as inputs for taking decisions on developing policies relating to pollution control and prevention. The pollution diagnostic includes an

independent validation exercise of the list of contaminating industries as well as an analysis of the possible social impacts of activities relating to industrial pollution abatement within the M-R River Basin.

28. This sub-component will also support work planned by ACUMAR and AySA for the inspection and control of water consumption and flow discharge of industries connected to AySA's network with sampling for analysis of industrial effluents.

(ii) Improvement in the monitoring system for industrial discharges

29. This sub-component will enhance the system for monitoring the status of industrial waste discharge by supporting institutional pollution control infrastructure such as independent certified labs and training of their staff. While initially ACUMAR will use AySA labs for some water quality tests, later in the project it plans to use commercial labs or professional services. The purchase of some equipment may be required. This sub-component will also finance the hiring of technical audits for follow-up and verification to ensure the compliance of enterprises with their PRIs.

(iii) Support for adjustments to the strategy for industrial discharges

30. The diagnostic process in sub-component (i) will help determine current conditions in each industry relating to its production, operations, and facilities to control water consumption, flow discharge, and sampling to analyze industrial effluents. Based on these diagnostics, and within the framework of the general industrial discharge strategy, ACUMAR will be able to make adjustments to the strategy by defining (i) the specific targets and conditions to be fulfilled by individual enterprises, in compliance with the permissible limits for the discharge of pollutants into sewer networks as well as surface water bodies⁴⁰; and (ii) the volumes and quality of industrial discharges for determining the resulting level of treatment of industrial effluents or industrial restructuring required of the individual enterprises, and the design of the necessary treatment systems. This adjustment of the overall industrial discharge strategy for the M-R basin will be supported through this sub-component.

(iv) Design and implementation of a program for encouraging corporate environmental responsibility

31. This sub-component supports activities aimed at building a culture of corporate environmental responsibility, i.e. through the design and development of an environmental disclosure program to help build a culture of corporate environmental responsibility. Such disclosure programs have been successful in other countries in changing industrial behavior related to environmental compliance. Changes in corporate culture will be promoted through the annual publication by ACUMAR of firm compliance with their respective PRIs.

(v) Support for pollution control and cleaner production investments by SMEs

32. This sub-component aims to provide technical and financial support to SMEs for investments in pollution control and cleaner production processes.

33. The project envisions a grant program (on a matching basis to ensure commitment) for micro, small and medium enterprises in the M-R basin for cleaner production investments

⁴⁰ Permissible limits (for organic and toxic substances) for discharging effluents into sewer networks, surface water bodies, and soil absorption are articulated in ACUMAR Resolutions 1/2007 and 2/2008. This is in line with World Bank Group's EHS Guidelines relating to Wastewater and Ambient Water Quality.

to improve environmental compliance. These investments in cleaner production technologies will increase the efficiency of plants and processes, minimize their use of resources, eliminate toxic inputs, and reduce the quantity and toxicity of effluents and emissions. Associations of similar SMEs (such as the ACUBA industry association for tanneries) will also be eligible to apply for financing of common environmental infrastructure, such as common effluent treatment plants. Eligible expenditures under this matching grants program may include consulting services, equipment (goods), and small works.

34. The proposed scheme for the execution of this component, including the details of the institutional roles, is the following :

1. Approval of PRIs by ACUMAR (according to ACUMAR's resolution on the subject).
2. Prior determination of objective criteria for eligibility, selection, and financing decided within the framework of the Interjurisdiccional Commission of Industry (*Comisión Interjurisdiccional de Industria, CIJI*) and contained in the regulation.
3. Presentation of financing requests.
4. Based on pre-determined objective criteria, SAYDS determines the SMEs to be selected and the amounts to finance, and signs agreements (CRIs) with the beneficiaries.
5. Disbursements on the part of SAYDS in accordance with the timelines defined in the agreements.
6. SAYDS will monitor financial execution under the grant agreements (CRIs) while compliance with action plans contained in the PRIs will be monitored by ACUMAR.

35. Enterprises that are eligible for participation in the matching grants program should (i) be categorized as an SME⁴¹; (ii) have a signed PRI; and (iii) lack access to commercial credit. Matching grants will be allocated to eligible enterprises, prioritizing on basis of the relative pollution load contributed by the enterprise. Firms signing CRIs with SAYDS will therefore constitute a subset of firms signing PRIs with ACUMAR. Detailed eligibility criteria for assistance to SMEs, as well as the specific disbursement modalities will be clearly defined and disclosed in the project's Operational Manual to ensure transparency of resource allocation. Moreover, the specific level of matching grants required from SMEs will be determined on an industry (sector) basis using best practice lessons, with special dispensations for micro enterprises.⁴² These matching grants will not substitute for commercial credit, as they focus specifically on environmental investments for SMEs for whom bank financing is not available. The launch of this matching grants program will occur when SMEs in the basin have met the eligibility criteria, which includes having signed PRIs.

⁴¹ Under Article 1° of Title I of Law N° 25,300, companies are considered to be Micro, Small and Medium Enterprises according to the registered maximum annual sales, excluding value-added tax and internal tax, expressed in pesos in the table below:

Enterprise Size	Farming	Industry & Mineral	Commercial	Services	Construction
Micro	456.000	1.250.000	1.850.000	467.500	480.000
Small	3.040.000	7.500.000	11.100.000	3.366.000	3.000.000
Medium	18.240.000	60.000.000	88.800.000	22.440.000	24.000.000

⁴² Typically, matching grant packages require a 1:1 level of matching funds from SMEs. In the case of micro-enterprises, however, a 2:1 (project: enterprise) ratio is more commonly recommended.

Component 3. Environmental Territorial Management (US\$81.9 million total, US\$65.5 million loan proceeds)

36. The main objective of the Environmental Territorial Management component is to improve public policies and programs for environmental territorial management in the M-R River Basin and steer the basin's development trends in a more sustainable direction. This will be achieved through a two-fold approach of (i) providing technical assistance to improve a broad range of planning aspects associated with land use at the regional level and (ii) executing pilot investments in selected areas of the basin to improve the quality of life.

37. **Preparation strategy and pilot interventions.** During the first year of implementation, the project will strengthen ACUMAR's capacity to implement Component 3. Subsequently, key technical information will be updated to support decision making to select the priority area for pilot investments financed by the component in consultation with the basin's stakeholders.

38. **Institutional arrangements.** The nature of the activities proposed under this component requires decisive participation by provincial and municipal governments. This will be ensured through the participatory mechanisms that are part of ACUMAR's structure and through specific implementation arrangements proposed for the component.

39. **Legal Framework.** ACUMAR has the legal power to design/develop environmental territorial management programs, and to take the necessary measures to implement the PISA. The content of this component is aligned with the scope of the PISA and will contribute to its implementation. SAYS/ACUMAR will be responsible for the execution of this component. Additionally, in order to comply with Argentina's legal framework, and prior to the carrying out of any pilot investment, a separate agreement will be signed between the Borrower (through SAYS), ACUMAR and the different jurisdictional entities which have primary responsibility to carry out said investments (e.g. AySA, PBA and municipalities), whereby the relevant entity will provide its consent to permit SAYS/ACUMAR to carry out and finance the pertinent pilot investment, and to accept the obligation to operate and maintain the pilot investment thereafter.

40. Figure 4.1 illustrates the synergy among the proposed sub-components.

Figure 4.1. Synergy among sub-components of Component 3: Environmental Territorial Management

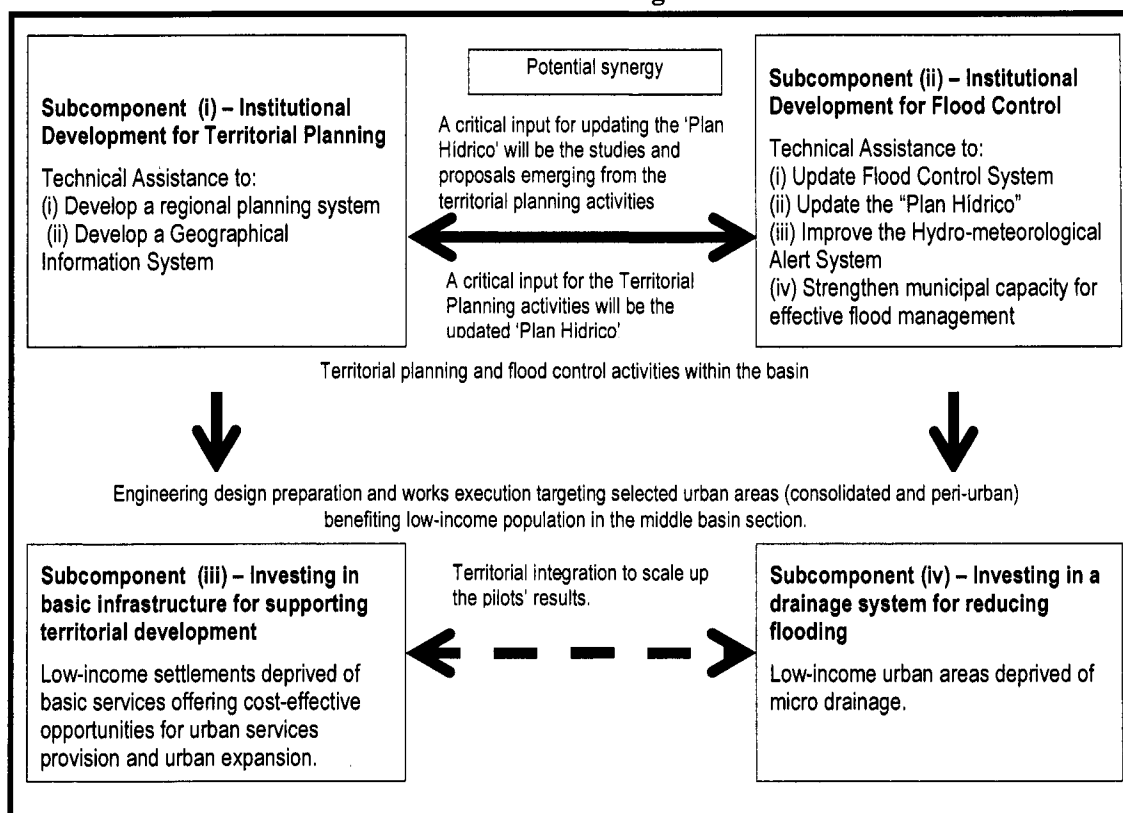


Table 4.4 : Activities and costs for Component 3: Environmental Territorial Management

Sub-component	Estimated costs without contingencies (US\$ m)	Estimated loan amount without contingencies (US\$ m)
(i) Institutional development for territorial planning	5.6	5.6
(ii) Institutional development for flood control	3	3
(iii) Investments in basic infrastructure for territorial development	21.4	17.4
(iv) Investments in basic infrastructure for flood control	33	27
Total	63	53

(i) Institutional Development for Environmental Territorial Planning

41. This sub-component will finance technical assistance activities that strengthen institutional capacity for environmental territorial planning with a comprehensive development perspective for the entire Matanza-Riachuelo River Basin. SAYDS/ACUMAR will execute the activities under this sub-component in close collaboration with the Province of Buenos Aires and the municipalities. The legal and institutional framework that created ACUMAR provides it with a broad coordination role. Therefore, ACUMAR's institutional

structure promotes the participation of the provincial and municipal governments. This sub-component includes the following activities:

42. *Develop a regional planning system for the Matanza Riachuelo River Basin.* To promote more integrated regional, sectoral, and local public policies that integrate water resource protection, land use trends, and socio-economic interests, this activity will (i) review and update the Territorial Development Study prepared for ACUMAR in October, 2007, develop a technical tool for enabling consensus building among basin key actors on priority issues and programs to create a more sustainable development model for the basin, (ii) finance workshops with key stakeholders on priority issues and programs related to sustainable development in the M-R River Basin, and (iii) develop specific studies proposed as a result of the agreements reached. Studies carried out on the basis of discussions with stakeholders will also identify social impacts of improved territorial planning and suggest strategies to mitigate any negative effects.

43. *Develop a GIS to support planning and decision making in the M-R River Basin.* This activity seeks to support the development and continuous update of a geographic information system (GIS) that will incorporate baseline data and information relating to the main variables driving territorial development, especially those related to the industries located in the basin. The following thematic areas will be covered and correlated: socio-economy; urban; physical-territorial; provision of services; land use; legislation; environmental conflicts and sources of organic and toxic contamination; and location of industries. Under the industrial contamination thematic area, the GIS will include baseline information on about 20 parameters relating to water consumption and pollution loads (organic and toxic) discharged by each of the industries. The information in this centralized database will be instrumental in evaluating the effluent loads from polluting agents discharged to the M-R, along with the capacity of the Rio de la Plata to accept these effluents discharges.

(ii) Institutional Development for Flood Control

44. This sub-component will provide technical assistance to improve and update hydrological studies and management tools to control flooding events. Activities will be executed by SAYDS/ACUMAR in close collaboration with the provincial and municipal governments and are summarized below:

45. *Design and implement a flood control system.* Activities will include (i) improving and updating the flood control database with detailed information about the urban areas affected by flooding (macro and micro drainage) in order to develop a planning and decision-making tool to prioritize the most critical areas. Preliminary criteria for supporting this prioritization include the number of people affected, poverty levels, the impact of flooding on living conditions, and synergy with other relevant investments in the area; (ii) undertaking a detailed technical assessment in the sub-basins where the priority areas are located, including topographical studies to support planning of pre-feasibility studies, assessing major existing infrastructure to identify expected interference and needed institutional coordination to construct a drainage system in the area, and developing pre-feasibility studies outlining technical solution alternatives and incorporating alternatives in the existing flood control investment program.

46. *Update the Hydrological Master Plan (Plan Hídrico) prepared for the Matanza-Riachuelo Basin.* This activity will focus on the priority sub-basins and alternative technical

solutions, taking into consideration climate-change induced changes in rainfall patterns and levels of La Plata River. It will also include studies to update information on hydrological sub-systems and determine flooding risk levels and mapping.

47. ***Improve the Hydro-Meteorological Alert System.*** To strengthen meteorological and hydrological forecasting in the Matanza-Riachuelo Basin, this sub-component proposes a comprehensive approach to optimize the M-R River Basin's hydro-meteorological infrastructure. Upgrades will include new and updated sensors, integrated processing, analysis and data display that supports information dissemination to all users, and meteorological forecasts useful to decision makers. The objectives are to unify the efforts of local authorities and national-level forecasting centers to obtain real-time emergency management information, rainfall flooding forecast and warning information, M-R River flow and *sudestada*⁴³ forecasts, precipitation quantity forecasts, hail forecasts, and early warnings. Based on information from the detection and forecasting stages, the decision to issue *warnings* should be taken in time to allow a proper *response* by relevant authorities and inhabitants to reduce damage through flood preparedness.

48. ***Support municipal capacity strengthening in flood management.*** This activity will include (i) organizing contingency plans to prepare for emergencies, as well as implementation modalities and simulation exercises involving provincial and municipal agencies and local communities, (ii) developing communication plans to provide timely information to communities on forecasted flood events, and (iii) preparing environmental programs and management plans and disseminating information to design the next steps of flood prevention investments.

(iii) Investments in Basic Infrastructure to Support Territorial Development

49. This sub-component will finance pilot interventions to improve infrastructure in low-income urban areas appropriate for urban expansion. The proposed intervention includes technical assistance for developing specific studies and engineering designs, as well as investments in basic infrastructure. The basic infrastructure primarily includes: (i) secondary networks for water and sanitation and the connection of those to the main existing water and sanitation systems; (ii) micro drainage and the section of macro drainage required to operate the micro drainage system; (iii) secondary streets; and (iv) recreational areas. SAyDS/ACUMAR will be the executing agency, and tripartite agreements will be signed between SAyDS, ACUMAR and the different jurisdictions which have primary responsibility to carry out these investments.

50. The approach proposed for the targeted investments will focus on developing intervention models for properly providing basic infrastructure and inducing urban expansion in low population density areas of the basin. The current situation shows a very low population density in the upper section of the basin, and increased expansion (increase in population density), already occurring in the middle section of the basin. The lower section of the basin already features very high population density. Consistent with this situation and with the characteristics of the urban expansion in the middle basin, two intervention models are envisaged:

- (i) "Consolidating" urban areas: provision of urban improvement and provision of basic infrastructure services in dense and low-income settlements in order to improve the

⁴³ *Sudestadas* (literally, "Southeast hits") are exceptionally high tides in La Plata River from strong southeasterly winds.

quality of life in areas that are appropriately located in urban environmental terms. Such settlements are common in the lower portion of the middle basin;

- (ii) Promoting urban expansion: provision of urban improvements and basic infrastructure services in not-yet-dense low-income settlements in order to improve the quality of life of residents, and steer urban expansion in appropriate directions within the basin. Such not-yet-dense low-income settlements are common in the peri-urban upper portion of the middle basin.

51. By consolidating the targeted areas, this subcomponent aims at promoting interventions that will substantially improve living conditions of low-income population segments, and simultaneously induce a more intense use of appropriate urban areas through actions that integrate urban and environmental aspects. Currently, the offer of appropriate and affordable urban sites in the basin is scarce, and the number of unregulated subdivisions of land often located in sensitive environmental areas is increasing, thus driving urban expansion toward the upper basin, where rural uses were once dominant. This dynamic creates enormous pressure for expanding high-cost public infrastructure to distant, sparsely occupied areas. It also creates socially and economically segregated communities characterized by extremely low living standards.

52. Nonetheless, as the urban area expands, already subdivided land in less recent settlements remains vacant, contributing to the rapid socio-economic and environmental degradation that is clearly visible in these areas. In addition, large polygons of land, adjacent to these urban areas, are not incorporated in the informal or formal land market. In consequence, isolated and sparsely populated urban areas emerge following the urban sprawl.

53. Two (or perhaps three) pilot areas are envisioned for subcomponent investments and will comply with the following criteria:

- a) Selection is made in agreement with the provincial and municipal governments involved;
- b) More than 50 percent of population lives below the poverty level;
- c) Proposed basic infrastructure and urban improvements are consistent with a local integrated urban development plan prepared in consensus with local and regional government authorities;
- d) Located relatively near to existing or planned main infrastructure (water, sanitation, drainage, roads) to permit connections in the area;
- e) Integration with other programs such as: Plan Federal de Vivienda; PROMESA; progressive social housing programs supported by non-governmental organizations; investments in basic services executed by ACUMAR with IADB resources.

(iv) Investment in Infrastructure for Flood Control

54. This sub-component will finance microdrainage systems and the macrodrainage works needed to make the selected microdrainage system operational. The areas to receive the investments in drainage systems will be selected through a comprehensive socio-economic cost-benefit analysis in consultation with provincial and municipal governments. Typical areas in need of investments to reduce flooding have low-income settlements located in low-lying areas with insufficient municipal drainage services. Only those drainage solutions unlikely to be impacted by later updates to the Drainage Master Plan (*Plan Hídrico*) will be eligible. It is anticipated that two low-income settlements encompassing approximately 20,000 households will benefit from this component. Once the construction is concluded, these systems will be operated by the respective municipal governments.

55. The areas to be selected for subcomponent investments will comply with the following criteria:

- a) Selection is made in agreement with the provincial and municipal governments involved;
- b) More than 40% of the population lives below the poverty line;
- c) Micro technical solutions must be functionally consistent with the regional hydrological system;
- d) Areas should be ready to connect to existing or planned drainage investments in order to maximize the benefits of the regional flood control system;
- e) Consistent with the resources available;
- f) A separate legal agreement will be signed with the different jurisdictional entities which have primary responsibility to carry out the proposed investments (i.e. PBA, municipalities or AySA in the case of water and sewers), whereby the relevant entity provides its written consent to permit SAYS/ACUMAR to carry out and finance the pertinent pilot investment, and assumes the obligation to operate and maintain the pilot investment thereafter.

Component 4. Institutional Strengthening and Project Management (US\$18.2 million total, US\$18.2 million loan)

56. In line with the development objectives, a separate TA component has been prepared to support the design and implementation of measures required in both the short and long term to establish the necessary institutional capacity for the new basin authority. This capacity would allow ACUMAR to (i) implement the proposed infrastructure works, (ii) implement the pollution control and enforcement program, and (iii) support the strategic planning capacities of the agencies involved. The component has been divided into five sub-components, described below.

(i) Institutional Strengthening of ACUMAR

57. ACUMAR has an institutional structure defined under Resolution 5/2007. Using a participatory process involving the main stakeholders, this structure will need to be adjusted to respond to its institutional needs. The methodology created by Alain Tobolem (Analysis System and Institutional Capacity Development) has been proposed. Consideration has also been given to using the same institutional strengthening strategy used in the Bank financed GEF project *Coastal Contamination Prevention and Marine Management* that was implemented by SAYS between 2003 and 2007. ACUMAR's institutional development needs will be incorporated into a plan that will be developed over the course of six months, where the objectives and activities will be identified to overcome problems and that will allow for the assessment and monitoring of progress.

58. This subcomponent will support three workshops with key decision makers, staff and representatives from other institutions within ACUMAR. Representatives from civil society that have experience using the Tobolem methodology will also be invited to the workshops. A series of interviews with institutional and other civil society stakeholders is envisioned.

59. As a result of the participatory process, an institutional strengthening plan for ACUMAR will be developed and will include key activities, the required institutional structure, and defined needs in terms of skilled personnel, office equipment, furniture, human resources and other necessary material. ACUMAR is currently supported by SAYS staff, and will be able to count on technical support from universities and research institutes

during the institutional strengthening process. This subcomponent will also support the necessary investments to provide ACUMAR with proper office space.

(ii) Communication Strategy

60. This sub-component will finance actions for the creation of a communication unit and implementation of a communications strategy during project implementation. The communication strategy is designed to inform key actors and stakeholder groups in the basin in a transparent manner about the project; strengthen the credibility and support of the project; and contribute to its overall consultation, participation and empowerment framework. (For further details, see Annex 11D).

(iii) Establishment of a public information office in ACUMAR

61. This subcomponent will support the necessary investments such as human resources, office equipment, infrastructure and logistic tools, for the proper functioning of a public information office within ACUMAR.

(iv) Technical and monitoring studies

62. This sub-component includes financing for additional studies required during the course of project implementation, and for providing expert support (including a panel of recognized independent experts in the fields of environment, engineering, biodiversity, modeling etc.) to provide periodic advice in conjunction with project implementation. Studies will include additional data collection and analytical work relating to impact and water quality monitoring in the M-R River and La Plata River, as well as the control of high groundwater tables in the basin area. This subcomponent may also finance other studies required for the clean-up of the M-R River Basin as agreed with the Bank.

(v) Project management

63. Management of the loan requires human resources, physical space and equipment within SAYDS' UCOFI and proposed UEP units to strengthen purchasing, contracting and procurement processes as well as technical project management. The Bank's procurement capacity assessment of UCOFI identified the following needs: (i) engaging a specialist in procurement as well as an archives assistant; (ii) acquiring additional equipment (furniture, computing equipment, and space for archives); and (iii) improving and adapting physical spaces. Other needs will be determined based on the actual work load once the program begins. This sub-component will finance (i) consultants to support SAYDS/UCOFI for fiduciary activities and technical expertise and SAYDS/UEP for technical project management, (ii) UCOFI's and UEP's incremental operational expenses; and (iii) office equipment/minor remodeling of physical space for UCOFI and UEP as required.

Annex 5: Project Costs

ARGENTINA: Matanza-Riachuelo Basin Sustainable Development Project

Project Costs for APL-1

Project Cost By Component and/or Activity	Local	Foreign	Total
	US \$million	US \$million	US \$million
Sanitation Component			
Left Bank and Costanera Bypass Collector	130	30	160
Right Bank Collector	248	27	275
Industrial Effluents Conveyor and Pumping Station	24	10	34
Preliminary Treatment Plant <i>Riachuelo</i>	92	50	142
Pumping stations into and out of the Treatment Plant <i>Riachuelo</i>	60	62	122
Sub-aquatic Outfall <i>Riachuelo</i>	219	55	274
Subtotal Component 1 APL-1 (*)	773	234	1007
Industrial Pollution abatement Component			
Pollution diagnostics and policy analysis (*)	4		4
Improvement of the monitoring system for industrial wastes (*)	4		4
Support for fine-tuning of the industrial discharges strategy(*)	2		2
Design/ implementation of program for encouraging corporate environmental responsibility (*)	1		1
Support for pollution control and cleaner production investments by small and medium enterprises (SMEs)	32	15	47
Subtotal Component 2 APL-1	43	15	58
Environmental Territorial Management Component			
Creation/update of a GIS based industrial pollution and territorial database (*)	3.35		3.35
Institutional development for environmental territorial planning (*)	2.25		2.25
Institutional development for flood control (*)	3		3
Investments in basic infrastructure for territorial development	21.4		21.4
Investments in basic infrastructure for flood control	33		33
Subtotal Component 3 APL-1	63	0	63
Institutional Strengthening and Project Management			
Institutional and physical strengthening of ACUMAR (*)	7		7
Communications program of ACUMAR (*)	4		4
Establishment of the public information office of ACUMAR	1		1
Technical studies and monitoring (*)	1.5		1.5
Project administration	0.5		0.5
Subtotal Component 4 APL-1	14	0	14
Total Baseline Cost	893	249	1,142
<i>Price Contingencies (20%)</i>	<i>179</i>	<i>50</i>	<i>228</i>
<i>Physical Contingencies (10%)</i>	<i>89</i>	<i>25</i>	<i>114</i>
Total Project Costs	1,161	324	1,485
Interest during construction			
Front-end Fee		2.1	2.1
Total Financing Required			1,487

* Including Consulting and/or supervision costs

Project Costs for APL 2

Project Cost By Component and/or Activity	Local	Foreign	Total
	US \$million	US \$million	US \$million
Sanitation Component			
Bajo Costanera expansion collector (first stage)	70	18	88
Effluent Pumping station <i>Berazategui</i>	68	17	85
Berazategui preliminary treatment plant, signed contract for 33.5 m3/s	120	28	148*
Sub-aquatic outfall <i>Berazategui</i>	180	48	228
Primary and secondary sewage networks	325	81	406
Primary and secondary sewage networks financed by users	325	75	400
Subtotal Component 1 APL-2 (**)	1,088	267	1,355
Industrial Pollution abatement Component			
Pollution diagnostics and policy analysis (**)	8		8
Pollution control activities	5		5
Implementation of environmental disclosure program	2		2
Expansion of matching grants for SMEs pilot program	30	35	65
Subtotal Component 2 APL-2	45	35	80
Environmental Territorial Management Component			
Institutional development for environmental territorial planning (**)	4		4
Institutional development for flood control (**)	6		6
Investments in basic infrastructure for territorial development	30		30
Investments in basic infrastructure for flood control	60		60
Subtotal Component 3 APL-2	100	0	100
Institutional Strengthening and Project Management			
Consulting services (**)	5	4	9
Equipment (for office and supervision)	0.5		0.5
Works (office improvements)	0.5		0.5
Subtotal Component 4 APL-2	6	4	10
Total Baseline Cost	1,239	306	1,545
<i>Price Contingencies (20%)</i>	<i>248</i>	<i>61</i>	<i>309</i>
<i>Physical Contingencies (10%)</i>	<i>124</i>	<i>31</i>	<i>155</i>
Total Project Costs	1,611	398	2,009
Interest during construction			
Front-end Fee		2.9	2.9
Total Financing Required			2,012

*Awarded contract AR\$458 million, not financed by the World Bank

** Including consulting and/or supervision costs

Project Costs Summary Table (APL-1 + APL-2)

Project Cost By Component and/or Activity	APL-1			APL-2		
	Local US\$m	Foreign US\$m	Total US\$m	Local US\$m	Foreign US\$m	Total US\$m
Sanitation Component	773	234	1,007	1,088	267	1,355
Industrial Pollution Abatement Component	43	15	58	45	35	80
Environmental Territorial Management Component	63	0	63	100	0	100
Institutional Strengthening and Project Management	14	0	14	6	4	10
Total Baseline Cost	893	249	1,142	1,239	306	1,545
<i>Price Contingencies (20%)</i>	<i>179</i>	<i>50</i>	<i>228</i>	<i>248</i>	<i>61</i>	<i>309</i>
<i>Physical Contingencies (10%)</i>	<i>89</i>	<i>25</i>	<i>114</i>	<i>124</i>	<i>31</i>	<i>155</i>
Total Project Costs	1,161	324	1,485	1,611	389	2,009
Interest during construction						
Front-end Fee			2.1			2.9
Total Financing Required			1,487			2,012
Total Financing Required APL-1 + APL-2 US\$	3,500					

Annex 6: Implementation Arrangements

ARGENTINA: Matanza-Riachuelo Basin Sustainable Development Project

Introduction

1. The National Government will be the Borrower for the proposed loan. SAYDS will be responsible for overall project coordination and will directly implement macro-component 2 together with ACUMAR (as the inter-jurisdictional entity). While ACUMAR has been created as an inter-jurisdictional entity with overall responsibility for implementing the PISA, the Government recognizes that ACUMAR currently has limited implementation capacity which will be reinforced and developed over the course of APL-1. Regarding macro-component 1, AySA will be the executing entity while ACUMAR, will ensure the consistency of project activities with the PISA in line with its legal mandate.

2. SAYDS has informed the Bank that it is moving to a new management model for internationally financed projects under which procurement, financial management and general administrative functions are centralized in a single (existing) unit (UCOFI) while the technical leadership for each operation is vested in separate project implementation units (UEP). In line with this strategy, SAYDS will exercise its overall project coordination role for the present project through two units: UCOFI (*Unidad de Coordinación para la Gestión Técnica y Financiera Internacional*) and UEP (*Unidad de Ejecución del Proyecto*). The establishment of UEP will be a condition for loan effectiveness. UCOFI will manage the project's US Dollar denominated designated account as well as a Peso-denominated project account from which it will make all payments for goods, works, and services procured under components 2, 3 and 4. UCOFI will also carry out all procurement for these components with technical support from UEP and ACUMAR, which will help produce terms of reference and technical specifications, participate in bid/proposal evaluation and support contract supervision. Even though UCOFI has experience implementing projects financed by international financial institutions, funds have been set aside under component 4 (sub-component 5) to strengthen it and UEP further to cope with the demands of the proposed project. Strengthening would include contracting of additional staff (particularly a procurement specialist and assistant for UCOFI and technical staff as needed for UEP), the purchase of additional equipment and furniture and upgrades to UCOFI's and UEP's physical space.

3. For component 1, UCOFI will transfer funds to AySA pursuant to the terms of a subsidiary agreement between the National Government and AySA on terms acceptable to the Bank. The signing of this subsidiary agreement will be a condition of effectiveness of the loan. AySA will contract all goods, works and consulting services and make all payments to consultants and contractors using its regular technical units. These units include AySA's procurement and legal departments (for the preparation of bidding documents) and AySA's planning department for technical aspects. The environmental, social relations and communications departments of AySA will also participate in project implementation. While AySA has experience with large works, the concurrent execution of a recently approved IADB loan, works financed by other development banks (e.g. BNDES) and the works foreseen under component 1 is expected to challenge its implementation capacity. For this reason, the Left Bank Collector/Bajo Costanero Bypass, Riachuelo preliminary treatment plant and Riachuelo Sub-Aquatic Outfall (to be tendered as a single bidding process) will be supervised by one or more engineering consulting firms specialized in tunneling, large civil works and marine infrastructure. Subsequent works (including the Right Bank Collector (or

alternative) and the Industrial Collector) would also be supervised by independent engineering consulting firms hired by AySA. ACUMAR will certify that the package of works proposed under component 1 as well as key environmental design parameters are consistent with the PISA.

4. Official communications with the Bank on disbursement issues will be channeled through the SAYDS/UCOFI. On procurement matters, AySA will communicate directly with the Bank for component 1 while UCOFI handles communications for components 2, 3 and 4.

5. UCOFI will submit withdrawal applications to the Bank and handle all disbursement issues. The Loan proceeds will be transferred from the Bank to a Designated Account (DA) in dollars managed by UCOFI in a commercial bank acceptable to the Bank. For component 1, UCOFI will transfer an initial deposit of an amount to be agreed with the Bank to a "project account" in pesos opened by AySA in a bank acceptable to the Bank, from which AySA will make payments for eligible expenditures. In order to monitor the uses of Loan funds, this bank account should not receive funds from other sources. In addition, transfers from UCOFI will be made solely on the basis of eligible expenditures incurred or advances supported by expected expenditure reports prepared by AySA. For payments under components 2, 3 and 4, UCOFI will open its own project account in pesos.

6. An Operational Manual with a detailed description of the agreed implementation arrangements is under preparation and is a condition of effectiveness.

A. Implementation Arrangements for Component 1 (Sanitation)

7. The sanitation component that will be implemented by AySA pursuant to the terms of the subsidiary agreement mentioned in paragraph 3 above consists of large infrastructure works whose implementation requires expertise, knowledge, and capacity. Various organic units of AySA need to participate in the implementation of this component, and the entire AySA institutional structure needs to support it. Therefore, AySA will constitute a coordination unit rather than a project implementation unit which will coordinate the implementation of the sanitation component and involve the organic units of the company in implementation processes as required.

8. The coordination unit will consist of a project coordinator, a procurement specialist, a financial management specialist, and a technical expert, as well as support staff. The procurement and legal departments of AySA will be involved in preparing the bidding documents, and the planning department of AySA will be in charge of technical aspects. When legal support is required, it will be provided by the legal department of AySA. The environmental, social relations, and communications departments of AySA will also participate in project outreach program.

9. AySA will be in charge of the conceptual and detailed designs required under the sanitation component. AySA will (i) prepare the bidding documents related to implementing the sanitation component, (ii) conduct bidding processes and propose contract awards; (iii) sign contracts with construction contractors and supervision firms and assume the role of "employer" during the supervision process; and (iv) make all payments due under these contracts.

B. Implementation Arrangement for Components 2 (Industrial Pollution Abatement), 3 (Environmental Territorial Management) and 4 (Institutional Strengthening and Project Management)

10. The SAYDS and ACUMAR (as the interjurisdictional entity) will be the implementing agencies for components 2, 3 and 4. SAYDS, through UCOFI, will prepare bidding documents and requests for proposals with technical input from UEP and ACUMAR. Bids and proposals will be jointly evaluated by the SAYDS and ACUMAR but SAYDS will sign all contracts for components 2, 3 and 4 and make all payments. There will be no transfer of loan proceeds to ACUMAR. Technical studies, engineering designs and supervision of works will be contracted out to “technical support agencies” including universities, technical institutes and – where warranted – to individual consultants or firms.

Annex 7: Financial Management and Disbursement Arrangements
ARGENTINA: Matanza-Riachuelo Basin Sustainable Development Project

A. Executive Summary and Conclusion

1. A Financial Management (FM) Assessment of the arrangements for the proposed project was carried out in accordance with OP/BP 10.02 and Bank specific guidelines.⁴⁴ It is considered that the financial management arrangements in place meet minimum Bank requirements. The Coordination Unit for International Technical and Financial Management (*Unidad de Coordinación para la Gestión Técnica y Financiera Internacional*, or UCOFI), the unit in the Secretariat of Environment and Sustainable Development (*Secretaría de Ambiente y Desarrollo Sustentable*, or SAYDS) will be responsible for undertaking the financial management functions of the project comprising budgeting, accounting and financial reporting, internal control, disbursements and documenting expenditures to the Bank and external auditing arrangements. UCOFI has qualified staff experienced in World Bank-financed activities. However, there is a moderate element of risk because of the large amount of funding to be managed by Argentina Water and Sanitation Utility (*Agua y Saneamientos Argentinos S.A.*, or AySA) under Component 1 of the project.

B. Country Issues

2. A Country Financial Accountability Assessment (CFAA)⁴⁵ for Argentina was prepared jointly by the Bank and the IADB in 2006/2007, and the final report delivered to the Government of Argentina (GoA) in March 2008. The main findings indicate that Argentina's national Public Financial Management (PFM) has well-designed and legally supported systems to control funds received by the GoA from all sources and to provide sufficient information to determine whether funds are used as planned. However, the CFAA concludes that the country's PFM performance is below expectations. The Country Assistance Strategy⁴⁶ (CAS) states that the FM portfolio risk at the federal level is Moderate.

3. ***Fiduciary Action Plan.*** The Argentina CAS includes a Fiduciary Action Plan (FAP) to help strengthen the operating environment for Bank projects in Argentina. The FAP basically consists of three components: raising public awareness, bolstering Bank fiduciary monitoring, and increasing transparency and competition practices in public procurement. Regarding FM, the Plan aims to (i) improve the timeliness of external audit compliance for Bank-financed operations, (ii) increase the strategic focus and coverage of supervision tools to assess fiduciary risk in operations, and (iii) support streamlining and harmonization of fiduciary processes and reliance on country systems when these meet adequate fiduciary standards.

4. ***Project fiduciary measures linked to the objectives of the FAP.*** The following measures are part of the project's FM arrangements and contribute to the FAP objectives:

⁴⁴ World Bank. 2005 Financial Management Practices in World Bank-Financed Investment Operations. Washington, DC: World Bank Financial Management Board. November 3.

⁴⁵ World Bank. Argentina Country Financial Accountability Assessment. Report No. 39228-AR. Washington, DC. March, 2008.

⁴⁶ World Bank. 2006. Argentina Country Assistance Strategy 2006–2008: Fiduciary Assessment. Financial Management. May 4.

- a. ***Use of the country system.*** The Federal Government system specially designed for the execution of multilateral finance operations, the System of Financial Administration and Control for Agencies Executing Foreign Loans (*Sistema de Administración y Control Financiero para las Unidades Ejecutoras de Préstamos Externos*, or UEPEX) will be utilized to maintain the project accounts. UEPEX provides a good ex-ante internal control framework and is partly integrated into the national budget execution process.
- b. ***Clear administrative procedures.*** The project OM acceptable to the Bank will clearly define the roles and responsibilities of UCOFI and AySA in relation to the FM procedures of the project.
- c. ***Continuous support to the Office of the National Auditor General (Auditoría General de la Nación, or AGN) to ensure timely audit compliance for the project.*** Upon receiving audit findings, follow-up on the implementing agencies' action plans to address the auditors' recommendations.
- d. ***FM supervision.*** To ensure continuous adequacy of FM arrangements, evaluate project internal controls, and update assessed risk, at least one on-site visit as part of the project team is planned for the first year.

C. Risk Assessment and Mitigation

5. The risk assessment process aims to identify FM risks in order to take appropriate measures to mitigate identified project risks. This enables the Bank make decisions on the appropriate level of supervision and allocation of FM resources in a manner consistent with assessed risks. Although projects of this type entail certain risks because of the large amount of money involved, every effort has been made to minimize risk stemming from FM. The simplest flow of funds has been agreed to with SAyDS and AYSA to avoid any additional complexity and reduce associated risk. FM risk is considered Moderate. Table 7.1. shows the risk assessment for the Matanza-Riachuelo Basin Sustainable Development Project and the measures proposed to mitigate those risks.

Table 7.1. Risk assessment and mitigation measures

Risk	Risk rating*	Risk mitigation measures embedded in project design	Condition
Inherent Risk: Moderate			
• Country level	Moderate	• Fiduciary Action Plan (FAP)	No
• Entity level: Coordination Unit for International Technical and Financial Management (<i>Unidad de Coordinación para la Gestión Técnica y Financiera Internacional</i> , or UCOFI)	Moderate	• UCOFI skilled and experienced staff in WB-financed operations	No
• Entity level: Argentina Water and Sanitation (<i>Agua y Saneamientos Argentinos S.A.</i> , or AySA)	Moderate		
• Project level	Moderate		
Control Risk: Moderate			
• Budgeting	Moderate	• Use of country's Integrated Financial Information system (<i>Sistema Integrado de Información Financiera</i> , or SIDIF)	
• Accounting	Moderate	• Use of government System of Financial Administration and Control for Agencies Executing Foreign Loans (<i>Sistema de Administración y Control Financiero para las Unidades Ejecutoras de Préstamos Externos</i> , or UEPEX) designed for multilateral-financed projects	
• Internal control	Moderate		
• Funds flow	Substantial	• Centralized flow of funds • Advances to AySA on the basis of eligible expenditures • Separate bank account in AySA to deposit advances from the loan and make payments for eligible expenditures • Monthly AySA reconciliation and reporting on uses of funds	
• Financial reporting	Moderate		
• Auditing	Moderate	• Follow-up through the FAP	
Overall Residual Risk Rating	Moderate		

* L = low, M = modest, S = substantial, H = high

D. Strengths and Weaknesses

6. **Strengths.** The major strength for risk mitigation is the fact that SAYDS/UCOFI has skilled staff capable of fulfilling project needs. Previous experience in the implementation of Bank-financed projects is also an asset.

7. **Weaknesses.** The large amounts of money that would flow through AySA under Component 1 may pose some additional fiduciary risk.

E. Implementing Entities

8. The sanitation component (Component 1) will be implemented by AySA making use of its own organizational structure and line departments including the relevant accounting, budget, and finance units of the company. A small coordination unit will be created within AySA to facilitate coordination of activities among the different departments and with the SAYDS. The coordination unit will consist of a Project Coordinator, a procurement specialist, a financial management specialist, and a technical expert, as well as support staff.

9. The Coordination Unit for International Technical and Financial Management (*Unidad de Coordinación para la Gestión Técnica y Financiera Internacional*, or UCOFI) in the Secretariat of Environment and Sustainable Development (*Secretaría de Ambiente y Desarrollo Sustentable*, or SAYDS) will undertake the FM functions of the project as a whole. UCOFI has qualified FM staff experienced in Bank-financed activities that is capable of undertaking the FM functions of the Project. AySA will receive funds to make payments for eligible expenditures under the Sanitation component, but UCOFI will have overall responsibility for the FM functions of the entire project, including accounting, reporting (e.g., preparing interim unaudited financial reports, or IFR), internal controls, and management of the Designated Account (DA) and external audit process. In time, UCOFI may need additional FM staff to handle the additional workload. It has been proposed that a professional accountant with experience in WB-funded projects who is currently working on the IDB project also managed by UCOFI will be reassigned to the project once the Loan is effective.

F. Budgeting and Accounting

10. Budget execution in Argentina is recorded in the Federal Government's integrated budget and accounting system (*Sistema Integrado de Información Financiera*, or SIDIF) and subject to control over the budgetary execution process. SAYDS will be required to create and maintain a specific budget line entry for the project in its annual budget to allocate budgetary resources and track project execution, specifying the sources of funds. The UEPEX system will be used to maintain project accounting records. This in-house information tool was developed by the Federal Government. Its use is mandatory for multilateral-financed operations at the federal level, and it is considered adequate for accounting purposes. The project chart of accounts will reflect disbursement categories, project components, and sources of financing. Cash basis accounting will be used to record the project transactions (to be confirmed).

11. AySA will also reflect the transactions for Component 1 in its own accounting books and annual financial statements. AySA will be required to submit to the Bank its annual audited financial statements as a revenue-generating entity.

G. Internal Control and Internal Auditing

12. UCOFI within SAYDS and the Office of the Cabinet of Ministers (*Jefatura Gabinete de Ministros*, or JGM) are subject to internal audit by the General Syndicate of the Nation (*Sindicatura General de la Nación*, or SIGEN), which is the Federal Government's internal audit agency under the jurisdiction of the executive branch. SIGEN supervises and coordinates the actions of Internal Audit Units (IAUs) in all federal agencies, approves their audit plans, conducts research and independent audits, and systematizes the information from its own reports and those produced by the IAUs. If necessary, the internal audit reports on project performance will be reviewed during project supervision.

13. AySA, as a revenue-generating company, has an Internal Audit Unit, the *Dirección de Auditoría Interna* (DAI), that is also under the oversight of SIGEN. The DAI is headed by a professional accountant with some auditing background and broad experience in the public sector. DAI staff also comprises two professional auditors: a supervisor and a senior auditor for administrative and commercial operations. In addition, DAI has technical and information technology staff. DAI prepares annual audit plans following a risk-based approach. Annual audit plans are submitted to SIGEN for review and approval. The internal audit function is supplemented by an Audit Committee composed of three members of the board of directors of AySA.

H. Funds Flow and Disbursement Arrangements

14. *Uses of funds.* All the project uses of funds will be supported by documentary evidence for the related works, goods, and services procured in line with Bank guidelines and procurement thresholds for the project.

15. The following disbursement methods may be used under the Loan: Advance, reimbursement, and direct payment.

16. Flow of Funds for Components 2; 3 and 4. To facilitate project implementation, the Coordination Unit for International Technical and Financial Management (*Unidad de Coordinación para la Gestión Técnica y Financiera Internacional*, or UCOFI) will operate a segregated DA in U.S. dollars to be opened at the National Bank of Argentina (*Banco de la Nación Argentina*, or BNA). Funds deposited into the DA as advances will follow the Bank's disbursement operating policies and procedures described in the Disbursement Letter. Withdrawals from the DA will be made solely for payments of eligible expenditures as incurred. The proposed ceiling for advances to the DA would be \$40 million until disbursements reach a total of \$100 million, and \$90 million thereafter, estimated to be sufficient for peak disbursement periods of project execution. UCOFI will also control an operational bank account in local currency (pesos) from which it will make payments for eligible expenses incurred under components 2; 3 and 4 and from which it will transfer advances in pesos to AYSA for payments under Component 1.

17. Flow of Funds for Component 1. UCOFI will submit withdrawal applications to the Bank and handle all disbursement issues. The Loan proceeds will be transferred from the Bank to the Designated Account (DA) in dollars managed by UCOFI. UCOFI will transfer an initial deposit of an amount to be agreed with the Bank to a "special account" in pesos opened by AySA in a bank acceptable to the Bank, from which AySA will make payments for eligible expenditures under Component 1. In order to monitor the uses of Loan funds, this bank account should not receive funds from other sources. In addition, transfers from UCOFI

will be made solely on the basis of eligible expenditures incurred or advances supported by expected expenditure reports prepared by AySA on the basis of signed contracts and planned certifications of works as determined by the Bank. AYSA will report to SAYDS/UCOFI on uses of funds on a monthly basis.

18. Flow of Funds for subprojects under the matching grant sub-component (\$41 million of Bank financing) the following procedure will be followed for the provision of grants for investment sub-projects to small and medium enterprises (SMEs). Under the matching grants program, the SME will purchase eligible consulting services, equipment (goods), and small works following Bank procurement guidelines and procedures. UCOFI will then pay directly to the consultant/provider/contractor. The percentage of eligible expenditures to be covered with Loan funds will not be transferred by UCOFI to beneficiaries (consultants, contractors, suppliers) until eligible expenditures are incurred and after SME's share of expenditures have been paid or satisfactory evidence that SME funds are available for matching loan funds has been submitted to ACUMAR/UCOFI.⁴⁷

19. The following supporting documentation shall be provided with each application for withdrawal from the Loan account as indicated below:

- ***For requests for Reimbursement :***

- Records for payments for: (i) Goods and Non-Consulting Services against contracts valued at US\$ 500,000 equivalent or more; (ii) Consulting Firms against contract valued at US\$ 200,000 equivalent or more; (iii) Individual Consultants against contracts valued at US\$ 100,000 equivalent or more; (iv) Civil Works, under Part 1 (a) of the Project, against contracts valued at US\$ 5,000,000 equivalent; (v) Works, except for Part 1 of the Project, against contracts valued at US\$ 500,000 equivalent or more.
- Statements of Expenditures for payments against all other contracts below SOE thresholds, including all Training and Operating Costs.
- List of payments against contracts that are subject to the Bank's prior review.

- ***For reporting eligible expenditures paid from the Designated Account:***

- Records for payments for: (i) Goods and Non-Consulting Services against contracts valued at US\$ 500,000 equivalent or more; (ii) Consulting Firms against contract valued at US\$ 200,000 equivalent or more; (iii) Individual Consultants against contracts valued at US\$ 100,000 equivalent or more; (iv) Civil Works, under Part 1 (a) of the Project, against contracts valued at US\$ 5,000,000 equivalent; (v) Works, except for Part 1 of the Project, against contracts valued at US\$ 500,000 equivalent or more.
- Statement of Expenditures for payments against all other contracts below SOE thresholds, including all Training and Operating Costs.
- List of payments against contracts that are subject to the Bank's prior review.

⁴⁷ Typically, matching grant packages require a 1:1 level of matching funds from SMEs. In the case of micro-enterprises, however, a 2:1 (project: enterprise) ratio is more commonly recommended.

- **For requests for Direct Payment:** records evidencing eligible expenditures, e.g., copies of receipts, supplier invoices.

20. All SOEs supporting documentation of expenditures will be retained either by AySA or SAYDS/UCOFI for post review and audit purposes for up to five years after the final withdrawal from the Loan account.

21. The proceeds of the Loan will be disbursed against the disbursement categories listed in table 7.2.

Table 7.2. Allocation of Loan Proceeds

Category	Amount of the Loan Allocated (expressed in US\$)	Percentage of Expenditures to be financed (inclusive of Taxes)
(1) Works under Part 1 (a) of the Project	600,000,000	80% of each Eligible Expenditure incurred by AySA until disbursements under this Category have reached an aggregate amount of \$400,000,000; and 40% of each Eligible Expenditure incurred by AySA thereafter
(2) Consultants' services, goods and training under Part 1 (b) of the Project	30,000,000	100%
(3) Works, goods and/or consultants' services under CRI Subprojects	40,000,000	100% of the amounts disbursed under the relevant CRI Grant
(4) Eligible Expenditures for Parts 2 (a) through (d) and 3 of the Project	55,000,000	80%
(5) Eligible Expenditures for Part 4 of the Project	14,000,000	100%
(6) Unallocated	98,900,000	
(7) Front-end Fee	2,100,000	Amount payable pursuant to Section 2.03 of this Agreement in accordance with Section 2.07 (b) of the General Conditions
(8) Premiums for Interest Rate Caps and Interest Collars	0	Amounts payable pursuant to Section 2.07 (c) of this Agreement
TOTAL AMOUNT	840,000,000	

Notes:

(a) the term "Training" means expenditures (other than those for consultants' services) incurred by AYSA, as approved by the Bank on the basis of an annual budget acceptable to the Bank, to finance reasonable transportation costs and per-diem of trainees and trainers (if applicable), training registration fees, training materials, and rental of training facilities and equipment under Part 1(b) of the Project;

(b) "Eligible Expenditures for Parts 2 (a) through (d) and 3 of the Project" means (without limitation to the definition of Eligible Expenditure set forth in paragraph 36 of the General Conditions) any expenditure for

goods, works, consultants services, Non-Consultant Services, and/or Training under said Parts of the Project (as the case may be). For purpose of this definition, the term 'Training' means expenditures (other than those for consultants' services) incurred by the Borrower, through SAYDS, and/or by ACUMAR (in connection with its participation in the carrying out of said Parts of the Project), as approved by the Bank on the basis of an annual budget acceptable to the Bank, to finance reasonable transportation costs and per-diem of trainees and trainers (if applicable), training registration fees, training materials and rental of training facilities and equipment under said Parts of the Project; and

(c) "Eligible Expenditures for Part 4 of the Project" means (without limitation to the definition of Eligible Expenditure set forth in paragraph 36 of the General Conditions) any expenditure for goods, works, consultants services, Non-Consultant Services, Training and/or Operating Costs under said Part of the Project (as the case may be). For purpose of this definition, the term: (a) 'Operating Costs' means reasonable recurrent expenditures, based on an annual budget previously approved by the Bank, that would not have been incurred by the Borrower, through SAYDS, and/or by ACUMAR (in connection with its participation in the carrying out of said Part of the Project) absent the Project, for Project administration costs (including office rent), operation and maintenance of office equipment, non-durable goods and salaries for civil servants assigned to carry out administrative (e.g. secretarial) responsibilities under said Part of the Project (but excluding any other type of civil servant salaries); and (b) 'Training' means expenditures (other than those for consultants' services) incurred by the Borrower, through SAYDS, and/or by ACUMAR (in connection with its participation in the carrying out of said Part of the Project), as approved by the Bank on the basis of an annual budget acceptable to the Bank, to finance reasonable transportation costs and per-diem of trainees and trainers (if applicable), training registration fees, training materials and rental of training facilities and equipment under said Part of the Project.

22. **Retroactive Financing:** The Bank will reimburse the expenditures for payments made up to one year prior to the expected date of the Loan Agreement. These expenditures will not exceed US\$6,000,000 equivalent.

23. The new Policy Framework on the Eligibility of Expenditures in World Bank Lending applies to this project, since the country financing parameters for Argentina were approved by the Regional Vice President on May 11, 2005.

24. The project will request access to the Bank's Client Connection web page to download the Withdrawal Application Form and to periodically reconcile its bank account and resources received from the different sources.

I. Financial Reporting

25. UCOFI will be in charge of preparing the annual and interim financial statements for the project as a whole in line with Bank requirements. For that purpose, AySA will produce financial information for Component 1 and send it to UCOFI, which will compile all the information in one financial report.

26. The UEPEX system will be used to produce the requisite financial statements following public sector accounting standards in Argentina. The public sector accounting rules are comprehensive and consistent with public international standards, in particular IPSAS # 1. Said standards are set by the General Accounting Office (*Contaduría General de la Nación*, or CGN).

27. A draft format of the annual financial statements prepared by UCOFI and reviewed by the Bank will be incorporated to the Operational Manual. UCOFI will also prepare semi-annual Interim Financial Reports (IFRs) for monitoring and evaluation that will be part of project progress reports. The IFRs will contain the following:

- (i) A financial section stating cash receipts by sources and applications by main expenditure classification for the period and cumulatively (project life), as well as

beginning and ending cash balances of the project and a statement of accumulated investments by project component with a comparison between actual and planned expenditures;

- (ii) An output monitoring section that sets forth physical progress in project implementation and explains variances between actual and forecast implementation targets.

J. External Auditing Arrangements

28. The annual financial statements of the project will be audited by an acceptable auditor following terms of reference and conducted in accordance with auditing standards acceptable to the Bank. It has been proposed that the AGN be the external auditor for the project. The annual audit will cover all funding and expenditures reported in the project financial statements and will be submitted to the Bank within 6 months after the end of the reported period. For audit purposes the fiscal year will be the calendar year. AGN audit services will not be financed through the loan.

29. As part of the assessment tasks, the audited financial statements of AySA for the years 2006 (including internal control letter) and 2007 were submitted to the Bank FMS. The external audit reports do not show any accountability issue but some reconciliation differences and administrative issues. Table 7.3. lists the audit reports that the project will be required to submit and the due dates for submission.

Table 7.3. Audit reports and due dates

Audit report	Due date
1. Project Financial Statements	June 30 of each year
2. Special Opinions	June 30 of each year
• Statement of expenditure (SOE) an opinion on the eligibility of expenditures reported and correct use of Loan funds	
• Designated Account	
• Entity financial statements (Argentina Water and Sanitation (<i>Agua y Saneamientos Argentinos S.A.</i>, or AySA))	

K. Supervision Plan

30. The initial supervision plan is shown on the table below. Scope of supervision will be adjusted during project implementation by the assigned FMS on the basis of FM performance and updated risk.

Table 7.4. Supervision plan

Type	Timing	Mechanism	Objective
Visit	Annual	Participation in project team's supervision missions	• Review financial management (FM) system and controls; • Update assigned risk; • Review Designated Account (DA) reconciliations of uses of funds; • Follow up on External Audit issues; • Review Statement of Expenditure (SOE) as needed.
IFR Review	Semiannual	Desk review of IFR	• Review IFR information consistency
Audit review	Annual	Over the Audit Report submitted to the Bank	• Review Audit Report.

L. Action Plan

31. Pending steps at the end of the FM assessment and the expected date of completion are presented in Table 7.5.

Table 7.5. Pending steps on FM

Action	Responsible entity	Completion date
1. Request a specific budgetary line in the Secretariat of the Environment and Sustainable Development (<i>Secretaría de Ambiente y Desarrollo Sustentable</i> , or SAYDS). Annual budget to follow project execution	Coordination Unit for International Technical and Financial Management (<i>Unidad de Coordinación para la Gestión Técnica y Financiamiento Internacional</i> , or UCOFI)	To be included in the first year of project implementation and maintained thereafter Dated Covenant
2. Prepare the FM Section of the Project Operational Manual comprising the following: a) Flow of funds for all project components; b) Administrative procedures regarding FM; c) Chart of accounts; d) IFR format agreed with the Bank; e) A Draft Format of the Annual Financial Statements.	SAYDS/UCOFI and AySA	A draft Operational Manual has been received

Annex 8: Procurement Arrangements

ARGENTINA: Matanza-Riachuelo Basin Sustainable Development Project

1. Procurement for the proposed project will be carried out in accordance with the World Bank's Procurement Guidelines ("Guidelines: Procurement under IBRD Loans and IDA Credits," May 2004, revised October 2006), Consultant Guidelines ("Guidelines: Selection and Employment of Consultants by World Bank Borrowers," May 2004, revised October 2006), and the provisions stipulated in the Loan Agreement. The items under different expenditure categories are described in general below. For each contract to be financed by the Loan, the Borrower and the Bank agree on the different procurement methods or consultant selection methods, the need for pre-qualification, estimated costs, prior review requirements, and time frame in the procurement plan. The procurement plan will be updated at least every 3 months or as required to reflect actual project implementation needs and improvements in institutional capacity.

2. **Procurement of works.** Works procured under this project will include the construction of large sanitation works in the Riachuelo wastewater system, which comprises wastewater and industrial collectors, a sewage pretreatment plant, pumping stations, and sub-aquatic outfall. These works will be constructed to collect domestic and industrial wastewater. Contracts for sanitation works will be procured by Argentina Water and Sanitation (*Agua y Saneamientos Argentinos S.A.*, or AySA) through the International Competitive Bidding (ICB) method, preceded by pre-qualification using the Bank's Standard Pre-qualification Document and further carried out using the Spanish version of the Bank's Standard Bidding Document (SBD) for the Procurement of Works, with the necessary amendments to make it fully consistent with the 2004 Bank Procurement Guidelines (revised October 2006). As discussed with AySA, sanitation works would be grouped in two large packages for bidding: one comprising three lots (Lot 1: Left Bank collector and Costanera bypass collector, Lot 2: Riachuelo wastewater pretreatment plant and pumping stations, and Lot 3: Riachuelo outfall), and the other comprising two lots (Lot 1: Right Bank collector, and Lot 2: Industrial collector). AySA is already preparing the Pre-qualification Documents for the first package. Designs are also being prepared by AySA and it is expected that they would be finished by May, 2009 for the first package, and by December, 2009 for the second. Works under Components 3 (infrastructure for territorial development and for flood control) will be bid through the Secretariat of Environment and Sustainable Development (SAyDS) following the International Competitive Bidding (ICB) method and using the Smaller Contracts SBD with post-qualification. National SBDs agreed with or satisfactory to the Bank will be used for small works costing less than US\$5 million. If needed, shopping procedures will be followed for works costing less than US\$ 350,000. Possible rehabilitation of ACUMAR's headquarters could eventually be financed under Component 4 using NCB or Shopping methods.

3. **Procurement of goods.** Goods procured under this project will include industrial equipment for cleaner production, environmental control and monitoring equipment, laboratory equipment, office furniture, computer hardware and software, office equipment and supplies, to be procured through ICB using the Bank's SBD. Additionally, contracts below US\$500,000 will be procured using National SBDs agreed with or satisfactory to the Bank. Shopping procedures might be followed for contracts worth less than US\$100,000.

4. **Procurement of non-consulting services.** All contracts for services not related to consultant services (hydrometeorological control system, geographic information system,

water quality control system, logistics, organization of seminars and workshops, office rental, laboratory analyses, and printing services) may be procured under the same methodologies and thresholds specified for goods. The specific bidding documents will be prepared on the basis of the World Bank's sample bidding document for the procurement of non-consultant services.

5. **Selection of consultants.** Consulting firms will be contracted in the following areas of expertise: design and supervision of works, environmental impact studies, information management, communication strategies, personnel search, public consultation, and training services. All contracts will be procured using the Bank's guidelines for hiring of consultants. For firms, Quality- and Cost-Based Selection (QCBS) will be the preferred method, but Least Cost Selection (LCS), Quality-Based Selection (QBS) and consultant qualifications (CQS) may be used as well. CQS and LCS will only be acceptable for assignments estimated to cost less than US\$100,000. Shortlists of consultants for services estimated to cost less than US\$500,000 equivalent per contract may be composed entirely of national consultants, in accordance with the provisions of paragraph 2.7 of the Consultant Guidelines. Individual consultants will be hired to complete the staffing of the implementing agencies to provide specialized advisory services. These individual consultants will be selected by comparing the qualifications of three candidates and hired in accordance with the provisions of paragraphs 5.1 through 5.3 of the Consultant Guidelines. The consultants should be selected and consultant awards authorized on the basis of the expected overall duration and cost of the assignment, but contracts may be signed for shorter periods if the Borrower prefers.

6. **Operating costs.** The loan would finance operating costs, which would be procured using the implementing agencies' administrative procedures that were reviewed and found acceptable to the Bank. Such procedures will be described in the Operational Manual.

7. **Training Expenses.** The loan will finance training expenses for strengthening the capacity of ACUMAR staff and for designing ACUMAR's communication strategy (such as expenditures for rental fees of training facilities, books, workshop and seminar fees, and domestic and international travel and per diems). These expenses will be procured using the implementing agencies' administrative procedures that were reviewed and found acceptable to the Bank.

8. **Grants for subprojects.** Subprojects aimed at private beneficiaries would be financed through matching grants under Component 2 of the project. The matching grants program will help small and medium enterprises improve their environmental compliance, and meet requirements under their Plans for Industrial Restructuring (PRIs). Procurement of works, goods and services would be carried out in accordance with the procedures set forth in this annex. The Borrower will decide which enterprises to finance based on specific eligibility criteria agreed with the Bank, which include (i) categorization as an SME; (ii) having a signed PRI; and (iii) lack of access to commercial credit. Matching grants will be allocated to eligible enterprises, prioritizing on the basis of the relative pollution load contributed by the enterprise. Detailed eligibility criteria for assistance to SMEs, as well as the specific disbursement modalities will be clearly defined and disclosed in the project's Operational Manual to ensure transparency of resource allocation. The grant application will contain a business plan for cleaner production investments including a list of the works, goods and services to be procured and their estimated cost. The hiring of consultant firms and specialized individual consultants would also be listed in the grant application and follow the procedures for hiring firms and individuals established in this annex.

B. Assessment of the agency's capacity to implement procurement

9. Because of the complex nature of the sanitation works, specific procurement responsibilities for the implementation of Component 1 (Sanitation), including official communication with the Bank, will be undertaken by AySA, the water and sanitation concessionaire in the City of Buenos Aires and 17 nearby urban districts, which has acquired the required implementation capacity in tendering, contracting, managing, and administering large civil works contracts as part of its responsibilities in providing water supply and sanitation services.

10. The procurement responsibilities for the Components 2 (Industrial Pollution Abatement), 3 (Environmental Territorial Management), and 4 (Institutional Strengthening and Project Management) will be undertaken by the Coordination Unit for International Technical and Financial Management (*Unidad de Coordinación para la Gestión Técnica y Financiera Internacional*, or UCOFI), a unit in the SAsyDS with knowledge of and experience in managing projects financed by international financing institutions. UCOFI will handle the official communication with the Bank regarding procurement under these components, and will carry out procurement activities in close coordination with the UEP and ACUMAR.

11. An assessment of the capacity of the Implementing Agencies to implement procurement actions for the project was carried out by the Bank's Procurement Specialist assigned to the project in August, 2008, and updated in January 2009. The assessment reviewed the organizational structure for implementing the project. Both UCOFI and AySA have project implementation experience in procurement under international financial institution procedures. Loan proceeds may support AySA's plan to hire an international procurement specialist, since its local procurement staff needs to be strengthened. This international procurement specialist experienced in large works contracts under Bank procedures was already hired by AySA with its own resources.

12. UCOFI was created in September 2007 to formulate, supervise, and coordinate environmental projects with international financing and provide management control, procurement, and financial management services to SAsyDS. UCOFI has project implementation experience from successive World Bank and Inter American Development Bank operations, which include coastal cleaning (US\$ 8 million), sanitation and sewerage works (US\$ 100 million), and industrial restructuring (US\$ 20 million). UCOFI is staffed by experienced managers (General, and Financial Coordinators) and a team of less than 20 people, which needs to be strengthened to assume the project's procurement responsibilities. Loan proceeds will also support UCOFI's plan to hire a procurement specialist with knowledge of Bank procedures who will specifically be assigned to the Project. Audits are carried out by the Office of the National Auditor General (*Auditoría General de la Nación*, or AGN).

13. Infrastructure planning for water and sanitation works in the concession area is periodically prepared on a 5-year basis by AySA through its Planning Office, and presented for approval to the Planning Agency (*Agencia de Planificación*, or APLA of the Ministry of Infrastructure. Once planning is approved by APLA, procurement for water and sanitation works is implemented by AySA through its Works Office, and procurement for goods and services is implemented by the Logistics Office. Projects and designs for maintenance of existing infrastructure and expansion of the system are also prepared by the Planning Office, which has 100 staff (80 engineers and 20 technicians) organized in 4 management areas for

preparing the designs on the basis of a project approach. The Planning Office has prepared approximately 300 projects since 2006, including large detailed projects such as the Paraná de las Palmas tunnel works. The Bank was informed by the Planning Office during the capacity assessment that designs for the main works to be carried out through the project are being prepared and will be finished at the minimum required level for bidding the works under ad-measurement type contracts. Construction drawings (detailed engineering) will be prepared by works contractors.

14. The Works Office of AySA is staffed by approximately 140 employees in the areas of drilling, contracts and bidding, works supervision (90 staff), and management control. This office receives the project documents, cost estimates, and finalized bidding documents from the Planning Office and handles the bidding process, signs contracts, and supervises the execution of works until they are completed and ready to be delivered to the operational area of the company. A commission designated by the Works Office Director prepares the bid evaluation reports, which are further reviewed by the legal department of the company. The Works Office supervises several large works, including the southwestern plant expansion, the Paraná de las Palmas tunnel works, and Berazategui sewage preliminary treatment plant.

15. The Logistics Office of AySA carries out approximately 1,200 contracts a year for goods and services (e.g., materials, general and production consumables, and networks maintenance) and is staffed by approximately 25 employees, 50 percent of whom are professionals. The office handles the procurement processes and the supervision of the contract execution.

16. AySA has an internal audit unit and is also audited by AGN and external auditors selected by the General National Audit, (*Sindicatura General de la Nación*, or SIGEN). The internal audit unit, staffed by a principal auditor, two professional assistants, and three engineers, carries out biannual audits of the Logistics Office and the Works Office on a sample basis. The company carries out its procurement on the basis of a procedures manual called Purchase and Contracting Regulation (*Reglamento de Compras y Contrataciones*) that was approved by SIGEN. Bidding processes for contracts that exceed 10 million Argentine Pesos are reviewed by a control commission including two representatives of SIGEN and one of AySA's union workers. APLA also audits AySA from a technical point of view.

17. The following key issues and risks concerning procurement for implementation of the project have been identified:

- (i) Risk inherent to major sanitation works
- (ii) Risk of inadequate timing for the different activities involved in Component 1 (pre-qualification, design preparation, bidding process, and supervision)
- (iii) Lack of a suitable procurement information system to properly manage the project and a modern and appropriate information system to monitor progress of major works as foreseen in the project
- (iv) Lack of resources of UCOFI and AySA, particularly of the Logistics Office, to absorb the additional procurement workload that the new project will imply
- (v) AySA permanent staff not sufficiently experienced and updated in handling ICBs under World Bank procedures
- (vi) UCOFI's lack of a procurement specialist with knowledge of Bank procedures
- (vii) Shared responsibilities between UCOFI and ACUMAR in the institutional arrangements for the implementation of Components 2, 3 and 4.

18. The following actions are recommended to reduce risk and facilitate project implementation:

- (i) Contracting supervision of large sanitation works to specialized consulting firms
- (ii) AySA hiring of international experts to support procurement, legal, and technical matters (design and works supervision)
- (iii) Bank approval of a detailed plan and timetable prepared by AySA for the different activities involved in the sanitation works (pre-qualification, design preparation, list of drawings and technical documents to be prepared, bidding process, and supervision) before launching any pre-qualification process
- (iv) UCOFI commitment to using the Procurement Plan Execution System (*Sistema de Ejecución de Planes de Adquisiciones*, or SEPA) to manage procurement planning, and AySA commitment provide UCOFI with timely and accurate information on the Procurement Plan for Component 1, and to develop modern works monitoring systems to be used in the project. SEPA is used in the Bank's projects in Argentina as a transparency tool for procurement processes.
- (v) Proper staffing of the procurement teams paid by loan proceeds, of UCOFI and AySA (as needed) to respond to the foreseeable procurement workload.
- (vi) AySA recruitment of an international procurement specialist experienced in conducting ICBs for works under Bank procedures to strengthen the AySA procurement team
- (vii) UCOFI recruitment of a local procurement specialist experienced in Bank procedures
- (viii) A detailed description of the roles and responsibilities of SAYS/UCOFI, SAYS/UEP and ACUMAR for procurement and contract monitoring under Components 2, 3 and 4 included in the procurement section of the Operational Manual that spells out technical, administrative, and contract management tasks and responsibilities, as well as expected timeframes
- (ix) Special procurement provisions, as described below, were agreed and will be included in the Loan Agreement.

19. The overall project risk for procurement is High.

C. Procurement Plan

20. The Borrower has developed a procurement plan for project implementation, which provides the basis for the procurement methods. This plan will be available to the public i.a. through the SEPA system. The procurement plan will be updated by UCOFI (including the contracts of Component 1 on the basis of the information to be provided by AySA), in agreement with the project team on a 3-month basis or as required to reflect actual project implementation needs and improvements in institutional capacity.

21. The quarterly updates of the procurement plan to be submitted for the Bank's approval shall reflect the progress achieved in project procurement up to the date of submission, actual contract costs, the differences compared to estimated contract costs and the updated planned dates for pending procurement. They should also incorporate the procurement processes that the Borrower plans to carry out during the following year.

22. The Borrower shall also submit an updated procurement plan whenever he wishes to substantially modify the approved procurement plan, including modifying the scheduled dates for the various steps of the procurement process.

D. Frequency of Procurement Supervision

23. In addition to the prior review supervision to be carried out from Bank offices, the capacity assessment of the implementing agency has recommended semi-annual supervision missions to the field for post-review of procurement actions.

E. Special Procurement Provisions

24. Legal aspects

- (i) National Public Procurement in Argentina is governed by separate and incomplete statutes. Supreme Decree No.1023/2001 and its subsequent modifications enacted through Supreme Decrees Nos. 666/2003 and 204/2004 establish the principles for public procurement and govern the procurement of goods and services. The applicability and enforceability of these Supreme Decrees is, however, limited, as not all the necessary regulations have been issued. Decree 436/2000, which regulated the procurement-related articles of the National Account Law 23354 of 1956, is still used despite the fact that those articles were repealed by Supreme Decree No.1023/2001. Public Works Law 13064 of 1947 (*Ley de Obras Públicas*) governs the specifics of procurement of civil works. Law 22460 of 1981 for the contracting and promotion of consulting services was partially repealed in 1994 and is in practice ignored. Decree 1184/2001 governs the hiring of individual consultants, whose approval process is regulated by Decree 577/2003. In addition, the 1992 Law 24156 on Financial Management and Control Systems (*Ley de Administración Financiera y de los Sistemas de Control*) contains provisions related to procurement, particularly in respect to reference prices.
- (ii) National laws only cover procurement by the central administration and decentralized agencies of the executive branch. Government-owned enterprises, the legislative and judicial branches of government, the provinces, and the City of Buenos Aires have their own public procurement rules and regulations, which apply to Bank funds that are on-lent by the Federal Government to the provinces and the City of Buenos Aires.
- (iii) Because national procurement regulations are not totally consistent with Bank guidelines, legal documents should include the Special Procurement Provisions listed in the section below. Although provincial procurement regulations were not assessed in all cases; practice has demonstrated that the proposed provisions are enough to mitigate risks of misinterpretation, as provincial regulations are modeled along federal rules in most cases.
- (iv) Because regulations concerning the hiring of individuals result in lengthy approval processes by government authorities and unrealistically low fees for consultants, implementing agencies often resort to procurement agents.

25. The following Special Procurement Provisions shall apply to procurement under the project:

General

- (i) All procurement shall be done using standard bidding documents, standard requests for proposals, model bid evaluation forms, model proposal evaluation forms, and contract forms previously agreed with the Bank. For consultant services contracts,

only the types of contracts listed in Section IV of the Consultant Guidelines may be used. *Convenios* will not be permitted.

- (ii) Foreign and local contractors, service providers, consultants, and suppliers shall not be required to register, establish residence in Argentina, or enter into association with other national or international bidders as a condition for submitting bids or proposals and/or for contract award.
- (iii) The invitations to bid, bidding documents, minutes of bid opening, requests for expressions of interest, and pertinent summaries of the evaluation reports of bids or proposals of all processes for the procurement of goods, works, and services (including consultant services), as the case may be, shall be published in the web page of the National Contracts Office (*Oficina Nacional de Contrataciones*) in a manner acceptable to the Bank.
- (iv) The Borrower shall (i) Provide data to include in the Bank's publicly accessible Procurement Plan Execution System (*Sistema de Ejecución de Planes de Adquisiciones*, or SEPA) within 30 days of loan approval with the information contained in the initial procurement plan, (ii) update the procurement plan at least quarterly or as required to reflect the actual project implementation needs and progress and include in the SEPA the information contained in the updated procurement plan immediately thereafter.
- (v) Bidders and consultants shall not be allowed to review or make copies of other bidders' bids or consultant proposals, as the case may be.
- (vi) The provisions set forth in paragraphs 2.49, 2.50, 2.52, 2.53, 2.54 and 2.59 of the Procurement Guidelines shall also be applicable to contracts for goods, works and Non-Consultant Services to be procured under National Competitive Bidding procedures.
- (vii) Consultants shall not be required to submit bid or performance securities.

Goods, non-consultant services, and works

- (i) A two-envelope system of procurement shall not be used for the procurement of goods, services (other than consultants services), or works.
- (ii) Contracts of goods, services (other than consulting services) and works shall not be awarded to the "most convenient" bid but to the bid determined to be substantially responsive and the lowest evaluated bid, provided that the bidder is further determined to be qualified to perform the contract satisfactorily.

Works

- (i) Price adjustment for civil works contracts shall follow the price adjustment methodology agreed between the Government of Argentina and the Bank.
- (ii) Where large works or complex items of equipment are involved, not less than 8 weeks shall be allowed for the preparation and submission of prequalification applications, and not less than 12 weeks shall be allowed for the preparation and submission of bids.

Goods

- (i) Inclusion of the bidder in lists published by specialized magazines shall not be an acceptable criterion for determining the qualification of such bidder.
- (ii) Compliance with ISO standards shall not be required as a condition for contract award.

Procurement-related covenants

- (i) In order to disseminate project implementation requirements and procedures and define roles, responsibilities, mechanisms, schedules, and accountability arrangements, the Borrower shall implement the project following the stipulations of an Operational Manual (OM) acceptable to the Bank. The OM will include, inter alia, the project's institutional arrangements and operational, accounting, procurement, and disbursement procedures.
- (ii) In order to disseminate project implementation requirements and procedures, the Implementing Agency will – not later than 6 months after effectiveness – establish and thereafter operate and maintain a web page to disseminate its most significant project information.⁴⁸

F. Details of the Procurement Arrangements Involving International Competition

26. Goods, works, and non-consulting services

(a) Table 8.1. Contract packages to be procured following ICB and direct contracting

Ref. No.	Contract (Description)	Estimated Cost \$ million	Procurement Method	P-Q	Domestic Preference (yes/no)	Bank's Review (Prior/Post)	Expected Bid-Opening Date
1	Lot 1: Left Bank collector and Costanera bypass	160	ICB	Yes	No	Prior	TBD
	Lot 2: Riachuelo outfall	274					
	Lot 3: Riachuelo wastewater pretreatment plant and	264					
2	Lot 1: Right Bank collector	275			No	Prior	TBD
	Industrial collector						
3	Lot 1: Territorial Management works in Area 1	10.0	ICB	No	No	Prior	2010

⁴⁸ For example, the Operational Manual, prequalification documents, bidding documents, and standard request for proposal, technical background, designs, specifications, and drawings of the works to be financed, terms of reference of the works to be executed with project funds, sub-projects to be financed, beneficiaries of such sub-projects, agreements with third parties, and progress and monitoring reports.

Ref. No.	Contract (Description)	Estimated Cost \$ million	Procurement Method	P-Q	Domestic Preference (yes/no)	Bank's Review (Prior/Post)	Expected Bid-Opening Date
	Lot 2: Territorial Management works in Area 2	10.0					
4	Flood Control works in Area 1	12.5	ICB	No	No	Prior	2009
5	Flood Control works in Area 2	18.5	ICB	No	No	Prior	2010
6	Non-consultant services for Hydro-meteorological Control System	2.0	ICB	No	No	Prior	TBD
7	Non-consultant services for Geographic Information System (GIS)	3.4	ICB	No	No	Prior	TBD

(c) The first two contracts for works, goods, and non-consulting services under each procurement method shall be subject to prior review, regardless of amount.

27. Consulting services

(a) Table 8.2. Consulting assignments with shortlist of international firms

Ref. No.	Description of Assignment	Estimated Cost \$million	Selection Method	Review by Bank (Prior/Post)	Expected Proposals Submission Date
	Supervision of the construction of Left Bank collector, Costanera bypass, Riachuelo outfall, and Riachuelo wastewater pretreatment plant and pumping stations	28.0	QCBS	Prior	TBD
	Supervision of the construction of Right Bank collector and Industrial collector	12.4	QCBS	Prior	2010
	Designs of Territorial Management Works (Area 1 and 2)	0.8	QCBS	Prior	2009
	Designs of Flood Control works (Area 2)	0.7	QCBS	Prior	2009
	Supervision of the construction of Territorial Management works (Area 1 and 2)	0.8	QCBS	Prior	2010
	Supervision of the construction of Flood Control works (Area 1)	0.5	QCBS	Prior	2009
	Supervision of the construction of Flood Control works (Area 2)	0.7	QCBS	Prior	2010

(b) Consultancy services provided by firms estimated to cost more than US\$200,000 per contract, consultancy services to be provided by individuals estimated to cost more than US\$100,000, and all single source selection of consultants will be subject to prior review by the Bank.

(c) The first two contracts procured by the agency under each selection method will be subject to prior review regardless of amount.

(d) Short lists of consultants for services estimated to cost less than US\$500,000 equivalent per contract may be composed entirely of national consultants in accordance with the provisions of paragraph 2.7 of the Consultant Guidelines.

28. Grants for subprojects

Thresholds set forth in under items 26 and 27 of this annex will be applicable to the procurement carried out by grant beneficiaries. The Operational Manual will establish the scope of the prior review of contracts that the agency would conduct with respect to subproject beneficiaries.

Annex 9: Economic and Financial Analysis

ARGENTINA: Matanza-Riachuelo Basin Sustainable Development Project

1. The Matanza-Riachuelo Basin Sustainable Development Program (APL-1 + APL-2) is economically feasible, with net benefits of US\$0.5 billion and an internal rate of return of 22 percent. From a financial point of view, the program shows a net loss of US\$480 million. The financial analysis of the program differs from the financial analysis of the entity (see Annex 11B) that operates the project, i.e. AySA; and hence the conclusions reached for the financial viability of the program do not apply to the financial viability of the company. A detailed analysis is presented in this annex, complemented by a distributional analysis and sensitivity and risk analyses.

A. Methodology

2. The objective of this comprehensive urban environment program is the clean-up of the Matanza-Riachuelo (M-R) Basin through several components aimed at upgrading social conditions and urban infrastructure in the basin. On the basis of this objective, a cost benefit analysis was carried out to determine the financial and economic feasibility of two components of the program, sanitation (Component 1) and industrial pollution abatement (Component 2). The evaluation was carried out to assess the accompanying benefits from (i) sewerage expansion, (ii) clean-up of the M-R River, and (iii) clean-up of the M-R River and improvement of the water quality of La Plata River.

3. The analysis was carried out from three perspectives: financial, economic, and distributional. From a financial perspective, each activity was appraised by measuring its costs and benefits at market prices. From an economic perspective, each activity was evaluated by converting financial cash flows into economic cash flows through (i) conversion factors applied to the investment and operating costs in order to eliminate market distortions, and (ii) measurement of the welfare impact on households within the program's area of influence. The fiscal impact was estimated by measuring the taxes and subsidies that occur in the transactions associated with the program and its financing. A distributional analysis was conducted based on the results of the economic and financial evaluation. This analysis assesses the impact of each activity on specific groups of stakeholders, such as government, customers, employees, and suppliers. Finally, the results of the economic and financial analyses were tested against real world uncertainties by conducting a sensitivity and risk analysis.

4. The net benefit of each activity equals the difference between the incremental benefits and the incremental costs of two scenarios: with and without the program. The "with program" scenario considers the proposed investment program implemented. Hence the program achieves environmental recovery of the M-R River as well as the associated benefits for La Plata River, sewerage expansion, and complementary investments. The "without" program scenario assumes that pollution levels and sewerage coverage remain at the current level. The activities were appraised by measuring the flow of costs and benefits for the lifetime of the program, estimated at 30 years. Costs and benefits were expressed in constant

prices as of October 2007.⁴⁹ The discount rate corresponded to the opportunity cost of capital in Argentina, estimated at 11 percent.⁵⁰

5. Benefits expected from the program consist of (i) an improvement in the water quality of the M-R River, (ii) an improvement in the water quality of La Plata River, and (iii) provision of sewerage by 2016 to about 1.2 million households⁵¹ that are currently not connected.

6. The program will have a large impact on the quality of life of households in the basin. The benefits from the improvement in water quality include positive health impacts, new recreation opportunities, and convenience from a more convenient sanitation solution, and the indirect impacts that will result from follow-on development in the basin that will become possible with the environmental improvement of the condition of the M-R River. Benefits from sewerage expansion will accrue to low-income households currently without sewage services that this program aims to cover. Poorer households tend to reside in close proximity to the M-R River and suffer more from the poor quality of the river than the better-off households that tend to live further away. The welfare effect of the extension of sewerage provision will be particularly pro-poor, given the comparatively lower current access to sewerage among poor households.

7. Economic benefits were estimated based on a combination of revealed and stated preference approaches. The revealed preference approach, which relies on data from observed transactions in the market or on expenditures by households or other agents, was used to estimate (i) the benefits from sewerage expansion through the avoided costs and hedonic pricing methods and (ii) benefits of water quality improvement in the M-R River through the hedonic pricing approach. The stated preference technique, which is based on survey methods—in this case, asking households how much they are willing to pay for an improvement in water quality—was used to estimate the benefits of the clean-up of the M-R River only, as well as the benefits of the clean-up of both the M-R and La Plata Rivers. Estimates of the benefits of different components of the program were obtained using different valuation methods and aggregated to yield total program benefits without double-counting.

8. Financial benefits were estimated based on current sewerage tariffs charged by the water and sewerage utility Argentina Water and Sanitation (*Agua y Saneamientos Argentinos, S.A.*, or AySA), the number of households connected to the sewerage network, and the volume of sewage discharged. Financial benefits were programmed for two scenarios: with and without the program. The “with program” scenario assumes that sewerage coverage will increase from the current level of 35 percent to 72 percent by 2016 for the 14 municipalities that lie (at least partially) in the M-R River Basin, including the upper basin municipalities that lie outside AySA’s concession area.⁵² The “without program” scenario assumes that current sewerage levels remain constant. The average bill for sewerage service per household every 2 months is approximately ARS13.

⁴⁹ The exchange rate used was Ar\$3.16 = US\$1.

⁵⁰ Jenkins, Glenn P. 1998. *Buenos Aires-Colonia Bridge Project, Financial and Economic Appraisal*. Cambridge, MA: International Institute for Advanced Studies. July 30.

⁵¹ Households located in and outside the basin, but within the 14 municipalities of the basin

⁵² It is assumed that the municipalities in the upper basin (General Heras, Canuelas, Marcos Paz, Presidente Peron, and San Vicente) will see their coverage increased from about 14 percent to 40 percent.

B. Costs

9. The flow of costs comprises investment as well as operating costs. Investment costs include all the works needed to attain the benefits expected from the program (i.e., works to be financed by the Bank in APL-1 and APL-2, as well as by AySA), as well as those investments in sewerage expansion to be carried out for AySA with financial sources other than the World Bank (estimated at US\$ 400 million in the second phase). Operating costs were estimated using the two scenarios, “without program,” which assumes that current unit costs remain constant, and “with program,” which includes all operating costs incurred when the program is implemented. The financial flow of costs includes taxes and excludes subsidies. Financial costs were transformed into economic costs using conversion factors⁵³ to eliminate market distortions created by taxes, tariffs, and subsidies. Financial costs included a 100 percent subsidy on investment costs to be funded by the Federal Government. A value added tax (VAT) of 21 percent was added to financial costs.

10. **Investment costs.** Total investment costs required for cleaning up the M-R and La Plata Rivers is US\$2.8 billion, including over US\$800 million for sewerage expansion. Details of investment costs can be found in Annex 5.

C. Benefits

11. The M-R and La Plata Rivers receive large quantities of domestic and industrial wastewater from densely populated centers, including Buenos Aires. Of the 60 m³/second of domestic wastewater generated by the Buenos Aires area, only about 30m³/second is currently conveyed to the Berazategui site, where it is discharged without treatment 2.5 km offshore through an outfall.⁵⁴ Another 3 m³/second is treated at the Sudoeste treatment plant and then discharged to the M-R River, while the remaining 27 m³/second is discharged at many points at tributaries of different rivers, including the M-R River, as well as at La Plata River.

12. **Benefits for La Plata River.** La Plata River has a “moderate” pollution level (i.e., the water quality is unsuitable for drinking without prior treatment, and only non-contact recreational activities are permissible). Through works proposed in the program (construction of a coastal collector along the banks of La Plata River, preliminary treatment at two new treatment plants, and discharge of treated wastewater farther away from the river banks), the program is expected to improve water quality in La Plata River. According to the results of water quality modeling done by The Matanza-Riachuelo Basin Authority (*Autoridad de Cuenca Matanza-Riachuelo*, or ACUMAR), the program will protect water quality at the source water intakes and open up a large stretch of La Plata River for non-contact recreational activities that are currently not recommended because of water pollution.

13. **Benefits for the M-R River.** The M-R River Basin is contaminated by sewage disposal from households with no sewerage connection and by discharge of untreated wastewater by the AySA sewerage system. Industrial discharge with inadequate or no treatment is also a major problem in the basin. The program is expected to bring about a significant improvement of water quality of the M-R River. ACUMAR’s water modeling shows that the program would result in the elimination of anoxic conditions and recuperation

⁵³ Conversion factors for the main inputs of the investment and operating costs were estimated according to the guidelines of the National Directorate of Public Investment (*Dirección Nacional de Inversión Pública*).

⁵⁴ Phillips, Robert, and Beatriz Villegas. Review of the Capital and Berazategui Wastewater outfalls Buenos Aires. January, 2008.

of oxygen in the water body. The river's bad odor will be eliminated along the entire course of the river as it becomes suitable for recreational activities with no direct contact with water.

14. ***Benefits from provision of sewerage connections.*** Currently only 35 percent of households in the 14 municipalities of the M-R River Basin have sewerage coverage. The remaining 65 percent of unconnected households have their own on-site solutions: 30 percent use septic tanks, 17 percent use pits, and 18 percent directly discharge to the streets, nearby streams, or drainage systems. The lack of sewerage connections causes serious problems for the population and the environment and results in costs associated with maintaining on-site systems, flooding, and health problems. Furthermore, on-site sewerage systems discharge wastewater into the aquifer, increasing its water level and contaminating it. In some parts of the 14 municipalities of the basin, this contributes to high groundwater tables, causing damage to people's homes. Finally, lack of access to sewerage increases the incidence of diarrhea and other water-borne illnesses.

15. The program aims to connect by 2016 approximately 3.6 million people currently without service in the AySA's concession area. Of these, 2.4 million live in the 14 municipalities of the M-R River Basin, and the remaining 1.2 million live along the banks of La Plata River. Through this program, coverage for the whole concession area is expected to increase from the current level of 58 percent to 83 percent by 2016, while coverage will increase from 35 percent to 72 percent in the 14 municipalities that lie (at least partially) in the M-R River Basin (including those municipalities outside AySA's concession area).

16. The following data sources were used to measure the economic benefits of the program:

- (i) The *Real Estate Survey* of 1,200 properties for sale (apartments, houses, and empty lots) generated data on sale prices and attributes of the properties included in the survey. These data have been used to estimate the hedonic price function and evaluate the following benefits of the program: elimination of odor of the M-R River and provision of sewerage.
- (ii) A *Contingent Valuation Survey* of 1,001 households residing in the M-R and La Plata basins generated data on willingness to pay (WTP) for the elimination of odor from the M-R River, recreational possibilities in the M-R and La Plata Rivers, and the protection of water quality at source water intakes. This survey separately elicited WTP for the environmental benefits for the M-R River alone and in combination with environmental improvement in La Plata River.
- (iii) Data on *avoided costs* of sewerage connection was collected from (i) ACUMAR, which provided geo-referenced data for all municipalities in the basin, including the number of households, type of sewage disposal system (sewerage network, septic tank, pit, or other), altitude, and water table level, (ii) a small sample of households that use on-site sewerage solutions to generate data for cost of operating all types of sewage disposal systems, (iii) operators of vacuum pumping trucks to get information on actual costs paid by households, and (iv) experts on hydrology and operators of sewage disposal systems to get accurate information of required maintenance for proper functioning of the disposal systems.
- (iv) AySA provided data on *avoided costs* of moving the Bernal drinking water intake further out into La Plata River.

- (v) Partial data permitting the estimation of the *costs of illness* for households without sewerage connections were collected through the contingent valuation survey (the incidence of diarrheal diseases among children under 5 years old from the survey sample). Supplementary data were also collected from other sources, such as treatment costs for the public healthcare system.

17. **Summary of benefits from different approaches.** Benefits were measured using the hedonic pricing, contingent valuation, avoided costs, and cost of illness approaches (table 9.2). In estimating the program's benefits using alternative methods, special care has been taken to ensure that there is *no double-counting*, to ensure robust assessment of the economic viability of the program. For example, the benefits from sewerage connections were measured separately through two different approaches: hedonic prices and avoided costs, the results were used for comparative purposes and they were not added. The same with the estimation of the benefits from cleaning up the M-R River, which were measured through hedonic prices and willingness to pay; both benefits were analyzed and they were carefully used avoiding double counting. The benefits of cleaning up of the River de la Plata were measured along with the benefits of cleaning up of the M-R River, given that the benefits along the River de la Plata would not occur without the investment for the M-R River. Moreover, The estimated benefits for each component were carefully added to calculate the total benefits of the project.

18. In addition to avoiding double-counting, the beneficiary population for each type of category of benefits has also been carefully chosen. For example this economic analysis assumes that the total beneficiary population from sewerage provision consists of both (i) households within AySA's concession area that will obtain sewerage connections under the program; as well as (ii) households outside AySA's concession area that could become connected in the future because of the project. Thus, a total population of approximately 1.1 million households would benefit from sewerage connections. The beneficiary population of the clean-up of the M-R River, on the other hand, includes population in GBA (about 10 million people); and the benefits will correspondingly be the highest for households within approximately 20 blocks of the river (about 11%), whose housing prices are notably affected by the proximity to the river and its quality.

Table 9.2. Revealed and stated preference approaches to the program's economic evaluation

Benefits	Methodology
Sewerage connection for households in the M-R River Basin	(i) Avoided costs (ii) Hedonic pricing (iii) Cost of illness
Elimination of odor and suitability for recreation without direct contact in the Matanza-Riachuelo River	(i) Hedonic pricing (ii) Contingent valuation
Improving water quality in the Matanza-Riachuelo River and Suitability for recreational activities without direct contact and protection of water quality at water intakes for drinking water in La Plata River	(i) Contingent valuation (ii) Avoided costs

19. **Contingent valuation survey.** During preparation a contingent valuation survey was carried out to evaluate the benefits expected from cleaning-up the M-R River as well as the Río de la Plata, that is: elimination of bad odor along the entire flow of the M-R River, the possibility of using the M-R River for recreation without direct contact with water, the possibility of using La Plata River for recreation without direct contact with water (but for a wider range of water-related sports, including rowing, than in the case of the M-R River), and the protection of the water intakes for the public water supply. Benefits from the provision of sewerage to households in the M-R River Basin were not evaluated in this survey or mentioned as an outcome of the program.

20. Results showed that the average WTP is approximately US\$2.6 per household per month for the M-R River and US\$ 3.13 per household per month for both rivers.⁵⁵ The WTP for the clean-up of the M-R River is expected to vary depending on the distance to the river, as households that live closer to the river are directly affected by the contamination and the bad odor of the river. In that sense, the WTP of these households reflects their “use value.” Moreover, the M-R River may have a symbolically important value for the population of Greater Buenos Aires, as there is great awareness of contamination in the basin as one of the most severe environmental contamination problems in the country. In that sense, the clean-up of the M-R River may benefit households that do not use the river or frequent the surroundings. The WTP of those households reflects their “non-use” value of the river.

21. **Hedonic pricing survey.** Hedonic models were used in Buenos Aires to estimate the affects of access to sewerage on property values. In the hedonic literature, several studies have estimated the impact of proximity to a water body (e.g., canal or river) and view on housing prices, but very few papers have estimated the effect of water quality on property prices (Leggett and Bockstael 2000).⁵⁶

22. Results of the 2008 Real Estate Survey⁵⁷ conducted as part of this economic evaluation revealed a statistically significant difference between the average prices of houses, apartments, and lots from the survey sample with and without a sewerage connection. The difference between the average prices of properties affected and not affected by the odor was also significant, particularly for odor from any source of contamination (e.g., M-R River, factories). Property prices were found to fall from an average of US\$662/m² to US\$591/m² for houses, from US\$889/m² to US\$757/m² for apartments, and from US\$415/m² to US\$259/m² for empty lots closer to the river odor.

23. The magnitude of the benefits from cleaning up the M-R River for specific properties varies. As expected, the results show that once the M-R River is cleaned-up, the closer the property is to the river, the higher the value of the benefits. The impact on property values varies with distance, type of property (house, apartment, or empty lot), and location (properties in Buenos Aires have higher impact than those in the province). These results indicate that the WTP is significantly lower more than 20 blocks from the river than it is for households within 20 blocks of the river. Table 9.3 presents the increase in property values predicted by the hedonic pricing model in percentage terms.

⁵⁵ Faria, Diomira. Estudio de Disponibilidad de Pago para la Mejora Ambiental de los Ríos Matanza Riachuelo y Río de la Plata. Buenos Aires. August, 2008.

⁵⁶ Leggett, Christopher, and Nancy Bockstael. 2000. Evidence of the Effects of Water Quality on Residential Land Prices. *Journal of Environmental Economics and Management* 39: 121–144.

⁵⁷ Juan, Andres. Estimacion de Funcion de Precios Hedónicos para la Mejora Ambiental del Río Matanza Riachuelo y la Extension de Redes Cloacales. Informe Preliminar. Buenos Aires. August, 2008.

Table 9.3. Impact on property value of the clean-up of the M-R River

Distance to M-R River	Houses	Apartments	Empty lots
Up to 5 blocks	14.0%	10.9%	18.3%
6–10 blocks	9.1%	6.3%	10.3%
11–15 blocks			4.6%
11–20 blocks	4.5%	2.8%	

24. The estimated benefits of clean-up the M-R River using the hedonic pricing method are conservative in a longer-term perspective. In the long run, environmental remediation of the M-R River, elimination of its bad odor, and potential recreational uses is expected to revitalize the real estate market in the M-R River Basin and encourage new types of real estate development that are currently not possible because of the river's poor environmental condition. Thus, the estimated short-term benefits from the elimination of odor are a lower bound on the longer-term benefits.

25. *Avoided costs.* This technique was used to estimate two types of benefits to be attained through the program: (i) sewerage connections and (ii) protection of source water intakes. Access to sewerage was evaluated through hedonic pricing and avoided cost approaches, the results obtained with both methodologies were not added and were used only for comparison purposes. Benefits from the protection of water intake were measured through the avoided cost approach, as well as with the WTP method; the results obtained with both methods were not added as the results from the avoided costs were implicitly included in the WTP. However, avoided cost results were added to the results obtained with the hedonic price for the clean-up of the M-R River.

26. *Avoided costs due to sewerage connection.* Currently 65 percent of households in the 14 municipalities of the M-R River Basin do not have access to a sewerage connection. They use different solutions for sewage disposal: 30 percent of households have septic tanks, 17 percent have holes with no septic chambers, and 18 percent have other solutions such as illegal discharge into the drainage system or direct discharge into the street. When households have access to a public sewerage connection, they will not have to operate these on-site systems, resulting in cost savings. On the other hand, households will have to pay the bill from the water and sewerage utility. The difference between these two costs is the lower bound on households' benefits.

27. Current costs of on-site sewage disposal systems vary depending on the frequency of cleaning, volume of discharge, and -- very important in this area -- altitude of the house location, as rising water tables cause frequent floods that cause septic tanks to overflow. The costs were calculated for proper maintenance of the systems according to the water table level in a particular location (table 9.4). The information collected from the willingness to pay survey and a small sample of households with on-site systems reveals that many of the households incur lower costs given improper system maintenance.

Table 9.4. Costs of on-site sewage disposal systems assuming proper maintenance

Investment and operations and maintenance (O&M) costs (US\$/household)	Septic tank	Pit	Manual pump	No disposal
Annual cost (US\$/household)				
For houses located at an altitude < 10 meters	292	290	63	47
For houses located at an altitude > 10 meters	216	214	63	47

28. Households that rely on on-site sewerage solutions incur an additional investment cost of an “in-house adjustment” to connect to their on-site sewerage system. A cost ranging from US\$150 to US\$250 was assumed depending on the type of adjustment. This cost was applied to about 40 percent of the households.

29. Apart from this gain (assuming the increase in the households’ bills is lower than the cost savings), households may incur additional health benefits when a connection to public sewerage is provided. Also, households incur increased risk of water-borne illness if the private systems are not optimally maintained and operated. Because the averting actions (pumping and maintaining sewerage systems) do not completely eliminate the health risks for all households—particularly with improper system maintenance—the residual health risks remain, and the costs of illness (averaged for the population) can be added to the averting costs (on average for the population) to obtain the total benefits of sewerage provision.

30. An additional benefit of providing sewerage connections is the value of time savings. Provision of public sewerage saves each adult approximately 5 minutes a day. If those time savings are valued at 75 percent of average wages (8 Argentine Pesos per hour), the net present value (NPV) of the total time savings reaches US\$871 with a discount rate of 5 percent and US\$1,325 million with a discount rate of 10 percent.⁵⁸ However, because the valuation of time savings is controversial, these were not added to total program benefits to avoid overestimating the program’s expected benefits.

31. *Avoided cost for water intake.* If the program were not implemented, AySA would have to move the Bernal water intake to a site further out into La Plata River. With the program, AySA would avoid the cost of moving the intake. All sewage discharge will have appropriate treatment and will be diluted by La Plata River through the outfalls at an adequate distance from the water intake. Currently the Bernal water intake is close to the point of the untreated sewage discharge, and even though at present AySA deems that the water quality at the intake is adequate, increasing pollution in the “without program” scenario may require moving the water intake further out into the river in the near future. AySA previously moved that water intake to ensure adequate water quality at the intake.

32. If the program is implemented, AySA will not have to incur this cost. Hence, this is a financial benefit generated by the program. This cost was calculated by AySA and estimated at US\$140 million in present value. It is purely an engineering cost that is useful from the financial point of view but does not provide an estimate or bound to the welfare impact of protecting water quality at the water intake. The actual benefit of protecting water quality was assessed by estimating the health cost caused by poor water quality at water intakes.

⁵⁸ Strukova, Elena. 2008. Health Benefits of Water-Sanitation-and Water Source Improvements in M-R and de la Plata Basins. Background paper for the economic evaluation of the project.

33. **Cost of illness approach.** The cost of illness approach was used to estimate (i) the additional benefits of sewerage provision after taking into account the avoided costs of individual sewage disposal solutions and (ii) the health benefits associated with protecting water quality at the Bernal intake. These two interventions result in health benefits through the reduction of diarrheal mortality and morbidity as well as the risk of other water-borne diseases. For this program, the value of risk reduction from diarrheal morbidity and mortality was estimated by applying a benefit transfer approach to the value of statistical life numbers from developed countries.⁵⁹ The expected risk reduction in mortality and morbidity from diarrheal diseases in children under 5 and in the rest of the population was derived from meta-analysis of studies of health risk reductions, and the data on the incidence and prevalence of diarrheal illness among children under 5 with different types of sanitation were estimated from the contingent valuation survey undertaken for the program.

34. The health benefits analysis provides two estimates of cost savings: health benefits resulting from the provision of sewerage to households that will obtain connection through the program and health benefits from the protection of drinking water quality. The latter estimation assumes that if the water intakes are not moved farther out into the river, then in the absence of the program, drinking water quality would deteriorate with an increased incidence and prevalence of diarrheal illnesses. The health costs analysis uses the average current incidence and prevalence of diarrheal illnesses—taking into account the fact that some households already undertake averting actions and incur averting expenditures to reduce their health risks in the absence of a public sewerage connection. For that reason, the health costs estimated using this approach can be added to the averting expenditures to obtain an estimate of total benefits from the provision of sewerage connections.⁶⁰

35. The estimated benefits from the reduction of diarrheal morbidity and mortality in children under 5 and the rest of the population are US\$29 per person for sewerage expansion and US\$6 per person for protection of water in the Bernal water intake (table 9.5).

Table 9.5. Estimated health benefits from improving sanitation and source water quality in the M-R River Basin

Interventions	Sewerage expansion	Protection of water intake	Total
Population affected (million)	2.1	5.7	5.7
Annual diarrheal illness reduction	40%	11%	
Annual diarrheal morbidity reduction (US\$ million)	23	17	38
Annual diarrheal mortality reduction (US\$ million)	38	15	49
Total annual health cost reduction, (US\$ million)	61	33	87
Annual health reduction per person (US\$)	29	6	15

⁵⁹ Strukova, 2008.

⁶⁰ It is crucial to point out that the averting expenditures incurred by the households do not fully mitigate the risk of diarrheal illnesses, either because the non-network sewerage systems are not perfectly maintained (not cleaned frequently enough) or because not all households undertake these maintenance measures.

D. Results of the Evaluation

36. The results of the evaluation are presented below for each component ((i) sewerage expansion, (ii) clean-up of the M-R River, and (iii) clean-up of both the M-R and La Plata Rivers) and for the entire program.

37. **Financial results.** The results show a net loss of the program of US\$480 million. The financial analysis of the program differs from the financial analysis of the entity (see Annex 11B) that operates the program, (AySA), and hence the conclusions reached for the financial viability of the program do not necessarily apply to the financial viability of the company. The program would require an increase of sewerage tariffs for financial viability of the program, but that it is not the required increase for the financial viability of AySA.

38. The main differences between these two analyses are as follows:

- (i) The financial evaluation of the program calculates the incremental financial benefits and costs generated by each activity of the program (sewerage component and cleaning up of M-R and La Plata Rivers component) on an incremental basis ("with" and "without" program scenarios). The financial assessment of AySA analyzes the viability of the company as it is most likely to evolve following the implementation of the proposed program ("with" program scenario only).
- (ii) The financial evaluation of the program addresses only the cash flow generated by the program, without including other services provided by the company. The financial evaluation of AySA, on the other side, includes all the services it provides, i.e., water and sewerage services.

Table 9.6. Financial results of the M-R River Basin Sustainable Development Program

Financial results	Net present value of cash flows (US\$ thousand)			IRR
	Benefits	Costs	Net benefit	
Sewerage	131,024	188,065	(57,040)	N/A
Cleanup of the M-R River	-	21,203	(21,203)	N/A
Cleanup of the M-R and La Plata Rivers (cleanup of M-R River is included)	-	423,184	(423,184)	N/A
Total program (Cleanup of M-R River is not included)	131,024	611,249	(480,224)	N/A

39. **Economic results.** The benefits of the program for the clean-up of the M-R and La Plata Rivers were assumed to occur 10 years after the program starts implementation and is fully completed. The benefits from sewerage connections were assumed to occur gradually, at the same pace as the investment of service expansion. The program's expected benefits were estimated using more than one valuation approach (table 9.7).

Table 9.7. Choice of valuation methods to account for the program's benefits

Benefit	Choice of valuation method
Access to sewerage	a. Hedonic pricing <u>OR</u> b. Avoided costs for sewerage connection plus health benefits
Elimination of odor and recreation possibilities in the M-R River	c. Hedonic pricing for households within 20 blocks of the M-R River <u>AND</u> contingent valuation for households farther than 20 blocks from the M-R River in Buenos Aires <u>OR</u> d. Contingent valuation for the M-R River for the whole population of Greater Buenos Aires (within and outside the M-R River Basin)
Elimination of odor in the M-R River, recreation possibilities in the M-R and La Plata Rivers, and protection of water intakes	e. Avoided costs (extending the Bernal water intake) <u>AND</u> contingent valuation for the M-R River <u>OR</u> f. Contingent valuation for the M-R and La Plata Rivers <u>OR</u> g. Hedonic pricing for households within 20 blocks of M-R, <u>AND</u> contingent valuation for the households located more than 20 blocks from the M-R River in Greater Buenos Aires

40. The results of the hedonic, contingent valuation, avoided costs, and cost of illness analyses were expressed in NPV terms for each type of benefit along with the associated investment costs and then aggregated using alternative approaches without double-counting any of the benefits (table 9.8).

Table 9.8. Economic results for each component

Component	Net present value of flows (US\$ thousand)			Internal rate of return (IRR)
	Costs	Benefits	Net benefit	IRR
Sewerage component				
Avoided costs (without health benefits)	925,660	1,340,975	415,316	18%
Avoided costs (with health benefits)	925,660	1,802,285	876,625	27%
Hedonic price for sewerage	925,660	2,019,987	1,094,328	76%
M-R River				
Willingness to pay (WTP)	466,019	454,880	(11,138)	11%
Hedonic prices for the M-R River	466,019	644,238	178,219	16%
Hedonic prices plus WTP for the M-R River (non-use value)	466,019	849,236	383,217	19%
M-R AND La Plata Rivers				
WTP (M-R and La Plata Rivers)	1,546,614	538,480	(1,008,134)	
Hedonic price plus WTP (M-R plus La Plata Rivers) non-use	1,546,614	986,654	(559,960)	
Hedonic price plus water intake	1,546,614	783,889	(762,725)	

41. Both program activities —sewerage provision and clean-up of the M-R River—are economically viable. The net gain for sewerage is between US\$415 million and US\$1 billion,

depending on the approach (avoided cost or hedonic pricing). The IRR varies from 18 percent to 76 percent. The clean-up of the M-R River is profitable when benefits from the hedonic prices method are included with returns between US\$178 million and US\$383 million and an IRR from 16 percent to 19 percent. However, the component for cleaning up both the M-R and La Plata Rivers requires that the additional benefits for the sewerage component become economically viable.

42. When benefits are added to obtain the benefit of all the components of the program, results vary depending on the approach used to aggregate the benefits (table 9.9). Five different combinations of approaches to measure program benefits are shown in the first column in table 9.9. For example, combination 1 combines the hedonic estimates of the impact of sewerage provision and eliminating odor for people living within 20 blocks of the M-R River with the contingent valuation estimates for cleaning up both rivers for the rest of the population of Greater Buenos Aires.

43. The rationale for combining these two sets of results is the following. The prices of properties in proximity to the river are depressed by the odor and environmental contamination, and the adverse effect of the M-R River on property prices disappears beyond 20 blocks from the river. Since revealed preference data are less likely to be affected by the hypothetical bias that is a common weakness of stated preference approaches, the hedonic estimates are likely to be closer to the true value of the benefits from removing the river's bad odor than the WTP estimates from the contingent valuation survey for households living in the proximity of the river. However, for the population living further than 20 blocks away from the river, the only way to estimate the value of the benefits from removing the bad odor of the river (i.e., the non-use value) is through the contingent valuation approach, since housing prices are not affected by the river at such a distance.

44. Combination 1 is also preferable to the other approaches that rely on averting expenditures and health benefit estimates to estimate the benefits of sewerage provision or the benefit from protecting water quality at the water intake given the methodological issues and the quality of the available data on health risks and the avoided costs. The measure of the benefits of sewerage obtained from the hedonic function in this program is more robust. The net returns estimated using this approach are US\$ 1 billion and the economic internal rate of return is 31 percent.

Table 9.9. Results for the whole program using different approaches

Approaches for measuring benefits	Benefits per component				Total program			
	Sewerage		M-R River plus La Plata River		Total benefits	Costs	Net benefits	IRR
	Approach	NPV (US\$ thousand)	Approach	NPV (US\$ thousand)	NPV (US\$ thousand)			
Combination 1: Hedonic price (HP) plus contingent valuation (CV)	HP	2,019,987	HP	644,238				
			CV (> 20 blocks from the M-R River)	342,416				
			Total	986,654	3,006,642	2,472,274	534,368	22%
Combination 2a: Avoided cost (AC) without health benefits (HB) plus CV	AC without HB	1,340,975	CV	538,480	1,879,455	2,472,274	(592,818)	
Combination 2b: AC with HB plus CV	AC	1,340,975	CV					
	HB							
	Total benefit	1,340,975		538,480	2,340,765	2,472,274	(131,509)	10%
Combination 3a: AC without HB plus HP	AC	1,340,975	HP (M-R River)	644,238				
			AC water intake	139,651				
			Total	783,889	2,124,864	2,472,274	(347,409)	
Combination 3b: AC with HB plus HP	AC	1,340,975	HP (M-R River)	644,238				
	HB	461,309	HB (protection of water intake)	185,389				
	Total benefit	1,802,285	Total	829,627	2,631,912	2,472,274	159,638	12%
Combination 4: HP plus AC	HP (sewerage)	2,019,987	HP (M-R River)	644,238				
			AC water intake	139,651				
			Total	783,889	2,803,876	2,472,274	331,603	23%
Combination 5: HP plus avoided CV	HP	2,019,987	CV (M-R plus La Plata Rivers)	538,480	2,558,467	2,472,274	86,194	13%

45. **Distributional impact.** The difference between economic and financial flows represents rents or monetary flows that accrue to an entity other than AySA. Taxes represent monetary flows accruing to the Government, while subsidies are transfers from the Government to AySA. Winners and losers from the program were identified by deconstructing economic and financial costs for each input of the program and then estimating the difference between economic and financial results. This analysis uses the economic results obtained with estimation 1, which was considered the preferred approach.

46. As table 9.10 shows, the big winners are the customers, with a return of US\$2.8 billion. If this benefit is distributed among the poor and non-poor, assuming that the poor are the 23 percent who lack basic services, according to geographical information on the beneficiaries and the 2001 census, results show that poor beneficiaries will reap approximately US\$645 million of this benefit. The earnings come from the benefits obtained with the increase in property values when sewerage service is connected. This benefit offsets the payment these households have to make to AySA when connected and billed, as well as the investment cost they incur to adjust in-house connections. The big loser is the Government because of the investment subsidy, which is not offset by the higher taxes received. The total loss for the Government is approximately US\$1.7 billion. The net gain for the economy as a whole is US\$1 billion, as shown in table 9.10).

Table 9.10. Distributional impact

Benefit	Net present value (NPV) of flows (US\$ thousand)			
	Sewerage	M-R River	M-R and La Plata Rivers	Total
Net fiscal impact for government	(618,381)	(431,021)	(1,075,389)	(1,693,770)
Net benefit for poor customers	418,608	148,175	226,930	645,539
Net benefit for non-poor customers	1,401,428	496,063	759,724	2,161,151
Total program	1,151,368	199,422	(136,775)	1,014,592

47. If sewerage tariffs were to increase to make the program financially viable, households that currently have access to sewerage would be adversely affected; and the provision of access to sewerage has a positive impact on household welfare. Households that did not have access to sewerage and who will gain it through the program will experience welfare gains; households that did not have sewerage and will not gain it will not be affected; and households that had sewerage service before the program will experience welfare losses if they have to pay higher tariffs for service as a result of the program.

48. The magnitude of the combined impact of a change in access and tariffs on welfare of households at different income levels can be assessed through a comparison of access rates and budget shares at different income levels. The welfare impact of access provision is an average of US\$ 40 per month per household for households living in apartments and US\$ 64 per month for households in houses, as estimated from the hedonic analysis.⁶¹ Access to sewerage tends to be

⁶¹ These figures were obtained by expressing the expected appreciation of housing values based on the hedonic pricing function in terms of monthly values assuming a 30 year annuity.

lower for the poor, so a disproportionate share of the welfare gains from new access will accrue to the poor, assuming the program eventually results in sewerage provision to all households that current have water service but lack sewerage connections (Table 9.11).

Table 9.11. Sewerage and other utilities: Access and expenditures by income level

Household Income level	Share of households connected to sewerage	Water		Water and sewerage		All utilities
		Argentine pesos (ARS)/mo	% of total income	ARS/mo	% of total income	% of total income
Poorest 20%	0.71	11.9	1.4	17.3	1.9	12.6
2	0.78	13.0	0.8	18.6	1.2	8.0
3	0.86	14.2	0.7	20.6	1.0	6.4
4	0.90	14.3	0.5	19.3	0.7	5.3
Richest 20%	0.92	20.8	0.4	23.0	0.5	4.3
Total	0.83	14.7	0.8	19.8	1.1	7.5

49. The welfare impact of an increase in water and sewerage tariffs is proportional to the magnitude of the tariff increase and to the share of water and sewerage expenditures in total household budgets.⁶² Thus the welfare impact will be higher for households that spend a higher share of their total income on water and sewerage. As estimated from the data on utility expenditures in the contingent valuation survey, on average, households spend 1.1 percent of total income on water and sewerage, ranging from 0.5 percent for households in the highest income quintile to 1.9 percent for the poorest households in the bottom quintile. Expenditures on all utilities (water, sewerage, street lighting, gas, electricity, and fixed telephone) average 7.5 percent of total income and range from 4.3 percent for households in the highest income quintile to 12.6 percent for those in the lowest quintile.⁶³ The relatively low share of water and sewerage expenditures in total household income means that a 50 percent increase or even a doubling of water and sewerage tariffs will on average have a limited impact on household incomes. However, the impact will be higher for the poorer households, as a result of the relatively higher share of water and sewerage expenditures in their total incomes.

E. Sensitivity and Risk Analyses

50. The results obtained in the analysis carried out so far assume certainty regarding all the variables. The sensitivity and risk analyses measure the impact of changes in assumed values for critical variables on the results. The sensitivity analysis measures the outcome if one of the variables changes while all others remain fixed. The risk analysis measures the outcome when all selected variables change at the same time, each one based on a probability distribution. Both

⁶² This is based on the first-order approximation of the loss of consumer surplus from a tariff increase. As most households pay a flat rate for water and sewerage service, depending on the square footage of their houses or apartments and the census radius where the homes are located, the quantity consumed before and after the tariff increase cannot be estimated.

⁶³ These shares are consistent with the total utility expenditures and with the water and sewerage expenditures estimated by Delfino and Casarin (2001) using the 1998 Household Budget Survey data (INDEC) for Argentina.

analyses were undertaken to test the robustness of the economic outcome. The financial results were not tested because the program is not financially viable without a sewerage tariff increase.

51. **Sensitivity analysis.** The variables selected as those conveying major risks were: (i) investment cost overruns; (ii) operating cost overruns; (iii) program delays; (iv) effective connection rate; (v) reduction of benefits from hedonic prices; and (vi) reduction of benefits from willingness to pay for the environmental improvements in the two rivers. Table 12 shows the impact on economic results when one of these variables changes while the others remain constant.

Table 9.12. Risk Variables and their Impact and Risk Significance

<i>Risk Variable</i>	Description of Impact
<i>Investment cost overrun</i>	Direct impact on investment costs. Largely under management control. Conveys a moderate risk to the economic results. The project would show positive returns for an increase in costs as high as 28 percent.
<i>Operating costs overrun</i>	Direct impact on operating costs. Largely under management control. Conveys a low risk to the economic results. The operating costs could overrun 68% and still shows positive results.
<i>Investment delay</i>	Direct impact on benefits. Largely under management control. Conveys a moderate risk to economic results. If the project is delayed two or more years, the returns would be negative.
<i>Effective sewerage connection</i>	It corresponds to the acceptance of population to effectively connect to the sewerage network and eliminate household on-site sewage disposal systems. It has a direct impact on benefits. It depends largely on the communication strategy of AySA and financial alternatives offered to the beneficiaries to make it easy to accept the connection and adjust households' in-house facilities. It carries an important risk for the project. The connection rate has to be higher than 72% for the project to still have positive returns.
<i>Reduction of benefits from increase in property</i>	Direct impact on benefits and beyond management control. It conveys a medium to low risk to the project. If the real estate market does not react as expected, and the magnitude of the property value increase when sewerage is connected is only 74% of the assumed benefit, or just 17% of the assumed increase for the clean-up of the M-R River, the project would still show positive returns.
<i>Reduction of benefits from WTP for environmental improvements in the two rivers</i>	Direct impact on benefits and beyond management control. It conveys a low risk to the project. If the willingness to pay were zero for the non-use value (for the households living more than 20 blocks away from the M-R River), the project would still generate benefits totaling US\$ 200 million and have a 18% internal rate of return.

Risk Analysis

52. To enhance the accuracy of the economic analysis, the risk analysis has been carried out using a Monte Carlo simulation. The analysis measures the extent of various risks and their impact on the results of the project by modeling a likely probability distribution that best describes the behavior of each of the selected variables. Based on a simulation of 1,000 trials the

model recalculated the results of the financial and economic analyses by simultaneously changing each of the selected risk variables according to their probability distribution.

53. For each variable a probability distribution was chosen with the following criteria:

- (i) For the WTP, a normal distribution with standard deviation of 2.76 was chosen for the M-R River and La Plata River, and 2.94 for the M-R River, based on the results of the WTP survey.
- (ii) For the benefits of hedonic prices the analysis assumes an average reduction of 10% with the normal distribution with a standard deviation of 3%.
- (iii) For costs overrun, the analysis assumes a triangular distribution with the most likely outcome of no increase, minimum value a reduction of 5%, and maximum increase of 15%.
- (iv) For project delays, the analysis assumes the following probabilities: no delays: 70%; one year delay: 15% probability; two year delay: 10% probability, and three year delay: 5%
- (v) For connection rate, the analysis assumes 10% probability of a 70% connection rate; 70% probability for an 80% connection, 15% probability for a 90% connection; and 5% probability for 100% connection.

54. Using the Monte Carlo simulation, which calculates 1,000 possible scenarios by repeatedly sampling values from the probability distributions defined for the uncertain variables, the result shows that the probability of having a positive outcome is 77%. This result underscores the robustness of the project's expected positive net benefits under a range of scenarios.

Annex 10: Safeguard Policies

ARGENTINA: Matanza-Riachuelo Basin Sustainable Development Project

A. Introduction

1. This annex summarizes the findings and recommendations of Environmental and Social Impact Assessments prepared for the proposed Matanza-Riachuelo Basin Sustainable Development Project. It provides a brief description of the assessment process and main documents prepared; the legal and institutional framework governing the project; project and strategic alternatives considered; the likely environmental impacts associated with project execution and operation; and the principal management and mitigation measures proposed for addressing those issues arising from the environmental assessments.⁶⁴ For a complete description of the project objectives, phasing of the APL program and the specific scope of project components refer to Annex 4 of the PAD.

2. *Environmental assessment process.* The environmental risks, impacts, and benefits of the project have been analyzed at two different levels using two closely related environmental assessment approaches. The environmental impacts (both positive and negative) of specific project components have been assessed through a detailed seven-volume EIA conducted for specific works proposed under Component 1 and through separate Environmental Management Frameworks (EMF) prepared for activities to be carried out under Components 2 (Industrial Pollution Abatement) and 3 (Environmental Territorial Management). In addition to the site-specific EIA and the EMF, an Integrated Environmental Assessment (IEA) has been prepared which serves two purposes: first, it summarizes the main findings of the more detailed EIA reports; second, the IEA presents the overall regional context and strategic goals for the long term recovery and management of the Matanza-Riachuelo basin in its entirety.

3. At the site-specific level, a detailed EIA report was prepared for the works in APL-1 under Component 1, for which the final technical solutions have been identified. For those works where the final technical solutions have not yet been identified, site specific EIAs will be prepared and will be reviewed by the Bank during project implementation. For those works at or before pre-feasibility stage (such as the right bank collector or a suitable alternative for the same function), site-specific EIAs, acceptable to the Bank will be prepared by AySA⁶⁵. A seven-volume EIA was prepared under the general direction of the AySA Environmental Management Unit using independent environmental engineering consultants for key aspects of the analysis. These assessments, which were coordinated with associated modeling and other technical evaluations, indicate the overall environmental viability of the proposed project activities. The EIA provides a thorough assessment of the project's environment impact during the construction

⁶⁴ This Annex is based on documents prepared by the borrower including the Integrated Environmental Assessment document and the references indicated in it, in particular those referring to the results of the numerical modeling of water quality for Matanza-Riachuelo and the Río de la Plata (*Modelacion Matematica de la Cuenca Matanza-Riachuelo para el Estudio de Alternativas de Saneamiento*, Informe Integrador, Junio 2008, carried out by the Universidad Tecnológica Nacional under the Direction of Dr. Angel Menendez), as well as the Environmental Impact Studies on specific works of the Plan Director conducted by AySA.

⁶⁵ As stated in para. 53 of the PAD, a legal covenant has set March 31, 2010 as the date for the government to present to the Bank the review of technical alternatives to the Colector Margen Derecha (Right Bank collector).

and operational stages for the project's main civil works. The EIA for Component 1 includes specific volumes on the general conditions and background of the project; a description of AySA's corporate Master Plan (Plan Director); the proposed Riachuelo preliminary treatment plant; the pumping stations; the Riachuelo sub-aquatic outfall; as well as the colector margen izquierda and colector baja costenera. In addition, the site-specific EIA includes baseline data, legal and institutional capacity assessments, an environmental management plan and site specific management measures. Furthermore, since the Government advanced the construction of the Berazategui preliminary treatment plant (BPTP)⁶⁶ (functionally an investment under APL-2), an EIA was prepared in July 2007 and has been disclosed in-country by AySA.⁶⁷ The Bank team has assessed the EIA and found it to be broadly acceptable. Additionally, a revised Environmental Management Plan for the BPTP, acceptable to the Bank, will be presented by AySA within 6 months of loan effectiveness.

4. The Integrated Environmental Assessment was prepared under the overall guidance of ACUMAR and corresponds to a regional, basin-wide assessment of environmental risks and benefits. The IEA describes the baseline conditions of the Matanza-Riachuelo Basin, the legal and institutional setting and the strategic challenges and options that underpin the choice of the specific project components and investment activities. This IEA was used to evaluate the regional strategic objectives of the project (including infrastructure works that belong to the PISA but that will not be financed through the proposed Bank loans such as the Berazategui preliminary treatment plant) and to provide a clear rationale for the project components that respond to these objectives. The IEA also describes the structure of the project as a whole, including a detailed description of the environmental risks and benefits of the three project components and the institutional arrangements for implementing the Environmental Management Plan (EMP) for the project.

5. Social assessment process: Given the complexity of the Project, two complementary social assessments were prepared for ACUMAR and AySA during project preparation. ACUMAR's social assessment was a basin-wide study, describing the social baseline conditions which exist in the M-R River Basin, identifying stakeholder perceptions, positions, and influence on the project, highlighting the existing communication channels, pinpointing the overall social impacts of the project, and assessing its various risks. AySA's social assessment, on the other hand, explored the opinion of stakeholders about the proposed infrastructure works to be carried out by the company.

⁶⁶ In the 1970's, *Obras Sanitarias de la Nación (OSN)*, the public water and sewerage authority for the Buenos Aires Metropolitan area that preceded *Aguas Argentinas*, constructed a 2.5 km sub-aquatic outfall in the municipality of Berazategui to discharge sewage from its service area to La Plata River. In 2004, the municipality of Berazategui and local residents initiated judicial proceedings against *Aguas Argentinas* for discharging sewage to the La Plata River without adequate treatment. In 2006, the case was elevated to Argentina's Supreme Court, which ruled in favor of the municipality and local residents and ordered the construction of a preliminary treatment plant. After *Aguas Argentinas'* concession was rescinded in March of 2006, AySA assumed responsibility for complying with the Supreme Court ruling and, in 2008, awarded a contract for construction of the Berazategui sewage preliminary treatment plant (BPTP). While work on the BPTP is expected to be carried out in parallel with APL-1, from a technical standpoint, the plant complements a Bank-financed civil work under APL-2 (i.e. the Berazategui sub-aquatic outfall) and has therefore been included as a counterpart work under APL-2.

⁶⁷ The executive summary of the EIA is on AySA's website; the complete report is available at AySA's library in Buenos Aires.

6. Both SAs were developed based on available project information and consultations with relevant stakeholders. The teams that conducted the SA carried out 45 meetings with stakeholders including representatives of community organizations, NGOs, the industrial sector, provincial and municipal officials, and members of academia. Stakeholders also participated in workshops, including those for the presentation of the TOR's for the EIAs and the draft EIAs themselves.

7. Regarding the BPTP, the World Bank team was recently informed (March 2009) that the construction site required resettlement of 18 families that had illegally occupied the site and erected precarious sheet-metal dwellings. In October 2008, families were relocated to new concrete houses approximately 2-3 km from the BPTP site. The Bank considers that the relocation of families from the BPTP site does not require application of OP 4.12 since construction of the BPTP is neither directly related to the project (APL-1) nor necessary to achieve its objectives. This conclusion notwithstanding, the Bank team conducted a preliminary assessment of the resettlement program and has found no evidence to indicate worsened living conditions. Families have been provided housing built from concrete blocks on 300 m² lots with running water. Schools and health facilities are nearby and transport services are accessible. The Bank will continue to monitor living conditions going forward.

B. Legal and Institutional Framework

8. The legal and institutional context for the management of the Matanza-Riachuelo Basin is complex. It involves 17 jurisdictions and 29 relevant agencies that co-exist in the Matanza-Riachuelo. Each of these agencies has jurisdiction over and/or responsibility for some aspects of water resources and/or their areas of influence. The institutional complexity stems from the need to create an environment of coordination that would allow legal-institutional consistency in order to implement the integrated management of water resources. Therefore, there is a risk that these various regulators may clash and fail to respond to an integrated management focus on water resources, making it impossible to achieve the objective of reversing the basin's polluted status. To respond to these challenges, Law 26.168, approved on November 15, 2006, created the Matanza-Riachuelo Basin Authority (Autoridad de Cuenca Matanza-Riachuelo, or ACUMAR) as an inter-jurisdictional public agency under the scope of the Secretariat of the Environment and Sustainable Development (Secretaría de Ambiente y Desarrollo Sustentable, or SAYDS) in the Office of the Chief of the Cabinet of Ministers. The duties of the Authority include (i) inter-jurisdictional coordination under a principle of subsidiarity, giving ACUMAR some preeminence over local authorities; (ii) execution authority to adopt measures necessary or relevant for the basin's sanitation and execute the budget of ACUMAR; (iii) regulation and control responsibilities, among others, to harmonize regulatory initiatives dealing with liquid effluents and gas emissions, environmental land use planning, rates for services, ACUMAR's organizational bylaws, establishment of a special administrative procedure, implementation of preventive measures, intervention on allocations, audits, monitoring, EIAs, sanctions, warnings, writs, inspections, closures, seizures, and cessation of activities. The manner in which these activities will be undertaken will also require that there is compliance with the law on access to environmental information.⁶⁸

⁶⁸ Law No. 25831 on Access to Public Environmental Information approved on 23 November, 2003.

9. Under Resolutions 1/2007 and 2/2008, ACUMAR has in place permissible limits (relating to organic and toxic loads) for discharges of effluents into sewage networks, surface water bodies as well as soil absorption. For effluent discharges, these limits are in accordance with the norms established by AySA for discharges into its sewage networks.⁶⁹ Additionally, the Government is in the process of finalizing norms/ regulations relating to uses and quality of the receiving water bodies in the short, medium and long terms. These regulations and norms will serve as the underlying basis for determining the treatment to be undertaken and targets to be met by polluting enterprises in the M-R River Basin.

C. Analysis of Alternatives

10. Approach to Alternatives Analysis. The Analysis of Alternatives carried out for the project has taken place both at the strategic level through the PISA and the PDSCMR and through the evaluation of alternatives for each of the proposed works with respect to their technical configurations such as size, flow volumes, and location. In addition to the no-project alternative, the project has identified a comprehensive set of alternatives for the management of the municipal wastewater of the M-R River Basin and analyzed these alternatives in conjunction with alternatives for management of the industrial wastes of the basin. A detailed description of alternatives analyzed and their implications for the environment is presented in Annex 11A.

11. Alternatives considered. Six strategic alternatives were considered in the development of the sanitation component of the project. These include: (i) a decentralized treatment option involving a series of treatment plants using different treatment levels, discharging their effluents into the Matanza-Riachuelo River, its tributaries, and the Rio de la Plata; (ii) a centralized treatment option with a single preliminary treatment plant in Berazategui and a sub-aquatic outfall discharging the effluent into La Plata River; (iii) a centralized treatment option with two points of discharge into La Plata River through sub-aquatic outfalls; (iv) different levels of wastewater treatment prior to discharge into La Plata River (these alternatives involve determining the lengths of the outfalls as a function of the treatment levels, as higher levels of treatment may require shorter outfalls); (v) secondary treatment options followed by chlorination of the effluent and discharge of the effluent into La Plata River, coupled with using a short outfall, and (vi) alternatives for the main wastewater conveyance system to ensure the collection of all the wastewater of the basin and of the entire Argentina Water and Sanitation (*Agua y Saneamientos Argentinos S.A.*, or AySA) concession area.

12. Among the conveyance alternatives analyzed was (i) a single left bank collector with the capacity to also convey all wastewater from the middle and upper basin of the Matanza-Riachuelo River toward the Riachuelo preliminary treatment plant and (ii) two collectors. Factors taken into consideration in selecting the preferred strategic alternative included

⁶⁹ This is in line with the World Bank Group's Environmental Health and Safety Guidelines applicable to the World Bank Group relating to Wastewater and Ambient Water Quality, which states that "when wastewater treatment is required prior to discharge, the level of treatment should be based on: · Whether wastewater is being discharged to a sanitary sewer system, or to surface waters; · National and local standards as reflected in permit requirements and sewer system capacity to convey and treat wastewater if discharge is to sanitary sewer; · Assimilative capacity of the receiving water for the load of contaminant being discharged wastewater if discharge is to surface water; · Intended use of the receiving water body (e.g. as a source of drinking water, recreation, irrigation, navigation, or other).

achievement of environmental objectives, avoidance of unwarranted environmental risks, total cost and accommodations of future expansion of the wastewater system.

13. In addition to the strategic alternatives identified above, a detailed description of site-specific alternatives was carried out by AySA through the EIA process for the works in APL-1, for which technical solutions have been identified. Alternatives were discussed for each of the specific civil works investments, including the Riachuelo preliminary treatment plant, the Riachuelo outfall, the Left Bank collector, the Baja Costanera bypass collector, and the Riachuelo preliminary treatment plan associated pumping stations. Alternatives evaluated included alternative site selection and routing, design capacity, and technology. These alternatives and their environmental implications are discussed in detail in the seven-volume EIA for AySA works.

14. The two alternatives considered for reducing the industrial pollution were (i) high-level treatment of all industrial wastes and discharge of all effluents except those containing toxic matter into the M-R River and (ii) lower-level treatment of all industrial wastes and discharge of all effluents except those containing toxic matter into the municipal sewerage network for further processing with domestic wastewater. In addition to the strategic alternatives, ACUMAR will evaluate alternative specific initiatives to be financed under the industrial pollution abatement component under the Environmental Management Framework (EMF) developed for this component.

15. Because sub-projects to be financed under Component 3 (Environmental Territorial Management) of the project have yet to be identified, it is not possible to identify site-specific alternatives. Under the terms of the EMF for this component, ACUMAR will be required to collaborate with local communities to identify specific alternatives through the planning and sub-project selection process.

16. The Selected Alternative. Based on the analysis of alternatives, Alternative 3 -- distributing wastewater flow to two sub-aquatic outfalls -- is the preferred option for the sanitation component. This alternative distributes the total collected wastewater flow into two preliminary treatment plants, one located on the southern bank of the M-R River at its confluence with La Plata River and the other in Berazategui. The effluents will be discharged into La Plata River through two long sub-aquatic outfalls. This alternative will result in lower and more affordable capital costs, the technical advantages of simple operation and maintenance, significant positive environmental impacts in the M-R River, and low, and broadly acceptable, adverse impacts with minimum negative social impacts in the Rio de la Plata. See Annex 4 of the PAD for a detailed description of the project options selected.

17. The industrial pollution abatement component of the selected alternative will incorporate (i) generation of databases on the contributions of industrial pollutants, (ii) definition of priorities and order of implementation of Industrial Restructuring Plans (*Planes de Reconversion Industrial*, or PRI), (iii) definition of discharge conditions and setting of permissible limits of pollutants in the short, medium, and long terms, (iv) identification of specific objectives or targets to be met in the PRIs of each industry, (v) prioritization of PRIs in industrial plants with higher pollutant loads and/or greater toxic effects, and (vi) implementation of real-time systems

to monitor water input flows, wastewater discharge flows, and concentration of polluting substances in the industrial effluents.

D. Environmental and Social Baseline Conditions

18. General. The Matanza-Riachuelo (M-R) is a tributary of the Río de la Plata and pollution of the river basin is one of the most visible environmental issues in the country. Over the past century, the M-R River Basin has been used as an open sewer for a large portion of the city of Buenos Aires. Pollution levels have increased steadily with urbanization in the greater metropolitan area of Buenos Aires and accompanying industrial growth within the basin. It is estimated that more than 4,000 industrial facilities are located in the lower and middle sections of the basin. Many of these industries discharge untreated effluents into the local drainage system or directly into the M-R River. In addition to high levels of organic pollution, these discharges contribute toxic contaminants such as heavy metals from petrochemical industries, tanneries, and meat processing facilities. The acute environmental and social degradation of the M-R River Basin has created serious health risks, especially for the highly vulnerable social groups.

19. Project Area of Influence. The environmental assessments (EA) prepared by ACUMAR took a broad approach in defining the project's area of influence. The area of influence includes the entire Matanza-Riachuelo Basin as well as the entire concession area of AySA. These two areas overlap but are not entirely contiguous. The project's EA also took into consideration three other zones: (i) the fluvial sections of the Lujan River from the mouth of the Reconquista River up to its confluence, (ii) the section of La Plata River between the coast and the outer limit of the so-called Parana de las Palmas flow corridor to the height of the channel entering the Port of La Plata, (iii) the coastal section between the bank of La Plata River and the neighborhood access road to the river approximately 14 km southeast of the boundary of the General Belgrano plant site. The area of influence includes coastal sections of La Plata River up to 50 km downstream from the treatment plants as well as approximately 30 km upstream.

20. Population and Development Context. The M-R basin is home to Argentina's largest concentration of urban poor. Of the approximately 3.5 million inhabitants in the basin, 1.2 million live below the poverty line. An estimated 10 percent of the total population in the M-R River Basin lives in informal settlements, often in flood-prone areas near open garbage dumps. The poorest populations living alongside the river are in constant contact with numerous contaminants ranging from untreated organic waste to toxic industrial chemicals. The M-R also floods frequently because of high flows in the rainy season and high levels of water in La Plata River. This flooding spreads highly polluted waters into informal settlements, exposing the inhabitants to further health risks from contaminated flood waters.

21. The M-R basin includes the jurisdictions of 14 municipalities and of the City of Buenos Aires, with an estimated population of 3.5 million (projected by the 2001 Population and Housing Census). The distribution of population in the basin responds to a progressive evolution of the location and not as a result of a logical distribution or planning of urban land uses and services. The population in the northern zone of Buenos Aires ranks in the highest socio-economic level, while the southern zone has a population in a lower socio-economic level.

22. Water Quality. The significant deterioration of surface water quality in the M-R River and most of the tributaries in the middle and upper parts of the river is a result of decades of poorly treated or uncontrolled pollutant discharge. These water courses have received organic discharges, mineral and inorganic substances, heavy metals, pesticides, fungicides, herbicides, and sediments at levels far exceeding the river's capacity for dilution and self-purification. Residents along the course of the river must endure both the severe health and aesthetic impacts of the water pollution. In some sections, the odor from the river can be overwhelming.

23. Land use. Land use within the lower M-R basin has a strong industrial profile with over 4,000 companies located within the basin. Industrial development along the banks of the water courses in the basin has had a direct and profound impact on the nature, levels, and frequency of liquid effluents disposed into the water bodies of the basin. The industrial sector is highly concentrated in the lower part of the basin near the Dock Sud Petrochemical Hub. Preliminary data show that the industrial sector is the major source of organic contamination, contributing about 73 percent⁷⁰ of the total load of organic pollution. These industries are also a significant source of other toxic pollutants, including heavy metals.

24. The composition of industries is highly varied with many different types of activities and industrial processes. Those with the greatest environmental impacts are in the chemical, petrochemical, food, tanning, cold storage, and electrometallurgical sectors. Many industries use arsenic and heavy metals such as chromium, mercury, and lead for their production processes, and all of these substances have been found in the waters of the M-R River in concentrations that exceed the levels permitted by relevant regulations

25. Groundwater. These domestic and industrial discharge problems also affect regional groundwater resources related to the basin. In zones lacking sewerage, pollution from discharges from domestic septic tanks can reach the water table. Aquifers are also affected by the inadequate treatment and disposal of industrial wastes. Frequent floods in the M-R River Basin produce considerable losses and damage to goods and people. Flood risks in all urban zones are a consequence of the low-lying terrain, insufficient drainage and deficient discharge from main pipelines, and unplanned urbanization that overtook natural drainage areas.

26. Critical Natural Habitats and Biological Diversity. The project's area of influence covers a large geographic area that includes a number of potentially sensitive terrestrial and aquatic sites of biodiversity importance. While the area as a whole has been subject to a number of threats and impacts from a wide range of development pressures, there remain some areas of biological interest. For example, there are two known terrestrial ecological reserves within the project area of influence.

27. The Reserva Natural Costanera Sur is located upstream of the planned treatment and outfall works near Puerto Madero. The Reserva Natural Punta Lara is a terrestrial reserve downstream of the planned works to the southeast of Quilmes and close to Gran la Plata.

28. There are no declared aquatic conservation areas within the project's area of influence, but there are two areas which have been identified as priorities for coastal and marine protection.

⁷⁰ From Modeling Report No 5.

FREPLATA has proposed two such potential areas of bi-national protection. One in the area known as Frente de Delta in the delta of the Rio Paraguay and the second in the area referred to as the Frente de Turbidez. Both of these areas are far from the impact zone of the project (see Section on Impacts). The Atlas of Sensibilidad de la Costa y El Mar Argentino identifies the region as providing habitat for marine birds, fresh water fishes and of habitat for dolphins and porpoise.

E. Expected Environmental and Social Risks Associated with component 1 (Sanitation)

Construction Phase

29. *General.* The environmental and social impacts and risks associated with this component will vary during the construction and operational phases of the project. During construction the main socio-environmental risks relate to: (i) the temporary construction-related impacts associated with the large-scale civil works. Among others these include the disruption to local traffic flows, and access to commercial areas, dust and noise control, handling and management of hazardous construction materials, emergency response planning; erosion and sediment control and disposal of construction debris, (ii) the potential impacts on bio-diversity and human health associated with disposal of spoil materials from tunneling; and (iii) the potential biodiversity and health impacts of re-suspension of dredge materials during dredging operations.

30. Impacts during construction stem from such activities as transportation of sand, concrete and construction materials, earth compacting, concrete pouring, and operation of construction equipment. Some of the specific impacts from the Riachuelo preliminary treatment plant and pumping stations, based on AySA's EIA, are discussed below.

31. *Tunnel spoil disposal.* The main collectors and sub-aquatic outfalls will be installed using trenchless tunnel technology methods. That is, pipes will be installed by tunneling using pipe jacking for smaller diameter (up to 3 m) pipes and segment installation for larger-diameter pipes. The trenchless pipe-jacking technology will significantly reduce the negative environmental impacts of the works by concentrating the impacts in the limited areas of the attack shafts, which are installed at distances of 1–1.5 km along the pipe route and for segment tunneling at distances of approximately 5 km.

32. *Air quality and odors.* During the construction stage, air quality may be affected by an increased concentration of suspended particles and carbon monoxide as a consequence of earth moving and machinery operation. Bothersome odors may be expected from the removal of dirt as a product of excavations. Another action that can generate odors is the temporary storage and disposal of construction waste. These impacts were characterized as negative with a low long-term environmental risk. They will generally be of medium or low intensity, brief in duration, and localized, appear immediately, and have a direct effect, ongoing while the operation that produces them lasts, and be reversible. These impacts are expected to be temporary and not significant, and they will be addressed through the contractor's site-specific management plans.

33. *Noise.* Temporary or continuous elevation in noise levels in the areas directly affected by the construction work may occur as a result of the movement and operation of trucks and mechanical equipment. The main sources of noise and vibrations will be (i) operation of heavy equipment and manual tools, (ii) movement of personnel and light vehicles, and (iii) mobile

equipment and machinery such as electric generators. Noise impacts will be temporary during the construction phase. They are likely to be negative and have a medium or moderate effect, a low to medium intensity, an immediate effect of short duration, a local scope, and intermittent occurrence while the works that generate them continue. The contractor's site specific management plan will include measures to address these effects.

34. Soil quality and stability. Soil quality may be affected by leaching, dumping, and hauling of solid or liquid materials that are in temporary use or transported to their final disposal site. Possible impacts on soil quality in these cases could be of medium or high intensity depending on the type of material involved and the local context. Overall, direct effect and will be temporary in duration and strict controls on dumping will be included as part of the contractors site specific management plans.

35. Compacting and/or settling of soils around construction sites may result from a variety of project construction activities. These include excavation, movement of heavy machinery, temporary disposal of large volumes of materials, residues, and debris, depression of the water table, large-scale, heavy installations, and demolition. These possible impacts will be negative, of medium or high intensity, of local scope, with a direct effect and of an occasional nature, and of temporary duration.

36. During earth moving and excavations, the walls of ditches may cave in, causing the loss of soil stability and potentially affecting the foundations of buildings and public facilities or existing pipelines and underground infrastructure (e.g., water network, gas network, electrical and phone lines). The impacts that may occur in these cases will be negative, of medium or high intensity, of local scope, with direct effect, of an occasional nature, and of temporary or permanent duration. Although such impacts are highly unlikely if the works are properly designed and executed, all necessary measures to avoid these risks must be taken into account. The contractor's site specific management plans will include specific references to contingency measures in the event of unexpected construction accidents.

37. Surface water quality. Construction activities that may affect the quality of water resources during the construction stage include (i) hauling of solids and/or liquids during the clearing of work sites, (ii) leaching, dumping, and hauling of solids that have been in temporary use or are transported to their final disposal site, and (iii) sediments from earth movement may reach surface waters. The negative impacts that these activities may generate will be direct, of low to medium intensity, of short duration, of local scope, and occasional. No long-term negative impacts were identified in this aspect, taking into account the current baseline conditions of the water bodies in the project area.

38. Groundwater levels and quality. Available data on the relationship between the behavior of the groundwater table and the activities carried out during construction (e.g., depression of the water table, disposal of extracted water) are not adequate to evaluate their effect. However, based on AySA's experience in this type of works, the activities associated with the project are not expected to modify the aquifer's behavior in the zone.

39. Re-suspension of sediments during dredging operations in the rivers. One of the impacts associated with dredging operations is the movement and re-suspension of sediments that may

contain significant concentrations of pollutants. In this regard, two types of impacts can be identified for the project: (i) increased turbidity from the re-suspension of solids and increased concentration of suspended solids in the water body, with a consequent decrease in transparency, and (ii) transfer of polluting substances between solids in suspension and water, producing an increase in the concentrations of these contaminants in the aqueous phase.

40. This impact is considered moderate to high, depending on the factors affected (especially turbidity) and the zone where dredging operations will be conducted. Especially important will be the impact of works in La Plata River and their potential impact on the turbidity of water in the intakes for the Buenos Aires drinking water treatment plants. To evaluate potential impacts from dredging, a mathematical simulation model was developed for solids behavior in La Plata River. Its results are presented in the AySA EIA study. This model takes into account extreme scenarios of current velocities and winds, identifies the risks, and proposes mitigation measures. A mitigating effect is attributed to the fact that the outfalls will be constructed using trenchless technology, with the majority of tunneling works located below the riverbed.

41. Impacts on flora and fauna. No significant construction impacts on terrestrial flora or fauna are expected during the construction phases of the project. Project construction works will not affect any known sensitive or critical terrestrial natural habitats in the project's area of influence. All project works are either located underground or in highly degraded or heavily industrialized areas. Conservation areas are located within the project's area of influence, but these areas will not be affected by project construction. Plant cover and small shrubs may be affected by the establishment of work staging sites and storage areas, the temporary disposal of surplus earth and work debris, and the movement of vehicles and heavy machinery. These impacts are considered to be temporary and insignificant.

42. The construction of the Riachuelo preliminary treatment plant will occur in the Dock Sud area on a parcel of land given to the national government by the municipality of Avellaneda under a 99-year lease. Based on the EIA and confirmed by site visits, it is apparent that the selected site for the Riachuelo preliminary treatment plant is highly degraded and has been heavily impacted by previous industrial development of the petro-chemical industry. AySA has confirmed that it has already taken possession of the site.

43. The construction of the Riachuelo outfall is expected to have temporary but limited impacts on aquatic biodiversity since these works are mainly underground. Re-suspension of sediments during dredging operations may contribute to some temporary incremental degradation of the water quality but this must be viewed in a context of high incidence of maintenance dredging which is already occurring throughout the major shipping lanes throughout project area of influence.

44. It is possible that construction site accidents could lead to some unexpected impacts on vegetated areas or urban parks. Special care will be taken to avoid spilling construction site contaminants that may be harmful to vegetation. The impacts resulting from such accidents, though negative, would be likely be localized as no construction site is near any important vegetated areas.

45. Occupational health and public safety. Because of the magnitude and extent of civil works, a large number of skilled and unskilled workers will be employed during project construction. Strict occupational health and safety measures will be applied to avoid accidents affecting workers and people living in the project construction areas. The movement of heavy equipment may pose some risk to local residents and all construction contractors' site management plans will include specific reference to public safety concerns.

Operational Phase

46. General. During project operations the principal environmental risks relate to (i) the environmental and human health effects of long-term discharges of wastewater into the Rio de la Plata through the two sub-aquatic outfalls; (ii) the environmental and human health risks associated with management and disposal of continuously generated solids from the Riachuelo preliminary treatment plant; and (iii) localized nuisance issues relating to operations at the Riachuelo plant such as odor and noise. The main social risks are associated with (i) pollution on the coast resulting from the discharge into the Rio de la Plata, affecting recreational activities and navigation, (ii) sewerage tariff increase, and (iii) displacement of low income groups which cannot afford higher land costs (gentrification).

47. Consideration of the potential adverse operational risks should be balanced with a consideration of the very important potential environmental and human health benefits of the project which include: reduced exposures to existing high-levels contaminants in the Matanza-Riachuelo and improved environmental conditions on the shoreline of the Rio de la Plata including the interface with conservation areas. In addition to the health benefits of the project, significant quality of life benefits are expected to result over the long term including improvements in recreational value of the basin and improved livability in the impact zones.

48. Long-term positive impacts on water quality. Dissolved oxygen (DO) concentrations in the M-R River system were forecast during the low flow period after project implementation. The simulation rendered positive impact in water quality in various sections of the basin showing positive concentrations of dissolved oxygen in previously anoxic conditions (Menéndez, Angel. UTN. 2008).

49. Figure 10.1 shows the Rio de la Plata zones which are inadequate as a water source for human consumption under current conditions, prior to project implementation. It shows that most of the inadequate zone is affected by pathogenic organisms. Pathogens are a critical pollutant because organic matter discharges do not significantly modify the DO levels of La Plata River. Metals and toxic substances are also not the main problem, because of the pre-existing load in the river arriving from its upstream sections. Figure 10.2 illustrates the expected improvement of water quality after project implementation, showing the alleviation of the risk of inadequate water quality for human consumption in the intake zones of Bernal and San Martin, as well as the major positive impact of the project in terms of reducing pathogenic contamination along the coast of La Plata River.

50. During the operational phase of the project, there will be a redistribution of zones currently considered unsuitable for recreation with direct contact, alleviating pollution in the coastal zone and permitting its use for recreation. A significant part of the coastal zone will also be recovered

for recreation without direct contact. The unfavorable status of chronic toxicity for aquatic life in La Plata River will not change because the contribution of pollutants from the Paraná de las Palmas River (upstream from Buenos Aires) continues and is independent of the project. The reduction of the zone unsuitable for aquatic life because of acute toxicity is insignificant, because Sarandí and Santo Domingo creeks continue to contribute significant amounts of chromium.

Figure 10.1. La Plata River zones inadequate as a water source for human consumption, current situation without project

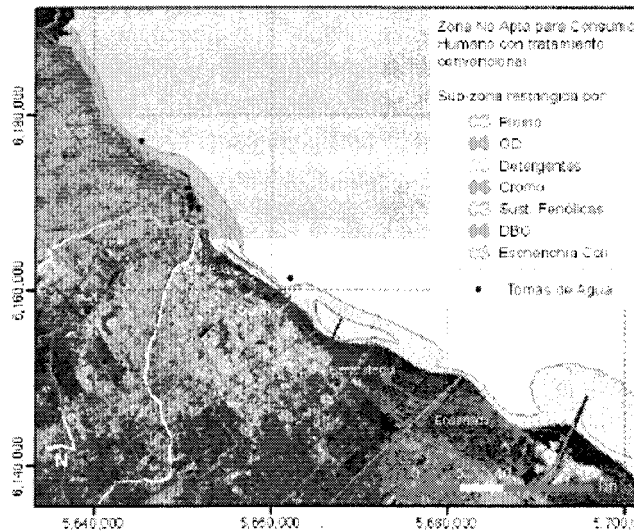
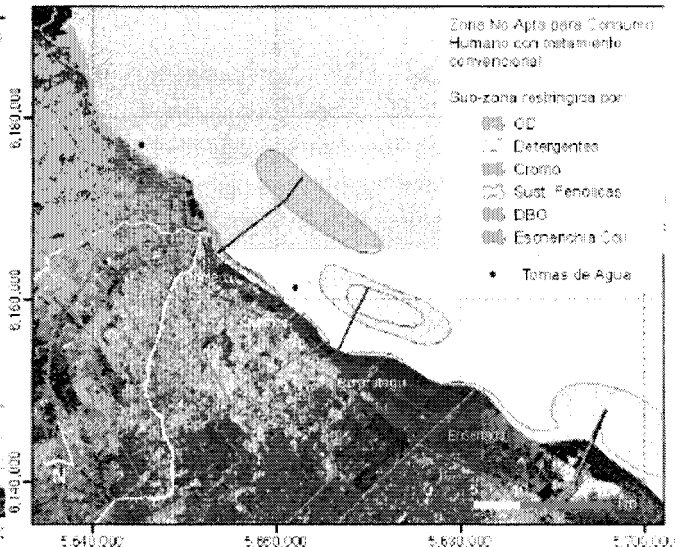


Figure 10.2. La Plata River zones inadequate as a water source for human consumption, predicted situation after project implementation (APL-1 and APL-2)



51. Impacts on Critical Natural Habitats and Aquatic Biodiversity. A conclusive assessment of project's incremental impacts on critical natural habitats and on the aquatic biodiversity of La Plata River is difficult due to the complexity of the project context and because detailed information on the status of regional bio-diversity is incomplete. The disposal of wastewater in the Rio de la Plata will have some adverse impacts on fish and other aquatic biodiversity in the immediate disposal zone surrounding the outfall diffusers. In the immediate vicinity of the diffusers the likely implications for aquatic life are likely to be negative as a result of direct impacts leading to lower dissolved oxygen levels, increased turbidity and settling of sediments. Under current project design, these risks are largely unavoidable.

52. However, determining whether these risks are significant or not is difficult because the region is already under significant pre-existing pressure from contamination, shipping, and ongoing channel dredging. According to FREPLATA -- a project sponsored by GEF and jointly implemented by Argentina and Uruguay to assess and evaluate the principal threats to biodiversity in the Fluvial Zone of La Plata River and its Shoreline (*Zona Fluvial del Rio de la Plata y su Frente Maritimo* (RPFM)) -- the entire Rio de la Plata region is already under immediate pressures from a range of issues including over-fishing; incidental fisheries catch; cumulative pollution loads along the entire length of the river; introduction of exotic species; and harmful algal blooms. It is important to bear in mind that, while portions of the Zona Fluvial are considered important biologically, there are no existing aquatic biodiversity conservation areas currently declared in the project's area of impact.

53. FREPLATA had identified several major priority threats which are currently affecting the RPFM and it is important to consider how the proposed project may further contribute to or exacerbate existing threats. For example, among the most important issues affecting regional biodiversity identified by FEPLATA are the following: (i) incidental fisheries capture; (ii) invasive species; (iii) lack of tools for conservation of threatened species including the absence of a system of aquatic protected areas; and (iv) harmful algal blooms. Understanding these issues in a complex environment is hampered by lack of information on key considerations.

54. Considering these high priority issues it is clear that the project would not contribute to problems of over-exploitation of the fisheries resource through promotion of harmful fishing techniques and practices. Furthermore the project would not contribute to the problems associated with introduction of exotic or invasive species. Since there are no declared aquatic protected areas the project would have no impact on such areas. Two areas where the project could have impacts are in the area of incremental contributions to water quality decline and the problems associated with harmful algal blooms.

55. Potential for Harmful Algal Blooms (HAB). HAB are events which occur periodically as a result of the rapid multiplication of micro-algae in the Rio de la Plata. Not all algal blooms are harmful to human or ecological health but some may be depending on the species involved. According to FREPLATA, the incidence of these HAB has grown in recent decades and some have produced impacts such as fish mortality and alterations to the food chain. Human health impacts from such HAB can generate allergies or reactions of varying degrees of intensity.

56. The large inputs of nutrients from discharged wastewater may increase the risks of HAB but the timing, intensity and impact of such blooms will be very difficult to predict with any degree of accuracy. Also, the extent to which the nutrient loads from the project will be responsible for such blooms will be difficult to ascertain in practice. At present, it is very difficult to reduce or suppress these outbreaks when they occur. The EMP recommends that monitoring programs and early warning mechanisms would need to be set up to monitor outbreaks and to inform the public of appropriate measures to avoid contact and reduce the risks of contact

57. Nuisance impacts associated with treatment plant and pumping station operations. The Riachuelo preliminary treatment plant may generate odor and noise problems. Fortunately, the plant is located in an area well removed from residential neighborhoods, so potential negative impacts will be limited. Also, modern design takes care of noise and odor problems through the use of adequate technologies such as collection and treatment of odor generating gases, isolation of noise-generating equipment. The pumping stations that will be constructed under the project are located in the treatment plant sites, and odor and noise problems will be taken care of as part of managing the treatment plants.

58. Impacts Associated with disposal of solids generated in the treatment plants. The total quantity of solids of various types (grit and sand, oil and grease, solid residues) that will be generated by both treatment plants (Riachuelo and Berazategui) during the initial years of operation will oscillate in the range of 340–500 tons per day. These considerable quantities will require a sludge management plan to guide overall operations. Trucks will remove these solids and deposit them in the sanitary landfill of the State Company for the Ecological Coordination of

the Metropolitan Area (*Coordinación Ecológica Área Metropolitana Sociedad del Estado*, or CEAMSE). Handling and transporting such quantities of solids present environmental risks. AySA has experience in handling the solids generated in its wastewater treatment plants in a similar manner.

59. The project will take precautions to handle the Riachuelo preliminary treatment plant solids according to acceptable technical standards and as defined in the EMP. Installations for recovery of sand for reuse and generation of energy from the oil are included in the project, but investigations are required before finalizing specific actions. The impact of such installations will be assessed after decisions are taken.

F. Impacts Associated with component 2 (Industrial Pollution Abatement)

60. This component is designed to pilot investments that help reduce industrial wastewater discharges (organic as well as toxic effluents) that directly enter the M-R River, and over the next 15–20 years will contribute to the Government's PISA objective of overall environmental recovery of the river. As part of the Government program, ACUMAR will first focus on the 50 main industrial polluters contributing approximately 95 percent of the total industrial organic load (approximately 70 percent of the total organic load) discharged to the M-R River and the 170 industries with the highest toxic load that currently discharge to the M-R River. The component will finance technical assistance to improve monitoring, control, and enforcement by the M-R River Basin Authority and will help in collecting baseline data and carrying out diagnostics. Additionally, APL-1 will include the initial phase of a matching grants program to support private sector investments in pollution control and cleaner production processes, targeting SMEs with weaker financial and technical capacity to comply with environmental regulations.

61. The environmental impacts associated with investments to be supported under Component 2 will be mostly positive, with very few having some potential for adverse impacts. The initiatives contemplated under sub-component 2 of this component are intended to limit and control industrial discharges from a number of industries into the M-R River Basin and improve environmental compliance and, as a result, are expected have significant environmental and human health benefits. Building on the TA activities that will help identify, develop and sign PRIs with polluting industries in accordance with established environmental standards and permissible limits for discharge into the AySA sewage network, a matching grants program will be initiated to provide support to micro, small, and medium enterprises in the M-R River Basin area for investments in cleaner production processes. These cleaner production investments will increase the efficiency of the plants and processes, minimize the use of resources, eliminate toxic inputs, and reduce the quantity and toxicity of effluents and emissions.

62. What is not known at this time is whether any part of this sub-component would involve some physical interventions requiring environmental management since the exact scope of work and details such as sub-project location have yet to be determined. Furthermore, it is not known whether the implementation of the PRIs would reduce employment at the firms, or if the potential closure or relocation of industries would result in reductions in local taxes. Consequently, it is not possible at this stage to assess full environmental and social impacts.

63. The project has developed an Environmental Management Framework (EMF) which, among other things, describes the applicable institutional and legal framework, the types of eligible expenditures, and the overall screening mechanisms for sub-project activities. The EMF will be used to assess possible environmental impacts and to guide the implementation of this component, including an environmental management of activities with adverse impacts, which will be carried out once decisions are taken on respective works. A Social Management Plan is being developed in order to avoid, minimize or mitigate social adverse impacts.

G. Impacts Associated with component 3 (Environmental Territorial Management)

64. This component's main objective is to improve public policies on environmental territorial management within the M-R River Basin. Towards this end, the component seeks to improve territorial planning and develop pilots of integrated investments in basic infrastructure services and flood control to improve urban living conditions in selected low-income areas of the basin.

65. Environmental impacts of Component 3 are expected to be mostly positive, but there is potential for some adverse impacts associated with the construction of civil works for local flood control and drainage. These works are intended to resolve existing environmental problems through investment in improved infrastructure. Any impacts associated with these works are expected to occur during the construction phase.

66. Typical impacts associated with construction such as dust and noise control, waste management, construction site management are to be expected but these impacts are expected to be localized and manageable using known mitigation measures. Most of the works contemplated are of a small scale nature. An environmental management framework (EMF) has been developed to guide the implementation of this component. The EMF describes the measures to be taken with respect to environmental screening of proposed works, definition of eligible works, mechanisms for triggering additional assessment of proposed activities where necessary.

67. Social adverse impacts as a result of involuntary displacement are not foreseen but a Resettlement Policy Framework has been developed in case any actions require the application of OP 4.12. (See Section K, below)

H. Potential Regional and Cumulative Impacts

68. As mentioned previously, the impacts of the project are expected to contribute to significant improvements in the water quality and quality of life for those living in Matanza-Riachuelo basin and along the coastal areas of the Rio de la Plata. Adverse impacts will be associated with incremental degradation of the water quality in a limited part of La Plata River.

69. The water quality simulation modeling undertaken to predict the project's impact on La Plata River were based on wastewater flow values forecast for the next 20 years. The simulation assumes the full implementation of the infrastructure works under APL-1 and APL-2 and other works planned in the PISA and in AySA's Master Plan such as the Berazategui preliminary treatment plant and the full expansion of secondary networks. In addition, the simulations included the combined impacts of both outfalls, the impacts of the different project stages (one complete outfall and the current Berazategui outfall), as well as the available baseline data on

water quality which reflects the combined effects of development in the project's area of influence. In this respect, the assessment carried out takes into account potential cumulative aspects.

I. Stakeholder Consultation and Information Disclosure.

70. As part of project preparation, extensive stakeholder consultations were carried out using formal as well as informal methodologies. Information on the project was made available to the public through a series of public disclosure events and workshops. Informal focus groups discussions were conducted with nongovernmental organizations (NGOs), local government, affected people, and beneficiaries at various points during preparation and appraisal. The project EIAs were posted on the ACUMAR and AySA web sites and made available through the Bank's InfoShop. Other stakeholder consultation events are planned to disseminate information on the project during project implementation as part of the Communications Strategy of the project. The EIA and Social Assessment solicited stakeholder concerns and ensured an open process of debate and discussion about project objectives and goals.

71. ACUMAR and AySA sponsored a public consultation event in Buenos Aires on July 10, 2008 to present and discuss the detailed scope of work and terms of reference developed for the EIAs proposed for the project. This scoping workshop was attended by 140 participants representing a wide range of public and private stakeholders, including the Secretariat of the Environment and Sustainable Development (*Secretaría de Ambiente y Desarrollo Sustentable*, or SAYDS), ACUMAR, AySA, the Province of Buenos Aires, municipal governments, national and local environmental NGOs, national government agencies, community organizations, and professional associations. The terms of reference and proposed scope of work for the EIAs were discussed and agreed on by the workshop participants. This formal workshop built on informal dialogue and discussion with various partners and stakeholders including ACUMAR, AySA the Bank, technical specialists, and local government.

72. During the workshop many questions were raised by participants and answered by ACUMAR and AySA. Written comments received after the workshop were taken into account during project preparation. Important questions were raised relating to the project alternatives, specifically the level of wastewater treatment proposed, and the overall strategy for the basin, including concerns about the relative emphasis on industrial versus domestic wastewater controls.

73. A second workshop was conducted on November 7, 2008 to present the advanced drafts of the EIAs and to solicit stakeholder feedback on the findings, issues, and recommendations of the reports. During this workshop a number of issues were raised relating to modeling results on water quality, the technical scope of the project internal review and clearance procedures, and the importance of continued communications with stakeholders.

74. Additional consultations and project information dissemination were carried out as part of the preparation of the economic evaluation studies. A survey of more than a 1000 households and a large number of community meetings were held to collect information on a range of project design issues, including willingness to pay and perceptions and valuation of expected benefits. During this study, important project information on the project design, expected

benefits, and potential risks was shared with the sample populations selected for the survey. This process was a tool for information disclosure, dissemination, and stakeholder feedback.

75. As part of the outreach and disclosure program, two separate consulting teams conducted social assessment studies. AySA contracted a consulting team for a social assessment of Component 1 works, and a second team was retained to carry out a social assessment of the overall proposed project. As part of the preparation of these studies, the two consulting teams held a series of interviews and meetings with stakeholders including community organizations, industry representatives, and municipal governments. The main objective of these studies was to independently inform stakeholders and affected people about the project and collect feedback.

76. In addition to the project-specific consultation and disclosure efforts, both AySA and ACUMAR have social outreach units within their organizational structures. These teams have been established to liaise continuously with the local communities and facilitate communication tools. ACUMAR, for example, established a network of approximately 300 community and environmental NGOs as part of its broader communications strategy on the issues facing the M-R River Basin. This network provides an opportunity for information disclosure, not only on the project but also on a wide array of development issues facing the basin. AySA business practice for planning of civil works includes disclosure of information and stakeholder consultations which will be applied to the civil works in APL-1 which are at or before pre-feasibility stage (e.g., right bank collector).

77. A Communication Strategy (see Annex 11D) has been designed with ACUMAR, in order to inform key actors and stakeholder groups in the basin about the project in a transparent manner, strengthen the credibility of and support for the project, and to contribute to the overall consultation, participation and empowerment framework for the project. The Communication Program includes the creation of a Communication Unit in ACUMAR, which would be responsible for (i) providing adequate and timely information prior to consultation events; (ii) maintaining documentation on consultation processes and decisions taken; and (iii) providing feedback to concerned stakeholders. Staff in the new Unit will coordinate with those in AySA's Communication Unit.

78. In addition to general public disclosure, feedback on the project design was solicited from all international experts (including the four that reviewed the EIA, and the three technical experts). These high-level experts provided technical reviews in the fields of sanitation, engineering, and the environment were given an opportunity to comment and give advice.

79. Additional consultations and information-sharing workshops are planned as part of the project communication strategy and will be conducted at different project stages, including during project implementation.

J. Environmental Management Plan (EMP)

80. *Key Elements of the EMP.* As part of the EIA process, ACUMAR and AySA have developed an EMP that includes measures to avoid, minimize, mitigate, and monitor potential project impacts during construction and operational phases of the project. This EMP will be further strengthened during project implementation as described in paragraph 89.

81. The core programs of the EMP include: (i) Management of impacts during the construction phase; (ii) Management of impacts during the operational phase; (iii) Management of impacts associated with Component 2; (iv) Environmental monitoring in the project's area of influence; (v) Contingency planning and management; (vi) Consultation, communications and disclosure; and (vii) Capacity building, training and research.

82. Overall Institutional Responsibilities. ACUMAR will be responsible for overall project adherence to required environmental measures under all components of the project, with particular emphasis on Components 2 and 3. ACUMAR will assign specific qualified staff dedicated to the management and supervision of all aspects of the EMP during project implementation. ACUMAR will also be responsible for ensuring periodic audits and supervision of environmental performance and they will take the lead in coordinating information disclosure and consultations.

83. AySA will have specific institutional responsibilities to ensure that all construction contracts for works carried out under Component 1 include explicit requirements to ensure minimization and avoidance of all construction-related impacts. The costs associated with meeting these requirements will be included in such contracts. Through its existing environmental unit, AySA will ensure that competent core staff will be available at all times to supervise the implementation of environmental management measures directly under its control and to ensure compliance of contractual obligations of contractors. AySA will also have the responsibility to ensure that future operations of facilities under Component 1 adhere to the agreed EMP.

84. Construction contractors will be required to follow site-specific EMP that describe various measures to follow at each site. Each contractor will be required to assign an environmental engineer or other suitably qualified staff to oversee compliance with the site-specific plans on a day-to-day basis.

85. Monitoring. The monitoring program will be implemented to track indicators relating to the medium to long term needs of the basin thus ensuring sustainability after project completion. Environmental monitoring will be required for all three components during both the construction and operational phases with associated costs incorporated in the relevant contracts. AySA will have the primary responsibility for establishing baselines and monitoring specific works and facilities operations under Component 1 both during construction and operation of facilities. ACUMAR will be responsible for establishing baseline and monitoring programs associated with Components 2 and 3. ACUMAR will also establish a regional environmental trends monitoring program at the level of the project area of influence. These monitoring programs will be mutually supportive, and a large degree of coordination and cooperation is expected between project's implementing authorities and the various stakeholders. The water flow, quality and sediment monitoring program will be financed by the proposed loan under component 4.

86. Disclosure of Information. A high degree of public interest is expected in the progress of the project and the results of monitoring activities, especially during project operations. ACUMAR and AySA will work together to ensure close collaboration related to public disclosure of monitoring reports and data throughout the life of the project. ACUMAR will

establish a public web site where periodic status reports, monitoring data, and information will be available. Stakeholders will also be able to obtain information through the environment units of both AySA and ACUMAR.

87. Information will be made available to local government and local stakeholders through a comprehensive communications and outreach strategy, which will provide regular project updates and organize media events and other information campaigns at the local level. Both ACUMAR and AySA will establish communications links with concerned stakeholders including local government and NGOs working in the project area.

88. *Institutional Capacity Building.* The institutional capacity within AySA and ACUMAR to implement an EMP for a project of this type can be considered limited. AySA has a small Environment Unit with some experience in overseeing operations but lack sufficient technical skills to oversee a complex project of the scale envisaged under the proposed project. For this reason, AySA tends to outsource technical studies. At this time ACUMAR has no equivalent unit and in many respects will be executing the operational aspects of this project for the first time.

89. Because some of these activities will be new to both ACUMAR and AySA, the project will support specific activities to enhance their capacity to plan and execute the monitoring and information disclosure aspects of the project. The project will support specific training to enhance organization, staffing, and other resources to assist in building core capabilities in monitoring, communications, and outreach, as well as workshops, seminars and public information events. EMP activities, scope of work, institutional responsibilities and further details are available in the project file and in the EIA reports prepared by AySA and ACUMAR.

K. Social Safeguards

90. *Social analysis.* With regard to the social impact of the project, results of economic and social analysis studies indicate that the project will have a positive impact on the population in this basin and in the Buenos Aires metropolitan area. Preliminary results of a recent study that interviewed nearly 1,000 households about their perception of the current status of the M-R and La Plata rivers indicate that the project will help solve problems identified by the population. According to this study, "... the pollution of both rivers affects people mainly because of damage to the environment and risks of disease and because space for recreation is lost". For a detailed description on social impacts assessment see Annex 11C.

91. Because the project will contribute to restoring the condition of the M-R River Basin, improving the environmental situation will have a positive impact on the health of the affected population and on spatial use. Several stakeholders—groups that use the river for recreation (yachting), artisanal fishermen, La Plata River navigation authorities (because of potential interference with navigation), and municipalities downstream of Buenos Aires—may perceive that they are negatively affected by the project. In reality, all these stakeholders will benefit from the project's improvement of the quality of the water in La Plata River along the shoreline. The communication strategy that is being developed will include actions to convey this message to the referenced stakeholders.

92. *Involuntary Resettlement.* In addition, no resettlement of population is foreseen under Component 1 of the project. Selected pipe routes are located in public areas and will not affect private property. During construction, access to some small extensions of public areas may be temporarily restricted, but the trenchless methods to be used for pipe laying will minimize such restrictions. Maps, satellite photos, and draft designs suggest that no resettlement will be needed. As the physical interventions of Components 2 and 3 of the project have not been determined, a Resettlement Policy Framework was prepared for implementation if and when warranted.

L. Compliance with World Bank Safeguard Policies

93. The proposed project is categorized as an Environmental risk category “A” based on the expected long-term impacts of the large-scale sanitation civil works investments under Component 1. Works to be carried out under Components 2 and 3 are expected to have impacts that are mostly positive and where adverse impacts may occur they are expected to be small-scale, temporary, local in nature, and can be addressed with known mitigation measures.

94. Table 10.1 indicates which of the World Bank policies are expected to be triggered under the proposed operation. A description of the rationale and the works completed to comply with these policies is described below.

Table 10.1. Safeguard policies triggered by the project

Safeguard Policies Triggered by the Project	Yes	No
Environmental Assessment (OP/BP/GP 4.01)	[x]	[]
Natural Habitats (OP/BP 4.04)	[x]	[]
Pest Management (OP 4.09)	[]	[x]
Cultural Property (OPN 11.03, being revised as OP 4.11)	[x]	[]
Involuntary Resettlement (OP/BP 4.12)	[x]	[]
Indigenous Peoples (OP 4.10)	[]	[x]
Forests (OP/BP 4.36)	[]	[x]
Safety of Dams (OP/BP 4.37)	[]	[x]
Projects in Disputed Areas (OP/BP/GP 7.60)	[]	[x]
Projects on International Waterways (OP/BP/GP 7.50)	[x]	[]

95. *Environmental Assessment (OP/BP/GP 4.01).* Environmental assessments have been prepared for the project at two levels of analysis. At the overall project and river basin level, an Integrated Environmental Assessment has been prepared which describes the regional environmental issues facing the M-R River Basin, the baseline environmental situation, the applicable legal and institutional framework guiding the management of the basin, the principal management strategies considered, and alternatives evaluated for different basin management approaches. In addition, the Integrated Environmental Assessment (IEA) provides a comprehensive description of the overall project components, their main environmental risks and benefits, and the measures to be employed during project implementation to ensure management of adverse environmental risks.

96. A seven-volume Environmental Impact Assessment (EIA) for Component 1 describes AySA's strategic corporate vision, the Integrated Plan developed for its concession area, baseline environmental conditions in the project's area of influence, and the applicable legal and institutional framework guiding AySA's sanitation works. The EIA report from AySA considers realistic alternatives to each proposed work and evaluates the proposed works in the context of AySA's five year investment plan (Plan Director). A comprehensive EMP has been developed, spelling out institutional responsibilities for implementation of mitigation actions as well as supervision and monitoring of compliance with agreed mitigation measures. For example, all construction contractors will be required to follow site-specific management plans agreed on and incorporated into all construction contracts.

97. A seven-volume EIA describes AySA's strategic corporate vision, the Integrated Plan developed for its concession area, baseline environmental conditions in the project's area of influence, and the applicable legal and institutional framework guiding AySA's sanitation works. This documentation includes site-specific EIAs prepared for the civil works contemplated to be constructed during the early years of APL-1 under Component 1. Additional EIAs for other works to be constructed towards the end of APL1, for which design will be prepared during the course of APL1, will be reviewed by the Bank during project implementation.

98. A panel of four independent experts⁷¹ was hired by the Bank to review the drafts EIA reports and to provide comments and recommendations to ACUMAR and AySA in order to improve the quality of their EIA reports. The draft reports of the panel of experts were shared with AySA and ACUMAR and were used by these institutions to improve the final version of the EIAs.

99. The AySA EIA includes site-specific EIAs for works included in APL1 for which final technical solutions have been identified. For those works for which final technical solutions have not yet been identified, site-specific EIAs will be prepared in accordance with country requirements and the Bank safeguard policies. The Integrated Environmental Assessment included the Right Bank Collector as part of the analysis of alternatives for conveyance of wastewater and summarized the modelling of the project's overall impact on water quality. Also, since the Government advanced the construction of the Berazategui preliminary treatment plant (BPTP)⁷² (functionally an investment of APL 2), an EIA was prepared in July 2007 and has been disclosed in-country by AySA.⁷³ The Bank team has assessed the EIA and found it to be broadly

⁷¹ See annex 11 for the experts' names.

⁷² In the 1970's, *Obras Sanitarias de la Nación (OSN)*, the public water and sewerage authority for the Buenos Aires Metropolitan area that preceded *Aguas Argentinas*, constructed a 2.5 km sub-aquatic outfall in the municipality of Berazategui to discharge sewage from its service area to La Plata River. In 2004, the municipality of Berazategui and local residents initiated judicial proceedings against *Aguas Argentinas* for discharging sewage to the La Plata River without adequate treatment. In 2006, the case was elevated to Argentina's Supreme Court, which ruled in favor of the municipality and local residents and ordered the construction of a preliminary treatment plant. After *Aguas Argentinas'* concession was rescinded in March of 2006, AySA assumed responsibility for complying with the Supreme Court ruling and, in 2008, awarded a contract for construction of the Berazategui sewage preliminary treatment plant (BPTP). While work on the BPTP is expected to be carried out in parallel with APL-1, from a technical standpoint, the plant complements a Bank-financed civil work under APL-2 (i.e. the Berazategui sub-aquatic outfall) and has therefore been included as a counterpart work under APL-2.

⁷³ The executive summary of the EIA is on AySA's website; the complete report is available at AySA's library in Buenos Aires.

acceptable. Additionally, a revised Environmental Management Plan for the BPTP, acceptable to the Bank, will be presented by AySA within 6 months of loan effectiveness.

100. The EIA considers realistic alternatives to each proposed work and evaluates the proposed works in the context of AySA's five year investment plan (*Plan Director*). A comprehensive EMP has been developed, spelling out institutional responsibilities for implementation of mitigation actions as well as supervision and monitoring of compliance with agreed mitigation measures. For example, all construction contractors will be required to follow site-specific management plans agreed on and incorporated into all construction contracts. Mitigation for works will be guided by an environmental construction manual, which describes best practice methods for common environmental issues such as sediment and erosion control, solid waste management, and noise and safety concerns.

101. For Components 2 and 3, an Environmental Management Framework (EMF) has been prepared describing the procedures for screening and evaluating any proposed sub-projects. Where needed, an EMP will be developed for sub-projects that will address specific issues identified during screening. The frameworks also describe the procedures and expectations with respect to disclosure of information and consultation with affected communities.

102. EIA documentation has been made available for public review and comment. During project implementation, ACUMAR and AySA will carry out a strategic communications plan for the project that will define the methods for public disclosure of all relevant documents, including monitoring reports during implementation.

103. ***Natural Habitats (OP/BP 4.04)***. Within the project's area of influence, there are two known terrestrial ecological reserves and no existing aquatic reserves. Based on studies conducted by the GEF-funded FREPLATA project, priority areas are proposed for the future for protection of aquatic biodiversity have been identified but these areas are far from the immediate impact zone and not likely to be affected by the project.

104. Because the project aims to contribute to the significant improvement of water quality conditions in the M-R River Basin and in the coastal areas of the La Plata River, it is expected to have significant positive impacts for the natural riverine habitats found in the project area. An area of the receiving waters in La Plata River will be degraded as a result of the pollution discharge through the outfalls. Models developed for the outfalls indicate that the discharge plume is expected to be fully contained within a relatively small area of the receiving waters. The pollution is not expected to affect downstream coastal areas designated as conservation areas.

105. ***Pest Management (OP 4.09)***. The project will not procure nor will it lead to increased use of pesticides. Therefore no action is required under this policy.

106. ***Forests (OP 4.36)***. There are no forests affected by the project and there are no activities proposed which would promote improved forest management. Therefore no action is required under this policy.

107. ***Safety of Dams (OP 4.37)***. The project does not finance the construction or rehabilitation of dams therefore no action is required under the policy.

108. ***Physical Cultural Resources (OP 4.11)***. Preliminary site evaluations for proposed civil works construction do not show any evidence of cultural property that will require mitigation or management measures. However, since many of the proposed works are underground or will require some excavation, all construction contracts will include a "chance finds" clause that will require contractors to halt construction if any underground cultural property sites are encountered during construction. Site specific management plans for any cultural property identified by chance finds procedures will be developed before construction can start up again.

109. ***Involuntary Resettlement (OP 4.12)***. Because the vast majority of construction will take place underground or on sites that do not require land acquisition or resettlement, the project expects minimal need for resettlement. However, given the complexity of works, and since the physical interventions of components 2 and 3 of the project have not yet been fully determined, a Resettlement Policy Framework has been prepared for the project as a precaution. Among other things, the framework spells out the eligibility criteria, forms of compensation, and dispute resolution mechanisms that will be available to potentially affected people.

110. ***Indigenous People (OP 4.10)***. There are no indigenous people affected by this project as defined under the World Bank's policy. Therefore, no action is required under this policy.

111. ***Projects in Disputed Areas (OP/BP/GP 7.60)***. The project is not being implemented in any area known to be the subject of international dispute as defined under the policy. Therefore, no action is required under the policy.

112. ***Projects on International Waterways (OP/BP/GP 7.50)***. Since both proposed sub-aquatic outfalls (the "Riachuelo" outfall at Dock Sud during APL-1 and the "Berazategui" outfall in APL-2), discharge to the Rio de la Plata, a bi-national waterway shared with Uruguay, OP 7.50 is triggered. Notification to Uruguay is made through the bi-national "*Comision Administradora del Rio de la Plata*" (CARP). On 13 Aug, 2001, Argentina's then Ministry of Infrastructure and Housing submitted a request to CARP to authorize construction of an outfall at Dock Sud. CARP replied on 23 May, 2002, that it had "no objections" to the works, as long as several conditions were met (analysis of possible algae blooms in the diffuser area, establishment of an environmental monitoring & control program, determination of the types of wastewater that would be discharged and definition of pre-treatment standards for industries, etc.). On Sept 11, 2008, AySA (through the Sub-secretariat of Water Resources in the Ministry of Planning, Public Works and Services) presented an updated submission to CARP. Subsequently, on 27 January, 2009, Argentine representatives gave a presentation of the project before CARP and answered questions from Uruguayan representatives. The Argentine government has received a formal written response from CARP that did not indicate any objection to the project.

113. ***Disclosure***. As part of the project preparation process, a series of public disclosures, workshops, and focus group meetings were conducted. Others are being planned during implementation in order to disseminate information on the project, solicit stakeholder concerns, and ensure an open process of debate and discussion about project objectives and goals. A major public consultation event organized by ACUMAR and AySA was held on 10 July, 2008, in

Buenos Aires. The objective of the workshop was to present and discuss the terms of reference for the EIAs proposed for the project. This scoping workshop was attended by 140 participants representing a wide range of public and private stakeholders, including representatives of SArDS, ACUMAR, AySA, the Province of Buenos Aires, municipal governments, national and local environmental NGOs, national government agencies, community organizations, and professional associations. A second public workshop to review the findings and recommendations from the advanced drafts of the EIAs was held on 7 November 2008 in Buenos Aires. The final EIAs document the range of meetings held and the issues raised. EIA documents have been disclosed in-country (through ACUMAR's and AySA's websites) and through the Bank's InfoShop.

Annex 11: Project Preparation and Supervision
ARGENTINA: Matanza-Riachuelo Basin Sustainable Development Project

	Planned	Actual
PCN review	02/04/08	02/04/08
Initial PID to PIC	02/12/08	09/05/08
Initial ISDS to PIC	02/12/08	09/05/08
Appraisal	11/25/08	01/26/09
Negotiations	04/09/09	04/20/09
Board/RVP approval	05/19/09	
Planned date of effectiveness	11/15/09	
Planned date of mid-term review	11/15/12	
Planned closing date	3/31/16	

Key institutions responsible for preparation of the project: SAYDS, ACUMAR, AySA

Bank staff and consultants who worked on the project included:

Name	Title	Unit
Menahem Libhaber	Lead Sanitary Engineer, co-TTL	LCSUW
Franz Drees-Gross	Sector Leader, co-TTL	LCSSD
Horacio Terraza	Sr. Environmental Specialist	LCSSEN
Paula Pini	Sr. Urban Development Specialist	LCSUW
Carter Brandon	Lead Environmental Specialist	EASCS
Glenn Morgan	Lead Environmental Specialist	LCSSEN
Anjali Acharya	Environmental Specialist	LCSSEN
Marcelo H. Acerbi	Environmental Specialist	LCSSEN
Irina Klytchnikova	Young Professional	LCSSEN
Maria Catalina Ramirez	Junior Professional Associate	LCSSEN
Ricardo Schusterman	Social Specialist	LCSSO
Ricardo Lugea	Procurement Specialist	LCSPT
Alejandro Solanot	Financial Management Specialist	LCSFM
Reynaldo Pastor	Sr. Counsel	LEGLA
Yanina Budkin	Sr. Communications Officer	LCC7C
Jimena Garrote	ET Consultant	LEGLA
Maria Emilia Sparks	Team Assistant	LCC7C
Santiago Sandoval	Language Program Assistant	LCSSEN
Juan Carlos Gimenez	Consultant	LCSSEN
Luz Maria Gonzalez	Consultant	LCSSEN
Nils Janson	Consultant	LCSSEN
Fernando Troyano	Technical Expert, Consultant	LCSSEN
James Thompson	Technical Expert, Consultant	LCSSEN
Phillip Roberts	Technical Expert, Consultant	LCSSEN
Elena Strukova	Consultant	LCSSEN
Beatriz Villegas	Consultant	LCSSEN

Name	Title	Unit
Alvaro Orozco	EIA Panel of Experts, Consultant	
Manuel Olivera	EIA Panel of Experts, Consultant	
Andres Carsen	EIA Panel of Experts, Consultant	
Marco Zambrano	EIA Panel of Experts, Consultant	

Bank funds expended to date on project preparation:

1. Bank resources: US\$ 1,084,016.04
2. Trust funds: US\$ 180,437.84
3. Total: US\$ 1,264,453.88

Estimated Approval and Supervision costs:

1. Remaining costs to effectiveness: US\$ 200,000
2. Estimated annual supervision cost: US\$ 200,000

Annex 11A: Analysis of Alternatives

ARGENTINA: Matanza-Riachuelo Sustainable Development Project:

A. Conceptual Framework for the Development of Alternatives

1. The original concept for managing the wastewater of the City of Buenos Aires consisted of intercepting all its wastewater, conveying it to the Berazategui District to the southeast, treating it at that point, and discharging the effluent to the La Plata River through a sub-aquatic outfall. On the basis of this plan, three main large-diameter sewers (*cloacas máximas*) were constructed to convey an average flow of approximately 18 m³/second of wastewater to the existing outfall located in Berazategui. The Berazategui outfall, 3.8 m in diameter and 2.5 km long, discharges the wastewater, without prior treatment, to La Plata River. The plan to construct a fourth main sewer to the Wilde pumping station, a treatment plant in Berazategui, and a longer outfall at this site with capacity to discharge the entire wastewater flow of the Buenos Aires metropolitan area, was never completed.

2. Based on the original concept for managing the metropolitan area's wastewater and the fact that the existing main conveyance of wastewater toward Berazategui forms only part of the system originally planned, it is necessary to seek solutions for the environmental problems of the Matanza-Riachuelo (M-R) Basin that are compatible with the existing system. This means treating and disposing of the wastewater in the Argentina Water and Sanitation (*Agua y Saneamientos, S.A.*, or AySA) concession area (i.e., the entire City of Buenos Aires and most of the metropolitan area) along with that of the entire basin. The raw wastewater from the northern part of the city and nearby districts contributes to and influences the wastewater of the basin even though these parts are not part of it geographically.

3. To comply with its commitment to expand the coverage of sanitation services in its concession area, AySA has developed a five year investment plan (Plan Director). AySA's concession area does not cover the entire M-R River Basin, but covers most of the basin's urbanized area and other areas of the city north of the basin. The AySA Management Plan modified the original concept of conveying all the wastewater of the concession area to Berazategui, but it retains the principle of intercepting the rest of the wastewater in the concession area and providing adequate disposal for the entire wastewater flow.

4. The AySA Management Plan does not provide for the evacuation of all wastewater from the basin to La Plata River outfalls. Instead, the wastewater of part of the middle and upper basin located in the concession area would be diverted to various inland wastewater treatment plants. Some of these plants already exist and need to be expanded (Sudoeste and Jagüel), while new plants will be built at Laferrere and Fiorito. According to AySA's Management Plan, their effluents will discharge to the Matanza-Riachuelo River.

5. In addition to domestic wastewater, the basin has a large amount of liquid industrial waste, most of which is currently not treated. Part of the waste is discharged directly to the M-R River and its tributaries, and part into AySA's existing sewerage network. Because the AySA Management Plan does not include a comprehensive solution for the basin's sanitation problem, a broader program is required.

6. The magnitude of the difference between the AySA Management Plan and the basin's needs can be illustrated by the fact that the estimated future flow of wastewater in the entire M-R River Basin, which according to the AySA Management Plan will not reach the outfall systems of La Plata River, is approximately 13.5 m³/second. All previously mentioned flows already include the contributions from industrial liquid wastes.

7. Because the goal of the Matanza-Riachuelo Basin Sustainable Development Project is to achieve the comprehensive sustainable development of the entire Matanza-Riachuelo Basin, not only to expand sanitation in part of the basin, alternatives were sought to the solution proposed in the AySA Management Plan. The alternatives that were analyzed cover the entire basin area and take into account the impact of industrial waste as well as domestic wastewater.

B. Alternatives Studied

8. **Sanitation alternative.** To meet the requirements of improving the quality of the watercourses in the M-R River Basin and achieving sustainable development in the basin, a series of viable project alternatives were developed. Each aimed to avoid environmental damage such as contamination of the potable water intakes in La Plata River (the Palermo intake for the San Martín plant and the Bernal intake for the Belgrano plant) and support the environmental recovery of the basin without causing other negative impacts on the receiving water bodies. The following alternatives were developed for analysis:

- (i) Decentralized treatment by various plants within the basin with different levels of treatment, discharging their effluents to the M-R River, its tributaries, and La Plata River.
- (ii) Centralized treatment with a single treatment plant in Berazategui and a sub-aquatic outfall at this site discharging the effluent to La Plata River. This was the original alternative adopted for the sanitation of the City of Buenos Aires, the construction of which was initiated but never completed.
- (iii) Centralized treatment with various points of discharge into La Plata River through sub-aquatic outfalls. Although a solution with several discharge points is theoretically feasible, only two treatment sites—one in Berazategui and one in Dock Sud, have been approved, and it would be difficult to find additional treatment sites for such large-scale plants on the banks of La Plata River. An analysis of the coastal land strip within the City of Buenos Aires and between the mouth of the M-R River and current Berazategui outfall, taking into account the location of the Bernal water intake for the Belgrano potable water purification plant, the recreational area on the Quilmes coast, and the State Company for the Ecological Coordination of the Metropolitan Area (*Coordinación Ecológica Área Metropolitana Sociedad del Estado*, or CEAMSE) sanitary landfill sector in Villa Domínico, showed that there are no other sites for a wastewater treatment plant with areas large enough for treatment installations.
- (iv) Alternatives based on different levels of treatment before discharge to La Plata River. These alternatives involve determining the lengths of the outfalls as a function of the treatment levels, as higher levels of treatment may require shorter outfalls. The outfall lengths were determined using a simulation model for the water quality in La Plata River. The development of these alternatives took into account their impact on the uses

of the river water (e.g., source of water supply for human consumption with conventional treatment, recreational uses with and without direct contact, preservation of aquatic life, and impact on navigation), as well as the environmental and social impact of the operation of different treatment systems (e.g., odor control, amount of sludge generated, truck traffic for sludge removal, and necessary space for the plants).

- (v) Secondary treatment followed by chlorination of the effluent and discharge of the effluent to La Plata River, coupled with using a short outfall.

9. **Main conveyance system alternatives.** The following alternatives were defined for the main wastewater collection and conveyance system in conjunction with the treatment and disposal alternatives in order to ensure the collection of the wastewater of the entire M-R River Basin and AySA concession area.

- (i) A separate left bank collector and right bank collector with capacity to convey all wastewater from the middle and upper basin of the M-R River to the Riachuelo pretreatment plant;
- (ii) A single left bank collector with capacity to also convey all wastewater from the middle and upper basin of the M-R River toward the Riachuelo preliminary treatment plant.

10. **Industrial pollution abatement alternatives.** The following alternatives were considered for reducing industrial pollution:

- (i) High-level treatment of all industrial wastes and discharge of all effluents except those containing toxic matter into the M-R River;
- (ii) Lower-level treatment of all industrial wastes and discharge of all effluents except those containing toxic matter into the municipal sewerage network for further processing with domestic wastewater.

11. **Comparison of alternatives.** The information developed for each of the alternatives was compared using technical, economic, environmental, and social criteria. As a result of this comparison, the most favorable alternative was selected and its stages of execution determined.

12. On the basis of water sampling at various points along the M-R River and its tributaries and along the coastal strip of La Plata River and an analysis of the water quality parameters of the samples drawn, mathematical models were developed to simulate the water quality of these rivers using the water quality data to calibrate the models. Mathematical models simulate the processes of transport, dilution, and transformation of pollutants in a receiving water body system. From a practical standpoint, the mathematical model has been used as a virtual laboratory that makes it possible to evaluate the receiving water body system's response (distributions of the concentrations of pollutants in the water body) to discharges of pollutants.

13. The mathematical models developed were an important instrument for technical analysis of the alternatives studied, i.e., as assessing the current state of contamination of the water of the M-R and La Plata Rivers, estimating the levels of contamination that will result from applying each alternative, and assessing the project's environmental impact after the favorable alternative is implemented.

14. The most suitable alternative found was Alternative 3: distributing the total collected wastewater flow to sub-aquatic outfalls in Berazategui and in Dock Sud near the confluence of the M-R River with La Plata River. This alternative offers a combination of lower cost, technical advantages, simple operation and maintenance, more positive environmental impacts, fewer negative impacts, and minimum negative social impacts.

15. The other proposed solutions were rejected for the following reasons:

- (i) *Alternative 1* (decentralized treatment with a variety of treatment plants with different levels of treatment that discharge their treated effluents into the M-R River, its tributaries, and La Plata River) was rejected for technical reasons. The modeling results showed that the M-R River has a low capacity to receive large amounts of organic loads. That means that the river would remain anoxic even if treated effluent were discharged into it. During the early stage of APL-1, AySA will continue to study technical alternatives that might involve retaining some inland treatment plants in the basin while offsetting their discharges with additional oxygenation of the M-R River. Any design alternative that might eventually be presented, however, needs to be consistent with the goal of achieving at least minimal levels of dissolved oxygen even in the M-R River's lower sections.
- (ii) *Alternative 2* (centralized treatment with a single treatment plant and a single outfall in Berazategui) has more environmental and economic disadvantages than the alternative of two outfalls and is not acceptable for legal, social, and political reasons, as it discharges all the wastewater of the Buenos Aires metropolitan area into La Plata River in an area located in the jurisdiction of a downstream municipality.
- (iii) *Alternative 4* (in fact, a series of alternatives of different levels of treatment prior to effluent discharge into La Plata River) involves establishing the lengths of the outfalls as a function of the level of treatment using the water quality simulation model developed for La Plata River. Simulation results demonstrated that treatment to a level higher than preliminary does not provide an advantage in terms of water quality. This finding is consistent with the recommendation of the World Health Organization (WHO) that treatment beyond preliminary treatment prior to discharge does not reduce the public health risk of human exposure to wastewater discharged to the sea through effective outfalls.⁷⁴ That is, the risk of preliminary treatment followed by an effective outfall is identical to the risk of secondary treatment followed by an effective outfall. The WHO report refers to risk related to human exposure through bathing, which is not exactly the risk in the case of discharge to La Plata River. However, because pathogenic organisms are the main risk from effluent discharge, which is also the case in the La Plata River system, the exposure risk referred to in the WHO report is a good proxy for the risk in La Plata River. Higher than preliminary treatment levels also imply enormous additional investment costs, as well as high operation and maintenance costs, which are not affordable at this time for Buenos Aires given the large population to be served and large wastewater flows to be treated. Plants with higher treatment levels also generate very

⁷⁴ WHO (2003): Guidelines for Safe Recreational Water Environments. Volume 12, Coastal and Fresh Waters, World Health Organization, Geneva, Switzerland, p. 80. Available at http://www.who.int/water_sanitation_health/bathing/srwe1/en/.

large quantities of biological sludge. Thus, the combination of preliminary treatment plants followed by long and effective outfalls was the preferred solution for the project because it is scientifically solid and economically favorable. However, sufficient area was reserved at the sites for the Riachuelo and Berazategui preliminary treatment plants to expand the treatment plants and increase the level of treatment if required.

- (iv) *Alternative 5* (secondary treatment followed by chlorination of the effluent and its discharge to La Plata River through a short outfall) is costlier than the one selected and environmentally very risky because of the need to handle enormous amounts of chlorine in close proximity to the city and the negative impact of the chlorine and its byproducts on aquatic life in the river. In this alternative, based on a short outfall, the effluent can be discharged into the river only if it is disinfected (i.e., chlorinated). Otherwise it will cause heavy pathogenic contamination along the riverbanks and threaten the water supply intakes and all other uses of La Plata River. Chlorination, however, is not reliable because the continuous provision of such large amounts of chlorine cannot be assured. Other disinfection methods are also impractical because of the huge effluent flow generated in Buenos Aires. Ozonation would be very expensive and is not practiced on such a large scale. Ultraviolet disinfection is also not practiced on such a large scale and would probably not be successful. The use of nanofiltration membranes to remove pathogens from the effluent would require increased investment for pretreatment and is an untested technology that is not yet recommended for such a large-scale project.

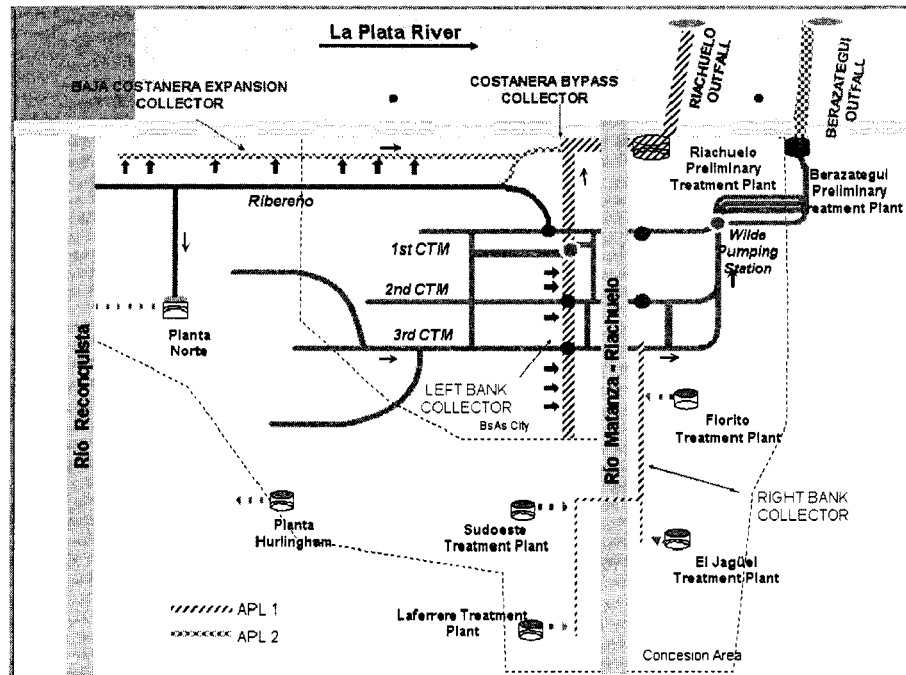
16. Comparison of the alternatives for industrial pollution abatement resulted in rejection of Alternative 1 (high-level treatment of all industrial wastes and discharge of the effluents to the M-R River) because mathematical modeling showed that the discharge of industrial effluents to the M-R River would render the river anaerobic even after a high level of treatment. The only way to induce aerobic conditions in the river and recover its water quality is the adoption of Alternative 2 (a lower level of treatment of all industrial wastes and discharge of all effluents except those containing toxic matter into the municipal sewerage network for further processing with the domestic wastewater). This alternative was coupled with the alternative selected for the sanitation component in the sense that the simulation model confirmed that the organic matter loads, including those of industrial origin, that would be discharged to La Plata River through the two outfalls, can be absorbed by the river and will not generate water quality problems. Similarly, the conveyance and treatment system were dimensioned to include all the industrial waste flows generated in the M-R River Basin.

17. The main wastewater collection system and the conveyance and distribution of the wastewater between the two preliminary treatment plants are an important part of the project. The following alternatives were studied for this system:

- (i) The first consists of two main collectors along the banks of M-R River, one on each side (figure 12.1). The Left Bank collector is short and has the capacity to collect only the AySA concession area wastewater (domestic and industrial). The Left Bank collector is 12.5 km long and 0.8- 2.9 m in diameter, followed by the 8.5 km Baja Costanera collector with a 4.5 m diameter, which ends at the Riachuelo preliminary treatment plant site. The Right Bank collector is approximately 37 km long. The Right Bank collector starts at the upper part of the basin with a small diameter of 2.3 m and increases in diameter as it moves along the middle and lower parts of the basin and

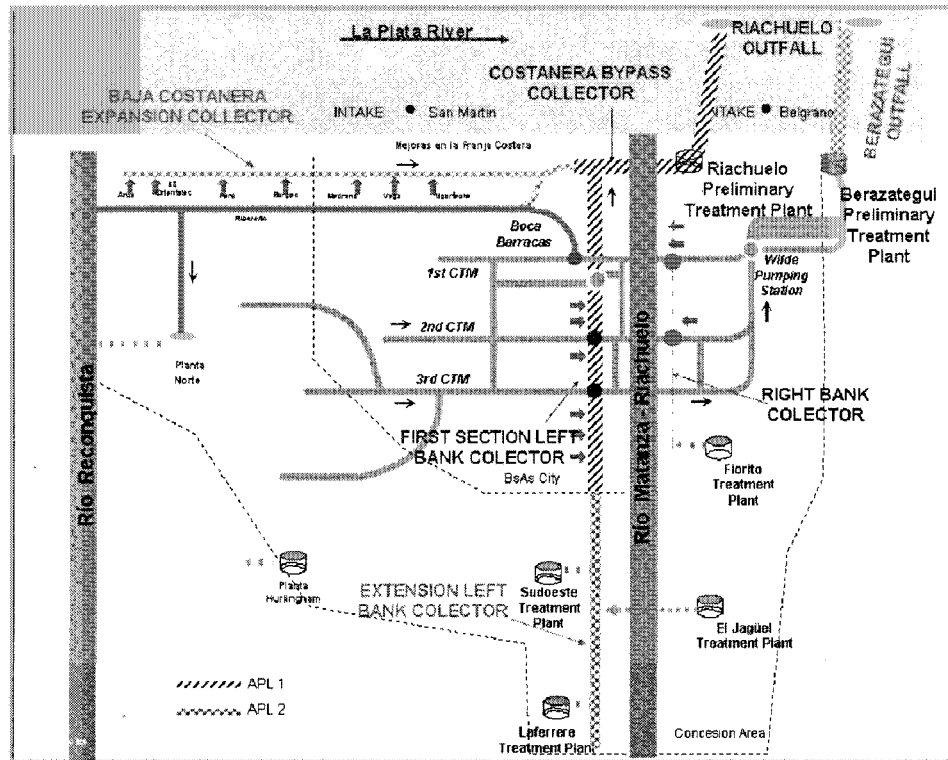
collects more and more wastewater. This collector conveys the wastewater generated in the middle and upper parts of the M-R River Basin, both domestic and industrial.

**Figure 12.1. Scheme of conveyance alternative 1
(two main collectors along the banks of the M-R River)**



- (ii) The second alternative consists of only the Left Bank collector, which would also be used to convey the middle and upper basin wastewater flows on top of the flows generated in the AySA concession area (figure 12.2). This collector would increase in diameter, as mentioned above and be extended by approximately 27 km to reach the upper part of the basin. The pipe diameter along the 27 km section would vary between 2.2 m in the upper basin and 3.5 m in the middle basin. In the section of the lower basin, the Left Bank collector is 12.5 km long and 0.8-2.9 m in diameter, followed by the 8.5 km Baja Costanera collector 4.5 m in diameter, which ends at the Riachuelo preliminary treatment plant site. A preliminary analysis of these two alternatives showed that the costs were similar, with similar environmental and social benefits. However, alternative 1 was selected, as it provided more flexibility to the GoA in terms of the design of future investments to expand coverage in the upper and middle basin.

**Fig. 12.2. Scheme of conveyance alternative 2
(one main collector along the left bank of the M-R River)**



18. **Selected industrial pollution abatement alternative.** The selected alternative is therefore based on intercepting most of the domestic and industrial wastewater of the AySA concession area (except that diverted to the North and Hurlingham treatment plants, which discharge their effluents to the Reconquista River) and the municipalities in the upper Matanza-Riachuelo Basin and diverting this entire wastewater flow to La Plata River through two new sub-aquatic outfalls. The Riachuelo outfall will be located on the left bank of Sarandí Creek. It will be 3.8 m in internal diameter and 11.5 km long, of which the final 2.3 km are the diffuser zone. The new Berazategui outfall will have an internal diameter of 3.8 m and a length of 7.5 km, of which the final 3.5 km are the diffuser zone. The maximum design flows of each outfall are 25 m³/second at Riachuelo and 33.5 m³/second at Berazategui.

19. Prior to its discharge through each outlet into La Plata River, the wastewater will be subjected to preliminary treatment in two plants, one in Dock Sud and the other in Berazategui near the inlet to the existing 2.5 km long outfall. The preliminary treatment plants consist of coarse screens to retain heavy material, followed by milliscreens with 6 mm openings to remove finer material (the milliscreens opening to the Riachuelo preliminary treatment plant might be smaller than 6 mm), followed by aerated grit chambers to remove sand and floating material such as oil and grease.

20. **Selected sanitation alternative.** The selected sanitation alternative is also coupled with a conceptual alternative selected for managing the industrial wastewater generated in the basin. The selected industrial pollution abatement alternative consists of providing a relatively low

level of treatment (meeting quality standards in effect) of all industrial wastes except those containing toxic mater and discharging the effluents to the municipal sewerage network for further processing with the domestic wastewater. Industries that discharge toxic wastes will have to provide full treatment at the source before discharging the effluents to the sewerage network.

21. ***Selected conveyance alternative.*** The selected main conveyance alternative consists of a separate left bank collector and right bank collector with a combined capacity to convey all wastewater from the middle and upper basin of the M-R River to the two pretreatment plants. It includes the (i) Left Bank collector, with a total length of approximately 12 km, in the lower part of the basin (the diameter varies between 0.8 and 2.9 m), and the Right Bank collector, with a total length of about 37 km with an internal diameter ranging from 2.3 m to 3.5 m. Additional complementary works will include the (i) the Baja Costanera bypass collector with a length of 5 km and a diameter of 4.5 m; (ii) the Baja Costanera expansion collector (Radio Antiguo section); and (iii) the Baja Costanera expansion collector (Capital Federal–Vicente Lopez section).

22. The selected alternative allows connecting the entire population and all industries in the Matanza-Riachuelo Basin to the sewerage network, thereby eliminating most of the discharges to the M-R River. This will enable recovery of the water quality of this river and consequently contribute to the sustainable development of the M-R River Basin.

23. As currently designed, APL-1 includes the Right Bank Collector (CMD), a key piece of sewerage infrastructure that would receive and transport wastewater collected in the coverage area of the Southwest, Jagüel, Fiorito and Laferrere treatment plants as well as wastewater generated in the municipalities of the upper M-R River Basin located outside the AySA concession area. Initial water quality modeling during project preparation has identified the withdrawal of these discharges to the M-R River as a key enabler of improved water quality. Given the size of the proposed investments (estimated at about US\$275 million (without contingencies) based on preliminary designs), AySA will conduct a pre-feasibility study of alternatives to the CMD. AySA is considering, for example, whether side stream elevated pool aeration (SEPA) stations, being used in Chicago, Illinois, could be used -- possibly in conjunction with higher levels of treatment at existing and future treatment plants – to meet water quality objectives established by ACUMAR. Water quality modeling of various alternatives is already underway in conjunction with ACUMAR. Should a better alternative in terms of economic, environment and social aspects emerge, the latter would need to be submitted to SAyDS and ACUMAR for approval before it would be presented to the Bank for review. A covenant has been added to the Loan Agreement requiring that the study of alternatives to the CMD be completed by March 31, 2010. Any alternative to construction of the CMD would need to be approved by the Bank and may need to be reflected in an amendment to the Loan Agreement. Given this uncertainty, the Bank and the Borrower initially included the CMD under APL-2. Subsequently, however, the Bank and the Borrower agreed to advance the CMD to APL-1 to provide the Government with an instrument for accelerated basin clean-up in the short and medium-term. Including the CMD in APL-1 also has the important advantage of concentrating all major trunk infrastructure works needed to remove contaminant flows from the M-R River Basin in a single phase over the next 6 years. On the other hand, construction of the CMD during APL-1 increases the size of the project's first phase to nearly US\$1.5 billion, meaning that the Bank's US\$840 million loan would, on average, finance only about 56% of APL-1 activities.

24. **Panel of technical experts.** The following high-level international technical experts were hired during project preparation to review the studies undertaken and the selected alternative:

- (i) **Prof. Philip Roberts**, Professor of Environmental Engineering at Georgia Tech. University and Beatriz Villegas, an expert in numerical modeling, reviewed the modeling work undertaken by AySA and the conceptual validity of the selected alternative. They were positive about the alternative selected but recommended an extensive campaign of hydro-meteorological data generation for the design of the outfalls and for more detailed modeling. AySA accepted the recommendations and hired consultants to carry them out.
- (ii) **Mr. James C. Thomson**, a consultant on tunneling and author of many books on the subject, reviewed AySA's plans and conducted field visits along the routes of the planned tunnels. He confirmed that the concepts in the selected alternative are viable and provided recommendations. He also conducted a cost estimate of the proposed tunneling works.
- (iii) **Mr. Fernando Troyano**, a top-level engineering consultant thoroughly familiar with the water and sanitation system of Buenos Aires, participated as a consultant and lecturer in the first project workshop conducted in Mar del Plata in December 2007. He reviewed the selected alternative and confirmed its viability. He also provided recommendations related to engineering and institutional aspects of the project.

In addition, the following four experts provided technical guidance during preparation of the EIA for the project:

- (iv) **Marco Zambrano Chavez**, an Ecuadoran national with extensive experience in the application of Environmental Impact Assessments for large infrastructure projects in Latin America. He holds a Bachelors degree in Civil Engineering from *Pontificia Universidad Católica del Ecuador (PUCE-Q)* and a Masters Degree in Natural Resource Management from the *Instituto Centro Americano de Administración de Empresas (INCAE)* and has more than 20 years of international experience throughout Latin America. Mr. Zambrano has been a consultant to the World Bank, the GEF, the Inter-American Development Bank, CAF and the GTZ among other international organizations.
- (v) **Andres Carsen**, an Argentine ecologist who has extensive knowledge and expertise in the freshwater and estuarine ecology of the La Plata River. Mr. Carsen holds a Bachelors Degree in Ecology from the *Universidad CAECE* in Argentina and a M.Sc. Degree in Biology from Dalhousie University in Halifax, Canada. He has more than 20 years of professional experience in applied ecology. He has recently held the position of General Director of the FREPLATA program (*Environmental Protection of the La Plata River and its Maritime Front*) focusing on the management of the ecological priorities of the La Plata River. Mr. Carsen has held a variety of technical positions over the span of his career including consulting tasks on fresh water ecology and environmental impact analysis.

- (vi) **Alvaro Orozco**, a Colombian Civil Engineer who has extensive experience with the applications of EIA within the private sector, academia, and in private consulting. He holds a Bachelors Degree in Civil Engineering from the *Facultad de Minas* of the National University in Medellín and a Masters Degree in Sanitary Engineering from Pennsylvania State University in the US. Mr. Orozco has extensive international experience in advising in the design, management and operations of wastewater treatment facilities. He has taught engineering classes as a Distinguished Lecturer at the University of the Andes and has held executive positions with numerous industry association groups.
- (vii) **Manuel Olivera**, a Colombian environmental planner with more than 25 years of experience in sustainable development, clean production, environmental policy, environmental impact assessment and planning. Mr. Olivera holds a Bachelors degree in Biology from the *Universidad de los Andes* and a M.Sc. in Natural Resources Management from the University of E. Anglia in the U.K. He has worked as a technical consultant to international development agencies such as IDB and the World Bank. UNDP, and FUNDES and has served as a recognized national expert on numerous projects in his home country of Colombia. Since 1999, Mr. Olivera has been the Director for Bogota of the Climate Change Initiative of the Clinton Foundation. He is also the director of the Hybrid Test Bus System Program for Latin America.

Appendix to Annex 11A
Wastewater Treatment Prior to Disposal in the Sea through an Effective Submarine Outfall

1. Approximately 50 percent of the world's population lives in or near coastal areas. The oceans and seas are the bodies of water most likely to receive the sewage and treated effluents generated by this population. Clean disposal of sewage and its treatment prior to discharge are therefore issues of major importance. For developing countries, where resources for wastewater treatment and disposal are limited, low-cost solutions are critical. In October 2003, WHO published guidelines for recreational water quality protection. Table 12.1 presents the WHO findings on the relative health risks to human health of exposure to effluents discharged from submarine outfalls, classified according to various schemes combining sewage pretreatment and submarine outfall length.

Table 1. Relative risk to human health of exposure to sewage through outfalls

Treatment	Discharge type		
	Directly on beach	Short outfall	Effective outfall
None	Very high	High	NA
Preliminary	Very high	High	Low
Primary (including septic tanks)	Very high	High	Low
Secondary	High	High	Low
Secondary plus disinfection	—	—	—
Tertiary	Moderate	Moderate	Very low
Tertiary plus disinfection	—	—	—
Lagoons	High	High	Low

Source: WHO. 2003. Guidelines for Safe Recreational Water Environments. Volume 12, Coastal and Fresh Waters, Geneva, p. 80. Available at http://www.who.int/water_sanitation_health/bathing/srwe1/en/.

2. The methods to discharge sewage or effluent into oceans can be classified into three principal types: (i) discharge directly onto the beach, (ii) “short” outfalls, where sewage or effluent is likely to contaminate recreational waters, and (iii) “effective” outfalls, designed so that the sewage or effluent is efficiently diluted and dispersed to ensure that it does not pollute recreational water areas. While the terms “short” and “long” are often used, outfall length is generally less important than proper location and effective diffusion. An effective outfall is assumed to be one that is properly designed, with sufficient length and diffuser discharge depth to ensure that the sewage does not reach recreational areas.

3. According to WHO, if an effective outfall is provided to discharge effluent to the sea, the level of pretreatment of domestic sewage has practically no bearing on the risk to human health from the discharged effluent. The risk to human health of secondary effluent discharged through

an effective outfall is classified as low. The risk of primary effluent discharged through an effective outfall, and even the risk of preliminary treated effluent discharged through an effective outfall, is classified as low. The WHO findings lead to a conclusion that if an effective outfall is provided, preliminary treatment is sufficient prior to discharge of domestic sewage to the sea. Conversely, if a short (ineffective) outfall is provided, even secondary treatment is not sufficient to reduce the high risk to human health of effluent discharged through that type of outfall. These findings are important for developing countries as they imply that only preliminary treatment may be required prior to discharge of effluent into the sea through an effective outfall. Preliminary installations as pretreatment prior to discharge of effluents into the sea can save a lot of money in developing countries, whose financial resources are usually scarce. The investment costs, as well as the operation and maintenance costs of preliminary treatment, are approximately one-tenth of those for secondary treatment, while the use of preliminary treatment yields results comparable to those of secondary treatment.

4. It can also be concluded from the WHO findings that stabilization lagoon effluents discharged into the sea directly on the beach or through a short outfall constitute a high risk to human health, and that this effluent needs to be discharged into the sea through an effective outfall to reduce this risk. This is an important conclusion because developing countries commonly treat wastewater in stabilization lagoons, and it is wrongly believed that effluents of such lagoons pose no risk to human health and can be discharged directly on the beach.

5. The conclusions above refer to typical municipal sewage that does not contain toxic wastes originating from industrial wastewater.

6. For coastal cities in developing countries, the strategy of domestic wastewater disposal by preliminary treatment followed by an effective submarine outfall is an affordable, effective, and reliable solution. It is simple to operate and free of negative health and environmental impacts. Many outfalls of this type are successfully functioning and have a proven track record in many coastal cities all over the world. This strategy should be coupled with a monitoring program beginning before and continuing after construction of the treatment installations and outfall to verify the performance of the outfall and help determine whether higher treatment levels are necessary. This is a logical strategy that would prevent unnecessary investments in costly treatment installations.

7. A submarine outfall system, which includes the outfall itself and the near field zone, should in fact be considered a treatment plant. This treatment plant provides a high level of treatment, higher than what any conventional land-based plant can reach. Land-based plants can reach, in extreme cases, removal levels of biochemical oxygen demand (BOD) and total suspended solids (TSS) of up to 95 percent. If the effluent is not disinfected, they remove 50–80 percent of pathogenic organisms, a very low removal level, leaving the effluent with practically the same level of risk to human health as that of raw sewage. On the other hand, a well-designed outfall system removes over 99 percent of all contaminants, including pathogenic organisms. Removal of 99 percent of pathogenic organisms might not be sufficient, but the mentioned value refers to physical dilution. Taking into account the biological decay of pathogenic organisms in the marine environment, a well-designed outfall can ensure their overall removal to levels that pose no risk to human health. Tables 2 and 3 show the average operating results of 5 years of measurements near the outfall discharge points of two submarine outfall systems in the

Concepción region of Chile, each consisting of preliminary treatment followed by an effective outfall.

**Table 2. Marine water quality in the vicinity of the Tome Outfall, Concepción, Chile
(average values of 5 years of monitoring)**

Parameter	Effluent discharged into outfall	Maximum value according to local standards	Value at discharge point	Value 100m from the discharge point	Value in natural marine water in the region (background)
PH	7.3	5.5–9.0	7.600	7.70	7.800
Temperature (°C)	17.8	-	13.100	14.10	14.200
Oil and grease (mg/l)	46.3	150.0	4.600	9.20	2.400
Total suspended solid (TSS) (mg/l)	206.0	300.0	7.600	1.80	3.000
Biochemical oxygen demand (BOD) (mg/l)	348.0	-	2.300	2.20	2.400
Total organic carbon (TOC) (mg/l)	203.0	-	2.100	3.40	2.000
Dissolved oxygen (DO) (mg/l)	0.5	-	4.800	7.50	7.200
Detergents (mg/l)	9.2	15.0	0.070	0.06	0.050
Nitrogen (mg/l)	72.0	-	0.520	0.41	0.570
Nitrites (mg/l)	0.01	-	0.020	0.02	0.020
Nitrates (mg/l)	0.17	-	0.250	0.28	0.180
Phosphorus (total, mg/l)	12.8	-	0.200	0.21	0.200
Phosphate (mg/l)	38.0	-	1.000	1,00	1.000
Sulfur (mg/l)	1.0	5	0.200	0,15	0.240
Phenol	< 0.002	1	< 0.002	<0,002	< 0.002
Fecal coliform (most probable number, or MPN/100 ml)	3.5×10^7	-	2,546.000	30	11.000
Total coliform (MPN/100 ml)	5×10^7	-	3,450.000	41	18.000

Source: Leppe, A. Z., and L. B. Padilla. *Emisarios Submarinos de Penco y Tome, 5 Años de Vigilancia y Una Evaluación Global de Sus Efectos Ambientales*, XIII Congreso Chileno de Ingeniería Sanitaria y Ambiental, AIDIS Chile, Antofagasta, October, 1999.

**Table 3. Marine water quality in the vicinity of the Penco Outfall, Concepción, Chile
(average values of 5 years of monitoring)**

Parameter	Effluent discharged into outfall	Maximum value according to local standards	Value at discharge point	Value 100 m from the discharge point	Value in natural marine water at the region (background)
PH	7.400	5.5–9.0	7.600	7.600	8.000
Temperature (° C)	18.400	-	11.500	13.200	13.80
Oil and grease (mg/l)	48.800	150.0	4.000	5.600	3.200
Total suspended solid (TSS) (mg/l)	216.000	300.0	3.500	3.700	3.800
Biochemical oxygen demand (BOD) (mg/l)	236.000	-	2.500	2.800	2.500
Total organic carbon (TOC) (mg/l)	142.000	-	2.500	3.500	1.400
Dissolved oxygen (DO) (mg/l)	1.100	-	5.100	7.900	7.900
Detergents (mg/l)	15.400	15.0	0.050	0.070	0.060
Nitrogen (mg/l)	49.300	-	0.500	0.360	0.400
Nitrites (mg/l)	0.010	-	0.020	0.010	0.010
Nitrates (mg/l)	0.150	-	0.180	0.210	0.160
Phosphorus (total, mg/l)	12.900	-	0.200	0.250	0.200
Phosphate (mg/l)	39.600	-	1.000	1.000	1.000
Sulphur (mg/l)	0.800	5.0	0.180	0.250	0.220
Phenol	< 0.002	1.0	< 0.002	< 0.002	< 0.002
Fecal coliform (most probable number, or MPN/100 ml)	1.1×10^8	-	5,420.000	172.000	3.000
Total coliform (MPN/100 ml)	1.4×10^8	-	7,317.000	223.000	5.000

Source: Leppe, A. Z., and L. B. Padilla. *Emisarios Submarinos de Penco y Tome, 5 Años de Vigilancia y una Evaluación Global de Sus Efectos Ambientales*, XIII Congreso Chileno de Ingeniería Sanitaria y Ambiental, AIDIS Chile, Antofagasta, October., 1999.

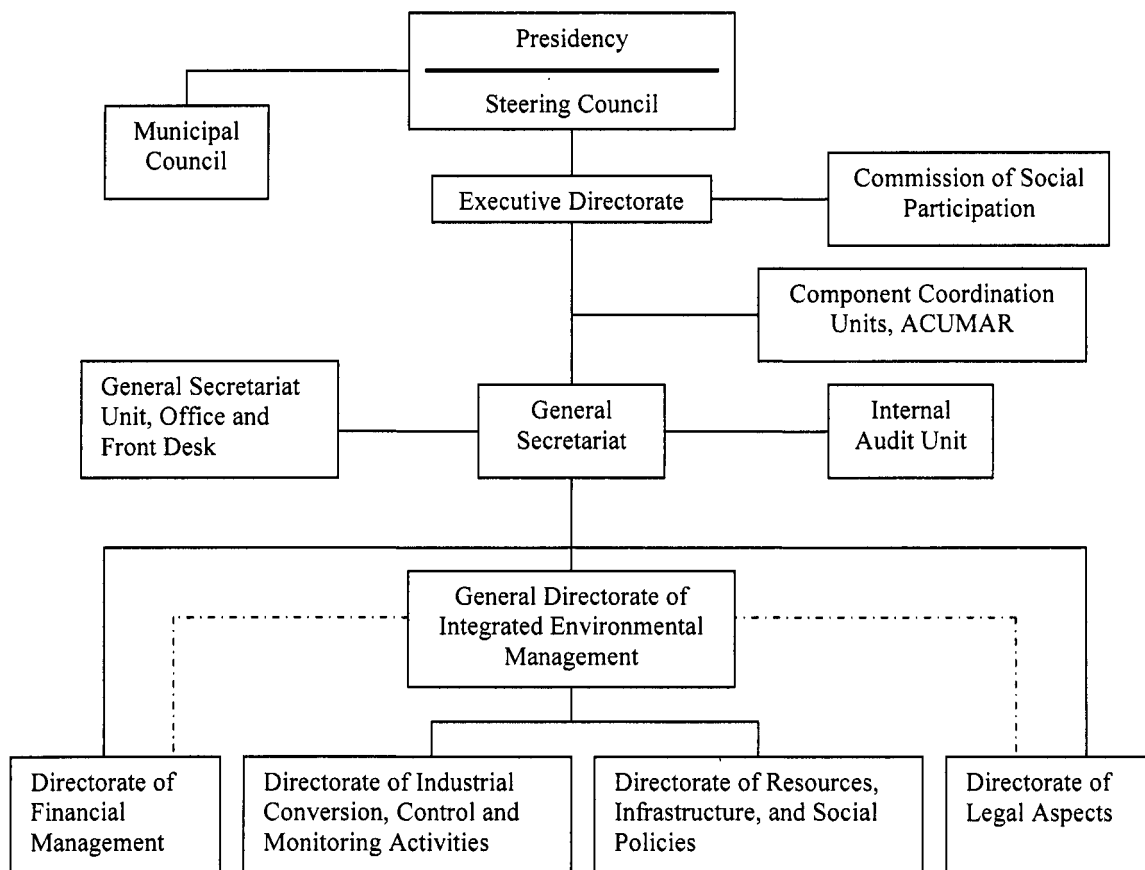
Annex 11B: Background on ACUMAR and AySA
ARGENTINA: Matanza-Riachuelo Basin Sustainable Development Project

A. ACUMAR

1. Law 26.168 was approved on November 15, 2006, creating ACUMAR as an interjurisdictional public agency. ACUMAR's presidency is exercised by the Secretary of the Environment and Sustainable Development (*Secretaría de Ambiente y Desarrollo Sustentable*, or SAYDS) under the Office of the Chief of the Cabinet of Ministers. ACUMAR is a unique institution for the environmental management of the basin.
2. Some of the triggers for the creation of the ACUMAR are listed below.
 - (i) Ombudsman Report (2003), (Office of the National Auditor General or *Auditoría General de la Nación*) Report (2005); and
 - (ii) Supreme Court lawsuit against national, provincial, and city of Buenos Aires governments and 44 industries, the *Causa Mendoza* (2004).
3. While there are more than 200 regulations issued by different jurisdictions that co-exist side-by-side in the basin these do not add up to an integrated management focus on water resources, making it impossible in the past to achieve the objective of reversing the basin's high levels of pollution. ACUMAR's legal attributions are expected to contribute to a harmonized set of regulations and to integrated basin management that takes into account the M-R river's very limited dilution capacity for organic pollutants.
4. ACUMAR's legal attributions include:
 - Regulatory functions: harmonize regulatory frameworks, parameters, etc.
 - Control functions: set standards, monitor, audit, inspect, and take actions against those outside the norms;
 - Promotion: implement plans.
5. ACUMAR's responsibilities may be grouped as follows: (i) Coordination: based on the principle of subsidiarity and pre-eminence over local authorities; (ii) Execution: Authority to adopt any measure necessary or relevant for the basin's clean-up and execution of the budget; and (iii) Regulation and control: unify schemes dealing with liquid effluents and gas emissions; conduct environmental land use planning; create fees for services, issue organizational bylaws (for ACUMAR); establish a special administrative procedure; implement preventive measures; intervene by means of audits, monitoring, environmental impact assessments, and sanctions; issue warnings, writs, and inspections; and order closures, seizures, and cessation of activities.
6. According to Article 2, ACUMAR's Board of Directors is composed of eight members: the Federal Secretary of the Environment and Sustainable Development who acts as president, three other representatives of the National Government, two representatives from the Province of Buenos Aires, and two from the City of Buenos Aires. The Authority issues its own organizational and operational bylaws. Article 3 creates a Municipal Council (composed of one representative of each municipality in the basin), whose objective is to cooperate with, assist, and

advise the entity. Article 4 creates a Social Participation Commission with the objective of communicating civil society organizations' comments and proposals to ACUMAR's management. This Commission has a consultative function, focused on promoting consultation and exchange with civil society organizations. Finally, Article 5 states that the Matanza-Riachuelo Basin Authority has functions including those of regulation, control, and promotion regarding industrial activities, provision of public services, and any other activities with an environmental impact on the basin. It may intervene administratively in the areas of prevention, sanitation, remediation, and the rational use of natural resources. ACUMAR's current internal structure is illustrated in figure 13.1.

Figure 13.1. ACUMAR's organizational structure (new structure)



7. To date, ACUMAR has issued the following eight resolutions:
 - Resolution N° 1: Permissible Limits for the Discharge of Liquid Effluents (8/31/2007);
 - Resolution N° 2: Parameters for the Measurement of Air Quality (8/31/2007);
 - Resolution N° 3: Operational Regulation of the Basin's Social Participation Commission (8/31/2007);
 - Resolution N° 4: Operational Regulation on Access to Information and Receipt of Complaints (8/31/2007);
 - Resolution N° 5: Internal Bylaws of the Matanza-Riachuelo Basin Authority (9/12/2007);

- Resolution N° 6: Appointment of the General Secretary of the Matanza-Riachuelo Basin Authority (10/25/2007);
- Resolution N° 7: Organizational structure of the General Secretariat of the Matanza Basin Authority (11/22/2007);
- Resolution N° 8: Approval of the Executive Summary of the Integrated Environmental Clean-up Plan (PISA) of the Matanza-Riachuelo Basin (11/22/2007).

8. Of the eight resolutions, the first, second and eighth are directly related to the Integrated Clean-up Plan for the Matanza-Riachuelo Basin (*Plan Integral de Saneamiento de la Cuenca Matanza-Riachuelo*, or PISA). Resolution No. 8 contains significant definitions regarding the objectives of the environmental restoration plan.

9. Resolution No. 1, empowers ACUMAR to establish basin-specific regulations which replace existing norms for the control of liquid effluents. This implies that ACUMAR has unified regulations within the basin. In contrast to previous regulations that set specific standards, ACUMAR's liquid effluent norms consider potential uses and quality objectives for the receiving water bodies.

10. This is in line with the objectives of PISA, one of whose key targets is to recover and preserve the quality of surface water bodies in the Matanza-Riachuelo Basin and for La Plata River, in terms of uses and water quality objectives to be proposed, agreed, and set for the various sections and zones of the basin and river in the short, medium, and long term.

11. The strategy contained in PISA consists of (i) proposing and reaching agreement on the uses and quality objectives for receiving bodies in the short, medium, and long term; (ii) evaluating their implications and correlation with AySA's next five year investment plan (*Plan Director*) for raw sewage of domestic and industrial origin through the application of water quality models; and (iii) verifying the baseline of water bodies and implications of sanitation efforts and works through monitoring programs carried out in collaboration with of the National Water Institute (*Instituto Nacional de Agua*, or INA), Naval Hydrology Service (*Servicio de Hidrografía Naval*, or SHN), and Dr. Raúl A. Ringuelet Institute of Limnology (*Instituto de Limnología Dr. Raúl A. Ringuelet*, or ILPLA) in the Matanza-Riachuelo Basin and La Plata River; and (iv) assessing the implications of AySA's Sanitation Management Plan on the basin's aquifers based on the results of the study of hydrogeological conditions and groundwater quality in the basin being conducted in cooperation with the National University of La Plata (*Universidad Nacional de La Plata*, or UNLP).

12. ACUMAR Resolution No. 1/2007 -- established to address control efforts based on common limits -- constitutes a transitional stage (Phase I) in a process of migration toward the new regulation (Phase II), based on a rational, coherent methodology with a comprehensive basin-level focus that considers the mass load of pollutants discharged to the receiving water body and its capacity to receive polluting substances in terms of the uses assigned to it.

13. ACUMAR Resolution No. 1/2007 will not currently be applied to the system operated by AySA or to the evaluations of the implications of the AySA Sanitation Management Plan for raw sewage for water bodies in the Matanza-Riachuelo Basin and in La Plata River. What is applicable are the provisions contained in the AySA Regulatory Framework, in conjunction with

the mass loads of polluting sources and the uses and quality objectives proposed for receiving bodies in the Matanza-Riachuelo Basin and in La Plata River.

14. Thus, ACUMAR's significant authority over environmental management of the Matanza-Riachuelo Basin has simplified the basin's management, since ACUMAR encompasses national and provincial jurisdictions (Province of Buenos Aires and City of Buenos Aires) in a single Basin Authority.

15. **Institutional Strengthening** ACUMAR has a central role in the sustainable development of the M-R River Basin and in the implementation of the project. To ensure that ACUMAR is able to perform its tasks it needs to be strengthened. A series of activities is included in the project aimed at institutional strengthening of ACUMAR within a relatively short period of time (see description of component 4).

16. Close coordination between the Social Participation Component, environmental staff and the proposed Communications Unit within ACUMAR will be key to ensure that engagement with key stakeholders throughout the implementation of the environmental management plan and consultations on social issues remain a two-way dialogue. To this end, terms of reference will be developed for the heads of key units emphasizing the need for coordinated action. In addition, ACUMAR (including its Social Participation, Industries and Environmental Territorial Management Components) will receive training on management of resettlement issues, including the ability to screen for impacts on livelihoods that go beyond physical resettlement and the ability to propose and assess alternatives that minimize any negative social impacts. A covenant under the Loan Agreement has been included which reflects the Argentine government's commitment to make available such national resources as are required to carry out any resettlement actions in accordance with Bank guidelines. The coordination between the units and staff within ACUMAR that will manage social, environmental and communications issues will be complemented by coordination between ACUMAR and AySA around the management of environmental and social issues related to the implementation of component 1.

17. As part of component 4, ACUMAR will undergo a 6-month participative planning exercise to identify weaknesses in its current organigram and to propose strengthening measures going forward. Loan proceeds under component 4 will allow ACUMAR to be strengthened with additional staff, equipment and the refurbishment/rental/construction of new office space.

B. AySA

18. Water and sanitation services in Argentina are highly decentralized: over 1700 public and private service providers (public and private utilities, cooperatives, municipal and provincial entities) provide services in 23 provinces and the autonomous city of Buenos Aires. In 14 provinces, regulatory entities nominally oversee these services, though in practice regulatory oversight has weakened considerably since the 2002 economic crisis. The Buenos Aires Metropolitan area represents a special case. The Government of Argentina (GoA) created AySA in March 2006 to provide water and wastewater services in this area⁷⁵. At the same time, the

⁷⁵ Decree No. 304/2006, subsequently ratified by Law No. 26.100/2006, created AySA to take over the service area previously served by Aguas Argentinas through its concession contract. Article No. 1 of this decree defines AySA's service area as the Autonomous City of Buenos Aires and the municipalities of Almirante Brown, Avellaneda,

GoA unilaterally terminated the concession contract with *Aguas Argentinas S.A.*, (AASA)⁷⁶ for the area now served by AySA. Shortly thereafter, the Government established a new regulatory framework⁷⁷ for AySA's service area which :

- (i) calls for a concession contract between AySA and the Government that expires on December 31, 2020 (it can be extended beyond this date by mutual agreement)
- (ii) delegates the monitoring and supervision responsibility for the concession to the Sub-secretariat of Hydrological Resources (*Subsecretaría de Recursos Hídricos*, or SSRH), a dependency of the Ministry of Federal Planning, Public Investment, and Services (*Ministerio de Planificación Federal, Inversión Pública y Servicios*, or MPFIPS), designating the SSRH as the enforcing authority (*autoridad de aplicación*).⁷⁸
- (iii) creates the Regulatory Entity for Water and Sanitation (*Entidad Reguladora de Agua y Saneamiento*, or ERAS) and a Planning Agency (*Agencia de Planificación*, or APLA) for the concession area.

19. The implementation of this regulatory framework is underway—ERAS and APLA have only recently begun functioning—but it is still incomplete. The concession contract has not yet been signed but is expected to be in place by loan effectiveness. AySA was created as a private company—with the Government as the controlling shareholder—to take over the concession of the service area held by Aguas Argentinas until April 2006. The SSRH, by delegation from MPFIPS, exercises authority both as majority owner and primary regulator of AySA.

20. ***Ownership and governance of AySA.*** As a private company with majority ownership by the Government, AySA is subject to controls mandated for private and government-owned entities. The government created AySA as a private company⁷⁹ under the jurisdiction of the MPFIPS to provide water and wastewater services to approximately 9.5 million people living in the Federal Capital and 17 other municipalities in the metropolitan area of Buenos Aires. In 2006 the Government capitalized AySA with an initial equity contribution of ARS150 million. In exchange for this equity contribution, the Government owns 130,000 of the company's 150,000 shares. Decree 304/2006 specifies that all of the Government's shares are denominated as Class A shares and represent 90 percent of the company's social capital. The other 20,000 shares,

Ezeiza, Esteban Echeverría, La Matanza, Lanús, Lomas de Zamora, Morón, Quilmes, San Fernando, San Isidro, San Martín, Tres de Febrero, and Tigre y Vicente López for drinking water and sewerage services; the municipality of Hurlingham e Ituzaingó for drinking water services; and the municipalities of Berazategui and Florencio Varela for sewage effluent services. Said Decree also approved the articles of the corporation and bylaws of AySA.

⁷⁶ Aguas Argentinas S.A. (AASA) was a company incorporated in Argentina. Shareholders were Suez, Vivendi Universal S.A., Sociedad General de Aguas de Barcelona S.A., AWG Group Ltd., and other minority shareholders including IFC. On April 28, 1993, Aguas Argentinas formally concluded a 30-year concession contract with the Government, and on May 1, 1993, took over the management of the designated water distribution and wastewater systems.

⁷⁷ In October 2006, the Ministry of Federal Planning, Public Investment, and Services (*Ministerio de Planificación Federal, Inversión Pública y Servicios*, or MPFIPS) signed a tripartite agreement with the Province of Buenos Aires and the Autonomous City of Buenos Aires regarding the provision of water and sewerage services by AySA. This tripartite agreement was approved by Congress in February 2007 through the enactment of Law No. 26.221, which approved the regulatory framework for AySA.

⁷⁸ Article No. 5 of Law No. 26.221 of 2007 defines the Sub-secretariat of Hydrological Resources (*Subsecretaría de Recursos Hídricos*) in MPFIPS as the *Autoridad de Aplicación* of the Regulatory Framework.

⁷⁹ AySA was created as a private company under Law No 19.550 on Commercial Companies.

denominated as Class B shares and representing 10 percent of the company's social capital, are owned by the ex-employees of the National Sanitation Works (*Obras Sanitarias de la Nación*), who had purchased shares in *Aguas Argentinas*.

21. AySA's Board of Directors consists of five members, four of whom are designated by the Government as the Class A shareholder and one of whom is designated by the Class B shareholders. Decree 304/2006 provides a key role for MPFIPS in the governance of AySA. According to this decree, MPFIPS is responsible, directly or through its representatives, for the following:

- (i) Exercising the Government's representation of the Class A shares (Article 2);
- (ii) Designating the Board members on behalf of the Class A shares (Article 5);
- (iii) Designating the members of on Oversight Commission (*Comisión Fiscalizadora*) (Article 5);
- (iv) Supervising the development of AySA and approving its action plan and budget (Article 9);
- (v) As the *autoridad de aplicación* of Decree 304/2006, establishing any pertinent norms for clarifying or complementing the provisions in the decree (Article 15).

22. Decree 304/2006 also specifies a key role for the SSRH in exercising the majority shareholder's oversight of AySA. Article 38 of AySA's bylaws states that AySA will act in accordance with the plans, programs, and policies developed by the SSRH. This is in addition to SSRH's designation as primary regulator, a dual authority which has potential to give rise to a conflict of interest.

23. *AySA's capital structure.* When the company was created, no debt was transferred to the company, and its fixed assets were transferred to its balance sheet with a value of zero. Since no new loans have been contracted by AySA, at the end of 2007 the company had close to zero debt. Its fixed assets were equivalent to approximately US\$90 million, with a net book value to customer ratio of about US\$31. Moreover, in 2007 AySA had no interest or principal payments, and its total depreciation equaled less than 0.5 percent of its operating revenues.

24. An important result of this capital structure is that the company's revenue requirements are significantly lower than the full cost of providing the service. This will continue to be the case, since the Government intends to cover the company's capital costs—capital expenditures and loan and interest payments—for the foreseeable future.

25. From 2008 to 2012 the Government intends to provide substantial subsidies to AySA, through budgetary transfers, to cover the company's operating costs and a large capital expenditure program. Recently the National Congress approved an ARS1250 million (approximately US\$412 million) capital improvement budget for CY2008, including transfers covering operating costs.

26. *AySA's financial indicators.* AySA's financial model projects that its EBITDA⁸⁰ will be negative in 2008 and will average -\$523 million pesos during 2008-2012 (see Figure 13.2). This

⁸⁰ EBITDA: earnings before interest, taxes, depreciation and amortization. The term is arrived at by subtracting operating expenses from operating revenues.

deterioration of EBITDA is due to a large rise in operating costs while operating revenues remain stagnant. The company projects that in 2008 its operating costs will exceed tariff revenues by about 45 percent and by an average of 67 percent during 2008-2012. Operating costs are projected to rise based on larger volumes of wastewater pumped and treated, increasing salaries and the increasing unit costs of inputs (especially electricity and chemicals), while tariffs have been frozen at the same level since January 2002. AySA's management has calculated the tariff increase it would need to reverse this trend in the short and medium term. The last graph in figure 13.2 shows the resulting cash flow taking into account this increase in tariffs.

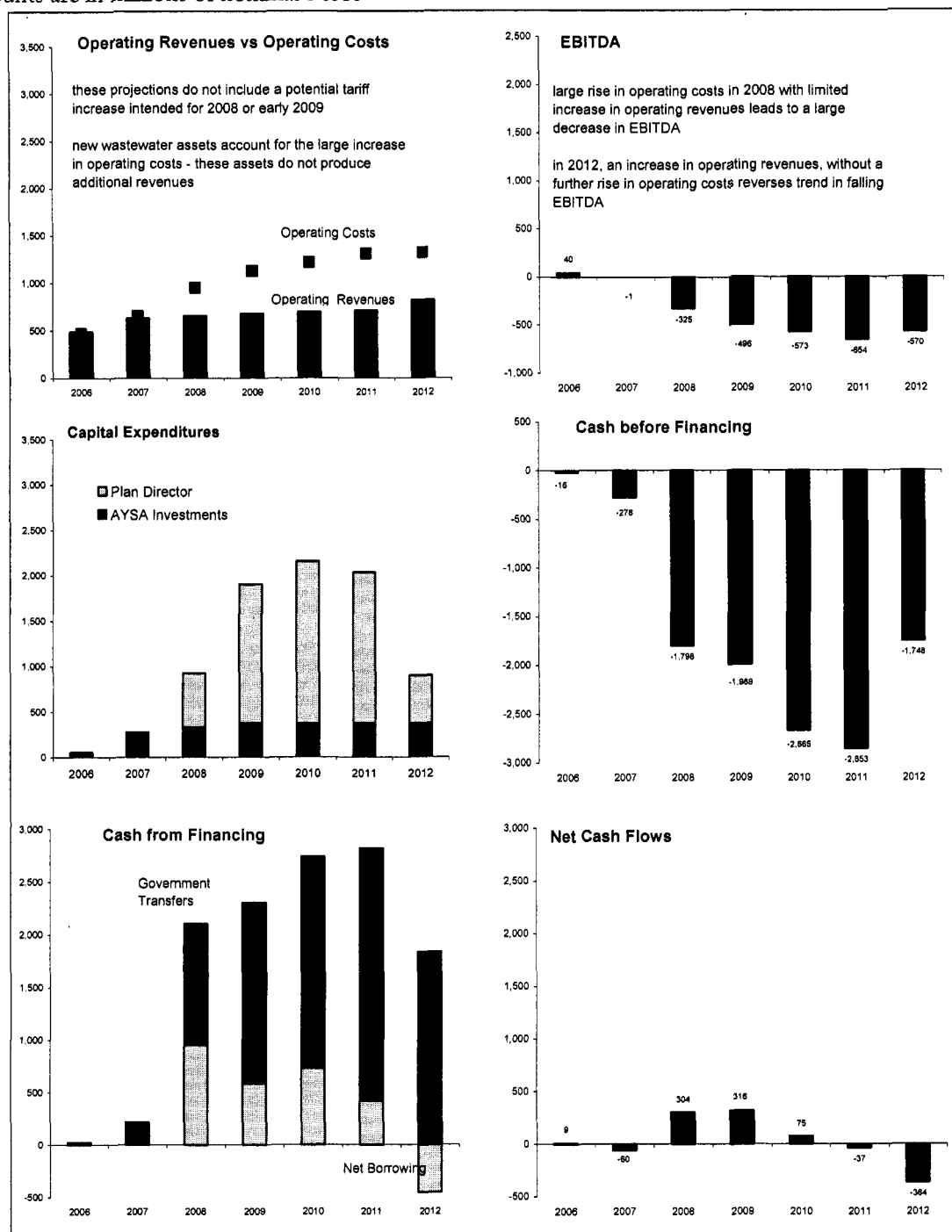
27. AySA intends to undertake a capital expenditure plan that will total about \$7.9 billion pesos from 2008 to 2012 (see Figure 13.2). These capital expenditures will fall into two broad categories. The first category is referred to by the company as "AySA Investments"; these are investments for maintenance and renewal of the company's existing assets. The second category corresponds to investments contemplated in AySA's Master Plan ("Plan Director") comprising new infrastructure investments. The proposed APLs (APL-1 as well as APL-2) include a number of key sanitation works identified under AySA's Master Plan which have been brought into agreement with ACUMAR's Integrated Cleanup Plan for the Matanza-Riachuelo Basin (PISA) to ensure that they will serve the dual purpose of expanding sewerage coverage while improving water quality of the M-R River Basin.

28. The government intends to provide AySA with the majority of the funds required for this large capital expenditure program on a grant basis (see below). The SSRH indicated that the Government intends to provide subsidies to AySA through annual budgetary allocations to cover any operating deficit and all capital expenditures. The government will borrow from the Bank to finance assets that will be operated and maintained by AySA.

29. AySA will borrow directly from the Brazilian Development Bank (*Banco Nacional de Desenvolvimento Econômico e Social*, or BNDES) and the National Social Security Administration of Argentina (*Administración Nacional de la Seguridad Social*, or ANSES) to finance the capital expenditures that will not be met by transfers from the Government (including transfers financed through loans to the national government from the World Bank and IADB).

Figure 13.2: AySA's Principal Financial Indicators, 2006 to 2012

Amounts are in millions of nominal Pesos



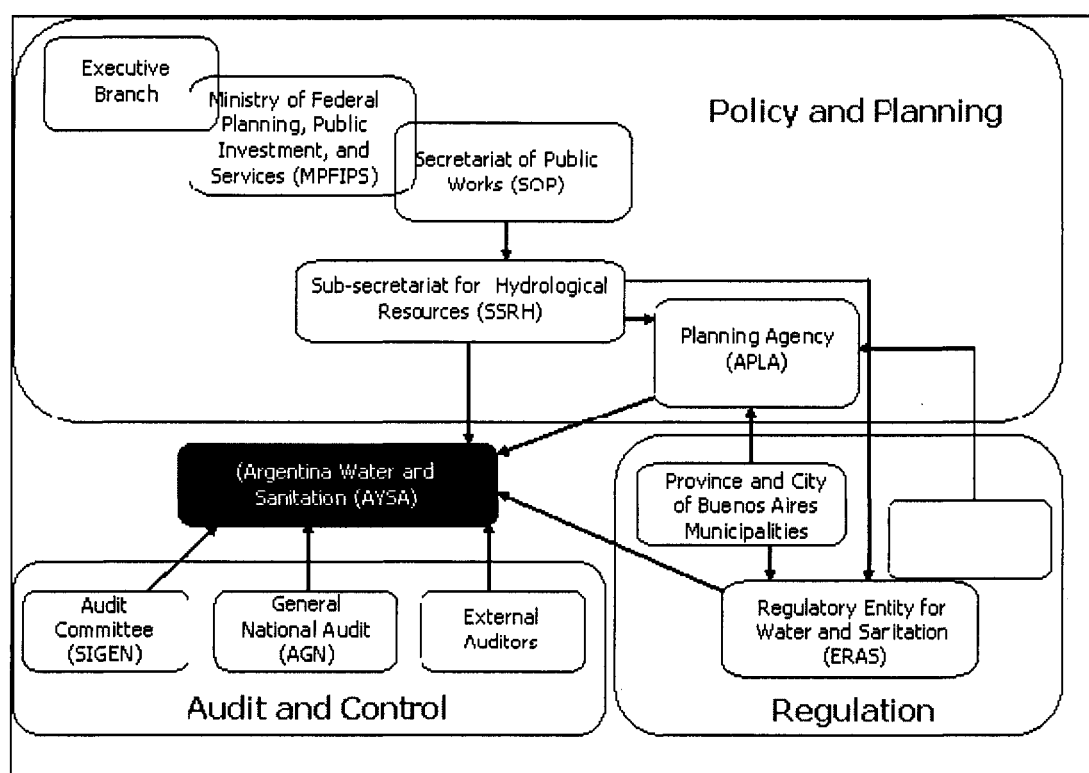
Amounts for 2006 and 2007 are from AySA's audited financial statements. The company began operations on 21 March 2006 so the values for 2006 are for 8 months of operations.

Projections for 2008 to 2012 are from the company's preliminary financial model and are subject to change.

30. **AySA's regulatory framework.** AySA's regulatory framework was created by Law No. 26.221/2007 and comprises the following three main components: (1) concession contract, as yet unsigned, between AySA and the Secretariat of Public Works (*Secretaría de Obras Públicas*, or SOP) in MPFIPS,⁸¹ (ii) an *autoridad de aplicación* with primary regulatory authority for the concession comprising the SSRH, a dependency of MPFIPS, and (iii) a regulatory entity (Regulatory Entity for Water and Sanitation (*Entidad Reguladora de Agua y Saneamiento*, or ERAS) and a planning agency (*Agencia de Planificación*, or APLA), both created out of the previous regulatory agency,⁸² that are effectively advisory bodies to SSRH.⁸³

31. SSRH's role as the primary regulatory authority is in addition to its policy and planning responsibilities as the delegated majority shareholder of AySA. AySA's dual nature as public and private entity means it is also subject to both government and external audits. The apparently complex inter-relationships of all the entities involved in planning, control, and auditing AySA is illustrated in figure 13.3.

Figure 13.3. AySA's new regulatory framework



⁸¹ The regulatory framework and concession contract for AySA are modified versions of those that were in place for the concession held by Aguas Argentinas.

⁸² Law No. 26.221/2007 dissolved the previous regulatory agency for Aguas Argentinas, *Ente Tripartito de Obras y Servicios Sanitarios*, or ETOSS) and created the Regulatory Entity for Water and Sanitation (*Entidad Reguladora de Agua y Saneamiento*, or ERAS) and Planning Agency (*Agencia de Planificación*, or APLA) in its stead. In practice, most former ETOSS staff were allocated to either ERAS or APLA.

⁸³ For example, according to Article No. 38 Annex 2 of Law No. 26.221, SSRH has the authority to set tariffs. This article also specifies ERAS's responsibility for the content of the tariffs established by the *autoridad de aplicación*.

32. This regulatory regime is still in the early stages of development and contains a number of inherent weaknesses (most notably the dual role of SSRH as executor of ownership and regulatory functions). The concession contract has been drafted but not yet signed. ERAS and APLA have only recently been established and are still developing their functions. Most importantly, the SSRH may not have the staff in place to carry out its broad responsibilities as the primary regulatory authority and shareholding authority even if it can steer clear of the conflict of interest embedded in its dual mandate.

33. **Tariff regime.** Setting tariffs is one of the most important functions of the regulatory process. The SSRH, in its role as the *autoridad de aplicación*, is entrusted with the authority to set tariffs. In contrast with the regulatory framework in place for the concession held by Aguas Argentinas, which mandated full cost recovery, the section of Law No. 26.221 that deals with AySA's economic and tariff regime (Chapter IX. Economic and Tariff System) does not appear to require that AySA's tariffs cover its costs. The draft concession contract between AySA and the Government makes some additional provisions regarding AySA's tariff regime, but it is still unclear what portion of its cost of service AySA is supposed to recover through its tariffs and how the Government intends to ensure the financial sustainability of AySA through budgetary transfers. Water and sewerage tariffs in AySA's concession are low by the standards of neighboring countries (Chile and Uruguay in particular) and represent less than 1.5% of household income in the Buenos Aires Metropolitan area. Since few residential customers are metered, most consumers pay flat tariffs that are independent of actual consumption but vary by location (a zonal coefficient that is high in wealthy areas and lower in low-income neighborhoods adds a strong element of pro-poor cross-subsidy to the regime). Both the national government and AySA are keenly aware of the need to restore and then maintain AySA's ability to cover its O&M costs through its operating income.

34. Claims related to AySA's conduct can be submitted to the company. According to the Regulatory Framework, users can file claims related to service quality level or non compliance with plans. If AySA doesn't reply, users can file a claim before ERAS. Any other irregular conduct by the company that can affect users' rights, service provision or the environment can be sent to ERAS directly.

Strengthening AySA, Sector Policies and Institutions

35. As part of project preparation, the Bank contracted a Due Diligence assessment of the Corporate and Regulatory Framework of Agua y Saneamientos Argentinos S.A. (AySA).⁸⁴ The assessment found AySA to be a relatively efficient operating company whose financial sustainability was undermined by persistently low tariffs rather than operational inefficiencies. AySA reports a non-revenue water (NRW) level of 36% (not bad, but could be improved), labor productivity of about 1.5 employees per thousand customers (good) and a collection rate of about 95% (very good).

36. Under its US\$200 million loan in support of AySA's Master Plan (approved in November, 2008), IADB has included an institutional strengthening component designed, i.e., to (i) improve personnel management and provide training in key areas; (ii) improve bill collection

⁸⁴ Castalia Strategic Advisors, April, 2008

in low-income areas; (iii) support the efficient incorporation of new assets into AySA's management systems and increase awareness among new customers about appropriate water use and payment for service. As part of the proposed Matanza-Riachuelo Basin Sustainable Development project, the World Bank's team has also engaged AySA on the topic of information campaigns to promote rational water user, given that AySA's historically low levels of household water metering provide little incentive for reducing water consumption. At a wider sector level, the World Bank has engaged the Government and civil society with a Progress Note on the Water and Sanitation Sector⁸⁵ covering topics from sector institutions, financing strategies to meet the Millennium Development Goals (MDGs) to tariffs.

Compliance by AySA with the Operator Eligibility Criteria established under the Buenos Aires Infrastructure Sustainable Development Project (APL-2)

37. In June, 2007, the Bank's board approved a US\$270 million loan in support of the second stage (APL-2) of the Buenos Aires Infrastructure Sustainable Development Project which finances transport, drainage and water/sanitation (WSS) services (among other activities) in eligible areas of Buenos Aires province. As part of the PAD for this project, the Bank established four eligibility criteria for WSS operators that would be eligible to receive financing under the operation:

Criterion No. 1: *The operator has been granted or assigned the responsibility to operate such services by the corresponding authority (municipal, provincial or federal) under a verifiable procedure that establishes the conditions to guarantee the sustainability of these services.*

Criterion No. 2: *There is in effect a regulatory framework that ensures the provision of water and sewerage services in a context of economic/financial sustainability, with conditions acceptable to the Bank, such framework includes mechanisms for the control of the regulatory institutions.*

Criterion No. 3: *The operator is in compliance with said regulatory framework.*

Criterion No. 4: *The operator has the capacity to operate efficiently and within a framework of financial sustainability.*

38. For AySA to be in substantial compliance with these criteria, it would need a signed concession contract as well as some mechanism to provide greater certainty of its future financial sustainability. For the purposes of the proposed operation, the Government's **development program letter** (see Annex 11E, which explicitly reiterates the Government's commitment to AySA's financial sustainability) as well as clause 4.11 of **IADB's loan agreement** with Argentina for its November 2008 loan for AySA (which commits the Government to achieving operation, maintenance and administration cost recovery by year 4 of IADB's operation) are deemed to satisfy the second requirement.

⁸⁵ "Argentina: Nota de Progreso Agua y Saneamiento", June, 2008.

Annex 11C: Social Assessment and Public Consultation

ARGENTINA: Matanza-Riachuelo Basin Sustainable Development Project

1. The basin is home to Argentina's largest concentrations of urban poor. Of the approximately 3.5 million inhabitants in the basin, 1.2 million live below the poverty line. An estimated 10 percent of the total population in the M-R River Basin lives in informal settlements, often in flood-prone areas and/or near open garbage dumps. The poorest populations live alongside the river and are in constant contact with numerous contaminants ranging from untreated organic waste to toxic industrial chemicals. Frequent flooding of the M-R River – because of high flows in the rainy season and high levels of water in La Plata River – spreads highly polluted waters into informal settlements, exposing the inhabitants to further contamination.
2. With the worsening environmental conditions in the M-R River Basin, and rising social awareness of health problems, the complaints and legal claims filed by civil society have been growing. Failed attempts by governments to address the situation led to families in several communities filing a high-profile Supreme Court class action lawsuit to force government action. In 2006, the Supreme Court called on the Government to put into place a concrete action plan to improve environmental conditions in the basin. The Matanza-Riachuelo Basin Sustainable Development Project constitutes part of the Government's response.
3. As part of economic analyses carried out for the project, nearly 1000 households were asked through surveys about their perception of the current status of the M-R River and La Plata River. Results from these surveys indicate that the project will contribute to addressing the problems expressed by the respondents.
4. **Social Assessments (SA).** Given the complexity of the Project, two complementary social assessments were prepared for ACUMAR and AySA during project preparation. ACUMAR's social assessment was a basin-wide study, describing the social baseline conditions which exist in the M-R River Basin, identifying stakeholder perceptions, positions, and influence on the project, highlighting the existing communication channels, pinpointing the overall social impacts of the project, and assessing its various risks. AySA's social assessment, on the other hand, explored the opinion of stakeholders about the proposed infrastructure works to be carried out by the company.
5. **Methodology.** Both SAs were developed based on available project information and consultations with relevant stakeholders. The teams that conducted the SA carried out 45 meetings with stakeholders including representatives of community organizations, NGOs, the industrial sector, provincial and municipal officials, and members of academia. Stakeholders also participated in workshops organized by Fundación Metropolitana and by Universidad de La Matanza on the Matanza-Riachuelo Basin, and in workshops for the presentation of the TOR's for the EIAs and the draft EIAs themselves.
6. **Main Findings** The main finding of the SAs was that all stakeholders were in favor of the project, considering the present scenario whereby given the intervention of the Supreme Court and with the assistance by the World Bank, the environmental improvement of the M-R River

Basin becomes achievable. None of the interviewed stakeholders spoke out against the overall project. However, critical views about diverse aspects of the project were expressed which need to be considered in order to ensure the social viability of the project. A complete list of positive and negative impacts is included below, by component:

7. SA's findings relating to component 1: The proposed infrastructure works are perceived as bringing about a number of positive impacts: improved environmental and health conditions, especially for the poor and vulnerable groups in the M-R River Basin; reduction of the contamination of the M-R River and groundwater; improved reliability and flexibility in the management of the sewerage system in the Buenos Aires Metropolitan Region; increase in the capacity of AySA's trunk sewerage infrastructure which will enable AySA to expand service to 1.2 million households; mitigation of groundwater contamination due to the reduction of effluents discharged by latrines and cesspits; likely employment opportunities and population density increases associated with the improved environmental conditions; and reduced costs of living associated with the closure of latrines that require costly sludge removal services. These benefits are considered significant for achieving social inclusion, mostly for the un-served poor and other vulnerable groups.

8. Some stakeholders canvassed by the SA expressed concerns that the proposed level of treatment (preliminary only) prior to discharge via the outfalls would simply displace contamination from the M-R River to La Plata River. In response, the technical justification of the proposed solution supported by mathematical modeling and an analysis of alternatives (presented in Annex 11A) was presented in detail in workshops and meetings. Other stakeholders raised the possibility of negative impacts during construction, operation, and maintenance: possible temporary disruption of access to residences and commerce due to the implementation of sanitation works; the concern that large underground works might structurally affect existing buildings in the area; intense traffic of trucks transporting soil removed from underground works and waste from the Riachuelo preliminary treatment plant which might cause traffic disruption; the challenge of managing and properly disposing of solid waste, sand, grease and oil from the plant. These impacts have been addressed by the Environmental Management Plan (Annex 10). In terms of a possible tariff increase, the impact on low-income populations would be mitigated by the cross-subsidies embedded in AySA's tariff structure (see Annex 11B for further discussion).

9. SA's findings relating to component 2: The reduction of industrial contamination, by means of the implementation of the PRIs and the enforcement of appropriate pollution control policies, is perceived by stakeholders as essential to improve environment and health conditions in the M-R River Basin. Other identified benefits were: greater public sector involvement to prevent and enforce pollution control policies; improved industrial territorial distribution; attraction of new population segments associated with new employment opportunities; improved road quality and reduction in maintenance costs faced by local governments due to increased control of unregulated industrial waste disposal on roads and in other public areas; and improved local industrial competitiveness associated with the adoption of clean production mechanisms.

10. However, doubts were expressed on the industrial component in terms of ACUMAR's capacity to carry out appropriate pollution control policies; unemployment increases and a

reduction in local government revenues which could accompany the possible shut down of industries or their transfer to other areas; a reduction of non-skilled jobs and an increase of skilled jobs due to the change in industrial technology. As part of the pollution diagnostics under component 2, a validation exercise of the list of contaminating industries will also include an analysis of the possible social impacts of activities relating to industrial pollution abatement within the M-R River Basin. It is worth noting that Argentina has unemployment insurance and training programs available to support former workers, in part financed by a US\$ 200 million World Bank loan approved in 2007 (Loan 7474).

11. SA's findings relating to component 3: The improvement in living conditions is associated with the proposed urban upgrading and infrastructure provision. Improvements perceived by stakeholders include the reduction of flooding, loss of assets and healthcare costs; reduction of informality in land use and increase in urban standards. The provision of basic infrastructure to un-served vacant pieces of land and the social housing programs to be located there are considered opportunities for low-income groups to gain social inclusion.

12. The main concern on Component 3 is the possible displacement of low-income groups settled in areas affected by physical intervention, being unable to adapt to formal housing and livelihoods.

13. SA's findings relating to overall project implementation: With respect to perceived shortcomings during implementation, many stakeholders pointed to the shortage of easily-accessible, coordinated and updated information; the lack of opportunities for stakeholder participation, including civil society and municipalities; and the weaknesses of the Social Participation Commission (Comisión de Participación Social) of ACUMAR.

14. Some stakeholders contrasted these shortcomings with the role that civil society organizations (including 5 entities recognized in the lawsuit as "interested third parties") is exercising under the Supreme Court's 8 July, 2008 ruling requiring clean-up of the M-R River Basin. As a group, these entities have been very active in following up on government actions and the advance of the Bank's loan and it is expected that they will closely follow up on implementation issues (including procurement actions) as well.

15. Measures to address SA findings. Several initiatives have been developed and are incorporated in project design to address the above findings. A Communication Management Program (see Annex 11D) has been designed with SAyDS and ACUMAR, in order to significantly improve information flows to key actors and stakeholder groups in the basin about the project in a transparent manner, strengthen the credibility of and support for the project, and to contribute to the overall consultation, participation and empowerment framework for the project. The Communication Management Program includes the creation of a Communication Unit in ACUMAR to assume the responsibility for (i) providing adequate and timely information prior to consultation events; (ii) maintaining documentation on consultation processes and decisions taken; and (iii) providing feedback to concerned stakeholders. In parallel, AySA's communications efforts are improving as demonstrated by its outreach to NGOs and improved internal harmonization of project information among its central and regional offices.

16. In addition, in response to the institutional weaknesses of ACUMAR including the limited activity of its Social Participation Commission (Comisión de Participación Social), component 4 will focus on the institutional strengthening of ACUMAR and the revitalization its Social Participation Commission. Close coordination between the Social Participation Component, Environmental Unit and Communications Unit will be key to ensure that engagement with key stakeholders throughout the implementation of the environmental management plan and consultations on social issues remain a two-way dialogue. To this end, terms of reference will be developed for the heads of all three units emphasizing the need for coordinated action. The Social Participation Component will receive training on management of resettlement issues, including those that go beyond physical resettlement and to propose and assess alternatives to minimize negative social impacts. The Loan Agreement includes a requirement that the Government prepare and implement Resettlement Plans on the basis of the Resettlement Policy Framework should actual cases of resettlement arise and to ensure funds are available to carry out any resettlement actions in accordance with Bank guidelines. In addition, as shown in Annex 11D (Communications Strategy), US\$100,000 have been budgeted to support further diagnostics and consultation around social impacts as needed during project implementation. Institutional strengthening will be an ongoing activity throughout the project's lifespan, and will be supported through Component 4.

17. Social as well as environmental considerations will guide the finalization of engineering designs under component 1 as well as the preparation of specific interventions under components 2 and 3. The project's social assessment process, combining a diagnosis of social context with transparent consultation, will be coordinated and integrated with technical, financial, environmental and other inputs in further preparing and implementing the project's components and activities. In particular, the social assessment will constitute an important element in finalizing the designs for components 2 and 3. ACUMAR will develop a Social Management Plan that will integrate and coordinate the approaches suggested in the two social assessments and the project's communication strategy for addressing identified social impacts, risks and opportunities under the project as well as stakeholder engagement and communications efforts. This plan will comprise of activities, budgets, responsibilities, timelines and indicators, for integration of social development issues into the project, on the basis of the following subjects: (i) the resettlement framework; (ii) management and mitigation of impacts in the construction phase of Component 1; (iii) management and mitigation of impacts in the operation phase of Component 1; (iv) management and mitigation of impacts associated with industrial conversion of Component 2; (v) monitoring; and (vi) consultation, communication and disclosure.

18. Public Consultation and Communication. Consultation and communication activities have been carried out during preparation and will be continued during project implementation. As part of the project preparation process, extensive stakeholder consultation has been carried out using formal as well as informal methodologies (see Annex 10 for more details). Additional activities are planned for project implementation. Funding for these activities has been considered as part of component 4.

19. Information on the project has been made publicly available through public disclosure events and workshops. Informal focus group discussions have been conducted with NGOs, local government, affected people and beneficiaries at various points during preparation and appraisal.

It is worth mentioning some of these events: surveys and focus groups during the “willingness to pay” study for the economic analysis (see Annex 9), workshops for the consultation of the terms of reference for the EIAs on 10 July, 2008, and dissemination of the results of the EIAs on 7 November, 2008, AySA meetings previous to the EIA results workshop with 11 NGOs, World Bank meetings and exchanges of correspondence with several NGOs, and multiple interviews with stakeholders carried out during the two social assessment studies. The project EIAs have been made available through the ACUMAR and AySA websites as well as through the Bank’s InfoShop.

20. In addition to the project specific consultation and disclosure efforts, it is important to recognize that both AySA and ACUMAR have also established social outreach departments within their organizational structure. These teams have been established to be in constant contact with the local communities, and are used to facilitate communication tools. ACUMAR, for example, has established a network of approximately 300 community and environmental NGOs that are part of its broader communications strategy about the issues facing the M-R River Basin. This network provides an opportunity for information disclosure not just on the project but on a wide array of development issues facing the basin. AySA business practice for planning of civil works includes disclosure of information and stakeholder consultations which will be applied to the civil works in APL-1 which are at or before pre-feasibility stage (e.g., right bank collector).

21. **Involuntary resettlement (OP 4.12):** No resettlement is expected as a result of these project components. The design of the sewage collector routes has been conceived to minimize adverse impacts and avoid resettlement, and no displacement is projected regarding industrial and urban improvements. The Left Bank collector and the Bajo Costanero By-pass collector will be constructed by tunneling techniques (rather than open-trench construction) which limit above-ground works to a relatively small number of access shafts. Moreover, the Riachuelo pre-treatment plant is located in Dock Sud on uninhabited land. However, as a precaution, since the physical interventions of components 2 and 3 of the project have not yet been fully determined, a Resettlement Policy Framework has been prepared. As pilot and initial phase interventions for components 2 and 3 are selected, minimizing resettlement will be one of the selection criteria.

22. Additional reputational risks to the Bank could materialize if ACUMAR, subnational governments or private firms in the basin carry out actions broadly in line with basin clean-up under the PISA entirely with their own funds without full consideration of social and environmental impacts (including relocation or restoration of livelihoods).

23. **Mechanisms for mediation, conflict resolution, and redress:** There are several alternative mechanisms at various levels available to citizens who wish to seek information or complain if they feel the project does not adequately address their needs or concerns. At the lowest level, there are project consultation mechanisms through the Social Participation Commission (created within ACUMAR, with the objective of channeling comments and proposals from civil society organizations to ACUMAR’s management), as well as submissions to the specific administration unit or institution (ACUMAR, SAYDS, or AySA). While some civil society organizations have expressed concern about the efficacy of ACUMAR’s Social Participation Commission, ACUMAR’s new management (in place since December, 2008) is committed to strengthening its outreach.

24. At the next level, claims related to the implementation of the PISA can be presented to the Collective Body ("Cuerpo Colegiado) created by the Supreme Court's final decision in the Mendoza⁸⁶ case. This body, is lead by the National Ombudsman and composed of the 5 NGOs participating in the case as interested third parties. This body is in charge of monitoring ACUMAR's compliance with the Supreme Court decision. As this Monitoring Body has only recently been created, uncertainties about its capacity to handle these submissions remain. For the same reason, it is also still uncertain if claimants will still have the possibility to go to court directly or if they will have to rely on this Body to make the decision. On the other hand, according to the Supreme Court decision in the above-mentioned case, any claim submitted before this judge can be appealed directly to the Supreme Court.

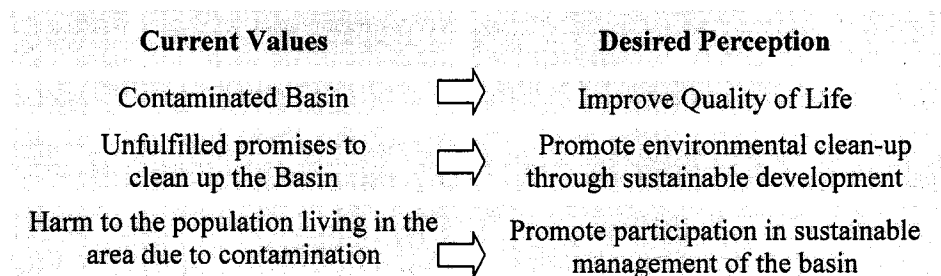
25. Alternatively, people can directly file "*amparo*" claims (restraining orders) before local, provincial or national courts, which may eventually be redirected to the Federal Judge of Quilmes.

26. Claims related to AySA's conduct will be submitted before the company using different ways of communication (telephone, office visits or email). According to AySA's Regulatory Framework, users can present claims related to service quality levels or non-compliance with plans. If AySA doesn't reply, users can file a claim with the Regulatory Entity for Water and Sanitation (*Entidad Reguladora de Agua y Saneamiento, or ERAS*). Any other irregular conduct of the company that can affect users' rights, their service or the environment can be directed to ERAS directly.

⁸⁶ 2003 lawsuit: Mendoza Cause, Beatriz Silva and others against the National Government and others.

Annex 11D: Communication Strategy and Implementation Program
ARGENTINA: Matanza-Riachuelo Basin Sustainable Development Project

1. The communication strategy for the Matanza-Riachuelo Basin Sustainable Development Project will be designed to inform key actors and stakeholder groups in the basin in a transparent manner about the project; strengthen the credibility of and support for the project; and contribute to the overall consultation, participation and empowerment framework for the project.
2. The main objectives of the strategy are to:
 - Improve communication delivery by ACUMAR
 - Establish direct and indirect two way communication channels with the community
 - Ensure the dissemination of information regarding the project and its execution to key audiences
 - Promote transparent, direct and pro-active communication with relevant stakeholders
3. The main audiences that need to be reached and the main stakeholders that need to be engaged by the communication strategy are:
 - Institutions involved in project implementation such as the 14 local governments participating in ACUMAR together with the Governments of the City and Province of Buenos Aires with their different audiences;
 - Residents of the project area, especially the project-affected people;
 - Key opinion leaders;
 - General public (mainly through mass media).
4. In order to ensure that these objectives are met, the following activities will be developed under the project:
 - a. Establishment of a Communication Unit in ACUMAR to design and implement the strategy and to act as a focal point and source of information for all stakeholders;
 - b. Implementation of a communication strategy targeting key audiences in 3 phases:
 - Phase 1 – reach key opinion leaders to share the Government’s plan regarding the management of the basin and to provide specific information about the project to be financed by the World Bank.
 - Phase 2 – reach local communities and beneficiaries with information about the project and its implementation progress.
 - Phase 3 – Once project execution has advanced, the strategy focus will expand to reach a mass audience by launching a communication campaign to present the vision of a clean river and its impact on quality of life in Buenos Aires City and the Province.
 - c. Design and production of communication and advertising materials.
5. In terms of the message, based on the social analysis done for the project and on independent research by a consultant who developed the draft communication strategy for ACUMAR, the proposed positioning of the project is as follows:



6. These messages will be adjusted after further diagnostics and consultations around social impacts are done by the team during the initial phase of implementation.

7. Today, people living in the basin area perceive it as highly polluted, and the first association that comes to their mind when asked about the Matanza-Riachuelo River is “contamination”. With the implementation of the strategy, the project will act to promote a perception (once the contamination level in the M-R River is reduced) that the project contributes to improving the quality of life of the Basin’s residents.

8. Secondly, the idea of clean-up of the Matanza-Riachuelo River is not new in Argentina and there is a history of failed attempts to do so. This has created a feeling of mistrust and lack of confidence towards any initiative that promises a clean river. To manage this perception, the communications team in ACUMAR will base the communication actions on facts rather than on promises. The strategy is to slowly gain credibility among the beneficiaries and other key audiences by promoting the concept that sustainable development of the basin is being achieved through concrete actions and works of reducing contamination from domestic sources thereby improving the quality of the water in the river, as well as reducing industrial contamination and improving solid waste management in the basin.

9. Thirdly, in view of the reality of high contamination affecting the quality of life of the basin’s residents, the project will promote participation mechanisms to include the different key stakeholders in the sustainable management of the basin.

10. In terms of implementation, the communication strategy for the project has initially identified 4 clear stages that will take place throughout the project’s life. These stages will be further adjusted once the Communications Director for ACUMAR is hired.

- a. **Re-positioning.** In this initial phase, the priority is given to comply with the ruling of the Supreme Court, to organize the communication department of ACUMAR and to execute the initial phase of the communication strategy for the period July 2009 – February, 2010.
- b. **Positioning.** Start activities through three communication axes:
 - Improve quality of life in the basin
 - Promote environmental clean up through sustainable development,
 - Promote participation in sustainable management of the basin
- c. **Leadership.** Profile ACUMAR as leading the implementation of the Project. Define communication strategy for year the 2010.

- d. **Consolidation of Leadership** – after the main works have been initiated and the project is delivering on its promises, from the communication perspective it is time to launch the vision of a cleaner basin and a strong commitment to promoting the environment and sustainable development. It is in this phase when the project will take the leadership in terms of communication, to generate a vision of the clean basin which will include active participation of the neighbors and more massive ways of communications will be used.

11. In terms of resources to execute the 6-year communication program, the following activities have been identified as ones that will be financed to complement the financing for the communications department of ACUMAR (which will be part of ACUMAR's structure and will obtain its financing from the ACUMAR budget).

12. The activities contemplated in the communication strategy include (i) financing of consulting services to reinforce ACUMAR's communication department to help ACUMAR strengthen its capacity regarding informed consultation, information management and communication, and in particular to help ACUMAR carry out the activities under the proposed project; (ii) purchase of office equipment for the communication department; (iii) financing the production of communication materials; (iv) financing for advertising spots on TV, radio and in print media; (v) financing of advertising inserts directed to communities. Advertising material will be inserted in local newspapers or sent out together with utility bills; (vi) investments in capacity building through field visits to similar projects, which may include those in Cartagena (Colombia), Montevideo (Uruguay), and several cities in Brazil and Chile; (vii) consulting services for surveys; (viii) specialized consulting services for design and advertisement; (ix) consulting services for the design and implementation of a campaign to promote the rational use of water as well as the appropriate use of sanitation infrastructure.

	US\$
Support to Creation of Communications Department of ACUMAR	
Consulting Services	\$350,000
Office equipment	\$45,000
Production of Communication Tools	
Video, brochure, web, promotional materials, stickers, posters, PISA + PDS, games, image bank, etc	\$300,000
Advertising	
TV, Radio, print and street media	\$1,050,000
Direct communication and informed consultation with key stakeholders & community	
Insert in newspapers and service utilities, etc	\$995,000
Further diagnostics and consultation around social impacts	\$100,000
Training, study tour and Capacity building	
Neighbors, journalists, other key stakeholders	\$220,000
Surveys	
Focus groups and surveys	\$200,000
Consulting	
Artis gra gn	\$240
Educational campaign on water consumption and sanitation	
Design and implementation of a campaign to promote the rational use of water	\$500,000
Total	\$4,000,000

Annex 11E: Letter of Development Program
ARGENTINA: Matanza-Riachuelo Basin Sustainable Development Project

(Translation)

Buenos Aires, 16 April 2009
Reference: Matanza Riachuelo Basin
Sustainable Development Project (P105680)

Mr. President
International Bank for Reconstruction and Development
Robert Zoellik

I write to you with regards to the loan operation referenced (above) that has been negotiated over the past year and a half between the technical teams of the Bank and our government, so that it might contribute to the implementation of policies that the Argentine Nation is undertaking.

One of these policies is the recovery of the Matanza Riachuelo Basin, through the Integrated Environmental Clean-up Plan (*Plan Integral de Saneamiento Ambiental, PISA*) which requires prompt implementation.

For this reason, our President, Ms. Cristina Fernández de Kirchner, held a meeting in November 2008 with the World Bank's Vice-President, Ms. Pamela Cox, in which she shared this concern and expressed her decision to undertake such actions, elaborating in detail the policies that have been instrumental for the sector.

It is worth remembering that the Argentine Republic has achieved one of the goals imposed by this government: grow in order to honor our obligations.

It is worth mentioning that over the last four years, the Argentine Republic has paid to the International Bank for Reconstruction and Development, an amount of approximately US\$5,131 million in interest and capital, for the operations with this organization, such that today there are no arrears of any kind.

In addition, it is worth mentioning that over the last five years Argentina achieved a fiscal surplus as well as a surplus in its balance of payments, thereby reaching a unique milestone in the modern history of our country.

With regard to the loan request presented by the National Government to the Bank for the financing of the "Matanza Riachuelo Basin Sustainable Development" Program, overall responsibility for which will lie with the Secretariat of Environment and Sustainable Development (*Secretaria de Ambiente y Desarrollo Sustentable, SAyDS*) of the Chief of Cabinet Office (*Jefatura de Gabinete de Ministros*), we would like to inform you of the guiding principles that will allow the expected results to be achieved.

First of all, we inform you that the National Government enacted Law No. 26.168 that created the Matanza Riachuelo Basin Authority (ACUMAR) as an overarching interjurisdictional body that brings institutionality and a new management tool to the area of the Basin. In the articles of this Law, there is a special emphasis on the creation of a Social Participation Commission (*Comisión de Participación Social*) with consultative functions under the auspices of the Matanza Riachuelo Basin Authority. This commission will be comprised of representatives of organizations with interests in the area.

Among other things, this Basin Authority has the mission of implementing the Integrated Environmental Clean-up Plan (PISA), that was developed by the Secretariat of Environment and Sustainable Development jointly with the Province of Buenos Aires and the Autonomous City of Buenos Aires. The program, to which this letter refers, will be partially financed by the Bank and has been conceived within this framework.

The main objective of this program is to improve the quality of life of the population as well as the environmental recovery of the Matanza Riachuelo Basin. At the same time, the program will offer a cost-effective solution that is sustainable over time for the safe disposal of sewage from a large part of the Buenos Aires Metropolitan Area which today lacks this basic service.

The National Government considers this program a high priority since it will directly benefit 3.5 million inhabitants of the Basin and – more generally -- the entire population of Greater Buenos Aires.

To finance the program, it was decided to use the APL (Adaptable Program Lending) tool since this allows for flexibility to carry out the investments in the short term and to prepare the conditions for future investments in the medium term. By virtue of this, we request US\$840 million from the Bank for the first stage APL-1.

The first stage of the program will contribute directly to the improvement of sewerage services in the Matanza-Riachuelo Basin, the reduction in untreated industrial discharges to surface water bodies, the improvement of the urban drainage system, land use in the basin, and the institutional strengthening of ACUMAR, given its fundamental role as basin authority and as an organization that coordinates the actions necessary to improve the basin in fundamental ways with the objective of increasing the social inclusion of its inhabitants and of restoring the environment.

Technically, for the environmental recovery of the Matanza-Riachuelo Basin, the strategy consists of the removal of pollutant loads and the progressive withdrawal of identified point source pollution discharges, as well as the industrial discharges with the largest relative contribution of pollutants, in support of a final objective of reaching dissolved oxygen levels higher than 2 mg/L along the length of the Matanza-Riachuelo River.

The program, as we mentioned before, will be under the responsibility of the Secretariat of Environment and Sustainable Development (*Secretaria de Ambiente y Desarrollo Sustentable, SAyDS*) of the Chief of Cabinet Office (*Jefatura de Gabinete de Ministros*), and divided into two Macro-components. Macro-component 1, with only one component titled component 1 -

Sanitation; and Macro-component 2, composed of three components identified as component 2 – Industrial pollution abatement, component 3 – Environmental territorial management, and finally component 4 - Institutional Strengthening. Macro-component 1 will implemented entirely by AySA and Macro-component 2 by SAYDS and ACUMAR as the interjurisdictional entity responsible for implementing PISA which will vet actions that will be developed under the program, determining in this way the legal, political and institutional framework for the same.

It is important to highlight that during the implementation of APL-1, part of the key structural works necessary for the transport, treatment and final disposal of the pollutant flows that are today channeled to the Matanza Riachuelo River will be built. These works include the Left Bank collector, the Costanera Bypass collector, the Riachuelo preliminary treatment plant (with its pumping stations), the Riachuelo outfall, the industrial collector, the Right Bank collector (or a possible technical alternative that could be agreed in its place). The above mentioned works will have the technical consent of ACUMAR and SAYDS and will be sent for consideration to the Bank in order to determine their ultimate feasibility.

During the second stage of the APL, these works are expected to complemented by others, also structural, that will allow the recovery of water quality on the shores of La Plata River (Baja Costanera Expansion Collector) and to complement the final disposal system (pumping station and Berazategui sub-aquatic outfall). In this second stage, it is planned to extend significantly the primary and secondary sewerage networks in AySA's operating area within the Matanza Riachuelo Basin.

Regarding component 2, the intervention strategy is based on the reduction of pollutant loads produced by industrial discharges to the basin, through the development and implementation of the Industrial Conversion Plans (*Planes de Reconversión Industrial*) for those industries that will be declared contaminating agents (*agentes contaminantes*).

Component 3 aims to improve public policies related to environmental territorial management of the basin and will be implemented through the design of a territorial master plan agreed with corresponding jurisdictions and direct actions on the ground.

Component 4, ACUMAR institutional strengthening, is fundamental to the development of the entire program, since it will allow for the implementation of institutional strengthening activities that structure the organization in a way that allows it to reach the objectives set-out in the PISA with efficacy and efficiency.

It is worth highlighting the importance the National Government attaches to having AySA carry out component 1 in its capacity as the water supply and sanitation utility for the Buenos Aires Metropolitan area and the City of Buenos Aires, where most human activity in the Matanza Riachuelo Basin are located. It is worth noting that 70% of all water supply and sewerage investments under AySA's master plan (*Plan Director*) are to be implemented in the Matanza Riachuelo basin.

It is the policy of the National Government to continue actions conducive to achieving AySA's operational sustainability, as well as, as you well know, to take on the program of works

necessary to connect the remaining population that today lacks water supply and sewerage connections -- the central objective recommended by all International Organizations, especially the United Nations.

In this context, the Government will ensure that AySA has the necessary resources for financing its operations and maintenance through its own revenues or through national government transfers.

Finally, we would like to draw attention to the support provided by the Inter-American Development Bank for the sanitation policies carried out by the National Government through AySA, given that, on November 5th its Board of Directors unanimously approved a credit of 250 million Dollars as part of a total loan of 900 million Dollars.

With nothing further to add, kind regards,

Dr. Homero Maximo Bibiloni
Secretary of Environment and Sustainable Development

Annex 12: Documents in the project file

ARGENTINA: Matanza-Riachuelo Basin Sustainable Development Project

Project Documents

- Project Concept Note
- Project Information Document Concept Stage
- Integrated Safeguards Data Sheet Concept Stage
- Project Information Data Sheet
- Minutes of the PCN Review Meeting
- Draft Project Appraisal Document

Project Reports and Studies

ACUMAR 2009. Executive Summary of the EIA Matanza-Riachuelo Basin Sustainable Development Project

ACUMAR 2009. Evaluación Ambiental Integral del Proyecto de Desarrollo Sustentable de la Cuenca Matanza-Riachuelo

ACUMAR 2009. Marco de Gestión Ambiental y Social: Componente de Reconversión Industrial (Componente 2)

ACUMAR 2009. Marco de Gestión Ambiental y Social: Componente de Manejo Territorial (Componente 3)

ACUMAR 2008. Marco de Política de Reasentamiento

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AySA. 2008. Estudio de Interacción Sólido – Líquido Cuenca Matanza-Riachuelo.

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Bombardelli, Fabian. 2008. Analisis de “Modelacion Matematica de la cuenca Matanza-Riachuelo para el estudio de Alternativas de Saneamiento”

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References:

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Boll, Jorge. 2006. Se requiere una solución integral para resolver los problemas de contaminación de la cuenca hídrica Matanza – Riachuelo.

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Annex 13: Statement of Loans and Credits
ARGENTINA: Matanza-Riachuelo Basin Sustainable Development Project

Project ID	FY	Purpose	Original Amount in US\$ Millions				Cancel.	Undisb.	Difference between expected and actual disbursements	
			IBRD	IDA	SF	GEF			Orig.	Frm. Rev'd
P101171	2009	AR SOC&FISC NTL ID SYS II	20.00	0.00	0.00	0.00	0.00	20.00	0.00	0.00
P106684	2009	AR 2nd Prov. Agric Dev	300.00	0.00	0.00	0.00	0.00	300.00	0.00	0.00
P106752	2009	AR Unleashing Productive Innovation	150.00	0.00	0.00	0.00	0.00	150.00	0.00	0.00
P110462	2009	AR Mining Environmental Restoration Project	30.00	0.00	0.00	0.00	0.00	30.00	0.00	0.00
P100806	2008	AR Sustainable Natural Res Mgt	60.00	0.00	0.00	0.00	0.00	60.00	0.00	0.00
P095569	2007	AR APL2 National Highway Asset Mgt	400.00	0.00	0.00	0.00	0.00	400.00	0.00	0.00
P090993	2007	AR-Essential Public Health Functions	220.00	0.00	0.00	0.00	0.00	151.82	23.29	0.00
P095515	2007	AR (APL2) Prov. Maternal-Child Health	300.00	0.00	0.00	0.00	0.00	257.79	68.79	0.00
P095514	2007	AR Lifelong Learning Project	200.00	0.00	0.00	0.00	0.00	193.00	18.00	0.00
P099051	2007	AR- SANTA FE ROAD Infrastructure	126.70	0.00	0.00	0.00	0.00	108.50	61.80	29.80
P099585	2007	AR-Cordoba-Road Infrastructure	75.00	0.00	0.00	0.00	0.00	48.45	43.12	0.00
P101170	2007	AR 2nd State Modernization	20.00	0.00	0.00	0.00	0.00	16.44	4.34	0.00
P105288	2007	AR APL2 Buenos Aires Infrastructure	270.00	0.00	0.00	0.00	0.00	270.00	23.00	0.00
P055483	2006	AR-Heads of Household Transition Project	350.00	0.00	0.00	0.00	0.00	45.06	45.06	0.00
P060484	2006	AR Basic Municipal Services Project	110.00	0.00	0.00	0.00	0.00	98.42	27.16	0.00
P070448	2006	AR Subnational Gov Public Sec Modernization	40.00	0.00	0.00	0.00	0.00	37.66	23.23	0.00
P070963	2006	AR Rural Education Improvement Project	150.00	0.00	0.00	0.00	0.00	129.33	60.46	-20.52
P093491	2006	AR (APL2) Urban Flood Prev. & Drainage	70.00	0.00	0.00	0.00	0.00	69.53	69.70	0.00
P092836	2006	AR Inst. Strengthening - ANSES II TA	25.00	0.00	0.00	0.00	0.00	20.28	11.28	0.00
P089926	2006	AR Solid Waste Management Project	40.00	0.00	0.00	0.00	0.00	38.38	17.78	0.00
P070628	2005	AR-Provincial Road Infrastructure Project	150.00	0.00	0.00	0.00	0.00	122.66	95.66	0.00
P088032	2005	AR(CRL1)Buenos Aires Infrastr SIDP(1APL)	200.00	0.00	0.00	0.00	0.00	66.27	61.59	0.00
P088220	2005	AR (APL1) Urban Flood Prevention & Drainage	130.00	0.00	0.00	0.00	0.00	121.90	102.06	0.00
P071025	2004	AR-Provincial Maternal-Child Health Inv Loan	135.80	0.00	0.00	0.00	0.00	32.39	20.27	0.00
P088153	2004	AR National Highway Asset Management	200.00	0.00	0.00	0.00	0.00	14.77	14.77	0.00
P006043	1999	AR RENEW.ENERGY R.MKTS	80.00	0.00	0.00	0.00	0.00	58.28	8.28	6.86
P006041	1998	AR SMALL FARMER DV.	120.00	0.00	0.00	0.00	0.00	45.00	0.00	0.00
P039584	1997	AR B.A.URB.TSP	300.00	0.00	0.00	0.00	0.00	99.21	-0.79	-0.79
P006010	1997	AR PROV AG DEVT I	162.00	0.00	0.00	0.00	0.00	26.27	-10.73	-10.73
Total:			4,434.50	0.00	0.00	0.00	0.00	3,031.41	788.12	4.62

ARGENTINA
STATEMENT OF IFC's
Held and Disbursed Portfolio
In Millions of US Dollars

FY Approval	Company	Committed				Disbursed			
		IFC				IFC			
		Loan	Equity	Quasi	Partic.	Loan	Equity	Quasi	Partic.
2000	ASF	3.76	0.00	0.00	3.85	3.76	0.00	0.00	3.85
1998	AUTCL	4.28	0.00	0.00	0.00	4.28	0.00	0.00	0.00
2004	Aceitera General	50.00	0.00	20.00	30.00	50.00	0.00	20.00	30.00
2006	Arcor	70.00	0.00	0.00	210.00	70.00	0.00	0.00	210.00
2000	BACS	0.00	6.25	0.00	0.00	0.00	6.25	0.00	0.00
2006	BACS	50.00	0.00	0.00	0.00	13.25	0.00	0.00	0.00
1999	Banco Galicia	57.79	0.00	0.00	40.91	57.79	0.00	0.00	40.91
2005	Banco Galicia	40.00	0.00	0.00	0.00	5.00	0.00	0.00	0.00
1997	Bunge-Ceval	0.00	0.00	5.00	0.00	0.00	0.00	5.00	0.00
2006	CAPSA	50.00	0.00	0.00	20.00	50.00	0.00	0.00	20.00
1995	CEPA	3.00	0.00	0.00	1.20	3.00	0.00	0.00	1.20
1998	F.V. S.A.	1.50	0.00	4.00	0.00	1.50	0.00	4.00	0.00
	Grupo Galicia	0.00	3.06	0.00	0.00	0.00	3.06	0.00	0.00
1998	Hospital Privado	8.40	0.00	0.00	0.00	8.40	0.00	0.00	0.00
1992	Huantraico	0.00	27.00	0.00	0.00	0.00	0.00	0.00	0.00
2004	Jumbo Argentina	0.00	39.12	0.00	0.00	0.00	39.12	0.00	0.00
	LD Manufacturing	0.00	0.00	5.00	0.00	0.00	0.00	5.00	0.00
	Milkaut	0.00	1.23	0.00	0.00	0.00	0.00	0.00	0.00
1997	Milkaut	5.33	0.00	9.44	1.44	5.33	0.00	9.44	1.44
1993	Molinos	0.00	2.00	0.00	0.00	0.00	2.00	0.00	0.00
1994	Molinos	0.00	0.57	0.00	0.00	0.00	0.57	0.00	0.00
1996	Neuquen Basin	0.00	26.40	0.00	0.00	0.00	0.00	0.00	0.00
1999	Neuquen Basin	0.00	5.00	0.00	0.00	0.00	0.00	0.00	0.00
2006	Noble Argentina	18.00	0.00	0.00	18.00	15.00	0.00	0.00	15.00
2005	PAE - Argentine	105.50	0.00	15.00	135.00	103.53	0.00	15.00	135.00
1998	Patagonia	1.76	0.00	1.00	0.00	1.76	0.00	1.00	0.00
1998	Patagonia Fund	0.00	8.54	0.00	0.00	0.00	1.65	0.00	0.00
1999	S.A. San Miguel	2.76	0.00	0.00	0.00	2.76	0.00	0.00	0.00
2005	S.A. San Miguel	20.62	0.00	0.00	10.00	17.29	0.00	0.00	8.33
1995	SanCor	8.70	0.00	19.89	0.00	8.70	0.00	19.89	0.00
	Socma	7.00	0.00	0.00	0.00	7.00	0.00	0.00	0.00
1995	Socma	0.94	0.00	0.00	15.00	0.94	0.00	0.00	15.00
1997	T6I	3.33	0.00	5.00	3.75	3.33	0.00	5.00	3.75
1997	Terminal 6	3.33	0.00	0.00	1.63	3.33	0.00	0.00	1.63
1995	Terminales Port.	0.50	0.00	0.00	0.00	0.50	0.00	0.00	0.00
2000	Tower Fund	0.00	0.85	0.00	0.00	0.00	0.00	0.00	0.00
1995	Tower Fund Mgr	0.00	0.05	0.00	0.00	0.00	0.05	0.00	0.00
1996	Transconor	4.20	0.00	0.00	157.58	4.20	0.00	0.00	157.58
2001	USAL	9.27	0.00	0.00	0.00	7.27	0.00	0.00	0.00
2005	Vicentin	20.00	0.00	15.00	0.00	0.00	0.00	15.00	0.00
1993	Yacylec	0.00	2.52	0.00	0.00	0.00	2.52	0.00	0.00
Total portfolio:		549.97	122.59	99.33	648.36	447.92	55.22	99.33	643.69

FY Approval	Company	Approvals Pending Commitment			
		Loan	Equity	Quasi	Partic.
2001	ITBA	0.01	0.00	0.00	0.00
2001	Gasnor	0.02	0.00	0.00	0.02
2006	Arcor Swap	0.00	0.00	0.00	0.00
2004	Banco Rio TFF	0.02	0.00	0.00	0.05
2005	Vicentin Exp.	0.00	0.00	0.00	0.05
Total pending commitment:		0.05	0.00	0.00	0.12

Annex 14: Country at a Glance

ARGENTINA: Matanza-Riachuelo Basin Sustainable Development Project

Argentina at a glance

3/27/09

Key Development Indicators

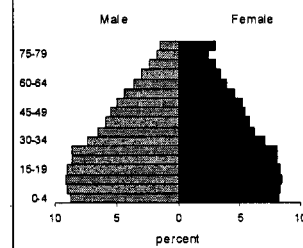
(2007)

	Argentina	Latin America & Carib.	Upper middle income
Population, mid-year (millions)	39.5	563	823
Surface area (thousand sq. km)	2,780	20,421	41,497
Population growth (%)	0.9	1.2	0.6
Urban population (% of total population)	92	78	75
GNI (Atlas method, US\$ billions)	238.9	3,118	5,750
GNI per capita (Atlas method, US\$)	6,050	5,540	6,987
GNI per capita (PPP, international \$)	12,990	9,320	11,888
GDP growth (%)	8.7	5.7	5.8
GDP per capita growth (%)	7.6	4.5	5.1

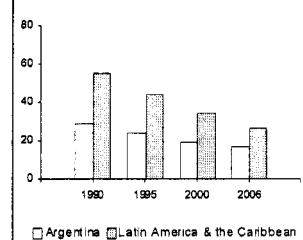
(most recent estimate, 2000–2007)

Poverty headcount ratio at \$1.25 a day (PPP, %)	..	8	..
Poverty headcount ratio at \$2.00 a day (PPP, %)	..	18	..
Life expectancy at birth (years)	75	73	70
Infant mortality (per 1,000 live births)	14	22	22
Child malnutrition (% of children under 5)	2	5	..
Adult literacy, male (% of ages 15 and older)	97	91	94
Adult literacy, female (% of ages 15 and older)	97	89	92
Gross primary enrollment, male (% of age group)	113	120	112
Gross primary enrollment, female (% of age group)	112	116	109
Access to an improved water source (% of population)	96	91	95
Access to improved sanitation facilities (% of population)	91	78	83

Age distribution, 2007



Under-5 mortality rate (per 1,000)



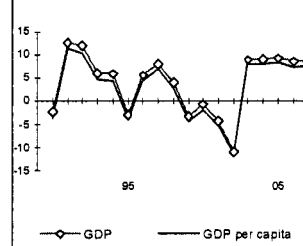
Net Aid Flows

	1980	1990	2000	2007 *
(US\$ millions)				
Net ODA and official aid	19	169	53	114
Top 3 donors (in 2006):				
Italy	1	82	-12	27
European Commission	0	2	10	25
France	1	6	8	16
Aid (% of GNI)	0.0	0.1	0.0	0.1
Aid per capita (US\$)	1	5	1	3

Long-Term Economic Trends

Consumer prices (annual % change)	100.8	2314.0	-0.7	8.6
GDP implicit deflator (annual % change)	90.8	2,076.8	1.0	14.1
Exchange rate (annual average, local per US\$)	0.0	0.5	1.0	3.1
Terms of trade index (2000 = 100)	..	85	100	114

Growth of GDP and GDP per capita (%)



Population, mid-year (millions)	28.1	32.6	36.9	39.5
GDP (US\$ millions)	76,962	141,352	284,204	262,331
	(% of GDP)			
Agriculture	6.4	8.1	5.0	8.4
Industry	41.2	36.0	27.6	35.6
Manufacturing	29.5	26.8	17.5	22.3
Services	52.4	55.9	67.4	56.0
Household final consumption expenditure	63.0	77.1	70.7	58.6
General gov't final consumption expenditure	11.0	3.1	13.8	12.4
Gross capital formation	25.3	14.0	16.2	23.5
Exports of goods and services	5.1	10.4	10.9	24.7
Imports of goods and services	6.5	4.6	11.5	19.2
Gross savings	22.9	13.4	13.0	26.8

1980–90 1990–2000 2000–07
(average annual growth %)

Population	1.5	1.2	1.0
GDP	-0.7	4.3	4.7
Agriculture	0.7	3.5	3.0
Industry	-1.3	3.8	5.3
Manufacturing	-0.8	2.7	5.0
Services	0.0	4.5	2.4
Household final consumption expenditure	..	2.7	2.2
General gov't final consumption expenditure	..	2.2	1.3
Gross capital formation	-5.2	7.4	9.2
Exports of goods and services	3.8	8.7	7.0
Imports of goods and services	-5.8	15.6	5.0

Note: Figures in italics are for years other than those specified. 2007 data are preliminary. .. indicates data are not available.

a. Aid data are for 2006.

Development Economics, Development Data Group (DECDG).

Balance of Payments and Trade

(US\$ millions)

	2000	2007
Total merchandise exports (fob)	28,341	52,238
Total merchandise imports (cif)	25,281	43,700
Net trade in goods and services	-1,832	12,937

Workers' remittances and compensation of employees (receipts)	86	602
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Current account balance as a % of GDP	-8.981	6.246
	-3.2	2.4

Reserves, including gold	25,147	14,153
--------------------------	--------	--------

Central Government Finance

(% of GDP)

Current revenue (including grants)	19.4	24.2
Tax revenue	11.3	17.3
Current expenditure	20.9	19.9

Overall surplus/deficit	-2.4	1.6
-------------------------	------	-----

Highest marginal tax rate (%)		
Individual	35	35
Corporate	35	35

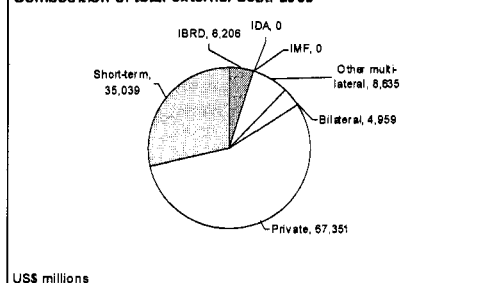
External Debt and Resource Flows

(US\$ millions)

Total debt outstanding and disbursed	144,126	122,190
Total debt service	26,966	18,994
Debt relief (HIPC, MDRI)	-	-

Total debt (% of GDP)	50.7	57.0
Total debt service (% of exports)	69.8	32.0

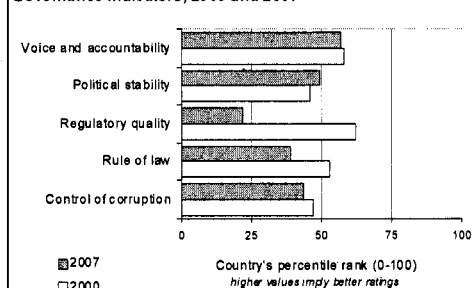
Foreign direct investment (net inflows)	10,418	4,840
Portfolio equity (net inflows)	-3,227	662

Composition of total external debt, 2006**Private Sector Development**

	2000	2008
Time required to start a business (days)	-	32
Cost to start a business (% of GNI per capita)	-	9.0
Time required to register property (days)	-	51

	2000	2007
Ranked as a major constraint to business (% of managers surveyed who agreed)		
Economic and regulatory policy uncertainty	..	16.5
Access to cost of financing	..	15.7

Stock market capitalization (% of GDP)	58.4	33.0
Bank capital to asset ratio (%)	..	13.7

Governance indicators, 2000 and 2007

Source: Kaufmann-Kraay-Mastruzzi, World Bank

Technology and Infrastructure

	2000	2007
Paved roads (% of total)	29.4	30.0
Fixed line and mobile phone subscribers (per 1,000 people)	39	126
High technology exports (% of manufactured exports)	9.1	6.8

Environment

Agricultural land (% of land area)	47	47
Forest area (% of land area)	12.3	12.1
Nationally protected areas (% of land area)	..	6.4
Freshwater resources per capita (cu. meters)	..	7,123
Freshwater withdrawal (% of internal resources)	10.6	..
CO2 emissions per capita (mt)	3.7	3.7
GDP per unit of energy use (2005 PPP \$ per kg of oil equivalent)	6.1	6.6
Energy use per capita (kg of oil equivalent)	1,679	1,644

World Bank Group portfolio

(US\$ millions)

	2000	2007
IBRD		
Total debt outstanding and disbursed	8,789	5,674
Disbursements	1,019	505
Principal repayments	538	1,037
Interest payments	701	350

IDA		
Total debt outstanding and disbursed	0	0
Disbursements	0	0
Total debt service	0	0

IFC (fiscal year)		
Total disbursed and outstanding portfolio of which IFC own account	2,305	1,078
Disbursements for IFC own account	1,196	628
Portfolio sales, prepayments and repayments for IFC own account	252	93
	74	138

MIGA		
Gross exposure	491	38
New guarantees	127	0

Note: Figures in *italics* are for years other than those specified. 2007 data are preliminary.

3/27/09

.. indicates data are not available. - indicates observation is not applicable.

Development Economics, Development Data Group (DECDG).

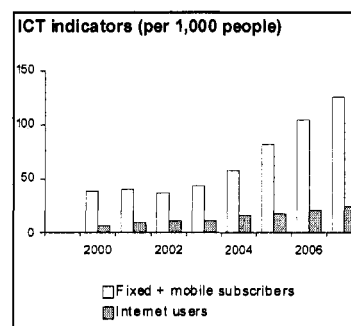
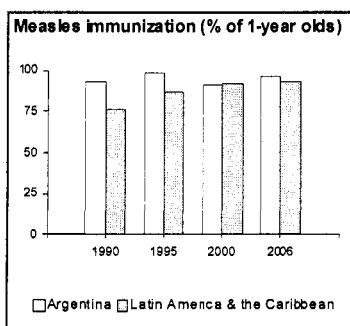
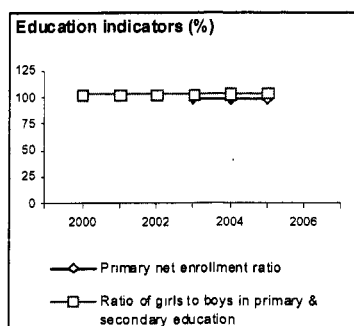
Millennium Development Goals

Argentina

With selected targets to achieve between 1990 and 2015
(estimate closest to date shown, +/- 2 years)

Argentina

Goal 1: halve the rates for extreme poverty and malnutrition	1990	1995	2000	2007
Poverty headcount ratio at \$1.25 a day (PPP, % of population)	..	..	..	..
Poverty headcount ratio at national poverty line (% of population)	..	..	..	..
Share of income or consumption to the poorest quintile (%)	4.6	3.9	3.2	3.1
Prevalence of malnutrition (% of children under 5)	..	..	..	2.3
Goal 2: ensure that children are able to complete primary schooling				
Primary school enrolment (net, %)	94	..	99	99
Primary completion rate (% of relevant age group)	..	98	99	97
Secondary school enrolment (gross, %)	72	..	97	84
Youth literacy rate (% of people ages 15-24)	98	..	99	..
Goal 3: eliminate gender disparity in education and empower women				
Ratio of girls to boys in primary and secondary education (%)	..	..	103	104
Women employed in the nonagricultural sector (% of nonagricultural employment)	37	40	43	45
Proportion of seats held by women in national parliament (%)	6	25	28	35
Goal 4: reduce under-5 mortality by two-thirds				
Under-5 mortality rate (per 1,000)	29	23	19	16
Infant mortality rate (per 1,000 live births)	25	20	17	14
Measles immunization (proportion of one-year olds immunized, %)	93	99	91	97
Goal 5: reduce maternal mortality by three-fourths				
Maternal mortality ratio (modeled estimate, per 100,000 live births)	..	..	..	77
Births attended by skilled health staff (% of total)	96	..	98	99
Contraceptive prevalence (% of women ages 15-49)	..	..	..	..
Goal 6: halt and begin to reverse the spread of HIV/AIDS and other major diseases				
Prevalence of HIV (% of population ages 15-49)	..	..	0.5	0.5
Incidence of tuberculosis (per 100,000 people)	73	60	49	39
Tuberculosis cases detected under DOTS (%)	..	4	31	71
Goal 7: have the proportion of people without sustainable access to basic needs				
Access to an improved water source (% of population)	94	95	96	96
Access to improved sanitation facilities (% of population)	81	85	89	91
Forest area (% of total land area)	12.9	..	12.3	12.1
Nationally protected areas (% of total land area)	..	..	..	6.4
CO2 emissions (metric tons per capita)	3.4	3.4	3.7	3.7
GDP per unit of energy use (constant 2005 PPP \$ per kg of oil equivalent)	5.3	6.1	6.1	6.6
Goal 8: develop a global partnership for development				
Telephone mainlines (per 100 people)	9.3	16.4	21.4	24.0
Mobile phone subscribers (per 100 people)	0.0	1.2	17.6	102.3
Internet users (per 100 people)	0.0	0.1	7.0	23.6
Personal computers (per 100 people)	0.7	3.6	6.9	9.0



Note: Figures in *italics* are for years other than those specified. .. indicates data are not available.

3/27/08

Development Economics, Development Data Group (DECDG).

MAP SECTION

ARGENTINA
MATANZA – RIACHUELO BASIN
SUSTAINABLE DEVELOPMENT PROJECT

- PROJECT COMPONENTS:
- RIACHUELO PRETREATMENT PLANT
 - LEFT AND RIGHT MARGIN COLLECTORS
 - - - RIACHUELO OUTFALL
 - · · · · OUTFALL SECTION WITH DIFFUSER INSTALLATIONS
 - - - MUNICIPALITY BOUNDARIES
 - URBAN AREAS

